The Pennsylvania State University

The Graduate School

College of Engineering

THE EXPERIMENTAL INVESTIGATION OF A ROTOR ICING MODEL WITH SHEDDING

A Thesis in

Aerospace Engineering

by

Edward W. Brouwers

© 2010 Edward W. Brouwers

Submitted in Partial Fulfillment

of the Requirements

for the Degree of

Master of Science

May 2010

The thesis of Edward W. Brouwers was reviewed and approved* by the following:

Jose L. Palacios

Post Doctoral Research Associate

Edward C. Smith

Professor of Aerospace Engineering

Thesis Adviser

Cengiz Camci

Professor of Aerospace Engineering

George A. Lesieutre

Professor of Aerospace Engineering

Head of the Department of Aerospace Engineering

*Signatures are on file in the Graduate School.

ABSTRACT

A critical operational problem for rotorcraft is flight in adverse weather conditions. Flight in icing conditions is fraught with operational hazards, including reduced vehicle performance and degraded handling qualities. Additionally, shedding of ice from blades due to centrifugal force poses a ballistics danger to the aircraft and creates large vibrations due to imbalanced rotors. Modeling the effects of accreted ice on rotorcraft flight performance has been a challenge due to the complexities of periodically changing conditions as well as spanwise variations of angle of attack, velocities and surface temperatures. Model validation has been complicated by the lack of available data due to the existence of only a few facilities designed to study the rotorcraft aspect of the icing problem.

As part of the development of the Adverse Environment Rotor Test Stand (AERTS), a new icing model was developed to predict ice shapes on a hovering rotor. The AERTS Rotor Icing, Shedding and Performance (ARISP) model has the goal of exploring rotor icing trends. The core feature of the model is the coupling with NASA's LEWis ICE accretion code (LEWICE). Rotor performance is predicted with a blade element momentum theory module, while increases in rotor torque due to ice accretions are empirically calculated. A shedding module predicts the shedding time and blade station based upon the accreted ice properties. The ARISP model was correlated with published ice shapes for both small-scale and full-scale rotor ice accretion results.

Further correlations were made with experiments in the newly developed Adverse Environment Rotor Test Stand (AERTS). These experiments constitute the first ice shape comparisons in the test stand. A total of 44 test cases were completed, of which the first 23 were designed to explore the limitations of the AERTS facility. The remaining 21 cases formed the test matrix for the ARISP model validation and were designed to investigate the influence of various icing parameters. Favorable comparisons have been made between ice shape, especially at inboard stations. Primary ice shape features were predicted well for over 81% of the experimental ice tracings. The largest discrepancies occurred at test cases that did not lie within the FAR Part 25/29 Appendix C icing envelope. Impingement limits were modeled well, with the discrepancies between the limit calculations and experimental values generally lower than 20%. Torque rise predications were generally within 20% of experimental values. Shedding behavior was also evaluated, but correction factors were required to improve test data correlation due to the general overprediction of tip ice accretions. These correction factors were derived from the difference in ice shape between predictions and experiments and reduced prediction error in shedding time to 25% and shedding station to 8%. Further investigations are required, but the icing model has laid the foundation for future research in the AERTS Facility.

TABLE OF CONTENTS

LIST OF FIGURES VIII

LIST OF TABLES XII

ACKNOWLEDGEMENTS XIII

1. INTRODUCTION 1

1.1 Background and Motivation 1
1.1.1 Icing Overview 1
1.1.2 Rotorcraft Challenges 2
1.1.3 Icing Conditions 5
1.2 Review of Related Work 14
1.2.1 Ice Accretion Facilities 14
1.2.2 Ice Accretion Prediction Codes 19
1.3 Objectives 23
1.4 Thesis Overview 24
1.4.1 Chapter 2: Ice Accretion and Shedding Model 24
1.4.2 Chapter 3: AERTS Facility Calibration 25
1.4.3 Chapter 4: Ice Adhesion Strength Calculation Fixture 25
1.4.4 Chapter 5: Experimental Exploration of ARISP Model in AERTS Facility 25
1.4.5 Chapter 6: Conclusions and Recommendations 25
1.4.6 Appendices 26

2. ICE ACCRETION MODELING – ARISP CODE 27

2.1 Objectives 27
2.2 ARISP Code Overview 28
2.2.1 Software 29
2.3 Analysis Concept and Implementation 31
2.3.1 Hover Performance – BEMT 32
2.3.2 LEWICE Integration 39
2.3.3 Ice Shedding 45
2.3.4 Iced Rotor Performance Degradation 51
2.3.5 Initial ARISP Model Validation 55

3. AERTS FACILITY DESIGN, CONSTRUCTION AND CALIBRATION 63

3.1 Requirements 63
3.2 CFD Modeling of the AERTS Chamber 64
3.2.1 Objectives 64
3.2.2 Model Setup 64
3.2.3 Model Results 66
3.2.4 Modeling Conclusions 74
3.3 Icing Cloud Calibration 75
3.3.1 Temperature 75
3.3.2 Droplet Size 76
3.3.3 Liquid Water Content (LWC) Calibration 78
3.3.4 Laser Cloud Density Instrument 83
3.4 Icing System Shakedown Tests 86
3.4.1 Test Rationale and Matrix 86

3.4.2 Data Recording *89*
3.4.3 Ice Crystals *89*
3.4.4 Test Chamber Sensitivity Studies *95*
3.4.5 Shakedown Test Conclusions *102*

4. ICE SHEAR ADHESION STRENGTH CALCULATION 104

4.1 BACKGROUND 104
4.2 BENDING BEAM ROTOR DESIGN 106
4.2.1 Overview *106*
4.2.2 Design Requirements *107*
4.2.3 Design Concept *107*
4.2.4 Design *110*
4.3 INSTRUMENTATION 121
4.3.1 Configuration *121*
4.3.2 Amplifier *122*
4.4 BEAM CALIBRATION 124
4.4.1 Rotating Sensitivity Study *124*
4.4.2 Beam Logic *125*
4.5 ADHESION STRENGTH CALCULATION TEST RESULTS 127
4.5.1 Test Coupon *127*
4.5.2 Test Matrix *128*
4.5.3 Test Results *128*

5. EXPERIMENTAL INVESTIGATION OF ARISP CODE 132

5.1 TESTING OVERVIEW 132
5.1.1 Objective *132*
5.1.2 Test Rotor *132*
5.1.3 Test Matrix *133*
5.2 RECORDED DATA 138
5.3 EXPERIMENTAL THICKNESS TRENDS 140
5.3.1 RPM Sensitivity *141*
5.3.2 Temperature Sensitivity *142*
5.3.3 LWC Sensitivity *143*
5.3.4 MVD Sweep *143*
5.4 LIQUID WATER CONTENT CALCULATIONS 144
5.5 ICE SHAPE CORRELATIONS 145
5.6 IMPINGEMENT LIMIT CORRELATIONS 149
5.7 PERFORMANCE DEGRADATION CORRELATIONS 151
5.8 SHEDDING CORRELATIONS 158
5.8.1 General Test Results *158*
5.8.2 Experimental Correlations *162*

6. SUMMARY, CONCLUSIONS AND RECOMMENDATIONS FOR FUTURE WORK 167

6.1 SUMMARY 167
6.1.1 BACKGROUND AND MOTIVATION *167*
6.1.2 ARISP MODEL *170*
6.1.3 AERTS FACILITY *171*
6.1.4 ICE SHEAR ADHESION TEST FIXTURE *172*
6.1.5 EXPERIMENTAL OF THE ARISP MODEL IN THE AERTS FACILITY *172*

6.2 CONCLUSIONS 174
6.2.1 ARISP MODEL 174
6.2.2 AERTS FACILITY *174*
6.2.3 ICE SHEAR ADHESION TEST ROTOR *176*
6.2.4 EXPERIMENTAL OF THE ARISP MODEL IN THE AERTS FACILITY *177*
6.3 FUTURE WORK 179
6.3.1 ICING MODEL *179*
6.3.2 AERTS LABORATORY *180*

7. WORKS CITED 184

A. APPENDIX A: AERTS FACILITY LIMITATIONS 191

B. APPENDIX B: AERTS FACILITY ICING SYSTEM 194

B.1. OVERVIEW 194
B.2. TEST CHAMBER 195
B.3. ICING SYSTEM 196
B.3.1. Icing Nozzles *196*
B.3.2. Air System *197*
B.3.3. Water System *200*
B.3.4. Manifold System *201*
B.3.5. Nozzle Freezing Problem *202*
B.3.6. Icing System Schematic *204*

C. APPENDIX C: AERTS DRIVE SYSTEM AND ROTOR HUB 205

C.1. ROTOR DRIVE COMPONENTS 205
C.1.1. Motor/Gearbox *205*
C.1.2. Auxiliaries *206*
C.2. INITIAL TESTING CONFIGURATION 207
C.2.1. Hub *207*
C.2.2. Stand *207*
C.3. NEW ROTORHEAD REQUIREMENTS 208
C.4. QH-50 ROTORHEAD SELECTION 209
C.5. DRIVETRAIN INTEGRATION 213
C.6. CONTROLS INTEGRATION 215
C.7. STRUCTURAL INTEGRATION 217
C.8. SLIP RING ASSEMBLY 218
C.9. ROTOR SENSORS 222
C.9.1. Six Axis Force and Torque Sensor *222*
C.9.2. Shaft Torque Sensor *225*
C.9.3. Stand Accelerometers *225*
C.9.4. Rotor RPM Sensor *226*

D. APPENDIX D: TEST ROTOR DESIGN 228

D.1. ARISP TEST ROTOR 228
D.1.1. Requirements *228*
D.1.2. Rotor Selection *228*
D.1.3. Blade Integration *229*
D.1.4. Blade Paint *237*
D.1.5. Tip Caps *238*

LIST OF FIGURES

FIGURE 1: RIME ICE ACCRETIONS ON AH-64 AADS 9
FIGURE 2: GLAZE ICE 9
FIGURE 3: GLAZE ICE IN AERTS CHAMBER 10
FIGURE 4: BEAK ICE 10
FIGURE 5: ICE TYPE LIMITS FOR LWC = 0.66 GR/M^3 11
FIGURE 6: TYPICAL BLADE ICE ACCRETION PROFILE 12
FIGURE 7: FAA PART 25/29 APPENDIX C ENVELOPE 13
FIGURE 8: COX ICING TUNNEL 16
FIGURE 9: UH-60 MODEL IN NASA GLENN (FORMERLY NASA LEWIS) IRT 16
FIGURE 10: MODEL UH-60 ROTOR ICE ACCRETION 16
FIGURE 11: NRC ROTORCRAFT ICING TOWER, BOEING VERTOL CH-46D UNDERGOING TESTING 18
FIGURE 12: HISS ARTIFICIAL ICING TESTING OF CH-47D 19
FIGURE 13: OHISS ARTIFICIAL ICING TESTING OF S-92A 19
FIGURE 14: TYPICAL ROTOR ICING MODEL PROGRAM FLOW 22
FIGURE 15: ARISP CONCEPT 31
FIGURE 16: ARISP CODE FLOW 32
FIGURE 17: CLEAN NACA 0015 C_L 34
FIGURE 18: CLEAN NACA 0015 C_D 34
FIGURE 19: PERFORMANCE CORRECTIONS PER NASA TM 86754 36
FIGURE 20: BEMT VALIDATION - BLADE LOADING COEFFICIENT VS COLLECTIVE 37
FIGURE 21: BEMT VALIDATION - PERFORMANCE POLAR 38
FIGURE 22: BEMT VALIDATION - PERFORMANCE POLAR 39
FIGURE 23: LEWICE INTERFACE 41
FIGURE 24: STANDARD LEWICE INPUT FILE 42
FIGURE 25: SAMPLE GEOMETRY INPUT INTO LEWICE 43
FIGURE 26: BATCH FILE 44
FIGURE 27: INPUT FILES 45
FIGURE 28: APPLIED FORCES 46
FIGURE 29: ICE TENSILE FORCE DETAILS AND SHEAR ADHESION 47
FIGURE 30: ICE CROSS SECTIONAL AREA SUMMATION 48
FIGURE 31: SHEDDING ELEMENT IMPLEMENTATION 51
FIGURE 32: ICED AIRFOIL PERFORMANCE ADJUSTMENT IMPLEMENTATION 55
FIGURE 33: TIME STEP SENSITIVITY STUDY SAMPLE 57
FIGURE 34: ARISP COMPARISON TO NASA TM 103712 CASE 34 58
FIGURE 35: ARISP COMPARISON TO NASA TM 103712 CASE 97 58
FIGURE 36: ARISP COMPARISON TO UH-1H HIFT FLIGHT E 59
FIGURE 37: ICE EXTENT CODE VALIDATION, UH-1H ROTOR 61
FIGURE 38: ICE EXTENT CODE VALIDATION, AERTS TEST ROTOR 61
FIGURE 39: CFD MODEL SETUP 66
FIGURE 40: XY PLANE TOTAL VELOCITY VECTORS AND CONTOURS 67
FIGURE 41: XY PLANE TOTAL VELOCITY VECTORS AND CONTOURS 68
FIGURE 42: NOZZLE EFFECTS AT ROTOR PLANE 69
FIGURE 43: FLOW TUBES 70
FIGURE 44: STREAM TUBES FROM A SINGLE INNER NOZZLE 71
FIGURE 45: STREAM TUBES FROM A SINGLE OUTER NOZZLE 71
FIGURE 46: DROPLET DISTRIBUTION IN AERTS CHAMBER 73

FIGURE 47: DROPLET DISTRIBUTION IN AERTS CHAMBER 74
FIGURE 48: NASA STANDARD NOZZLE CALIBRATION TABLE 77
FIGURE 49: FREEZING FRACTION VARIATION 79
FIGURE 50: EXPERIMENTAL LWC CALCULATION OVERVIEW 81
FIGURE 51: CYLINDER LWC CALCULATIONS 82
FIGURE 52: NACA 0012 LWC CALCULATIONS 82
FIGURE 53: LASER CLOUD DENSITY MEASUREMENT CONCEPT 84
FIGURE 54: POC LASER LOCATION IN AERTS 84
FIGURE 55: PROOF OF CONCEPT LASER SAMPLE TEST RESULTS 85
FIGURE 56: IMPROVED LASER UNDERGOING SHAKEDOWN TESTS 86
FIGURE 57: ICE TRACING PROCESS 89
FIGURE 58: CASE 1 ACCRETED ICE, NACA0012 ROTOR 90
FIGURE 59: CASE 1 ACCRETED ICE, NACA0012 ROTOR 90
FIGURE 60: CASE 11 ACCRETED ICE, NACA0015 ROTOR 90
FIGURE 61: CASE 11 THICKNESS COMPARISON 91
FIGURE 62: CASE 11 SHAPE COMPARISON 91
FIGURE 63: ICE CRYSTALS ON BLADE UPPER SURFACE 92
FIGURE 64: IRT ICE SHAPE EROSION 93
FIGURE 65: ANTI-CRYSTAL SCREEN LOCATION 93
FIGURE 66: ICE ACCRETIONS ON ANTI-CRYSTAL SCREEN 93
FIGURE 67: REDUCED ICE ACCRETION ON SCREENS 94
FIGURE 68: REFINED AERTS NOZZLE OPERATIONAL ENVELOPE 95
FIGURE 69: CASE 8 ICE ACCRETION, NACA0012 ROTOR 95
FIGURE 70: RPM SENSITIVITY 96
FIGURE 71: COLLECTIVE PITCH SENSITIVITY 97
FIGURE 72: TEMPERATURE SENSITIVITY 98
FIGURE 73: DROPLET SIZE SENSITIVITY 99
FIGURE 74: CASE 5 AND CASE 6 THICKNESS COMPARISON 100
FIGURE 75: CASE 2 SHEDDING EVENT 101
FIGURE 76: CASE 2 ICE TRACING 102
FIGURE 77: LR-350 TEST FIXTURE 108
FIGURE 78: SHEAR ADHESION STRENGTH FIXTURE THEORY 109
FIGURE 79: SHEAR ADHESION OPTIMIZATION 111
FIGURE 80: TEMPERATURE EFFECTS 111
FIGURE 81: SCHEMATIC OF ICE SHEDDING ROTOR BEAM WITH FAIRINGS 113
FIGURE 82: SINGLE BLADE ASSEMBLY WITH TEST FIXTURE MOUNTED AT TIP 113
FIGURE 83: TEST ROTOR IMPLEMENTATION, VIEW FROM HUB 113
FIGURE 84: FEM STRAIN FIELD ON THE TI BEAM (1200 RPM) 114
FIGURE 85: BEAM ASSEMBLY INTERNAL STRESSES (1200 RPM) 115
FIGURE 86: BEAM ASSEMBLY FOS (1200 RPM) 115
FIGURE 87: FAIRING DESIGN 116
FIGURE 88: FAIRING ICE PROTECTION HEATER MAT LOCATIONS (SHOWN IN RED) 117
FIGURE 89: SHEDDING ROTOR ANTI-ICE HEATERS 118
FIGURE 90: NOTCH ICE ACCRETIONS 119
FIGURE 91: EXTERNAL NOTCH HEATER 119
FIGURE 92: EIPSS DESIGN 120
FIGURE 93: EIPSS IMPLEMENTATION 120
FIGURE 94: ICE BRIDGING 121
FIGURE 95: SIDE FAIRINGS 121

FIGURE 96: FULL BRIDGE STRAIN GAUGE CONFIGURATION SCHEMATIC....121
FIGURE 97: STRAIN GAUGE LOCATIONS....122
FIGURE 98: STRAIN GAUGE AMPLIFIER SCHEMATIC (COURTESY RAETECH)....123
FIGURE 99: BENDING BEAM RESPONSE....124
FIGURE 100: COUPON SENSITIVITY STUDY LOADS....125
FIGURE 101: BEAM LOADING CONCEPT....126
FIGURE 102: TYPICAL TEST STRAIN GAUGE TRACE....129
FIGURE 103: AERTS SHEAR ADHESION STRENGTH RESULTS....130
FIGURE 104: AERTS SHEAR ADHESION STRENGTH RESULTS COMPARISON....131
FIGURE 105: AERTS CONFIGURATION FOR ARISP VALIDATION....133
FIGURE 106: ARISP VALIDATION TEST POINTS, TEMPERATURE VS LWC....135
FIGURE 107: ARISP VALIDATION TEST POINTS, MVD VS DROPLET SIZE....135
FIGURE 108: 15 μM DROPLET SIZE CASES....137
FIGURE 109: 20 μM DROPLET SIZE CASES....137
FIGURE 110: 25 μM DROPLET SIZE CASES....137
FIGURE 111: 35 μM DROPLET SIZE CASES....137
FIGURE 112: CASE 23 THICKNESS PROFILE....138
FIGURE 113: CASE 23 ICE ACCRETION....139
FIGURE 114: CASE 23 TIP ICE SHAPE....139
FIGURE 115: RPM SENSITIVITY....141
FIGURE 116: TEMPERATURE SENSITIVITY....142
FIGURE 117: LWC SENSITIVITY....143
FIGURE 118: MVD SENSITIVITY....144
FIGURE 119: SPANWISE LWC DISTRIBUTION....144
FIGURE 120: CASE 44 ICE TRACINGS....146
FIGURE 121: RED - CASE 29, 0.9R....148
FIGURE 122: YELLOW - CASE 24 0.9R....148
FIGURE 123: GREEN - CASE 25, 0.8R....148
FIGURE 124: SHAPE CORRELATION SUMMARY....149
FIGURE 125: CASE 43 GLAZE FEATHERS....150
FIGURE 126: IMPINGEMENT LIMIT CORRELATION RESULTS....151
FIGURE 127: TORQUE REQUIREMENTS....152
FIGURE 128: CASE 31 TORQUE INCREASE....153
FIGURE 129: CASE 31 C_D DISTRIBUTION....154
FIGURE 130: TORQUE CORRELATIONS....155
FIGURE 131: TORQUE CORRELATIONS....156
FIGURE 132: TORQUE CORRELATIONS....156
FIGURE 133: TORQUE CORRELATIONS....157
FIGURE 134: CASE 34 ICE ACCRETION, SHOWING SHEDDING....158
FIGURE 135: CASE 34 THICKNESS PROFILE, SHOWING SHEDDING LOCATION....159
FIGURE 136: CASE 39 SHEDDING, VIEW FROM HUB....159
FIGURE 137: SHEDDING DAMAGE....160
FIGURE 138: SHEDDING TIME....161
FIGURE 139: SHEDDING STATION CORRELATIONS....161
FIGURE 140: SHEDDING CORRECTION FACTOR....163
FIGURE 141: CASE 31 ICE THICKNESS, NO CORRECTIONS....164
FIGURE 142: CASE 31 ICE THICKNESS, CORRECTIONS....164
FIGURE 143: AERTS FACILITY OVERVIEW....194
FIGURE 144: CLOSE-UP OF EVAPORATOR FANS, SHOWING LARGE ICE ACCRETIONS....195

FIGURE 145: NASA STANDARD ICE NOZZLE 196
FIGURE 146: AERTS WATER AND AIR SYSTEM LAYOUT 204
FIGURE 147: AERTS DRIVE SYSTEM COMPONENTS 205
FIGURE 148: AERTS MOTOR CAPABILITIES 206
FIGURE 149: ROTOR STAND COMPONENTS (NEW SLIP RING MOUNTS NOT SHOWN) 208
FIGURE 150: QH-50D DASH 211
FIGURE 151: QH-50 ROTORHEAD ASSEMBLY AS RECEIVED FROM GYRODYNE 212
FIGURE 152: LOWER DRIVELINE INITIAL COMPONENT ARRANGEMENT 213
FIGURE 153: LOWER DRIVELINE COMPONENTS 214
FIGURE 154: REVISED LOWER DRIVELINE ASSEMBLY 214
FIGURE 155: ACTUATOR IMPLEMENTATION MODELS 217
FIGURE 156: WATERPROOF FAIRING 218
FIGURE 157: SLIP RINGS 220
FIGURE 158: SLIP RING CONFIGURATION 221
FIGURE 159: SLIP RING INSTALLATION IN AERTS CHAMBER 222
FIGURE 160: PREVIOUS BOEING RTS FORCE BALANCE 223
FIGURE 161: FLAPPING RESPONSE OF SCHWEIZER BLADES 231
FIGURE 162: SAMPLE FEA ANALYSIS OF MODIFIED SCHWEIZER ROTOR 232
FIGURE 163: FEA ANALYSIS OF MODIFIED SCHWEIZER ROTOR RESULTS 232
FIGURE 164: BLADE CROSS SECTIONAL PROPERTIES 233
FIGURE 165: SCHWEIZER BLADE MOUNTING PLATES 234
FIGURE 166: FEM MESH 235
FIGURE 167: FEM MESH, ROOT CLOSEUP 236
FIGURE 168: STRESS DISTRIBUTION IN SCHWEIZER 269 BLADE ASSEMBLY 236
FIGURE 169: FOS DISTRIBUTION IN SCHWEIZER 269 BLADE ASSEMBLY 237
FIGURE 170: BLADE STRIPES 238
FIGURE 171: ROTOR TIP TAPE COVER 238

LIST OF TABLES

TABLE 1: ICING SEVERITY DESCRIPTIONS....7
TABLE 2: CHARACTERISTICS OF DIFFERENT TYPES OF ICE CAUSED BY ATMOSPHERIC ICE ACCRETION....12
TABLE 3: FAA PART 29 ICING ENVELOPE SUMMARY (28)....13
TABLE 4: PARTIAL LIST OF ICING TUNNELS....15
TABLE 5: SAMPLE ROTOR ICE PREDICTION PROGRAMS....23
TABLE 6: PROUTY TAIL ROTOR PARAMETERS....37
TABLE 7: ICE DENSITY CALCULATIONS....49
TABLE 8: RANGE OF PARAMETERS USED IN CORRELATIONS DEVELOPMENT, FROM REFERENCE (51)....52
TABLE 9: PRIMARY AERTS FACILITY ICING SYSTEM REQUIREMENTS....63
TABLE 10: NOZZLE VELOCITIES....65
TABLE 11: TYPICAL TEMPERATURE RISE DURING TESTING (WEATHER STATION READINGS)....76
TABLE 12: ICING SHAKEDOWN TESTS....88
TABLE 13: SHAKEDOWN TEST RESULTS OVERVIEW....103
TABLE 14: SHEAR ADHESION STRENGTH....104
TABLE 15: BENDING BEAM EXPECTED STRAINS....115
TABLE 16: RATECH 1169-01-50-100 AMPLIFIER SPECIFICATIONS....123
TABLE 17: ICE ADHESION TEST MATRIX....128
TABLE 18: ARISP VALIDATION TEST MATRIX....134
TABLE 19: ARISP VALIDATION TEST MATRIX SUMMARY....134
TABLE 20: SHAPE CORRELATION RESULTS....148
TABLE 21: SHEDDING RESULTS....165
TABLE 22: SHEDDING PREDICTION ERRORS....166
TABLE 23: AERTS ROTOR SYSTEM LIMITATIONS....191
TABLE 24: AERTS ICING SYSTEM LIMITATIONS....192
TABLE 25: AERTS INSTRUMENTATION LIMITATIONS....193
TABLE 26: MANIFOLD POSITIONS....202
TABLE 27: ROTORHEAD REQUIREMENTS....209
TABLE 28: QH-50 ROTORHEAD PARAMETERS....211
TABLE 29: ACTUATOR REQUIREMENTS MATRIX....216
TABLE 30: PRIMARY SLIP RING SPECIFICATIONS....219
TABLE 31: FORCE BALANCE SENSING RANGE AND RESOLUTION....224
TABLE 32: VALIDATION ROTOR REQUIREMENTS....228
TABLE 33: SCHWEIZER MAIN ROTOR SPECIFICATIONS....229

ACKNOWLEDGEMENTS

I would like to acknowledge the guidance and support of Dr. Jose Palacios. His tremendous commitment to the development of the new AERTS icing facility and countless hours of troubleshooting the various highly customized systems in the laboratory made this thesis possible. By leading the transformation of a barely functioning motor and freezer into a highly capable unique icing facility, he has shown me that anything is possible with the proper combination of dedication and ingenuity.

The contributions of Andrew Peterson, the author's technical mentor at Boeing Philadelphia, are gratefully acknowledged. His five decades of experience with rotors in adverse environments has guided much of the icing model and facility development. He was an invaluable resource for verifying the accuracy of initial ice shapes from the new laboratory.

Thanks also goes to Dr. Edward Smith, who founded the icing laboratory at Penn State and secured the funding to build and maintain the laboratory. I am appreciative of all the help other members of the AERTS laboratory, including Yiqiang Han, Austin Overmeyer, Buck Poerio, Michael Jostes and Zachary Weyent, have provided while constructing the lab and running countless tests.

The donations of various companies have made the operation of the icing facility possible. Thanks goes to Boeing for much needed motor repair funds. Eric Kreeger of the NASA Glenn Icing Branch coordinated the donation of the critical icing nozzles and provided helpful calibration advice. Peter Papadakos, president of the Gyrodyne Historical Foundation, supplied the QH-50D DASH lower rotor hub and upper controls. Jason Given, a flight test engineer of Schweizer Aircraft Corporation, donated the test rotor blades. Much thanks goes to Ches Crognale, manufacturing engineer of Olympic Tool and Machine Corporation, for the donation of material and machining time for the torque cell mount and blade grips. In previous employment, he taught me all the drafting, and machining skills required to mate the QH-50 rotor head to the rotor stand.

I owe my deepest gratitude to my family for their steadfast support throughout my educational career. Special appreciation goes to my Dad for spending many hours teaching me the basics of engineering and design through many projects at home as a child. These experiences undoubtedly shaped my engineering interests.

Finally, thanks to my wife for allowing me to spend many hours in the lab and for proof reading all of my papers. Her support and encouragement was in the end what made this thesis possible.

1. Introduction

1.1 Background and Motivation

1.1.1 Icing Overview

> *"Strange as it may seem, a very light coating of snow or ice, light enough to be hardly visible, will have a tremendous effect on reducing the performance of a modern aeroplane."*(1)

Icing has long been a problem for aircraft. The dangers of icing did not greatly concern aviators in the first few decades of powered flight, as pilots flew by visual flight rules (VFR). The negative effects of icing were first observed in the 1920's, when expanding flight envelopes allowed longer range and higher altitude flights. Some of the first documented icing encounters occurred when aviators of the U.S. Air Mail Service attempted to maintain scheduled day-and-night operations between New York and Chicago (2). On these longer flights, aircraft could experience drastic changes in weather, as VFR flight could not be guaranteed. This first generation of instrument fliers noted that "the greatest of all our problems is ice." (3)

Ever since the first in-flight encounters with ice, engineers have sought ways to eliminate this hazard for aircraft. Some early attempts included coating critical surfaces with insoluble substances such as light and heavy lubricating oil, cup grease, Vaseline, paraffin, and simonize wax. Attempts with soluble compounds included glycerin, calcium chloride mixed with either glycerin and molasses, a hardened sugar solution, and a hardened glucose solution (2). Eventually, designers settled on heated air circulation, which was marginally effective due to the low power of engines in

the late 1920's. A major increase in safety was brought about with the advent of pneumatic boots in 1931. Since then, many other ice protection systems have been investigated and implemented on both civil and military aircraft, including electrothermal, fluid, electro-impulsive and electro-expulsive systems. These systems have improved flight safety.

Despite new technology, training and procedures developed in the past 60 years, accidents related to icing conditions continue to occur. Between 1990 and 2005, a total of 33,513 aircraft accidents and incidents in the United States were reported in the National Transportation Safety Board (NTSB) accident database (4). Of these, ice has been a factor in 1372 of the incidents (4). Carburetor icing was the largest problem, and was listed as a contributing factor in 571 of the events. Between the 1985 and 1999, a total of 255 icing accidents and incidents occurred with US Army aircraft (5). The highly publicized crash of an ATR-42 commuter aircraft in Roselawn, Indiana highlighted the fact that there continue to be shortcomings in the understanding of icing conditions. The accident generated significant research interest in freezing drizzle and freezing rain, collectively known as super large droplet (SLD) conditions.

1.1.2 Rotorcraft Challenges

Compared to fixed wing aircraft, rotorcraft have additional icing challenges. Ice accretion on rotors and its subsequent effect on helicopter performance cannot be analyzed in the straightforward manner used to explain ice accretion on the leading edges of a fixed wing aircraft (6). Rotor icing is made more complex by periodic changing of blade sectional angle of attack and velocity.

Furthermore, spanwise surface temperature differences and thinner airfoil thicknesses make all rotor blade icing analysis challenging.

The rotorcaft icing problem is compounded by the fact that helicopters are more susceptible to icing than a fixed wing aircraft of similar gross weight. Helicopter operations occur almost exclusively at low altitudes, where there is significant icing potential (7). At the typical cruise altitudes of helicopters, between 3,000 and 12,000 ft, the atmosphere contains supercooled water droplets (8). Furthermore, rotorcraft are typically time sensitive. The hovering capability of the vehicle allows it to participate in search and rescue, medical evacuation and critical supply missions, in which the urgency requires rotorcraft to fly in Instrument Meteorological Conditions (IMC).

1.1.2.1 *Rotor Performance in Icing Conditions*

The primary problem with flight into icing conditions is degraded rotor performance. The degree of performance reduction depends on the specific rotor characteristics, including airfoil profile, blade loading, Mach distribution, catch efficiency, blade flexibility, rotor/airframe reaction and control loads. Most rotorcraft rely exclusively on the main rotor for lift, thrust and control. These flight critical functions are dispersed on a fixed wing aircraft to the wings, engines and control surfaces. This provides a measure of icing tolerance such that all three functions are not adversely affected by local ice accretion. As with fixed wing aircraft, ice on the leading edge of the rotor blades induces a variety of negative effects. These affect the capability of the rotor system to lift, propel and control the vehicle. If any of these three flight critical functions are degraded, the aircraft is in serious danger of departing from controlled flight and potentially crashing. An overall loss of rotor

efficiency can be extremely detrimental to the performance of the vehicle since helicopters do not have the power margins typically afforded to jet transports.

The effects of a highly modified flowfield resulting from accreted ice include a slightly reduced sectional lift coefficient and modified pitching moment characteristics (9) (10) (11). The most significant effect of icing, however, is the increase in blade profile drag. Ice does not typically accrete in a uniform coating on the rotor; rather a rough and sometimes jagged structure is formed at the leading edge, causing premature flow separation. This, combined with its ensuing increase in collective pitch to maintain a flight condition due to decreased lift performance, makes the necessary torque required to increase with accreted ice. The torque increase can rapidly reach the transmission or engine limits, rendering flight at the current flight condition impossible (12). This trend has been observed both in flight and in experiments in icing wind tunnels (13) (14) (15).

A related major operational hazard of rotor icing is the deterioration of normal autorotational qualities. In the event of power loss, the minimum autorotational rpm may not be able to be maintained, for enough time to execute a safe landing. A study conducted by the US Army found that moderate ice accretion on inboard portions of a UH-1H Huey rotor caused a loss of 22 rpm during autorotation at 70 kts (6). Significant rpm degradation has the potential to increase the decent velocity to a point unsafe for landing (16).

1.1.2.2 *Ice Shedding*

The second major problem rotor icing is the due to the high rotational speed of the rotor. The system generates high centrifugal forces on accreted ice, especially near the tips. This provides a natural de-icing capability, which is theoretically beneficial for helicopters. The problem arises when the ice does not shed in a symmetric fashion from all blades. Uneven ice distribution causes rotor imbalance and subsequent severe vibrations in the fuselage. The transmission or engine torque limits can be exceeded due to the fluctuating drag, rendering flight at a given condition impossible for the pilot (12). Furthermore, researchers have identified problems with damage to the fuselage, engines and empennage from high velocity shed particles. Larger, multiengine vehicles, especially those of tandem rotor configuration, are more susceptible to this hazard (6).

1.1.3 Icing Conditions

1.1.3.1 *Ice Conditions*

In the atmosphere, water droplets condense on small soluble nuclei to form a cloud. The small size of the particles allows them to exist in the supercooled state down to -20°C and sometimes at temperatures as low as -35°C (12). These droplets, known as freezing drizzle, pose a danger to aircraft flying through the cloud, as they may accrete on vital surfaces such as wings, blades and control surfaces. The level of supercooling depends on a variety of factors, including water purity, droplet size and the presence of ice crystals. If ice crystals are present, they typically grow and cause the water droplets to evaporate. These crystals are not a particular concern for blades and wings, as they will typically bounce off of the surface without accreting. Ice crystals may pose a hazard to engines, as they may collect in an intake plenum (12).

Freezing rain conditions are different than those associated with freezing drizzle and can accrete significant amounts of ice on the aircraft. Freezing rain is formed when snow from glaciated clouds falls through a warm layer and melts (12). If the droplets fall through another sub freezing layer, they become supercooled. The droplets freeze when they come into contact with a surface.

1.1.3.2 *Icing Parameters*

In studies of aircraft icing, the primary atmospheric parameters are temperature, liquid water content (LWC) and droplet size distribution (17). The most important icing parameter is the temperature of the exposed aircraft surfaces. Since it is usually not possible to measure the temperatures of all aircraft surfaces, the outside air temperature (OAT) is used as a reference point. The surface temperature is important because it controls the degree of convective cooling and the dissipation of latent heat released from impinging droplets (12). This parameter controls whether or not ice accretes to the surface and the rate of icing. At lower temperatures, the potential for icing increases. As aircraft velocity increases, kinetic heating can increase the surface total temperature to a level that prevents ice from forming. Helicopter rotor tips typically are not susceptible to icing at warmer temperatures, which allows for limited flight into icing clearances without rotor ice protection systems (12).

Icing severity is typically indicated by the Liquid Water Content (LWC). LWC is normally expressed as the number of grams of liquid (supercooled) water per cubic meter of air (12). The collection efficiency of the surface must be taken into consideration, but in general, for a given icing condition, doubling the LWC will double the ice accretion rate. The severity is broken into four

categories, which are explained in Table 1 (18). The generic descriptions are biased towards fixed wing operation. Rotorcraft are more sensitive to icing, so studies need to be done for each vehicle configuration to determine limits.

Table 1: Icing Severity Descriptions

Icing Severity	LWC (gr/m^3)	Description	Pilot Action Required
Trace	0.0-0.1	Ice barely perceptible as rate of accumulation is slightly higher than rate of sublimation	Not hazardous even if encountered for over 1 hour without use of ice protection system
Light	0.1-0.3	Rate of accumulation can be hazardous only if encountered for over an hour	Periodic use of ice protection system is required
Moderate	0.3-0.5	Rate of accumulation over short periods can become hazardous	Continuous use of ice protection system and/or diversion
Severe	0.5+	High rate of accumulation is such that the ice protection system cannot keep up	Immediate diversion

The final major icing condition parameter is droplet size. Particle sizes typically range between 10 and 50 μm, although larger particles known as Supercooled Large Droplets (SLD) exist. Many different sizes of droplets may exist in a single cloud, so a mean volumetric diameter (MVD) is usually used to describe the condition. Typically, a Langmuir-D distribution is utilized for icing analyses to represent cloud particle spread, but other distributions may be used in artificial icing facilities.

The mass of supercooled water in each droplet is directly proportional to the cube of its diameter and the influence of airflow is proportional to the square of the diameter. This has an important effect on icing because larger particles will not be as effected by the flowfield around an object, such

as a wing (12). This phenomenon gives rise to the concept of collection efficiency, which describes where particles will impact on a given structure. Large bluff bodies at low speed will collect more water than small streamlined bodies at high speed (12). Also, larger particles will impact further aft on a body, potentially beyond the limits of the ice protection system.

The three major parameters are linked together. For example, the OAT controls the size of particles that exist. At low temperatures, smaller particles dominate the droplet distribution, which leads to a generally lower LWC. Large particles and high LWC are possible only at higher temperatures. The combination of parameters greatly affects the type of ice and growth rate.

1.1.3.3 *Ice Types*

Ice accretes differently to structures at different temperatures, LWC's and velocities (19) (20). Two physically distinct sub-processes control the ice accretion. The first is the inertial transport of liquid water in the form of cloud droplets, rain drops or mixed phase hydrometeors from the ambient environment to the aircraft surface (7). After the droplets impact the aircraft, thermodynamic processes control the freezing of particles. If the heat transfer from the surface is sufficient to remove all of the latent heat of freezing of impinging water, the droplets will freeze on impact (7). The ice surface is typically dry and ensuing ice formations are known as rime ice (21). Due to the conditions, small particles, low temperatures and low water concentrations are associated with the ice type. Rime ice is characterized by an opaque appearance and fine-grained structure. The ice effectively forms an additional smooth (as compared to other ice shapes) layer on the airfoil surface, but with a much higher surface roughness (22). Longer ice exposure times may lead to long and

pointed spear shape. Rime ice is relatively easy to remove with ice protection equipment (18), and is only encountered on inboard portions of rotors and fixed surfaces.

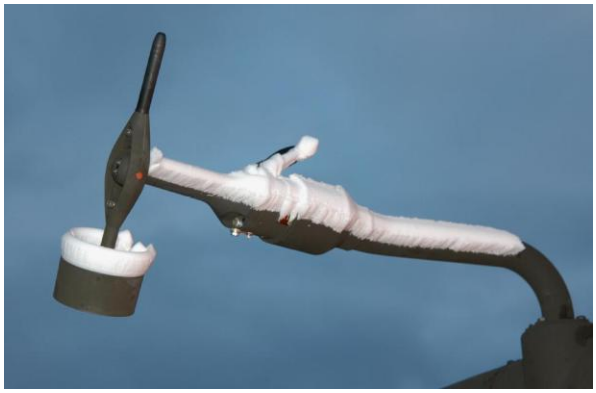


Figure 1: Rime Ice Accretions on AH-64 AADS
(adapted from Reference (23))

As the temperature and LWC increases, translucent glaze ice shapes dominate accretions (21). This type of ice is encountered in cumuliform clouds and freezing rain (18). The heat transfer on the surface is inadequate to remove all latent heat from impinging water particles and the ice surface becomes wet. Not all water freezes on impact; rather it runs back and freezes when contact with previously accreted ice (24). These runbacks may form two large horns if the iced object is an airfoil. Due to the irregular shape and higher surface roughness of the large ice accretions, glaze ice has a larger effect on airfoil performance than rime ice (7) (22). Glaze ice is harder to remove from aircraft surfaces.

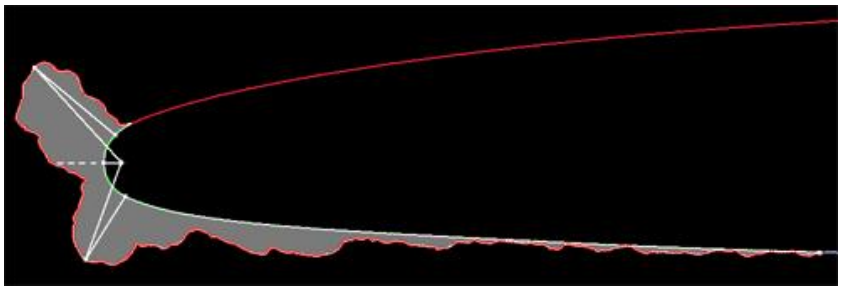


Figure 2: Glaze Ice
Note Large Ice Horns
Image from SmaggIce User Manual (25)



Figure 3: Glaze Ice in AERTS Chamber

Further increases in temperature and airspeed may lead to beak ice. This type of ice may form on rotor blade tips. The only place on the airfoil conducive to beak icing is a small area at the leading edge on the suction side (12). Cooling due to adiabatic expansion is balanced by kinetic friction, which leads to a slushy ridge as shown in Figure 4.

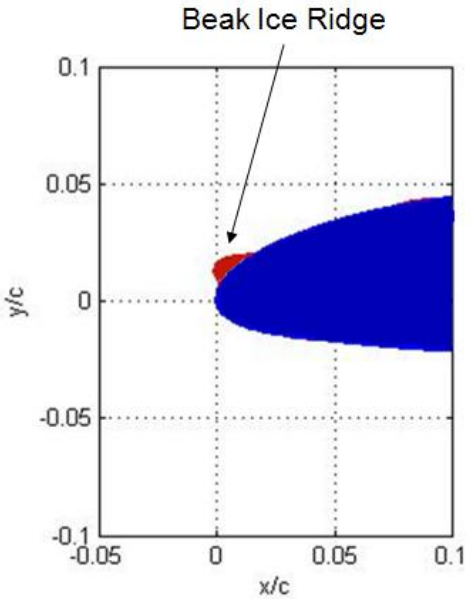


Figure 4: Beak Ice

Mixed ice occurs when glaze and rime ice are occur together (21). The resultant ice exhibits a combination of properties and is the most difficult to remove from a surface (18). Some authors

refer to this ice as 'glime' but this not a generally accepted term. Due to the variation in local velocity along the span of a rotor, combinations of ice can be encountered on the same blade. Furthermore, kinetic heating of the rotor surface varies with span. Rime ice tends to form on the inboard portion of the blade, while glaze ice forms on the outboard section and beak ice at the tip (12). This is shown in Figure 5, which indicates typical ice types as a function of velocity and temperature.

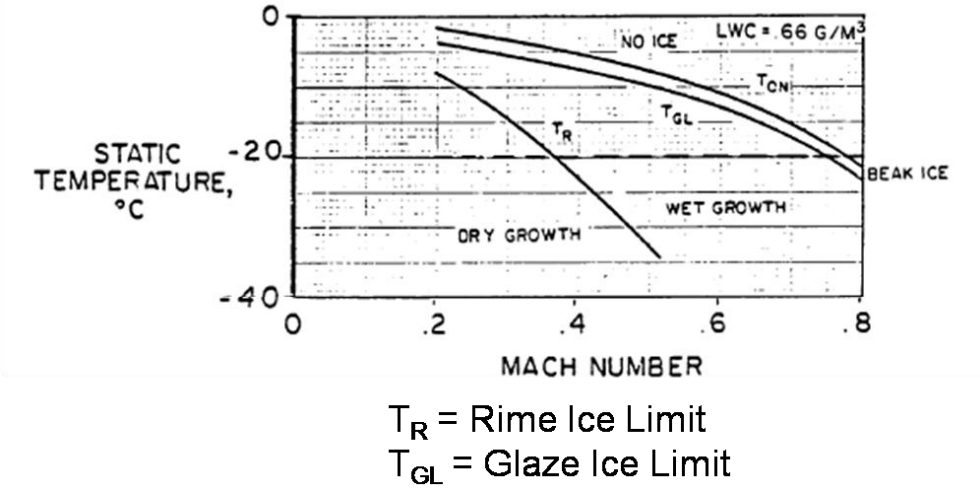


Figure 5: Ice Type Limits for LWC = 0.66 gr/m^3
Adapted from Reference (26)

A generic ice accretion profile is shown in Figure 6. Rime ice is only found at the inboard sections of the blade and is generally not a problem. Glaze ice accretes on the majority of the blade, while beak ice is found near the tip. At warmer temperatures, a significant portion of the blade tip may remain ice free due to kinetic heating. Ice accretes to the entire blade span at colder temperatures, typically below -15°C. It is important to note that the maximum ice thickness usually does not occur at the blade tip. For higher rpm systems, such as propellers or proprotors, the tip regions typically do not accrete significant ice due to shedding.

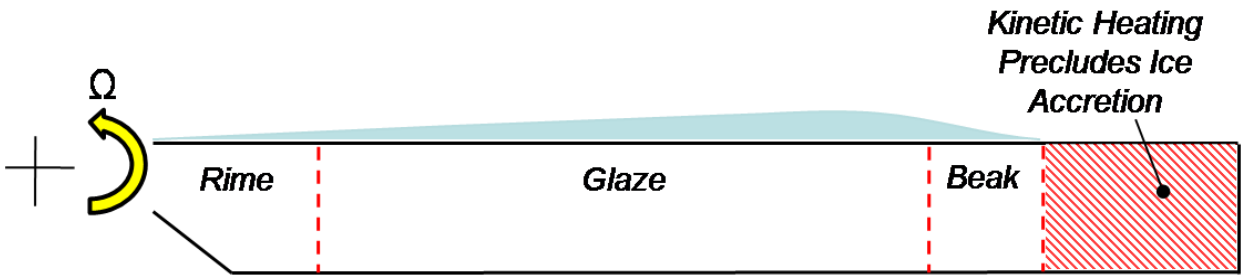


Figure 6: Typical Blade Ice Accretion Profile

The primary properties of various ice types are contained in Table 2. The visual properties of ice are important, as pilots primarily use color to distinguish between types.

Table 2: Characteristics of Different Types of Ice Caused by Atmospheric Ice Accretion

Adapted from Reference (27)

Ice Type	Characterization	Density (kg/m³)
Glaze	Hard, almost bubble free, clear homogeneous ice with a density close to that of pure ice	>900
Hard Rime	Rather hard, granular, white or translucent ice	500-900
Soft Rime	White or opaque ice with a loosely bonded structure, feathery appearance	< 500
Snow (ice crystals)	Snow that falls onto a structure and subsequently freezes *Rarely an in-flight icing issue; more of a concern for ground icing prior to flight*	500-900

1.1.3.4 *Icing Conditions*

The FAA defines icing conditions in Appendix C the Federal Aviation Regulations (FAR) Part 25/29. Icing is defined by the variables of cloud liquid water content, the mean effective diameter of the cloud droplets, the ambient air temperature, and the interrelationship of these three parameters (28). The conditions are separated into two categories: continuous maximum icing and intermittent maximum icing. The primary difference between the two is that intermittent icing generally is more severe than continuous maximum. Table 3 and Figure 7 detail these differences.

Table 3: FAA Part 29 Icing Envelope Summary (28)

Parameter	Value	Units
Cloud Length, Continuous Maximum Icing	17.4	nm
Cloud Length, Intermittent Icing	2.6	nm
Droplet Median Volumetric Diameter (MVD)	0 - 50	μm
Liquid Water Content (LWC)	0 – 3.0	gr/m^3

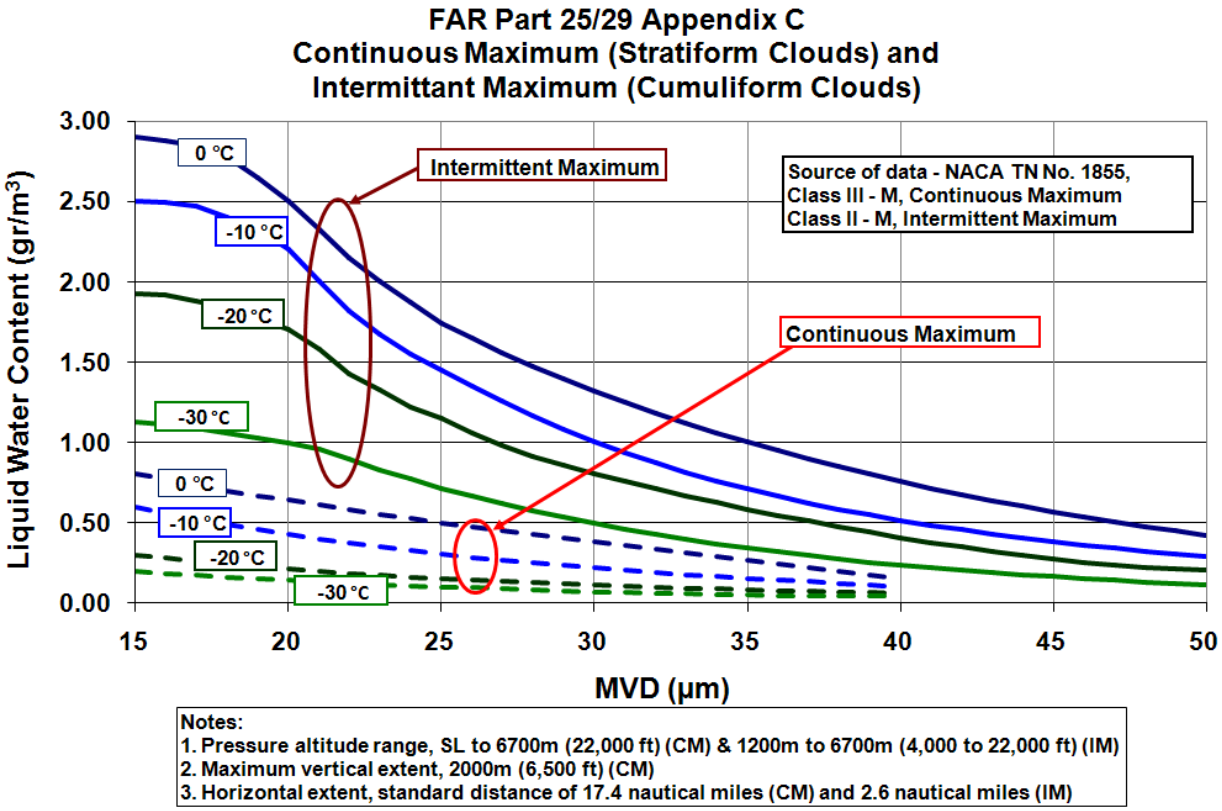


Figure 7: FAA Part 25/29 Appendix C Envelope

1.1.3.5 *Supercooled Large Droplets*

Supercooled large droplets (SLD) have generated a lot of attention in recent years because the condition has contributed to the crashes of recent turboprop commuter aircraft crashes, including an ATR-72 in 1994 (29). Recent work has targeted expanding the icing envelope established by FAR Part 25/29 Appendix C, which does not cover the effects of SLD (30).

SLD conditions occur when the droplet size is greater than 50 μm and a horizontal extent greater than 2.6 nm or conditions with a droplet size larger than 40 μm and a horizontal extent of more than 17.4 nm (31). These conditions are characterized by larger particle sizes (100 – 200 μm) which with higher momentum which can cross streamlines (17). This allows the particles to impact an aircraft structure further aft than smaller droplets, outside the range of the de-icing system. The most dangerous condition is when the accreted ice forms a ridge just aft of the de-icing boot. Airfoils with this ice accretion have shown a decrease in lift by more than 60% and an increase in drag as much as 200% (32). SLD are important to consider for rotorcraft operations in icing conditions primarily due to the danger posed to the engine intakes.

1.2 Review of Related Work

1.2.1 Ice Accretion Facilities

1.2.1.1 *Natural Icing Trials*

Ever since the negative effects of icing have been noticed in aviation, efforts have been made to understand the phenomena and predict meteorological conditions that pose the most danger to aircraft. Most early studies focused on flight testing since atmospheric icing conditions could not be reproduced in wind tunnels (2) (33). Aircraft were equipped with icing sensors and flown deliberately into areas of predicted icing. Either in flight or upon landing, the accreted ice thickness was measured to correlate accretion with meteorological conditions. Some extended campaigns were run with civilian and military aircraft to populate data sets for research purposes (34) (15).

Flight testing occurs to this day for vehicle certification purposes, as evidenced by the recent granting of icing clearances for the Sikorsky S-92 and Agusta Westland EH-101 rotorcraft for flight into (limited) icing conditions. Although flight tests are the ultimate verification of ice accretion and de-icing system performance, tests are expensive and are subject to the ability of the test aircraft to seek out icing conditions. In past testing, US Army rotorcraft spent an average of 0.15 hours in natural icing conditions per day on the test site (35), which is an inefficient use of resources.

1.2.1.2 *Icing Tunnels*

To mitigate the problem of constantly chasing natural icing clouds, icing tunnels were developed in the late 1920's to perform icing studies on sub scale models and aircraft components (2). Over the years, tunnels have successfully been implemented in the certification process of the de-icing systems as evidenced by their proliferation. Currently, there are many icing tunnels in the world (36) and some are detailed in Table 4.

Table 4: Partial List of Icing Tunnels

Tunnel Name	Tunnel Operating Organization	Test Section Size W x H x L	Location	Country
Icing Research Tunnel (IRT)	NASA Glenn	6' x 9' x 20'	Cleveland, OH	US
Boeing Research Aerodynamic Icing Tunnel (BRAIT)	Boeing Commercial Airplanes	4' x 6' x 3'	Seattle, WA	US
Goodrich Icing Tunnel (37)	Goodrich Inc	1.8' x 3.6'	Uniontown, OH	US
Cox Icing Tunnel (38)	LeClerc Icing Research Laboratory	2.3' x 3.8' x 6.5'	New York, NY	US
Icing Wind Tunnel (IWT) (39)	CIRA	11.8' x 7.7'	Florence	Italy

Even the smaller tunnels require significant cooling systems to provide low temperature flow over the test model. For example, the Cox tunnel requires an 80-ton cooling capacity, driven by a 250 HP motor (38). The much larger IRT requires a 2100-ton cooling unit, with a 5000 HP motor. Ice is introduced to the tunnel upstream from the test section with a series of nozzles mounted on spray bars. Many hours of calibration runs are required to ensure icing cloud uniformity at the test section (39).

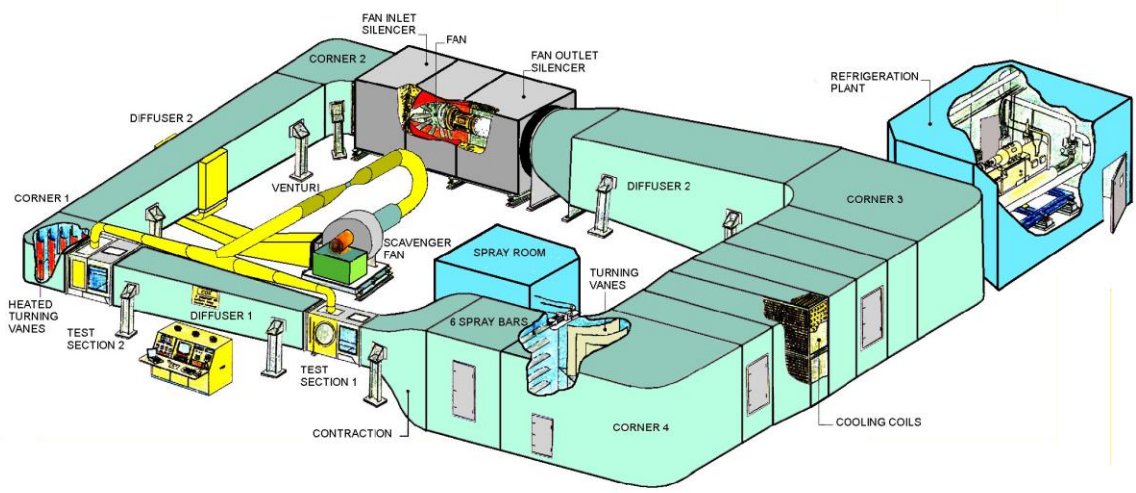


Figure 8: Cox Icing Tunnel
Adapted from Reference (38)



Figure 9: UH-60 Model in NASA Glenn (formerly NASA Lewis) IRT

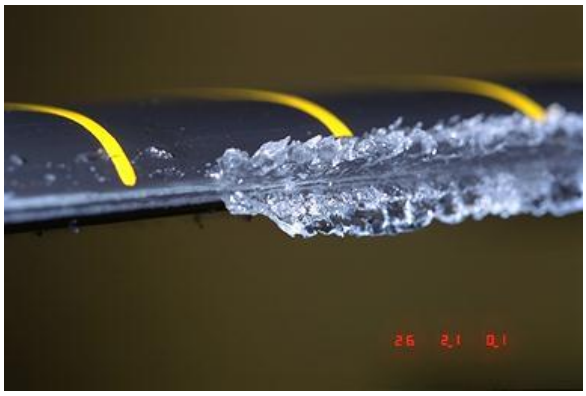


Figure 10: Model UH-60 Rotor Ice Accretion (Note Ice Shedding)

Icing tunnel tests of rotorcraft require small models to provide adequate wall clearance for the rotorblades. This leads to the requirement for very small rotorblades and the associated problems of scaling icing. Scaling electrothermal systems is not practical (40), so tests of the most common rotorcraft deicing system are not possible on rotating systems.

1.2.1.3 *Full Scale Testing Facilities*

These problems have led to the development of specialized facilities for full scale rotorcraft icing trials. The National Research Center (NRC) Spray Rig in Canada (now closed), and the Helicopter Icing Spray System (HISS) airborne tanker system are particularly suited for rotorcraft icing research and will be further explained. Large scale weather chambers, such as the McKinley Climatic Laboratory have been used for rotorcraft icing tests, but so far their application is limited to testing of full scale rotors at low collective pitch angles at low forward speed (14). Due to this limitation, their operation will not be further detailed.

1.2.1.3.1 *NRC Spray Rig*

The NRC Spray Rig was constructed in 1953 to create an artificial icing cloud to study rotorcraft performance degradation in hover. The facility consisted of a 161 steam atomized water nozzles positioned on a pole 60 ft above the ground (41). Combined with low ambient temperatures, the system created a large icing cloud in which a subject helicopter could hover as shown in Figure 11. A light wind (<10 mph) was required to move the cloud away from the tower, and helicopters with rotors up to 55 feet could be studied (41). Although the facility is now closed, it provided years of

valuable research. The design and operation of the icing tower is similar to the new PSU AERTS facility, the subject of this thesis.



Figure 11: NRC Rotorcraft Icing Tower, Boeing Vertol CH-46D Undergoing Testing
Image from archives of Andrew Peterson, Boeing Rotorcraft

1.2.1.3.2 *HISS System*

The Helicopter Icing Spray System was originally developed in 1973 to generate artificial icing clouds to test slow speed aircraft in icing conditions. The goal of the system is to provide a safe, reliable and controlled flight-test environment for evaluating de-icing technologies.

The system consists of a series of icing nozzles mounted on a retractable boom attached to a Boeing CH-47C Chinook heavy lift rotorcraft as shown in Figure 12 and Figure 13. Water for the icing system is provided from 1400 gallon tanks mounted in the aircraft fuselage, while air is supplied from the aircraft auxiliary power unit. It has the advantage of mobility, so the system can be moved to various locations for icing trials. Most operations take place near Duluth, MN.



Figure 12: HISS Artificial Icing Testing of CH-47D
Image from NASA Archives ID#: C-1983-03731



Figure 13: OHISS Artificial Icing Testing of S-92A
Image courtesy US Army

The system has proved so successful that the original aircraft has reached the end of its service life through heavy use. Consequently, the US Army has sponsored a series of improvements to the system. The Improved Helicopter Icing Spray System (IHISS) investigated improved icing nozzles (42), while the Objective Helicopter Icing Spray System (OHISS) under development will contain more advanced technology such as a new cloud generation system (CGS) and a cloud characterization system (CCS) (43).

1.2.2 Ice Accretion Prediction Codes

While icing tunnels and their rotorcraft equivalent whirl towers have demonstrated their ability to reproduce icing conditions, their costs have been ever increasing. Ice accretion codes have been written to predict ice accretion for various conditions on structures to reduce tunnel time and flight test entries. Several programs have been developed internationally, including LEWICE (US), ONERA (France), TRAJICE (UK), FENSAP-ICE, (Canada), CANICE (Canada) and MULTI-ICE (Italy) (19) (44). The codes are primarily two-dimensional in nature, although some have been expanded into three dimensions, such as LEWICE 3D. Currently, the use of the codes is widespread in the development and certification of ice protection systems (45). Their primary use is in determining the chordwise extent of ice accretion by using the impingement limits of the code's

droplet trajectory module. This information is used to size the de-icing system coverage limits. Aerospace companies and safety organizations are increasingly using the codes to verify the ice accretion on the flight vehicle and validate performance of de-icing systems.

1.2.2.1 *Ice Accretion Model Process*

Generally, icing prediction programs start by reading in a previously calculated flowfield, or by calculating it within the code. If the code calculates the flowfield, it is usually an inviscid potential flow solution for simplicity (46) (47). Particles are then released into the flow and their impact locations on the body under study using the methods of Reference (48) or (49). Ice accretions are governed by a balance of a heat flows, including air kinetic heating (Q_{ke}), water droplet impact heating (Q_{kew}), convective cooling (Q_c), evaporative cooling (Q_e), latent heat of freezing (Q_{Le}), sensible heat from the impinging water droplets (Q_{sw}) and sensible heat from runback water (Q_{srbw}) (12). Accurate determination of the boundary layer laminar to turbulent transition point is required for the proper calculation of the convective heat transfer coefficient. The result of the heat flow balance is the freezing fraction, which determines the percentage of water droplets that freeze to a differential area. The process is repeated for the entire surface of the body under study to generate the accreted ice shape. After the shape is established, the program moves to the next time step, and starts over by revealuating the flowfield.

1.2.2.2 *Accretion Model Issues*

The primary problem with the code implementation is verification. Due to the complexities of the icing physics, a comprehensive validation process is required before a code is accepted by industry for research or system development. This requires significant resources, as hundreds of data runs are required to cover a large icing condition envelope for a variety of airfoils. For example, the

LEWICE 2.0 validation program included 400 cases in the IRT, of which 231 were unique test conditions (50). The software proved to be highly accurate, with the predicted ice shapes differing from experimental results by 12.5% (50).

The problem, however, is that the validation process is limited by the facility in which the model is undergoing verification. For rotorcraft, the primary constraint is velocity. Also, from the rotorcraft perspective, validation runs are usually performed with static two-dimensional airfoils. This does not allow some of the complexities of the rotorcraft icing to be tested. For example, higher order particle dynamics, such as particle bouncing and particle splashing, are not usually well modeled. Also centrifugal effects and a non-uniform LWC inherent to rotating systems may lead to accretion prediction errors for rotor blades.

1.2.2.3 *Rotorcraft Specific Icing Models*

Despite many years of research on the problem, rotor icing remains a problem for both civilian and military operators. Traditional fixed wing icing analyses cannot be applied to a helicopter rotor due to the periodic changes in velocity and angle of attack for each blade station. Many authors have developed models that are designed specifically to explore various parts of the rotorcraft icing challenge. Common goals for models include predicting performance degradation of the vehicle and ice protection system sizing studies. A well validated model can assist with certification of rotorcraft for flight into icing conditions. Some models contain additional features, such as shedding analyses and shed ice tracking modules.

Most rotorcraft models take a two dimensional analysis approach, with all analysis taking place at discrete blade stations. At least three major steps are required at each station. First, the flowfield parameters, including sectional velocity and angle of attack, are calculated. The next step is accreting ice to the sections. This process may occur in a dedicated icing program such as LEWICE or calculated empirically based upon local collection efficiencies. Finally, the results of the ice accretion are collected and post processing occurs. Post processing may include performance and shedding analyses. An example program flow is shown in Figure 14. Similar methods can be applied to propeller analysis (51) and examples of these models are contained in the table below. Other proprietary models are used at various rotorcraft manufacturers for in-house analyses.

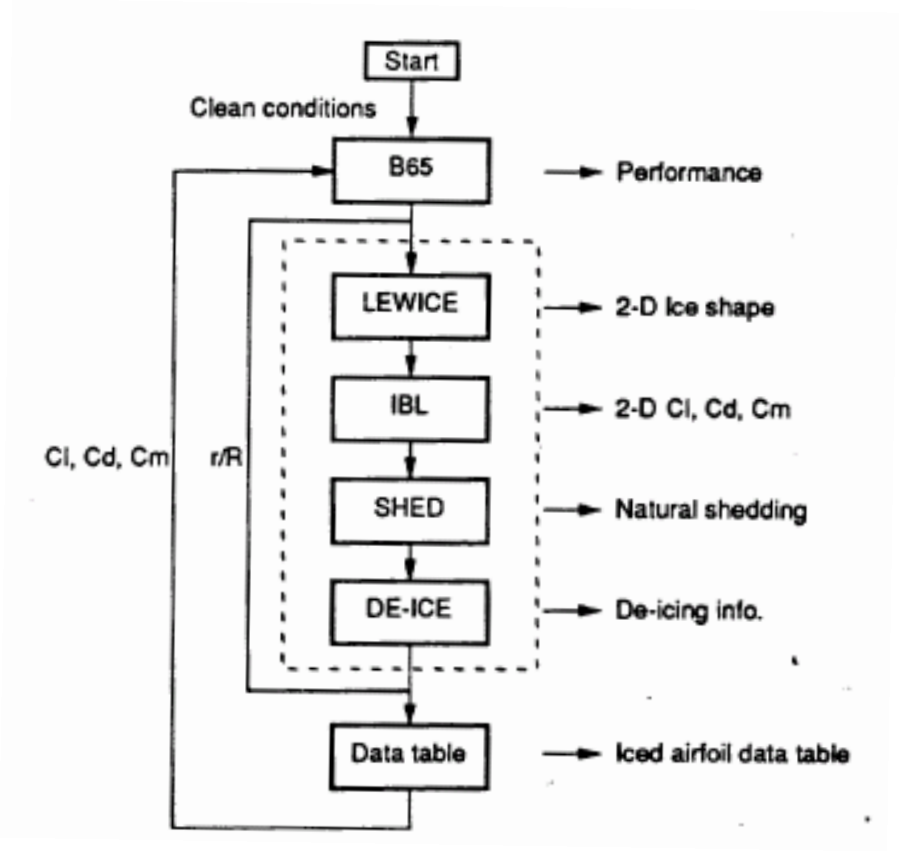


Figure 14: Typical Rotor Icing Model Program Flow
Adapted from Reference (52)

Table 5: Sample Rotor Ice Prediction Programs

Author/Year/Reference	Flowfield Calculation	Ice Accretion Model	Performance Degradation	Shedding
Miller, 1985 (51)	Inviscid Conformal Mapping	Empirical Trajectory Model	Empirical Coefficients	No
Flemming*, 1985 (26)	Unknown	Empirical Trajectory Model	Empirical Coefficients	No
Britton, 1992 (52)	BEMT	LEWICE	Interactive Boundary Layer (IBL)	Yes
Hartman, 2006 (53) (2D) Zanazzi, 2007 (54) (Rotor)	Various CFD	LEWICE FENSAP-ICE MULTI-ICE	Various CFD	No
Fortin, 2009 (55)	None	Empirical Model	Experimental	Yes
*Only rotor icing subroutines are detailed in paper				

1.3 Objectives

The objective of this research is to develop a rotor ice accretion model that is coupled with a new icing facility. This concept is designed such that the two entities can mutually support each other, which is important as they are both new and limits need to be established. The model can be used to learn about the facility and vice versa. For example, the model will be initially used for facility calibration, but later the facility can be used for extensive model validation. Once the model is validated, it will be able to be used for basic rotor icing research. While it is not as complex as many other rotor codes, it will be useful to uncover icing trends for particular rotor configurations and assist in the sizing/placement of ice protection systems.

To achieve these goals, the objectives of the present work are to:

1. Develop a hover performance calculation code based around the LEWICE ice accretion software to predict ice accretion on a rotor. This model will include the natural shedding of ice due to CF forces inherent to rotating rotors as well as a performance degradation module.
2. Construct and calibrate an icing test chamber capable of reproducing natural icing conditions in a controlled environment. Since the facility is unique, specialized validation processes need to be developed to determine the icing cloud parameters. Limitations of the facility need to be determined both computationally and experimentally.
3. Design, fabricate and calibrate a test fixture to measure the shear adhesion strength of ice to specific rotor materials. Ice accretion physics need to be properly replicated on the test fixture to improve data accuracy.
4. Correlate the predicted and experimental ice accretions for a series of test cases. These tests will occur at various icing conditions within the FAR Part 25/29 Appendix C icing envelope and include variations in temperature, icing severity and droplet sizes. Rotor torque degradation and shedding behavior will be compared with predictions.

1.4 Thesis Overview

The thesis is organized into the following remaining chapters:

1.4.1 Chapter 2: Ice Accretion and Shedding Model

The development of the AERTS Rotor Icing, Shedding and Performance (ARISP) icing model is presented. Each module is explained and the integration of the modules with NASA's LEWICE ice accretion code is detailed. Validation results from the individual modules and assembled model are described.

1.4.2 Chapter 3: AERTS Facility Calibration

Details of the AERTS Facility calibration are presented in Chapter 3. Computational Fluid Dynamics (CFD) modeling of the facility was used to understand the complex flow field inside of the test chamber. The effects of the flowfield on particle trajectories are explored. The specifically developed Liquid Water Content (LWC) calibration procedure for the AERTS facility is explained. Details of the icing system shakedown tests are presented, along with the subsequent generation of the facility operational envelope.

1.4.3 Chapter 4: Ice Adhesion Strength Calculation Fixture

The newly developed ice shear adhesion fixture and its benefits in measuring an important material property are described in Chapter 4. Details of the mechanical and thermal design are detailed, along with calibration results. Experimental results of the test fixture in the AERTS facility are presented.

1.4.4 Chapter 5: Experimental Exploration of ARISP Model in AERTS Facility

Correlations between the ARISP model and icing experiments in the AERTS Facility are described. The test process is described, along with general test observations. The effect of various parameters, including temperature, droplet size, and icing severity on accreted ice thickness, impingement limits, ice shape, shedding behavior and rotor performance degradation is explored.

1.4.5 Chapter 6: Conclusions and Recommendations

The final chapter summarizes the conclusions of the present work in experimentally investigating a rotor icing model in a newly developed icing facility. Recommendations for future work for improving the model and facility are described.

1.4.6 Appendices

Appendices explain the important components of the AERTS Facility, including the design and implementation of icing system. The integration of a new rotor hub with the drive system is described. The design and implementation of the mounting system for the current test blades to the rotor hub is presented, along with pertinent stress and dynamic analyses.

2. Ice Accretion Modeling – ARISP Code

2.1 Objectives

The primary goal of the AERTS Rotor Icing, Shedding and Performance (ARISP) Code is to accurately and quickly model the ice accretion on rotor blades. Once the ice profile is established, it can then be used to empirically determine the resultant performance degradation of the rotor system. The code is designed to investigate the effects of various conditions and various blade geometries on ice accretion. After validation, the code can be used in the AERTS facility to analyze various blade configurations.

To accomplish these goals, the code must meet the following objectives:

1. Model ice accretion across the blade, accounting for spanwise distributions in angle of attack, velocities and blade geometry
2. Account for natural ice shedding inherent to rotating systems
3. Empirically calculate the effects of ice on rotor performance
4. Be generalized such that various blade geometries and atmospheric conditions can be studied
5. Be able to quickly evaluate the effects of icing on a particular blade configuration

Within the context of this thesis, the ARISP will focus only on hovering flight. This is due primarily to limitations of the AERTS Facility, in which the model will be validated. This constraint was judged not to be an issue due to recent analysis by NASA Glenn. Experiments have shown that forward flight ice accretions can be generated by running the blade sections at an

average angle of attack and velocity (i.e. hover). If required, explicit analysis of forward flight is possible with relatively simple modifications to the code.

2.2 ARISP Code Overview

The AERTS Rotor Icing, Shedding and Performance (ARISP) model is based around a coupling of NASA's LEWis ICE Accretion program (LEWICE) ice accretion program with a variety of modules that predict ice shedding and rotor performance degradation as a function of rotor geometry and icing conditions. The overall process is similar to the model developed by Britton (52), but performance degradation is modeled empirically.

Like most rotorcraft icing models, the ARISP code differs from a standard fixed wing icing analyses in the fact that conditions are changing as time passes. The effective angle of attack may change due to ice accretion and/or ice may shed from the rotor. As such, the icing analysis must be broken up into a number of steps, with the ice profile and performance constantly being updated. A sectional approach is taken, with rotor performance and ice accretions calculated at various rotor stations. Rotor performance is determined with a BEMT routine, chosen for its speed of calculation and ease of integration with LEWICE. The model is focused on hover due to limitations of the AERTS Facility, but can be generalized to free forward flight. Corrections are made for the unique aerodynamic environment of the test chamber arising from its relatively small size. Based upon the BEMT module and icing conditions, automatic inputs are generated for LEWICE and icing analysis is run. LEWICE results are then read and post processors assess shedding criteria and update the sectional airfoil performance.

2.2.1 Software

The icing branch at NASA's Glenn (formerly Lewis) Research Center developed LEWICE ice accretion code to assist in the analysis of aircraft in icing conditions. The model contains an analytical ice accretion model that evaluates the thermodynamics of the freezing that occurs when supercooled droplets impact a body (46). The model uses details of the icing environment, including the liquid water content (LWC) and droplet size distribution, as well as atmospheric parameters such as temperature, pressure and velocity to determine the ice shape on a user specified body. In order to do this, the program has four major modules, which consist of the flowfield calculation, droplet trajectory calculation, thermodynamic and ice growth calculation and finally geometry modification due to ice accretion. The details of these modules are contained in the software user's manual, Reference (46).

First, the flowfield is calculated with a potential flow solver for the clean body. A user provided Navier-Stokes generated solution may be used in later versions of the code. Droplet trajectories are modeled with particle dynamics equations. The ice growth rate on each segment defining the surface is then determined by applying the thermodynamic model. The growth rate is correlated to a local ice thickness and the body coordinates are adjusted to account for the accreted ice. This process is then repeated at the next time step, starting with a recalculation of the flowfield over the iced geometry. The analysis ends when the user specified icing time has been reached. The length of each individual time step is automatically calculated or user defined.

LEWICE has additional features, including the modeling of ice protection systems in both de-icing and anti-icing modes. User inputs can be specified for electrothermal, hot air and pneumatic

systems and their effects on ice shedding and runback are included in the ice accretion calculations. The effects of super large droplets (SLD) on ice protection systems are included. These features are not used for the current application of LEWICE.

The decision was made to utilize LEWICE 3.0 due to its extensive validation in the NASA Glenn Icing Research Tunnel (IRT). Over 842 data runs with a total of 7 airfoil geometries were made. Shape comparisons with the LEWICE model indicated a 7.2% deviation from the experimental tracings, with an overall variability of 2.5% in the experimental data (50). This attribute is especially important due to the immaturity of the AERTS Facility, where the ARISP code would be verified. Empirical icing codes can be written based on the calculated airfoil collection efficiencies as in Reference (55), but extracting ice shapes for comparison to experiments would be difficult without extensive validation runs. Other ice accretion codes were considered, such as FENSAP, but the costs associated with purchasing them were not justified by higher accuracy or ease of use.

Another major decision made early in the ARISP model development was to write the code in a MATLAB script. The primary reason for this was for the ease of debugging and plotting in MATLAB. The script is not compiled like a Fortran or C++ generated executable, so it will generally run slower, but this is not an issue within the context of this thesis. The slower execution speed of MATLAB is not noticeable over the < 5 minutes required to analyze the rotor.

2.3 Analysis Concept and Implementation

The ARISP code differs from a standard fixed wing icing analyses in the fact that conditions are potentially changing as time passes. The effective angle of attack may change due to ice accretion and/or ice may shed from the rotor. As such, the icing analysis must be broken up into a number of steps. At the completion of each icing analysis, the rotor is checked for shedding and the local C_l and C_d are updated. A schematic of this process is contained in Figure 15.

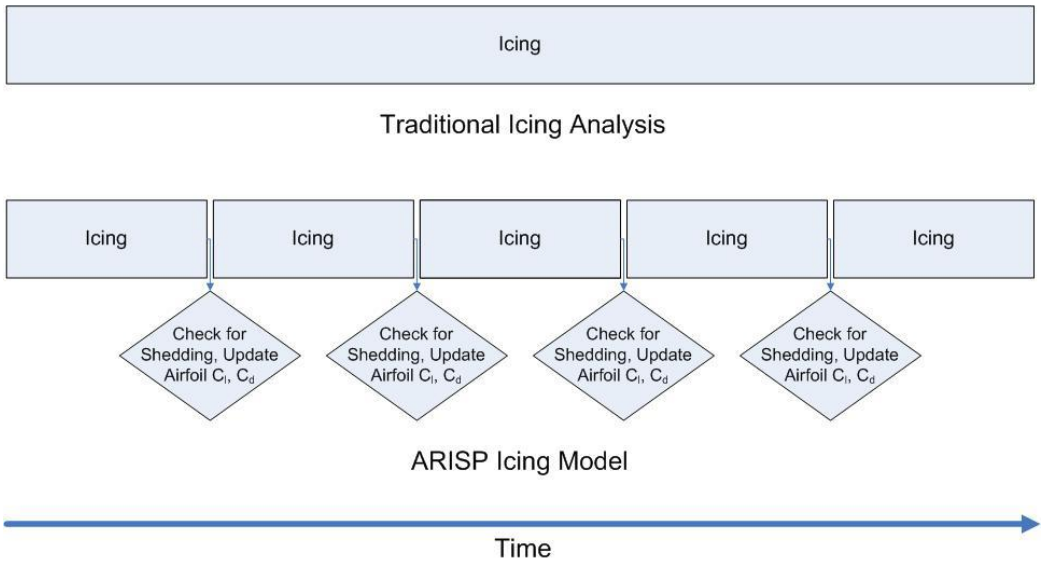


Figure 15: ARISP Concept

In the ARISP model, this process is handled automatically, as the code calls LEWICE and interprets the icing results a number of times for each data run. The specific number of times is user specified, but the code is usually set to check for shedding every 10 seconds of icing.

The code is based upon a two dimensional analysis of ice accretion and is broken up into four main modules. These are shown schematically in Figure 16. Each module is described in subsequent sections.

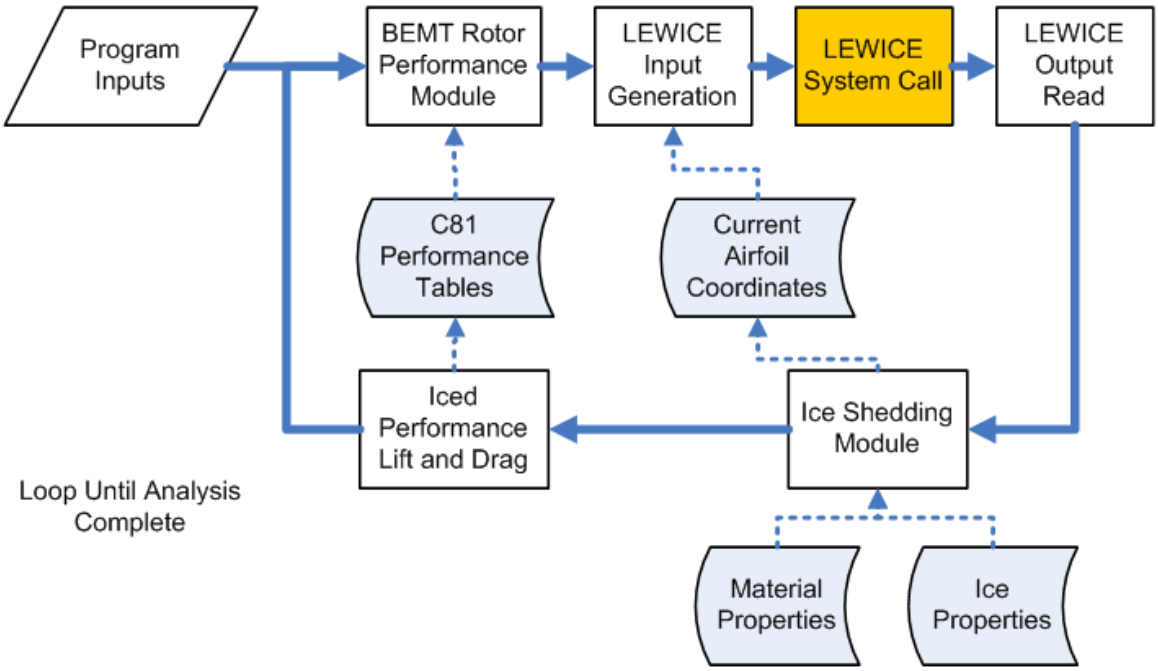


Figure 16: ARISP Code Flow

2.3.1 Hover Performance – BEMT

2.3.1.1 *Description*

The Blade Element Momentum Theory (BEMT) hover performance module has the goal of determining the rotor performance. The primary outputs are spanwise angle of attack and velocity distribution, which are input to the LEWICE module to calculate the sectional ice accretions. In addition, rotor total rotor forces such as torque and thrust are calculated.

The module was written based upon the method described in Reference (56), assuming Laminar flow directly upstream of the blade, Prandtl root and tip losses and smooth blade surfaces prior to icing. As with the rest of the ARISP code, the BEMT module is generic to accept inputs for various rotor geometries. All analysis is performed with NACA 0012 airfoil sections and clean airfoil performance is supplied by Mach number and angle of attack lookup tables. Additional airfoils can

be easily analyzed with the addition of the appropriate performance tables and properly linking them within the code.

The primary difference between a standard BEMT performance code and the one written for the ARISP code is the ability for accreted ice to modify the local lift, drag and pitching moment characteristics of the blade. The local C_l, and C_d are contained in dynamic libraries that are updated by the performance module, described in Section 2.3.4. All calculations are conducted at a constant user input collective pitch, regardless of the rotor thrust due to accreted ice. The AERTS control system currently does not have the capability to adjust collective pitch to maintain a certain thrust level. This capability can be added to the code to better model the effects of ice on a flight vehicle with another convergence loop.

2.3.1.2 *Airfoil Performance*

Clean performance for the NACA 0015 airfoil is taken from Reference (57) . Data is tabulated in standard C81 tables for Mach numbers from 0 to 1.0 and angles of attack from -180° to 180°. A sample of the data is contained in Figure 17 and Figure 18. The BEMT module uses an interpolation routine to obtain the proper coefficient. As airfoil performance is lowered due to ice accretions, changes in lift and drag are empirically calculated. Details on this procedure is contained in Section 2.3.4.

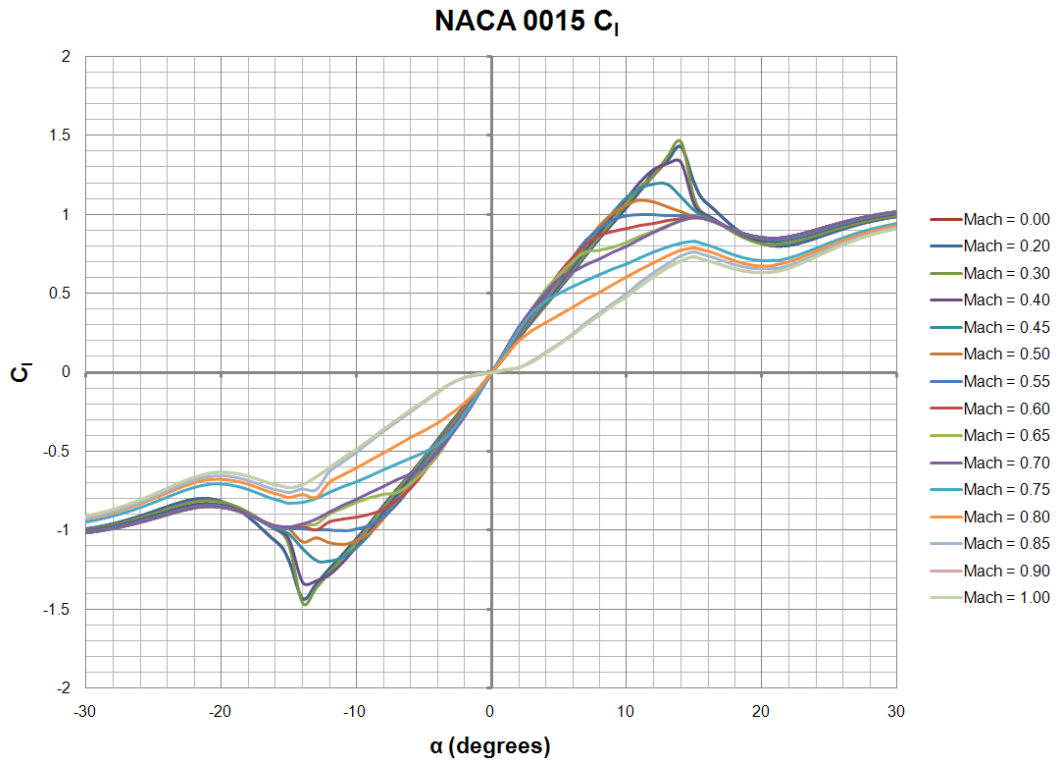


Figure 17: Clean NACA 0015 C_l

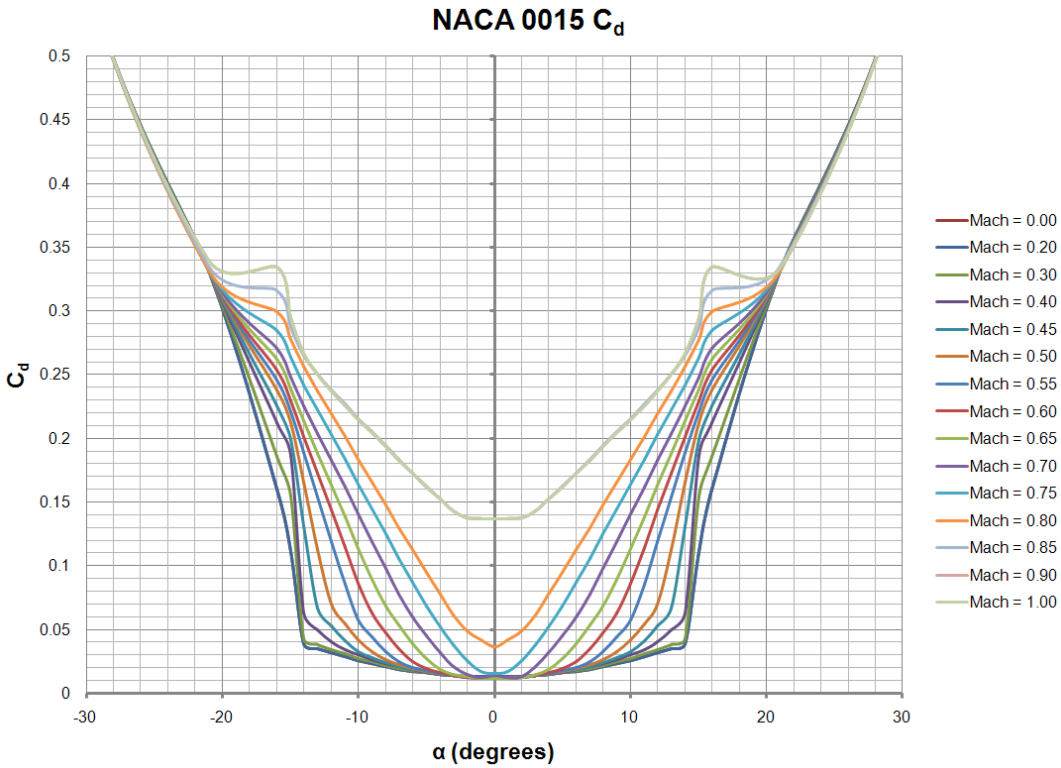


Figure 18: Clean NACA 0015 C_d

2.3.1.3 *Modifications for the AERTS Facility*

The unique nature of the AERTS chamber requires some modification of the code to accurately predict rotor performance. The test chamber is undersized for proper aerodynamic analyses. The chamber height is the most series issue, limiting rotor diameter to approximately 3 feet to avoid ground effect. All rotors in the AERTS chamber are larger than this to provide adequate area for ice shape trend analysis and to reduce the influence of the hub and support structure on accretions.

The first correction added to the ARISP model accounted only for ground effect. The following equation, from Reference (58), quantifies the increase in rotor thrust, incorporating the effects of blade loading. This correction essentially moves the model rotor from Hover Outside Ground Effect (HOGE) to Hover Inside Ground Effect (HIGE).

$$\frac{T}{T_{INF}} = \frac{1}{1 - \frac{\sigma C_{l\alpha}\lambda}{4C_T}\frac{(\frac{R}{4z})^2}{(1+(\frac{?}{\lambda})^2}} \qquad \textbf{Equation 1}$$

A better correction studied for use in the ARISP model is based on the developments of Rossow. As described in Reference (59), the research studied the effects of ceiling and ground planes on rotor performance with image cylinders. The primary intent of the studies was to develop performance corrections for the study of rotors in wind tunnels. Results of calculations compared to theory are shown in Figure 19.

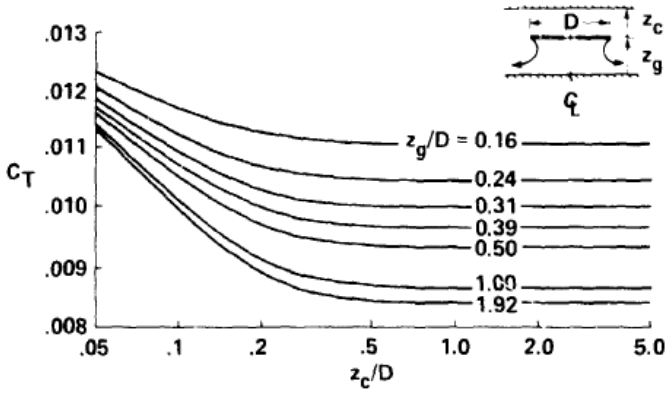


Figure 19: Performance Corrections per NASA TM 86754

No literature was found that discussed the effect of ground, ceiling and wall effects on rotor performance. A series of tests in the AERTS chamber were undertaken to verify the correction factors. After these tests, it was found that no performance corrections were required, as the rotor was operated at low thrust levels (C_T < 0.0004) to avoid distorting the icing cloud. Therefore, for all tests in the chamber, the corrections are disabled.

2.3.1.4 *BEMT Validation*

The BEMT module was independently validated from the rest of the ARISP code. First, it was tested without the corrections described in the previous section and compared to actual rotor test data. For this validation, the tail rotor of the generic sample helicopter as described in Reference (60) was run with the BEMT module. The tail rotor was selected because it was analyzed with the BEMT theory and the rotor utilized NACA 0012 airfoils. Its parameters are described in Table 6.

Table 6: Prouty Tail Rotor Parameters

Parameter	Value	Units
N_b	3	
R	6.5	ft
Chord	1	ft
V_{TIP}	650	ft/sec
Root Cutout	0.15r	
σ	0.146	
Twist (linear)	-5	deg
Airfoil	NACA0012	

Two primary parameters were compared. The first was the blade loading (C_T/σ) vs collective (θ_o), while the second was the performance polar. As shown in the following charts, there is a generally favorable comparison.

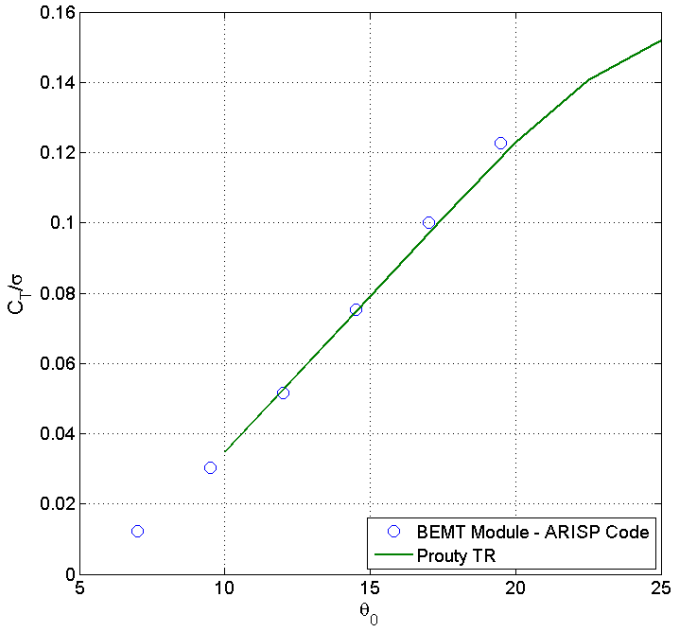


Figure 20: BEMT Validation - Blade Loading Coefficient vs Collective

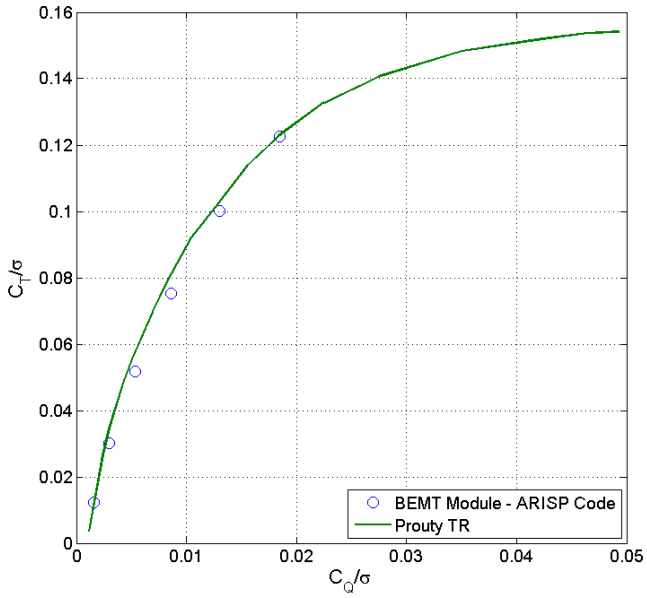


Figure 21: BEMT Validation - Performance Polar

Additional tests were made with the code, comparing results to isolated performance experiments. This step was critical, as the entire ARISP Code is dependent on the accuracy of the BEMT module. Performance data from Reference (61) was utilized because it was generated with rotors of the same size as those to be used in the AERTS facility. The correlation between the ARISP Code and the published data is generally good. As expected, the BEMT code slightly over predicts rotor performance because it does not explicitly account for 3D wake geometry and other higher order rotor effects.

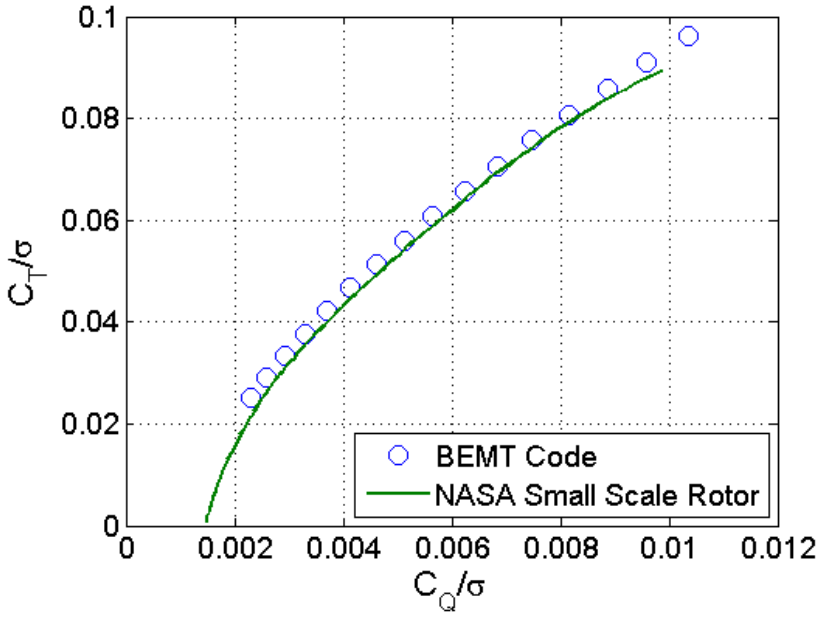


Figure 22: BEMT Validation - Performance Polar

2.3.2 LEWICE Integration

2.3.2.1 *LEWICE Overview*

Integration of LEWICE is the central feature of the ARISP icing model. LEWICE (LEWis ICE) is an ice accretion program written by NASA's Glenn (formerly Lewis) Research Center in Cleveland, Ohio. The program has been extensively validated in the NASA Icing Research Tunnel (IRT) to ensure that it accurately predicts ice accretions on a variety of airfoils (50). LEWICE 3.0, the most recent version, was chosen for this ARISP model due to its accuracy and ease of use. A comprehensive user's manual (46) makes it easy to interface with the program. Furthermore, the program structure is easy to integrate with and allows for some modeling simplification.

The advantage of using this program is that the flow field is calculated within the code. No CFD calculations are required for the code to determine the ice accretion. External calculations of the

flow field are possible in LEWICE, with the program just calculating the ice accretions based on the flow field calculation, but the ARISP code will be focused on the potential flow solver.

2.3.2.2 *LEWICE Limitations*

The primary limitation of the code from a rotorcraft perspective is the low speed limitation. The code is only validated up to Mach 0.45, the maximum velocity of the IRT. For a 6 ft diameter rotor, with a tip speed of 700 ft/sec, LEWICE would only be theoretically accurate up to ~70% radius. Analysis at higher mach numbers is possible, but the code is not validated. The heat transfer coefficient, particularly important for glaze ice shapes, does not behave as expected.

The next limitation of the current application of LEWICE 3.0 is the use of the potential flow solver. Potential flow is able to calculate flowfields at low (< 0.7) and high (> 1.3) V_∞ Mach numbers. It is not accurate in the transonic range, however, because the entire flow is not entirely subsonic or supersonic. The second issue with the potential flow solver is that it is unable to model separation at high angles of attack. This is important for rotorcraft, which may have local areas of high angle of attack. For the ARISP code, which is focused on hovering flight, this is not a large issue, but one that the user must be aware of.

2.3.2.3 *Integration Process*

Integration of LEWICE in the ARISP code is relatively simple and involves three primary steps as shown in Figure 23.

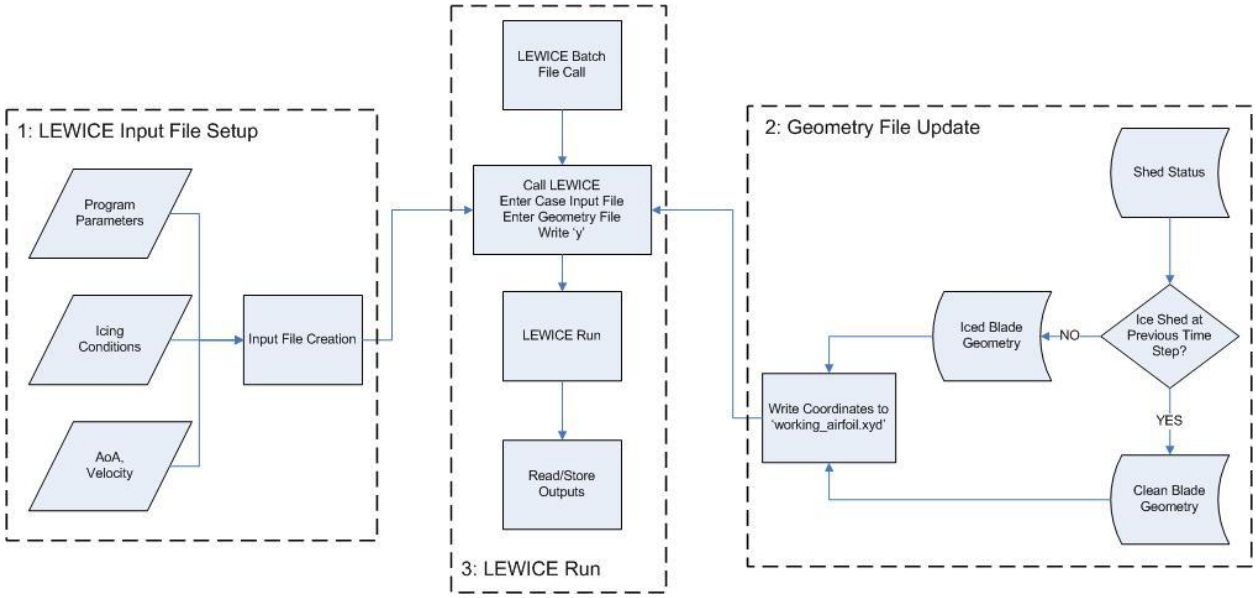


Figure 23: LEWICE Interface

2.3.2.4 *Lewice Input File Setup*

The main LEWICE input file defines program options, icing conditions, airfoil parameters and printing flags. This input file is setup according to Reference (46) and consists of a text file with the extension ".inp". The function "lewice_input_gen.m" formats and arranges the appropriate data into the proper form, and a sample input is shown in Figure 24.

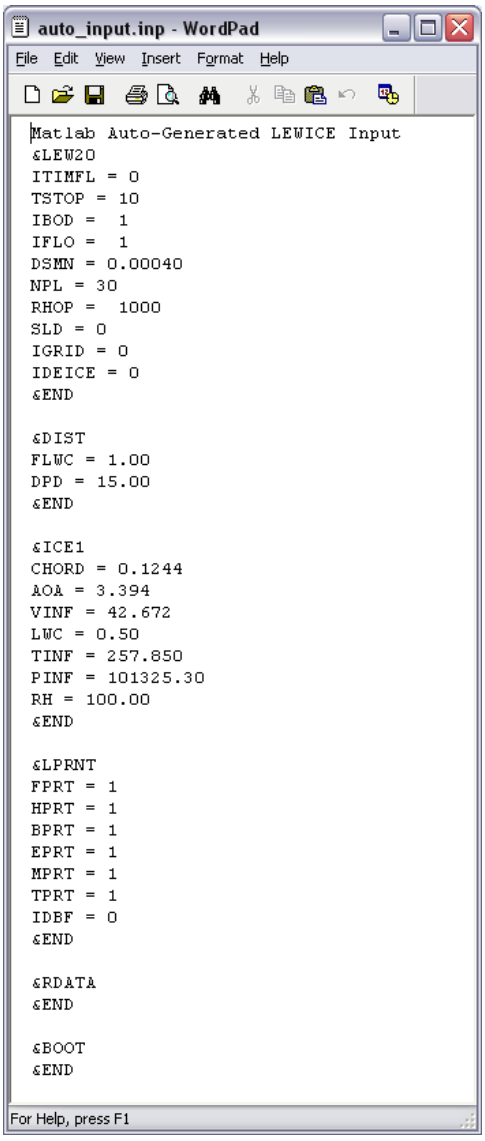


Figure 24: Standard LEWICE Input File

2.3.2.5 *Geometry File Setup*

The second required input file contains the coordinates of the geometry to be studied, non-dimensionalized by chord. Due to ice accretion, shedding and potential for spanwise variation in airfoil type, the airfoil geometry is different for each LEWICE run. Therefore, the coordinates are written to the generic geometry file "working_airfoil_xyd" before calling LEWICE by the function "geometry_input_gen.xyd". This file is constantly updated as the program moves through along the blade span and time.

The process of selecting which geometry to use is based on the shedding history at each blade station. At the program initialization, the clean airfoil geometry is used. Ice is then accreted on the airfoil, creating an iced profile. At the next time step, assuming that the ice is not shed, the iced profile is input into LEWICE to accrete more ice. This procedure is repeated until ice sheds, after which the clean airfoil geometry is used and the process repeats. A schematic of this is shown in Figure 25.

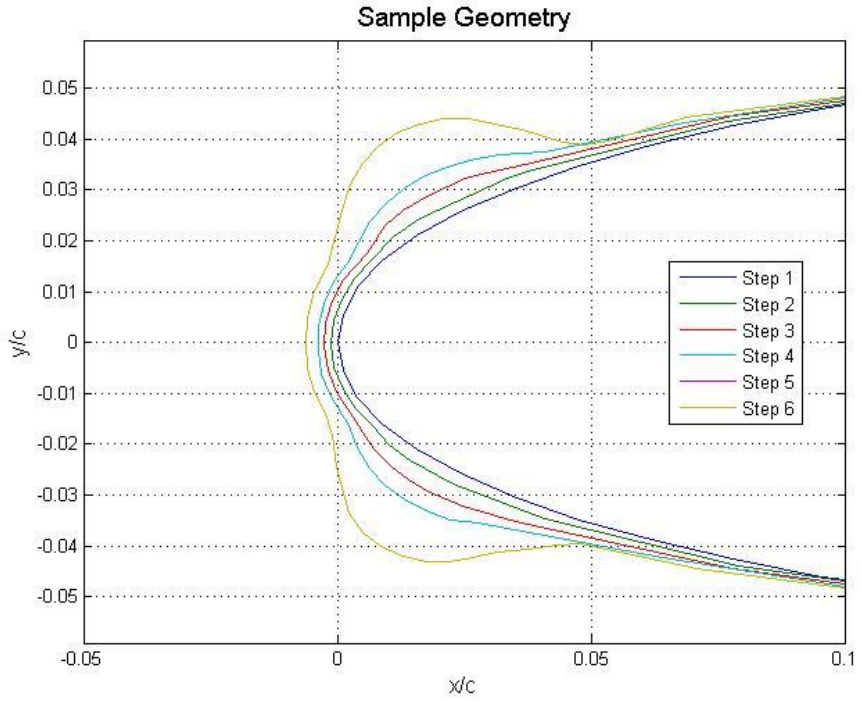


Figure 25: Sample Geometry Input into LEWICE
Closeup on Airfoil Nose

It is important to note that the same ice shapes are not simply repeated over and over until the time reaches the total icing time. The lift and drag coefficients are constantly updated, which modifies the angle of attack distribution across the blade. This subsequently modifies the accreted ice profile and shedding behavior.

2.3.2.6 *LEWICE Call*

Since LEWICE requires two input files, MATLAB cannot call just the executable. Instead, MATLAB automatically calls a batch file, which subsequently calls the LEWICE executable. The batch file, shown in Figure 26, also contains the location of a secondary file containing the input files.

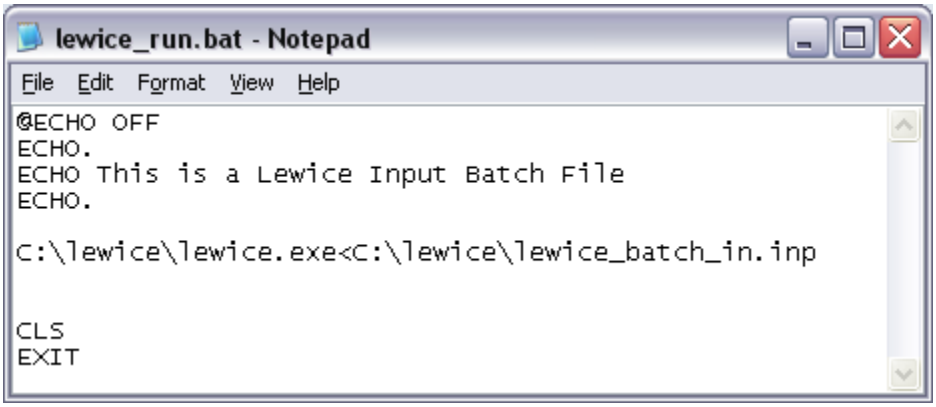


Figure 26: Batch File

The secondary file contains the locations of the automatically generated LEWICE input file and geometry file. The critical part of this file is the third line, which contains a 'y'. This allows the program to bypass the LEWICE warning messages which are generated whenever the user is attempting to run a case outside the validation envelope of the software or certain features of the program are not enabled. For the ARISP code, this is usually the case because high velocities, outside the Mach 0.3 validation envelope of LEWICE, are run. If a case is run within the normal parameters, the 'y' is ignored and the program proceeds as normal.

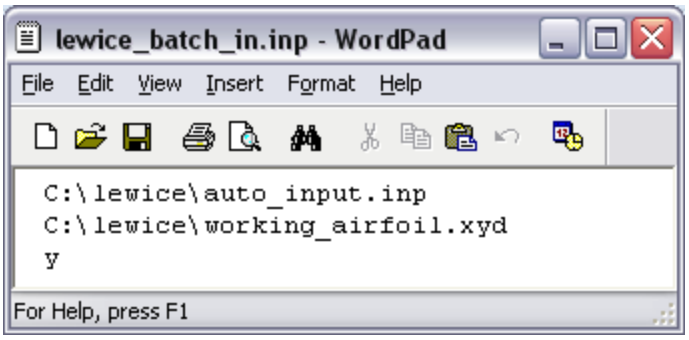


Figure 27: Input Files

After LEWICE runs, output files are updated and stored in the LEWICE directory. The MATLAB script reads and processes the data. A new LEWICE input file is written for the next blade station under analysis, and the process repeats.

2.3.3 Ice Shedding

As explained in Section 1.1.2.2, shedding of ice is an issue inherent to rotorcraft. High centrifugal forces place significant loads on accreted ice, causing it to release from the rotor and potentially impact the fuselage. More importantly for this research, shedding has an effect on rotor performance and ice accretion profile.

2.3.3.1 *Shedding Concept*

The ARISP model of shedding is generally similar to that contained in Reference (55). Accreted ice is subjected to many forces, with the most important shown in Figure 28.

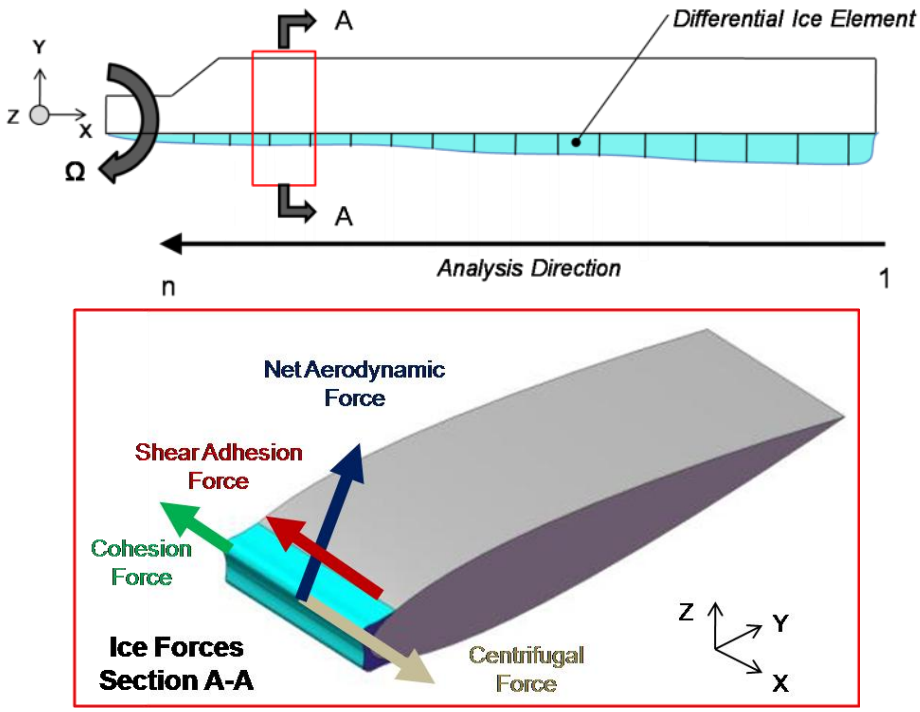


Figure 28: Applied Forces

Reference (62) details a method to quantify the aerodynamic forces in shedding. The aerodynamic forces may be as large as approximately 20% of centrifugal forces on the ice element. For simplicity, aerodynamic forces are neglected in the current analysis. The tip speed for the current research is less than 260 ft/sec, which is lower than the critical Mach < 0.46 as identified by Reference (62). Other forces, including blade vibratory and flexing loads, are also considered to be negligible. Therefore, for the current analysis, only the centrifugal force is considered.

2.3.3.2 *Shear and Tensile Force Calculation*

The centrifugal force is on a differential ice element is given by Equation 2.

$$F_C = m_{ice} r \Omega^2 \qquad \textbf{Equation 2}$$

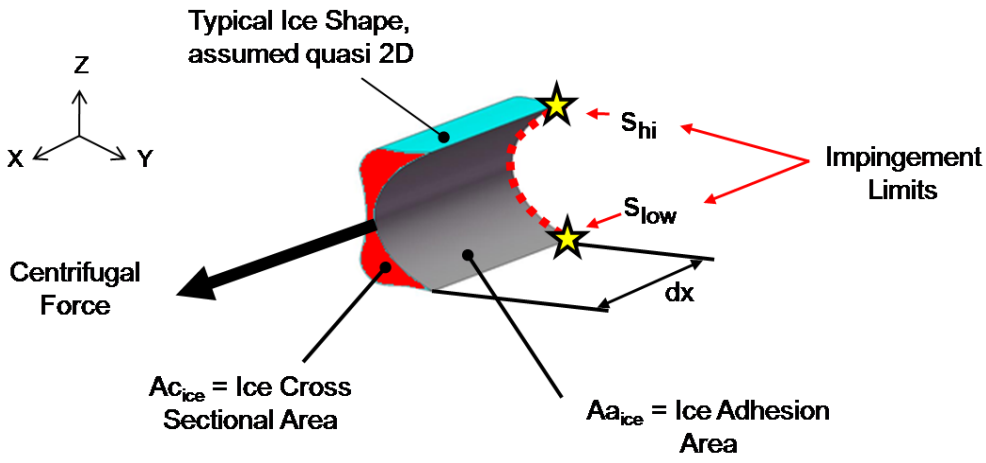


Figure 29: Ice Tensile Force Details and Shear Adhesion

The ice tensile strength (cohesion force) is given by:

$$F_C = \sigma A_c \qquad \textbf{Equation 3}$$

To determine the ice cross sectional area, A_c, results from LEWICE output file 'thick.dat' are read into MATLAB. The non-dimensional area is contained in the 4th column of the file and is rescaled to the dimensional area. Due to ice shedding and the piecewise analysis process, the cross sectional area is not always equal to that calculated by LEWICE. If ice did not shed at the previous time step, the newly calculated ice cross sectional area is added to the previously stored value. Details of this process are contained in Figure 30.

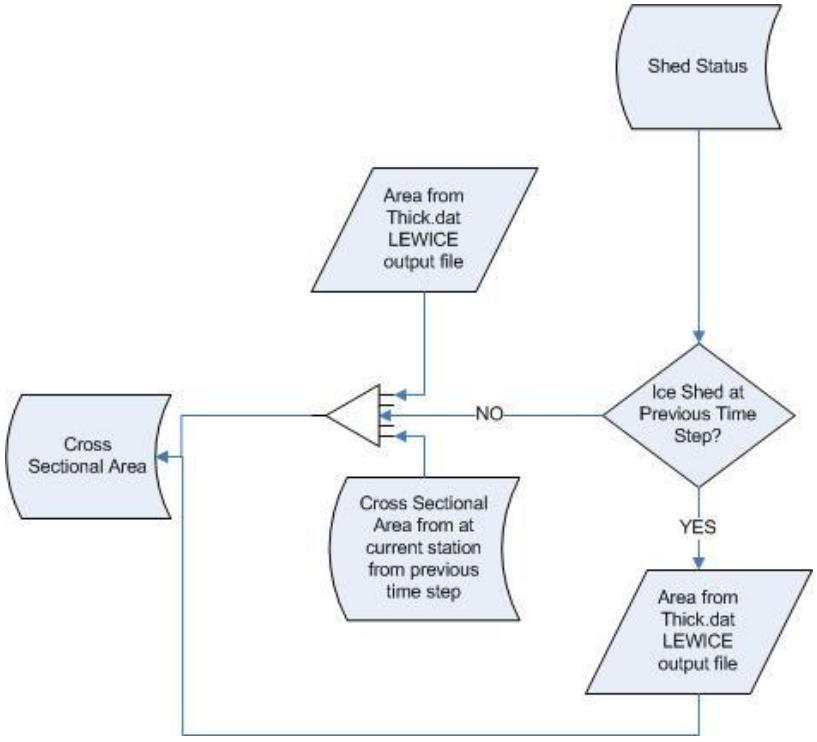


Figure 30: Ice Cross Sectional Area Summation

The ice shear force is given by:

$$F_C = \tau A_s \qquad \textbf{Equation 4}$$

As with the cross sectional area, the shear adhesion area A_s, is calculated from LEWICE output files. One output is the impingement limits, which are used to determine the surface length, s_{ice}, of the accreted ice. Both upper and lower limits are contained in the 'imp.dat' file, and are reported for a variety of coordinates, including non-dimensional absolute coordinates x/c, y/c, surface length coordinates based on the stagnation point, s_{hi}/c, s_{low}/c, and surface length coordinates based upon the leading edge, s_{hiLE}/c, s_{lowLE}/c. An impingement limit tracker is used to determine the correct limits, taking into consideration ice shedding. The result of tracking is used in Equation 5 to generate the iced length for the differential ice element.

$$s_{ice} = s_{hiLE} + s_{lowLE} \qquad \textbf{Equation 5}$$

This parameter is then used in Equation 6 to calculate the adhesion area.

$$A_S = s_{ice}\, l_{ice} \qquad \textbf{Equation 6}$$

2.3.3.3 *Ice Density Calculation*

One critical parameter in Equation 2 is the mass of ice. Since the volume of the accreted ice is known from LEWICE outputs and spanwise length of the differential element of ice, the only required parameter is the ice density. This density of impact ice is dependent on a variety of parameters, including temperature, particle size and velocity. The ARISP model includes a variety of models that the user can select from for analysis. These models were derived from References (63), (64), (65) and (66). For the application to the ARISP code, the results are generally similar.

Table 7: Ice Density Calculations
T = -10°C, LWC = 1.5 gr/m³, MVD = 20 μm

Author	Reference	Ice Density (kg/m³)
Ashton	(63)	918
Macklin	(64)	900
Stallabrass	(65)	884
Jones	(66)	910
LEWICE	(46)	917

2.3.3.4 *Ice Shear Adhesion Strength*

One critical parameter in shedding is the ice shear adhesion force. Many researchers have attempted to determine this parameter for a variety of materials. Research has been invested into the accurate determination of this parameter by a variety of authors, but data is characterized by significant scatter. Published results for the adhesion strength of a particular material may vary as much as 200% due to variations in test method and test facility (67), (68), (69). Details on the current research into the shear adhesion strength of ice is contained in Section 4.0.

2.3.3.5 *Ice Tensile (Cohesive) Strength*

As shown in Figure 29, ice tensile forces are also important to ice shedding calculations. As with shear adhesion strength calculations, a variety of authors have attempted to quantify the tensile strength of ice. As expected, it is a strong function of icing conditions. The ARISP model uses the relationship developed in Reference (70).

2.3.3.6 *Implementation*

Shedding is calculated in the ARISP model by starting at the rotor tip, as shown in Figure 28. At the beginning of the program, the shedding index for each element is set to 0, indicating that ice has not shed. A value of 1 indicates that ice is shed due to centrifugal forces, while a value of 2 is reserved for ice shed by an active ice protection system. This feature has not been fully enabled, but will include the effects of ultrasonic and electrothermal ice protection systems.

Elements are added together as the analysis continues, as shown in Figure 31. The red section indicates the active "super element." For each step, ice properties including the shear adhesion area, ice mass and center of gravity are updated. The furthest inboard cross sectional area of the super element under study is used as the cohesion area.

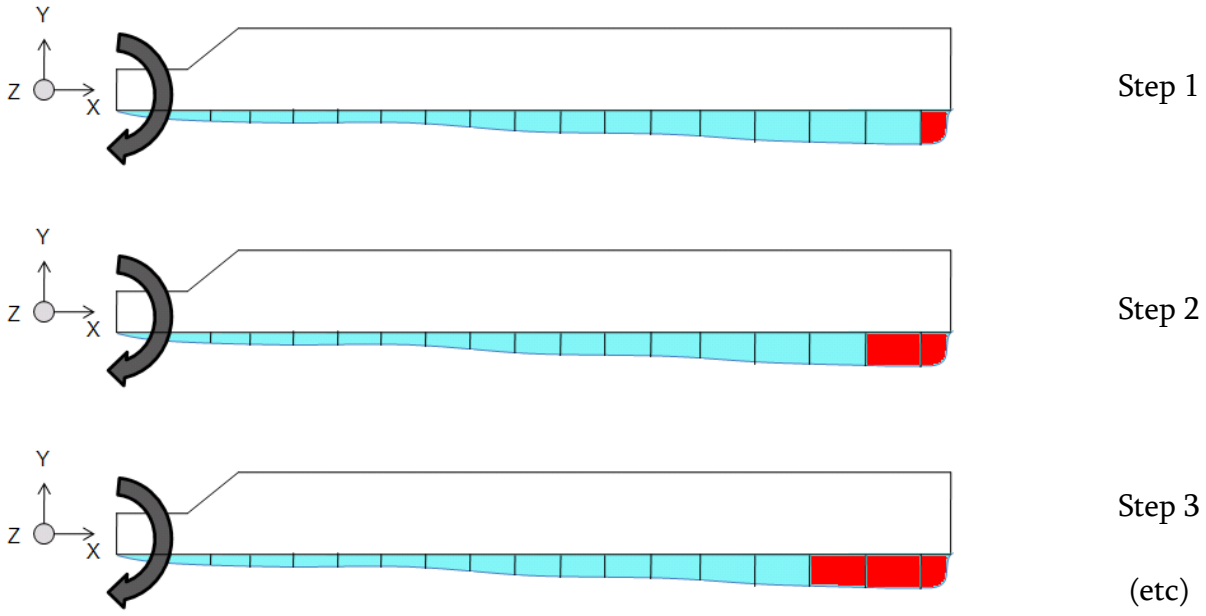


Figure 31: Shedding Element Implementation

At each element, the following parameters are checked:

$$Shear\ Shed = Rotor\ Shear\ Adhesion\ Strength - Ice\ Shear\ Force \quad \textbf{Equation 7}$$

$$Tensile\ Shed = Ice\ Tensile\ Strength - Ice\ Tensile\ Force \quad \textbf{Equation 8}$$

If both "Shear Shed" and "Tensile Shed" variables are less than zero, the entire ice block is assumed to be shed from the rotor. The shed index is set to 1 for the particular element and time step. The model will then move to the rotor element and/or time step.

2.3.4 Iced Rotor Performance Degradation

Accreted ice has a significant effect on the performance of a rotor. Airfoil performance is lowered, sometimes significantly, by increased C_d and decreased C_l. This has an effect on the rotor inflow solution from the BEMT module. This affects the local angle of attack, and therefore accretion of additional ice on the rotor.

2.3.4.1 *Icing Correlations*

Many authors have attempted to quantify the change in airfoil performance due to accreted ice. Three correlations are included in the 'Delta_Cl' and 'Delta_Cd 'functions of the ARISP code, and the user can select which one to employ in an analysis. All data was taken in icing wind tunnels in two dimensional airfoils. The range of parameters used in the correlation development is shown in Table 8, contained in Reference (51).

Table 8: Range of Parameters Used in Correlations Development, From Reference (51)

	Gray	Bragg	Flemming
Temperature (°C)			
min	-17.7	-17.7	-20.0
max	-3.8	-12.2	7.7
MVD (μm)			
min	11.3	7.0	11.0
max	19.0	19.0	50.0
LWC (gr/m^3)			
min	0.39	0.50	0.24
max	2.00	1.86	3.80
Velocity (ft/sec)			
min	183.3	256.6	287.7
max	403.3	403.3	745.1
Icing Time (min)			
min	3.0	1.0	0.3
max	17.7	27.0	5.0

2.3.4.1.1 *Gray's Correlation*

Gray's correlation is included in the ARISP model. The following equations, taken from Reference (51), indicate the complexity of the correlations. The integration of the correlations to the ARISP model has not been validated, so caution should be used when applying them.

$$\Delta C_D = \left[8.7 * 10^{-5}\left(\frac{\tau u}{c}\right)\sqrt{LWC * \beta_{max}}(32 - T_o)^{0.3}\right] *\ldots \quad \textbf{Equation 9}$$

$$(1 + 6\{[1 + 2.52r^{0.1}\sin^4(12\alpha)] *\ldots$$

$$\sin^2\left[543\sqrt{\mathrm{LWC}}\left(\frac{\mathrm{E}}{32-\mathrm{T_o}}\right)^{\frac{1}{3}} - 81 + 65.3\left(\frac{1}{1.35\alpha} - \frac{1}{1.35\alpha_i}\right)\right] -\ldots$$

$$\frac{0.17}{\mathrm{r}}\sin^4(11\alpha)\})$$

where

u = free stream velocity, mph

τ = icing time, s

β_{max} = maximum local collection efficiency

r = leading edge radius, %chord

To = freestream temperature, °F

α = angle of attack, °

α_i = induced angle of attack, °

2.3.4.1.2 *Bragg's Correlations*

Reference (71) details the experiments and results of Bragg's icing experiments. The following equations are incorporated in the ARISP model. The correction for larger accumulation parameters and total catch efficiencies is described in Reference (51).

$$\Delta C_d = 0.01[15.8 * \ln\left(\frac{k}{c}\right) + 28000 * \mathrm{Ac} * \mathrm{E} + \mathrm{I}, \quad Ac * E < 0.004 \quad \textbf{Equation 10}$$

$$\Delta C_d = 0.01[15.8 * \ln\left(\frac{k}{c}\right) + 1171 * \sqrt{\mathrm{Ac} * \mathrm{E}} + \mathrm{I}, \quad Ac * E \geq 0.004 \quad \textbf{Equation 11}$$

$$C_{d_{iced}} = (1 + \Delta C_d)C_{d_{clean}} \quad \textbf{Equation 12}$$

where,

$I = drag\ constant, 184\ for\ NACA\ 0012$

$$Ac = Accumulation\ Parameter = \frac{u\tau LWC}{\rho_{ice}\, c} \quad \textbf{Equation 13}$$

$E = Total\ Collection\ Efficiency$

$\frac{k}{c} = Particle\ Roughness\ Diameter\ to\ Chord\ Ratio$

$\tau = Icing\ Time, seconds$

2.3.4.1.3 *Flemming's Correlations*

Reference (26) contains details of the correlation generated by Flemming and Lednicer at the NRC High Speed Icing Wind Tunnel in Ottawa and the Ohio State University Transonic Airfoil Facility. Of the three studies, this is the most comprehensive, with over 4500 data runs, the correlations have the most applicability to rotorcraft applications due to the high speed ice accretions. Silicone rubber molds were made of ice accreted to a total of 10 airfoils. Later the molds were used to create a casting for wind tunnel studies. In addition to detailing the effect of ice on airfoil drag, the authors studied lift and pitching moments. Although they are small compared to the increase in drag, the lift degradation effects are included in the ARISP model to better predict rotor performance. Pitching moments are not required for any calculations and are therefore not included.

$$\Delta C_d = K_{DI}\left[0.00686 * K_o\left(\frac{k}{c}\right)^{1.5}(\alpha+6) - 0.313\left(\frac{k}{c}\right)^2 + K_D 0.006M^{2.4}\right] * \left[\frac{LWC\left(\frac{c}{0.1524}\right)^{0.2}\tau_c}{\left(\frac{c}{0.1524}\right)^{1.2}}\right], \text{ for glaze and beak ice} \quad \textbf{Equation 14}$$

$$\Delta C_d = \left[0.158 * \ln\left(\frac{k}{c}\right) + 175\frac{V}{\rho_{ice}\,c}LWC * \tau_c\right] * \left[\frac{\alpha+6}{10}\right] * C_{d_{clean}}\text{ , for rime ice} \quad \textbf{Equation 15}$$

$$\Delta C_l = \left[-0.01335K_o\left(\frac{t}{c}\right) * (\alpha + 2 + K_{LI}(0.00555)(\alpha-6)^2)K_L\right] * \left[\frac{LWC\left(\frac{c}{0.1524}\right)^{0.2}\tau_c}{\left(\frac{c}{0.1524}\right)^{1.2}}\right], \text{ for rime, glaze and beak ice} \quad \textbf{Equation 16}$$

where,

$$K_o = Modified\ Droplet\ Inertia\ Parmeter = K(1 + 0.0967R_u^{0.6397}) \quad \textbf{Equation 17}$$

$$\tau_c = corrected\ icing\ time$$

$K_D, K_{DI}, K_L, K_{LI}\ are\ f(T,\alpha)$, explained in Reference (72)

2.3.4.2 *Implementation*

The implementation of the airfoil performance adjustments is shown in Figure 32. All performance degradation is calculated relative to the clean airfoil performance. This is important, as the empirical ice performance equations are not accurate for small icing times. If shedding was not detected on the time step, the time since the previous shed at the station is determined for the degradation calculations. New sectional C_l and C_d coefficients are calculated and stored in dynamic databases for use at the next time step. If the ice was shed, the clean sectional performance coefficients are stored in the database for the next iteration of the BEMT module.

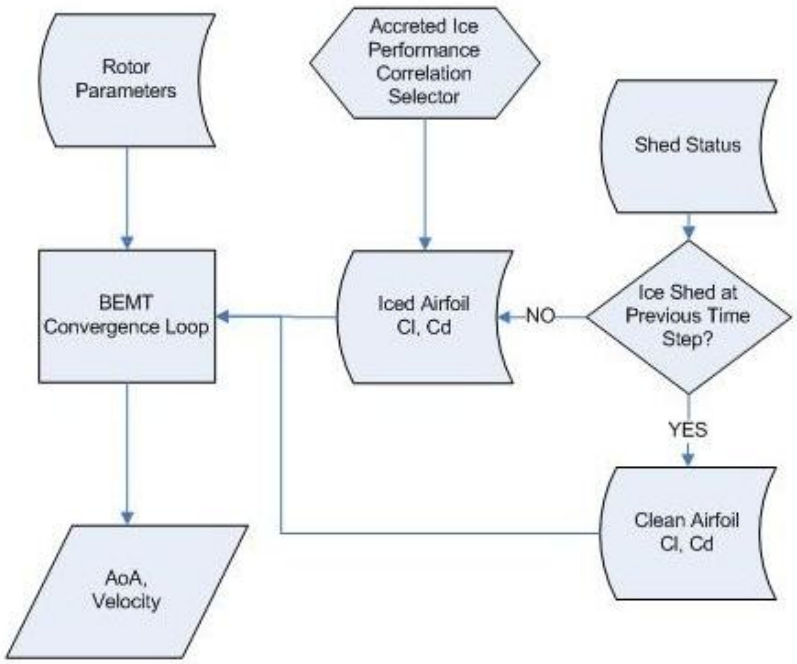


Figure 32: Iced Airfoil Performance Adjustment Implementation

2.3.5 Initial ARISP Model Validation

A validation effort was undertaken for the stepwise icing analysis procedure utilized by the ARISP to ensure that interface between LEWICE and MATLAB is properly handled. During an average

data run, 600 system calls of LEWICE are made from the MATLAB script, so proper passing of data between the two is vital. More importantly, the accuracy of the stepwise icing analysis, as illustrated in Figure 15, needed to be analyzed.

2.3.5.1 *Static Shape Prediction*

The first validation step focused on the stepwise icing analysis procedure. The results of the ARISP model various time steps were compared a standalone analysis with the same conditions using the LEWICE automatic time stepping procedure. Initially this parameter was set by the user in the ARISP code and studies were run to find the optimal step size. Due to significant variations in ice shape with numbers of steps, especially with glaze ice, the automatic selection function of LEWICE was integrated. The LEWICE automatic time step selection process is based upon LWC, velocity, total icing time, chord and ice density (46) and is shown below. For the ARISP application, the tip speed and average rotor chord are used in the calculation.

$$Preliminary\ Time\ Step = N = \frac{LWC * V * Time}{chord * \rho_{ice}} \qquad \textbf{Equation 18}$$

Where:

LWC = liquid water content (gr/m^3)
V = velocity (m/s)
Chord = airfoil chord (m)
ρ_{ice} = ice density (kg/m^3)

$$Second\ Time\ Step = N2 = \frac{Time}{60} \qquad \textbf{Equation 19}$$

$$Final\ Time\ Step = Max[Min(N,30), Min(N2,15)] \qquad \textbf{Equation 20}$$

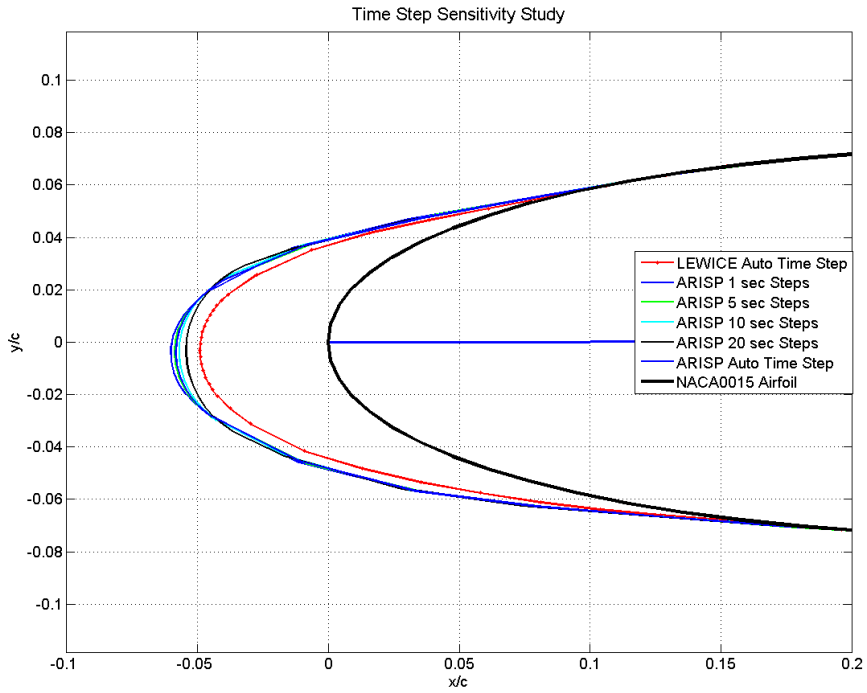


Figure 33: Time Step Sensitivity Study Sample
Time = 120 s, T = -15°C, LWC = 0.5 gr/m³, MVD = 20 μm, V = 300 ft/sec, AoA = 1.5°

The figures show that with the LEWICE automatic time step selection process in place, there is a difference of about 10% on the stagnation thickness between the ARISP model and the results obtained by running LEWICE independently. For the size rotors used in the AERTS test chamber, this result would be indistinguishable.

2.3.5.2 *Model Scale Icing Tunnel Shape Comparison*

The second icing model validation step was to evaluate the accuracy of cross sectional ice shapes to published literature. ARISP analytical results were compared to experiments undertaken in the IRT in 1988 (73). Since the test involved a model rotor in forward flight, a collective pitch setting was determined for each case to match the reported sectional average angle of attack. The sectional average velocities, equal to those in hover, were used in calculations. The concept simplifying forward flight icing analysis by averaging angle of attack and velocity has been tested in the IRT

with favorable results. As shown in Figure 34, the process yielded accurate results for the accreted ice shape for the inboard regions of the rotor. A comparison to an outboard ice accretion is provided in Figure 35. The stagnation thickness was correctly predicted, but the horns were not.

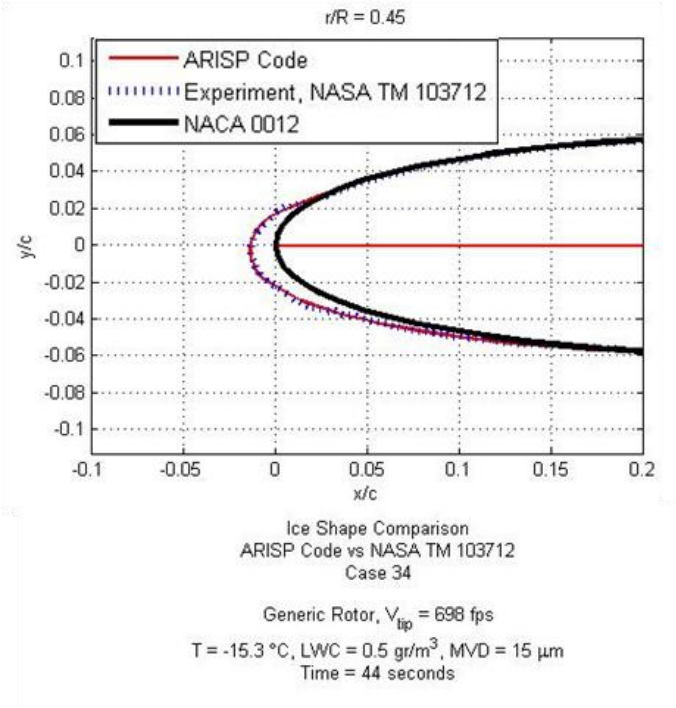


Figure 34: ARISP Comparison to NASA TM 103712 Case 34

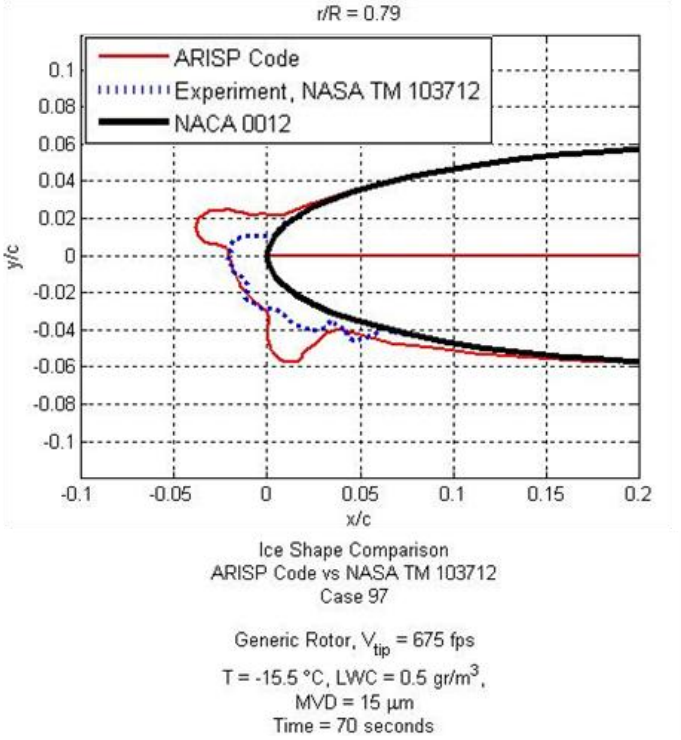


Figure 35: ARISP Comparison to NASA TM 103712 Case 97

2.3.5.3 *Full Scale Outdoor Hover Spray Rig Comparison*

Additional correlations were conducted with full scale icing flight trial data. The ARISP model was compared with results from Flight E of the Helicopter Icing Flight Test (HIFT) program, conducted with a UH-1H aircraft at the NRC Helicopter Spray Rig (74) with the ambient temperature corrected from -19˚C to -14˚C per Reference (75).

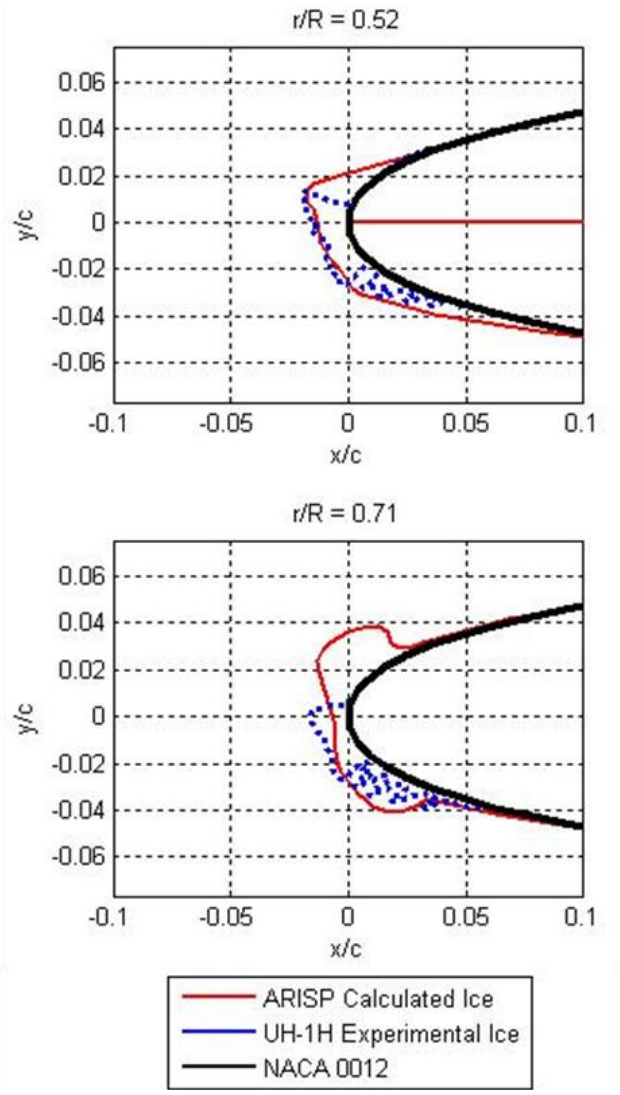


Figure 36: ARISP Comparison to UH-1H HIFT Flight E

As with the scale rotor correlations, the ARISP model provides relatively accurate results for the inboard regions of the rotor. The reason for ice over prediction at the tip is currently unknown, but

may be related to droplet bounce, centrifugal effects and a non-uniform LWC (75). A potential source of error lies in the fact that LEWICE uses a potential flow panel code to calculate ice accretion. As such, it assumes that flow is inviscid, incompressible and irrotational. The irrotational assumption may not allow the model to capture the complex rotating flow near the rotor tips. Finally, the flow velocity is higher than the maximum LEWICE validation envelope of 475 ft/sec (46). Therefore, for AERTS calibrations, ice shapes will be primarily compared at rotor stations with a velocity lower than this value (approximately 0.6r on a UH-1H).

2.3.5.4 *Ice Extent Calculation*

Further investigations of the model focused on the tip region of the blade. Predictions of ice extent were determined based upon Reference (26). A code was written that calculates the ram temperature rise at each blade section per Equation 21 with an empirically calculated recovery factor given by Equation 22. Both equations are taken from Reference (26). An iteration loop was then written to calculate the location of ice extent on the blade.

$$T_{corrected} = \left[(T_s + 273.15) * \left(1 + \frac{\gamma - 1}{2} rM^2\right) + 0.33(\alpha - 6)\right] - 273.15 \quad \textbf{Equation 21}$$

where:

T_s static temperature
γ ratio of specific heats
r thermodynamic recovery factor (see below)
M free stream Mach number

$$r = 0.0944 * M * LWC^2 + 0.285 * M * LWC + 0.98 * M^3 - 1.33 * M - 0.0283 * LWC^2 - 0.115 * LWC + 1.077 \quad \textbf{Equation 22}$$

The code was checked against icing flight trial data and published ice extent predictions with good results prior to implementation in the current research as shown in Figure 38. The validation data was from the HIFT icing trials. Once the code was verified, studies were run with a typical AERTS test rotor for a few different test conditions.

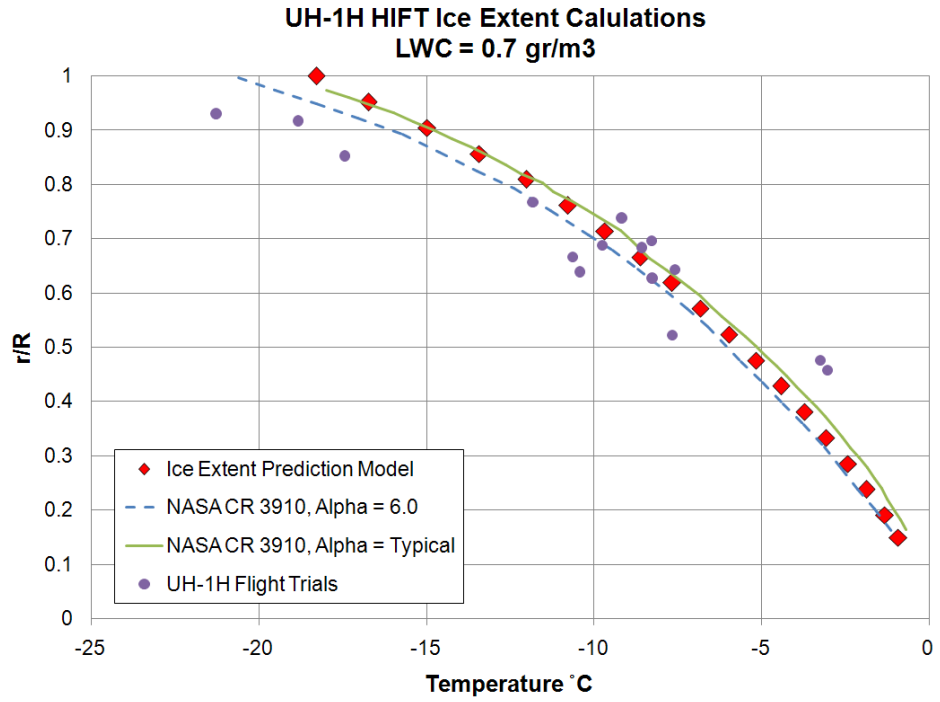


Figure 37: Ice Extent Code Validation, UH-1H Rotor

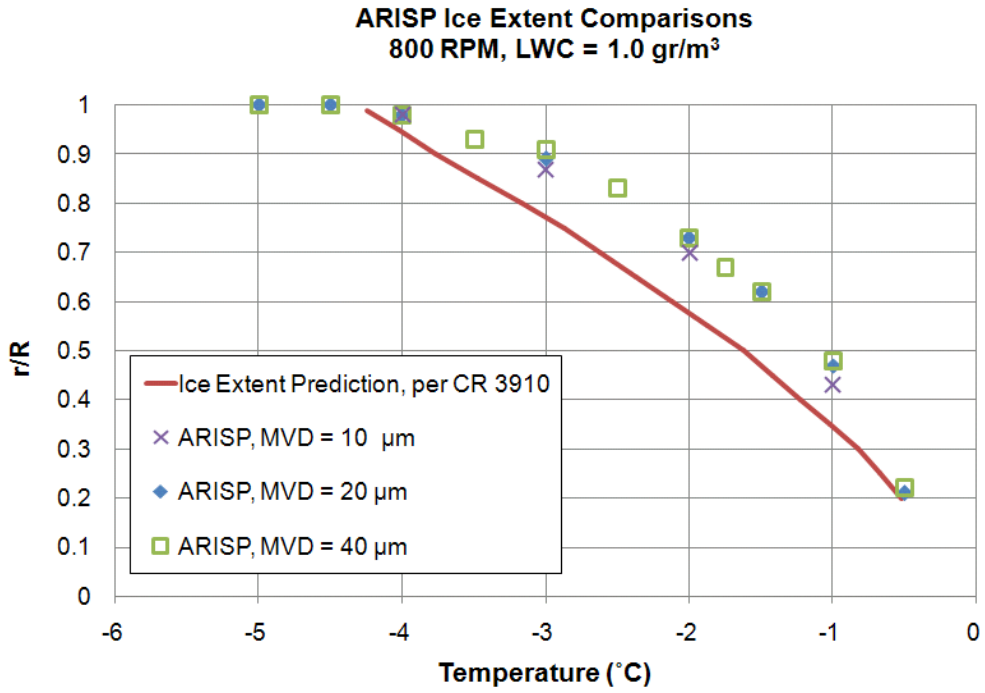


Figure 38: Ice Extent Code Validation, AERTS Test Rotor
(note x axis scale change relative to previous figure)

Overall correlation is good, especially for the full scale UH-1H. For the smaller scale AERTS test rotor, the ARISP model over predicted ice extent. This was expected based upon the shape correlations.

2.3.5.5 *Validation Conclusions*

The validation process has shown that the ARISP model has the ability to predict rotorcraft icing trends. Shape correlations and ice extent analysis for both full and sub scale rotors have shown that the ARISP model is relatively accurate. Issues arise in the outboard sections of the blade due to an over prediction of the accreted ice mass. Limits on ice shape correlations are not seen as major restrictions for the icing code. Since the AERTS Facility is primarily intended to evaluate for ice protection systems, the location of ice is more important than the exact ice shape. Accurate impingement limits help size ice protection systems, so if the ARISP model can predict them, it is useful. Studies will continue into improving outboard ice shape predictions.

3. AERTS Facility Design, Construction and Calibration

3.1 Requirements

Realistic icing tests require the design and construction of a facility with the capability of generating an artificial, but accurate icing cloud around a test rotor. During the design of the Adverse Environment Rotor Test Stand (AERTS) facility, consideration was given to rotorcraft icing test requirements and other test facilities in industry. The laboratory was designed to complement existing facilities in an attempt to further research in the field of rotorcraft icing while operating within a university infrastructure. As such, many components were donated from industry and were assembled in a unique way to create a niche facility designed for initial evaluations of ice protection systems and other rotor components. Table 9 describes the primary icing requirements that the facility set out to meet.

Table 9: Primary AERTS Facility Icing System Requirements

Parameter	Minimum	Maximum	Margin	Units
Temperature	-30	Ambient	± 1.0	°C
Mean Volumetric Diameter (MVD)	10	50	± 2.0	μm
Liquid Water Content (LWC)	0.5	2.5	± 0.2	gr/m^3

Details of the facility design, including subsystem configurations, are contained in Appendix A. A secondary rain erosion system was incorporated into the AERTS Facility, but the details of this will be not covered in this thesis.

3.2 CFD Modeling of the AERTS Chamber

3.2.1 Objectives

The AERTS Facility has a unique combination of an icing system and rotor in a relatively small freezer, which requires investigation. The rotor is mounted inside of an enclosed test chamber, which forces it to operate with ground, ceiling and wall planes. This affects rotor performance due to significant flow recirculation. More importantly, the recirculation affects the icing cloud. In most icing facilities, such as icing tunnels, supercooled droplets pass by the model and are removed from the test environment if they do not accrete on the model. In the AERTS facility, the droplets are not removed, and instead circulate around the test chamber. This has implications for the icing cloud, as the supercooled droplets may freeze-out into ice crystals, creating a non-uniform icing cloud. The crystals then erode ice shapes, negatively affecting attempts at accretion model validation (14).

In order to generate a basic understanding of the interaction between the rotor, walls and supercooled water droplets, some CFD modeling was undertaken. The studies were not intended to provide an exact solutions for the AERTS chamber. Rather, they were designed to enhance the understanding of the flowfield and droplet trajectories around the test rotor.

3.2.2 Model Setup

Computational Fluid Dynamics (CFD) was used to analyze the laboratory for some proposed rotor configurations. Solidworks Flow Simulation was chosen for the studies because the software is integrated with the Solidworks solid modeling package in which the AERTS chamber was designed.

The software uses a Farve averaged Navier Stokes solver with a κ-ε turbulence model and rotating systems can be modeled with a rotating fluid subdomain (76). The mesh is structured in the sense that it is a Cartesian code but arbitrary geometry is handled with linear facets to approximate the solid fluid boundary. It is generated by first meshing the fluid volume, and then going through each row of cells produced and approximating the solid/fluid boundary with linear facets on the rectangular cells. Coarse approximations are refined by dividing problematic cells by 8 and re-faceting. Small sliver faces or other CAD surface problems do not affect meshing.

Rotor geometry is specified inside of a rotating fluid subdomain in the model. The subdomain must be a body of revolution completely containing rotor components with clearance above, below and at the outboard radial edges of the volume. Cases with rotating subdomains are solved iteratively with relaxation to properly resolve the flow at the rotating/non-rotating boundary.

The icing nozzles are located in the ceiling of the model, per their actual installation. The nozzle flow was approximated with an appropriate inlet velocity, which was measured in the lab. Two nozzle settings were studied, and the parameters are detailed in

Table 10.

Table 10: Nozzle Velocities

Air Pressure Setting	Distance from Nozzle (in)	Velocity (ft/sec)
High (45 psi)	7	110.2
	20	48.9
Low (20 psi)	7	79.5
	20	32.2

Vents are provided to the simulated chamber to prevent problems with pressure rising in the control volume. The vents were placed at the bottom of the test chamber, in the same position as the AERTS drainage system. The drainage system is open to the remainder of the laboratory, so the system also serves a pressure release system in the real system. The initial mesh utilized a total of 1,152,000 elements, which was refined by the solver to over 1,800,000 elements. Important features are shown in Figure 39.

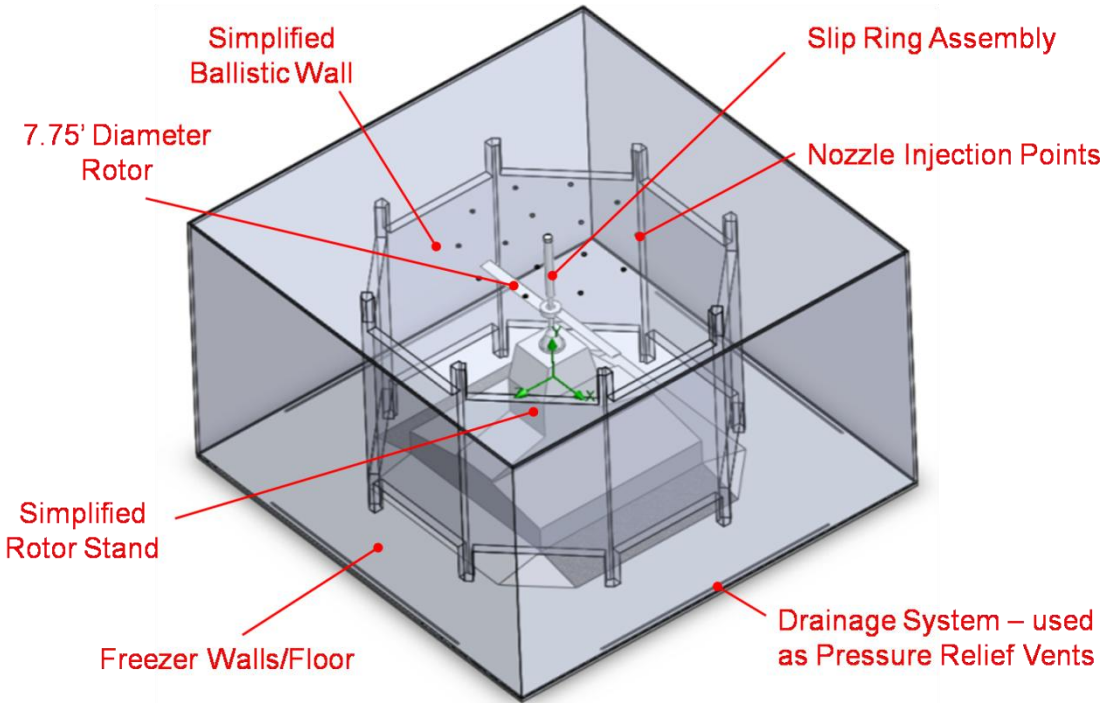


Figure 39: CFD Model Setup

3.2.3 Model Results

3.2.3.1 *Major Flow Patterns*

Initial studies focused on general flow patterns around the rotor. Significant recirculation was expected due to the small size of the test chamber and the ballistic wall configuration. Figure 40 shows the velocity distribution for a typical test case. In addition to the typical tip vortex structure, the system is fed by the by nozzle exit velocities of up to 110 ft/sec when operating at high air

pressures. The nozzle flow rapidly dissipates, but has a velocity on the order of magnitude as the rotor induced velocity.

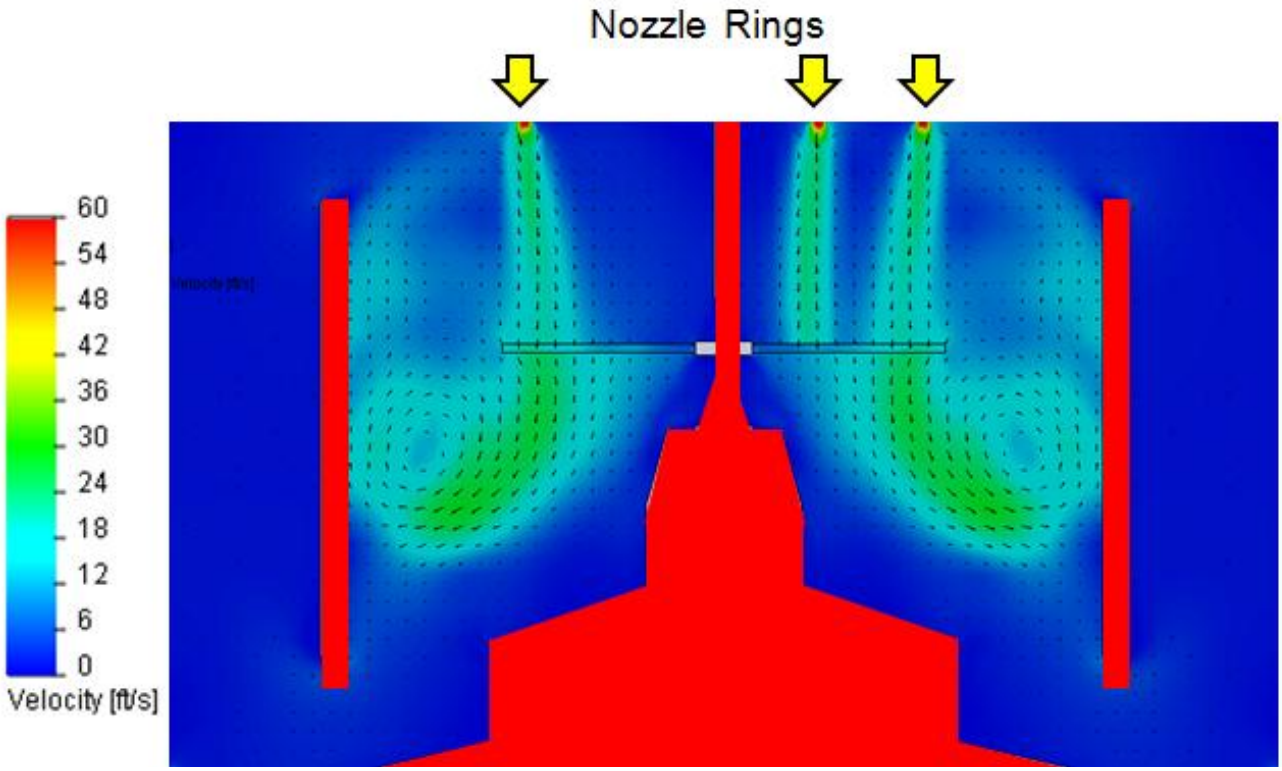


Figure 40: XY Plane Total Velocity Vectors and Contours
CFD Model Cross Section of AERTS
(7.75' diameter rotor, 600 RPM, 2.5° Collective)

The tip vortex structure strength is increased by collective pitch or rotor RPM. Results for calculations at 1000 RPM are provided in Figure 41. Discontinuities above and below the rotor plane are seen in the velocity profiles. This is due to the modeling of the rotating fluid sub domain, which requires significant relaxation for convergence.

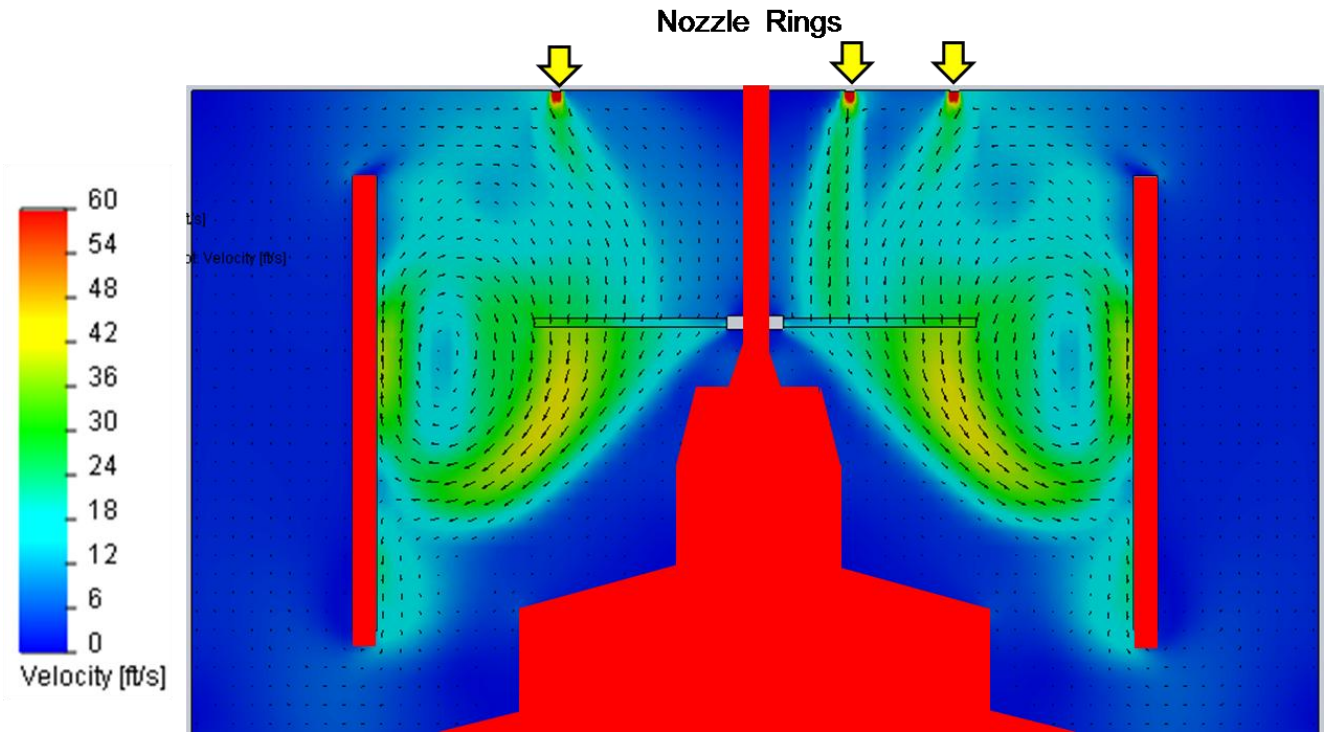


Figure 41: XY Plane Total Velocity Vectors and Contours
CFD Model Cross Section of AERTS
(7.75' diameter rotor, 1000 RPM, 2.5° Collective)

Another view of the nozzle effects on rotor plane aerodynamics is provided in Figure 42. The locations of the 15 nozzles are clearly seen. The effects on swirl can be seen, as the entire flowfield inside of the ballistic wall rotates in the direction of the rotor due to viscous effects. The flow outside of the ballistic wall has very little velocity.

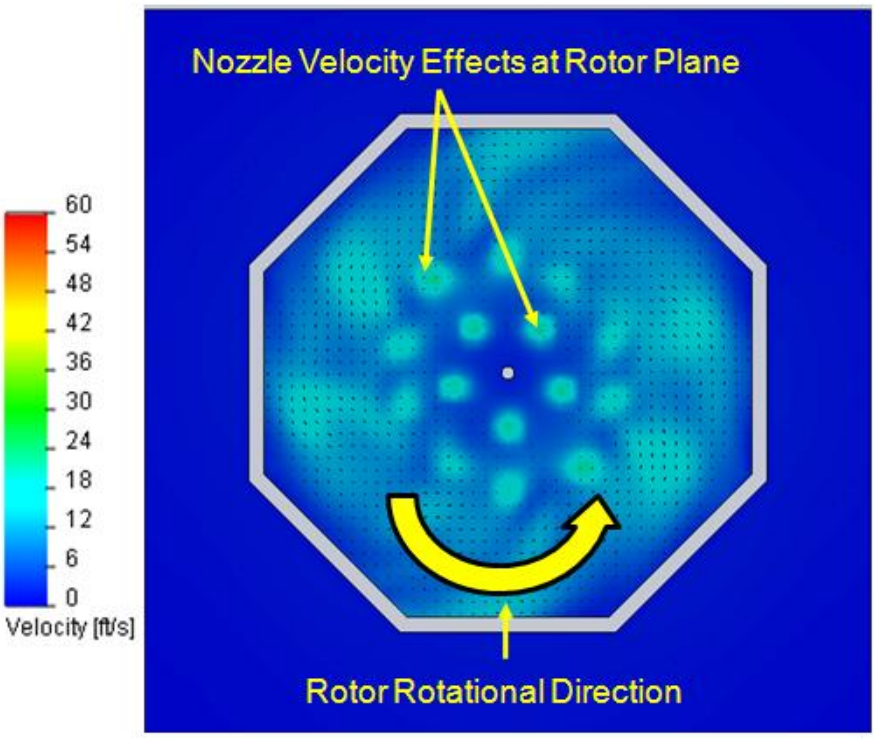


Figure 42: Nozzle Effects at Rotor Plane
CFD Model Cross Section of AERTS,
(7.75' diameter rotor, 600 RPM, 2.5° Collective)
showing total velocity vectors and contours

3.2.3.2 *Nozzle Flow Tubes*

The most important aspect of the modeling is the prediction of particle trajectories. For an icing facility, it is desired to have a uniform distribution of supercooled droplets throughout the test chamber. Since droplets are injected into the AERTS facility at 15 discreet nozzle positions, they may not evenly distribute themselves around the rotor system. Each droplet is small enough to be significantly influenced by the flow field, so they may get caught in the tip vortex system.

Initial studies were performed with flow tubes, which follow streamlines in the facility. A stream tube from each nozzle is shown in the Figure 48. The color of the stream tube indicates the velocity. It is evident that the tip vortex structure dominates controls the flow from the nozzles,

keeping it within the ballistic wall structure and preventing it from recirculating behind the walls. Regardless of whether the flow emanates from an inner or an outer nozzle, the flow predominantly remains inside the ballistic wall, as shown in Figure 44 and Figure 45.

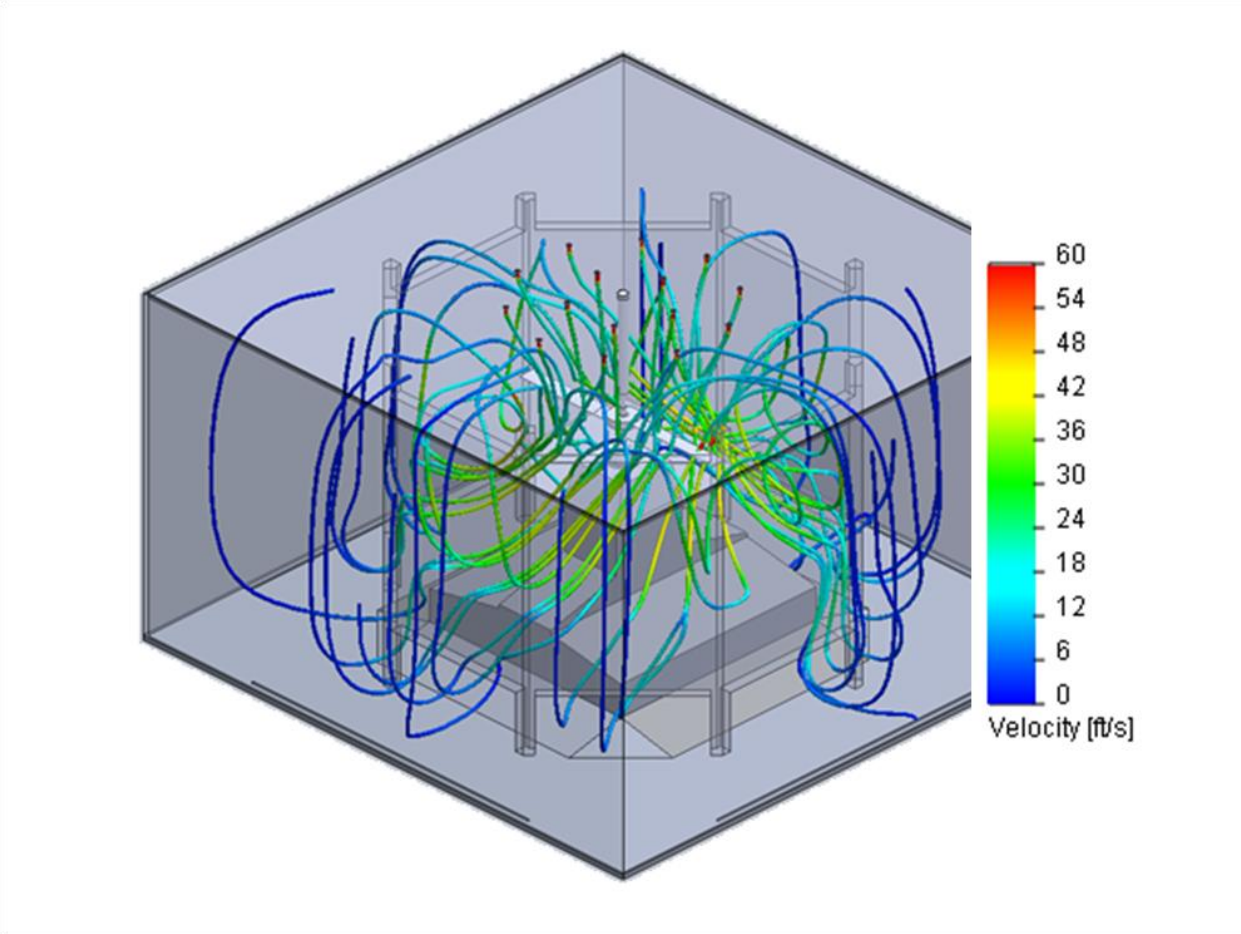


Figure 43: Flow Tubes
CFD Model of AERTS,
(7.75' diameter rotor, 600 RPM, 2.5° Collective)
showing recirculation around the ballistic wall and rotor support structure with stream tubes

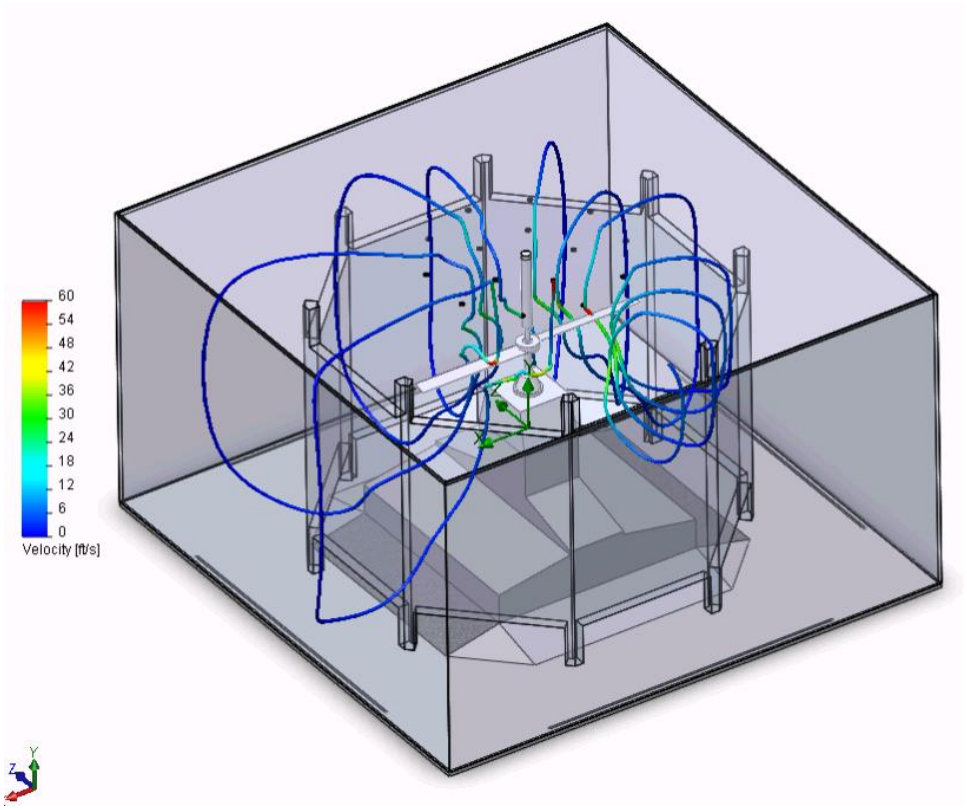


Figure 44: Stream Tubes from a Single Inner Nozzle
CFD Model Cross Section of AERTS,
(7.75' diameter rotor, 600 RPM, 2.5° Collective)

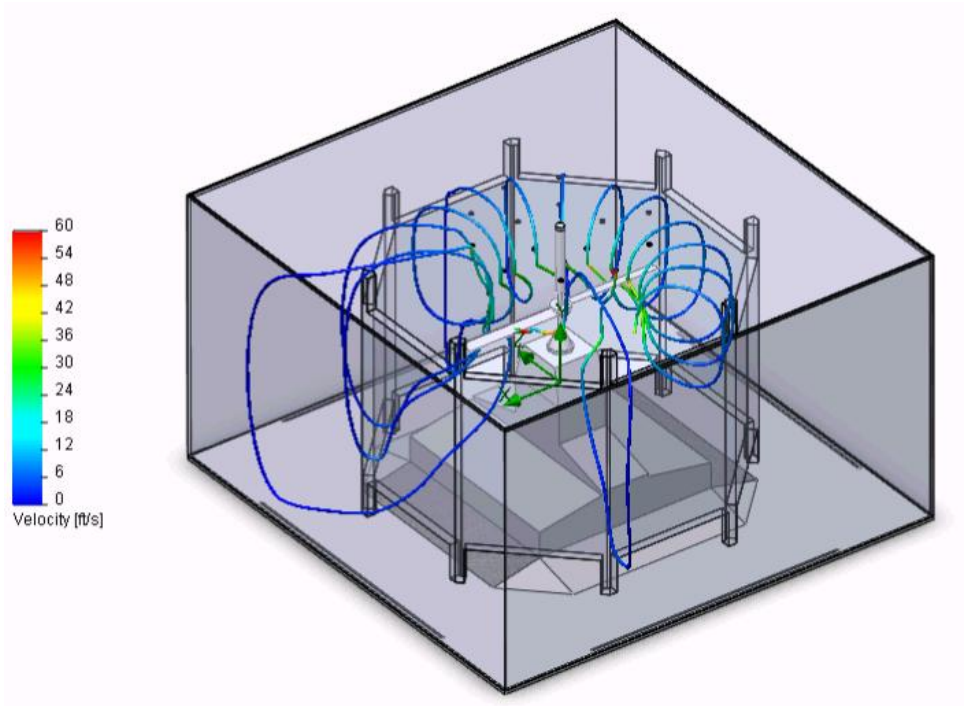


Figure 45: Stream Tubes from a Single Outer Nozzle
CFD Model Cross Section of AERTS,
(7.75' diameter rotor, 600 RPM, 2.5° Collective)

3.2.3.3 *Particle Trajectories*

Flow tubes only provide an approximation of droplet trajectories, as the droplets have small inertia. The software has a particle dynamics tracker, which is useful for the current studies. Particles are input into the system at the nozzle locations with a specified size, density and flow rate. For the NASA Standard nozzles, the flow rate in gallons/minute is given by Equation 23, which translates into roughly 0.00193 kg/second at the AERTS standard operating point.

$$Q = 0.004\sqrt{\Delta P} \qquad \textbf{Equation 23}$$

where ΔP *= Water Pressure – Air Pressure (psi)*

For the studies, a mono-dispersed droplet diameter of 20 μm is assumed and particles are input into the chamber at the nozzle positions with no relative velocity and are carried move through the chamber by via a particle dynamics model. The droplets have an assumed density of 917 kg/m^3, which allows for accurate trajectory modeling. The droplets can impact various surfaces and a mass loading is computed. However, the current modeling setup precludes an accurate assessment of mass collection on the blades.

A sample result of the study is shown in Figure 46, in which clear recirculations can be seen. The particles typically have approximately 0.15 seconds to cool between insertion in the chamber and passing through the rotor plane. This is similar to the IRT, in which particles have between 0.08 and 0.2 seconds to cool before passing through the test section.

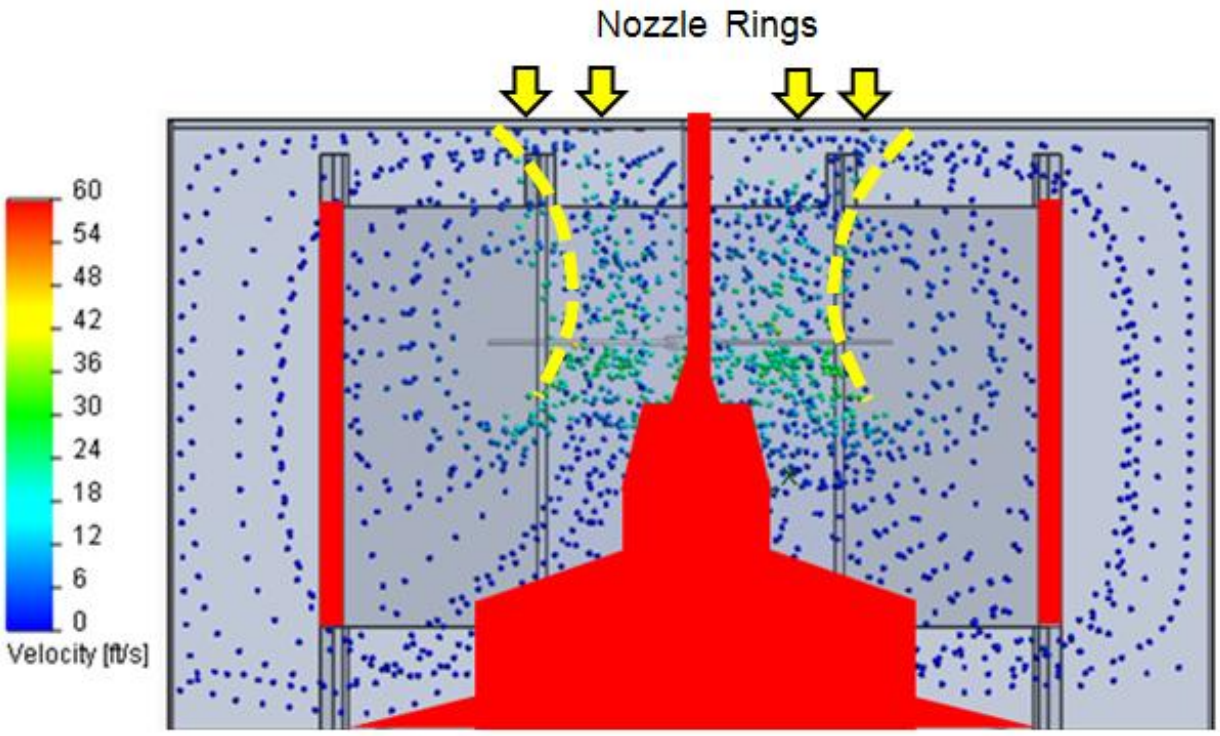


Figure 46: Droplet Distribution in AERTS Chamber
CFD Model Cross Section of AERTS,
(7.75' diameter rotor, 600 RPM, 2.5° Collective)
showing 20 μm droplet distribution in the chamber

The particles released from the nozzles tend to collect near the hub before passing through the rotor plane. Dashed lines superimposed on the figure to show the contraction of the injected particles. Most particles become trapped in the recirculation around the tips, regardless of which nozzle they are emitted from. Only a small fraction travel behind the ballistic walls.

The trajectories for the 1000 RPM case are shown in Figure 47. The strength of the tip vortex structure prevents particles from passing near the rotor tips. This was experimentally observed, as described in Section 3.4.4.2. The increased amount of particle circulation behind the ballistic wall can also be seen.

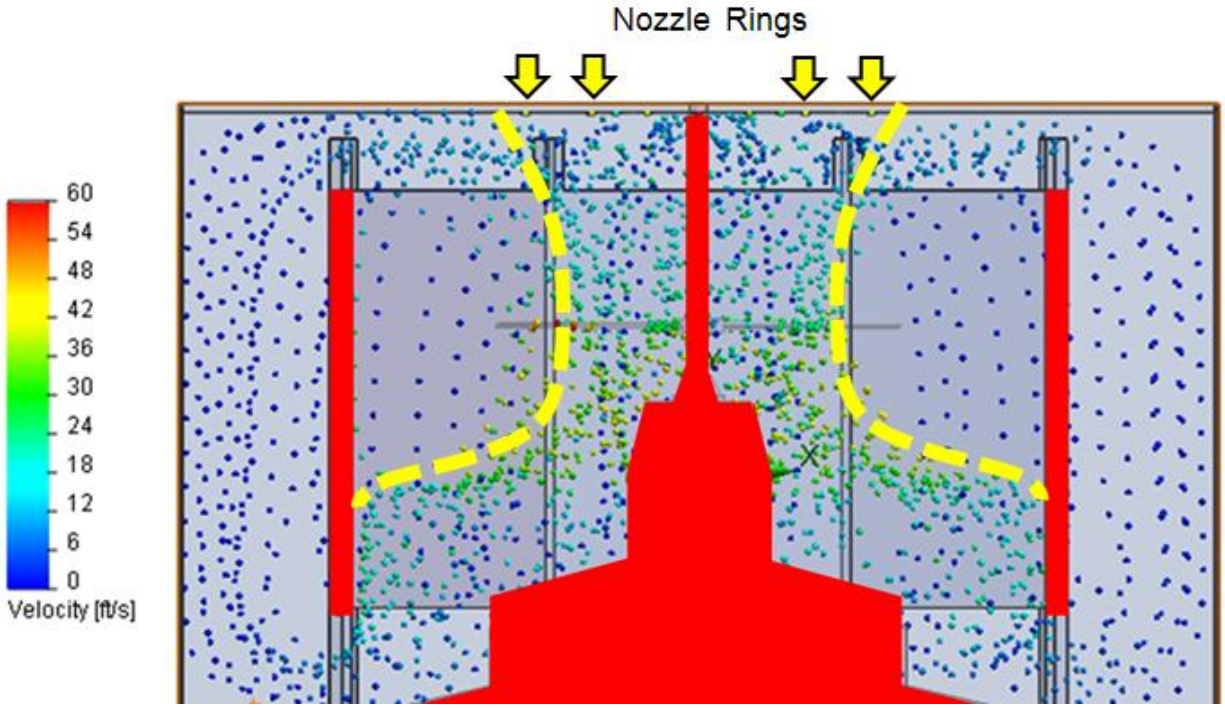


Figure 47: Droplet Distribution in AERTS Chamber
CFD Model Cross Section of AERTS,
(7.75' diameter rotor, 1000 RPM, 2.5° Collective)
showing 20 μm droplet distribution in the chamber

3.2.4 Modeling Conclusions

The important lesson learned from the general CFD results is that the potential for the particles to get trapped in the rotor tip vortex is high, especially at lower rotor tip speeds. Due to these recirculations, the potential for droplet collisions is higher. This in turn increases the possibility that the cloud in the vicinity of the rotor is mixed phase, containing both supercooled droplets and ice crystals. As explained in Section 3.4.4, this was encountered in preliminary testing in the AERTS Facility. Operating at higher tip speeds may force more droplets to recirculate behind the facility walls, but safety issues are encountered when operating at the high tip speeds.

3.3 Icing Cloud Calibration

3.3.1 Temperature

Temperature is arguably the most important parameter for icing tests (12). Three K type thermocouples were installed around the test chamber to monitor internal temperature. One sensor is mounted near the ceiling, one is mounted on the rotor stand just below the rotor head and the final is located beneath the floor. This allows the temperature profile of the test chamber to be monitored. The thermocouple mounted in the floor is isolated from the rotor wake, so the influence of the flow field and accreted ice on the temperature readings is eliminated. All thermocouple readings were verified with a calibrated temperature gauge to ensure accuracy. Additional control temperature readings come from a commercial weather station mounted inside of the chamber and a mechanical door thermometer. For all testing, the primary temperature measurement device is the weather station.

3.3.1.1 *Temperature Control Issues*

During a typical test, the temperature increases due to the fact that the cooling fans must be turned off. If the fans are not shut down, the ice cloud will be drawn through the heat exchanger, eventually clogging the inlet and causing large ice accretions on the fan blades. Stopping the fans prevents potentially damaging asymmetric shedding from the blades, but prevents air from flowing over the cooling lines. This effect, combined with the introduction of hot water through the icing nozzles to the chamber and kinetic friction on the rotating rotor blades causes the temperature in the test chamber to increase slightly. Per NASA IRT standards, this limits test time to approximately 3.5 minutes. At warmer temperatures, the temperature rise is less severe. Due to this temperature rise, vigilance must be exercised when running multiple tests in a day to ensure that all

data runs start at the appropriate temperature. Experience with the cooling system allows the establishment of procedures to limit the temperature rise. Table 11 lists typical test times and temperature increases.

Table 11: Typical Temperature Rise During Testing (Weather Station Readings)

Rotor	Test Time (seconds)	Rotor RPM	Temperature Start (°C)	Temperature End (°C)	Temperature Rise (°C)
Bat	180	500	-17.0	-14.0	3.0
	360	600	-17.0	-140	3.0
	600	600	-15.0	-9.5	5.5
NACA 0015	180	500	-11.1	-10.4	0.7
	180	600	-10.2	-9.9	0.3
	427	600	-5.9	-3.7	2.2
Shear Adhesion	468	350	-9.8	-5.9	3.9
	462	350	-8.9	-5.6	3.3
	478	350	-11.2	-8.2	3.0

Plans have been made to implement a secondary temperature control system in the facility to alleviate the temperature rise during testing. Pending funding, a series of pipes carrying a cooled refrigerant will be installed on the outside of the ballistic wall, forming a convective cooling system. The pipes will create a large surface area over which recirculatory flow generated by rotor downwash will pass over. The goal of this secondary system is to only maintain the temperature inside the chamber during testing, not necessarily cooling down the entire chamber from room temperature to test temperature. The advantage of the system over the current one is that ice can accrete on the pipes without serious consequence to the icing cloud or total system efficiency.

3.3.2 Droplet Size

The size of the droplets coming from the nozzles is an important parameter. In the AERTS facility, droplet size is not directly measured. Instead, droplet size is determined from a calibration chart,

provided by NASA Glenn. This chart is provided in Figure 48. Engineers in the Icing Research Tunnel (IRT) have used this chart for years for component level testing and certification. The nozzles have no moving parts, so there is a low chance of them going out of calibration, unless large particles clog the small central core. This is avoided with the use of reverse osmosis filtered water and in-line water filters. Also, nozzle o-rings are replaced each time the nozzles are serviced to preclude breakdown of the rings and potential clogging hazards. A feedback control system in Labview monitors the nozzle input pressures and adjusts multiple pumps as required to maintain particle size ± 2.0 μm (steady state) during testing.

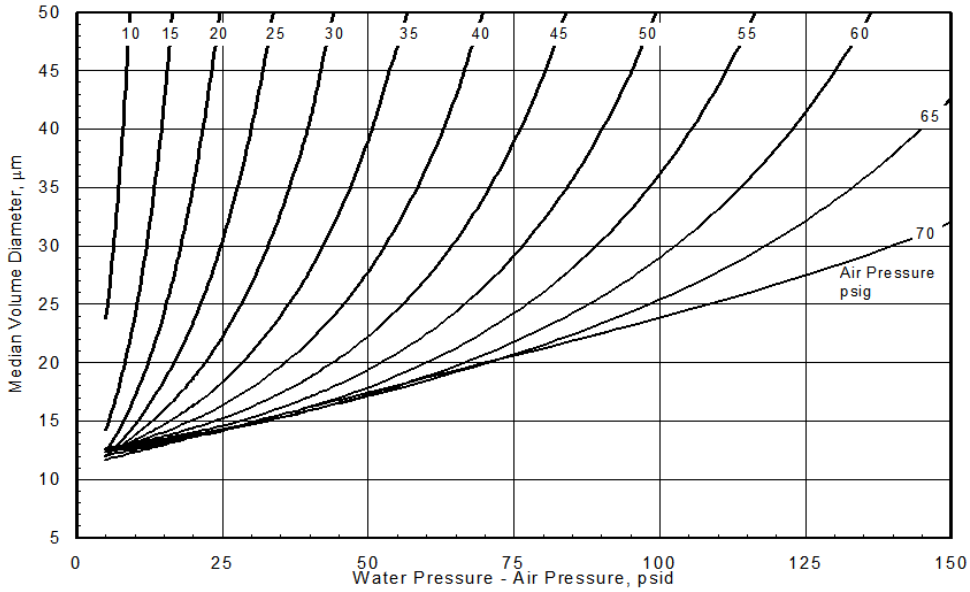


Figure 48: NASA Standard Nozzle Calibration Table
(Provided by NASA Glenn)

The AERTS facility has no way of validating the particle sizes of the nozzles due to lack of proper equipment, so the NASA chart is currently used as the absolute calibration. Funding requests have

been submitted for a Malvern MVD sensor to provide a direct measurement of particle size. In the meantime, a soot slide detector is under development.

3.3.3 Liquid Water Content (LWC) Calibration

The LWC calibration was the most difficult part of the AERTS Facility calibration due to the unique nature of the test chamber and icing system. Most LWC instruments require a minimum flow velocity over the sensor head. Due to this, and the cost of the instruments, no sensor is applicable to the facility. Therefore, a new calibration technique was developed to characterize the icing cloud. This technique was developed after consulting engineers at NASA's IRT and is specifically tailored to the AERTS facility.

3.3.3.1 *Calibration Theory*

In two dimensions, the LWC can be related to other icing parameters by Equation 24, which is derived from the accumulation parameter described in Equation 10.

$$LWC = \frac{\Delta S * SG}{n * \beta_o * V * t} \qquad \textbf{Equation 24}$$

Where:

$\Delta S/t$ = icing rate (in/min)
SG = specific gravity of the ice
n = freezing fraction
β_o = stagnation collection efficiency
V = flow velocity

The specific gravity of ice is determined from the ice density models described in Section 2.3.3.3. For all calculations, the Jones Model is used.

The freezing fraction, n, is determined by the Messinger Model. This parameter is one of the most important non-dimensional parameters in icing and it has a strong effect on ice accretions. The freezing faction varies from 0, where impinging water droplets do not freeze to a surface, to 1, where water droplets freeze immediately on impact. Rime ice is typically associated with $n \approx 1$, while glaze ice starts to appear at $n < 1$. A code, developed from Reference (77), calculates the surface heat balance and determines the freezing fraction based upon the properties of air and water at a particular test condition. The code was validated against test data from the IRT. Figure 49 shows the freezing fraction distribution over test conditions typically encountered in the AERTS facility.

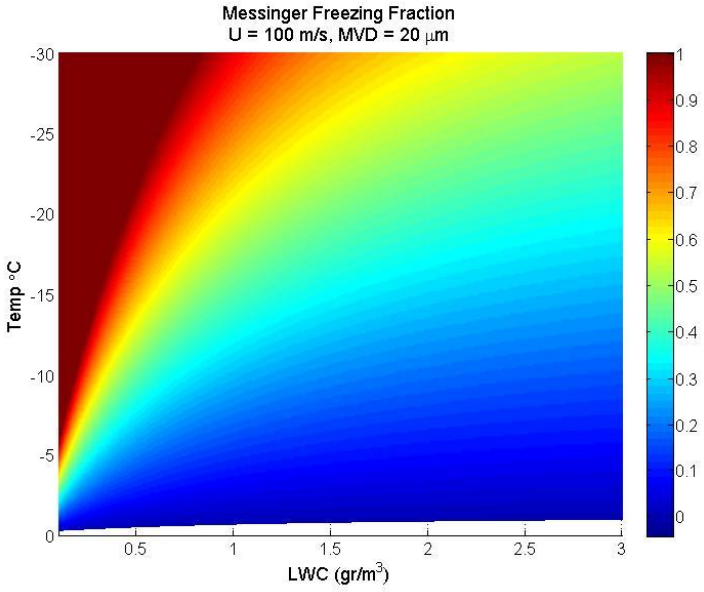


Figure 49: Freezing Fraction Variation

The stagnation collection efficiency, β_o, is based upon the geometry of the object under study. The analysis of Langmuir and Blodgett, described in Reference (48), is used in calculations. This equation was initially published for cylinders, but was later validated for airfoils. The following equations detail the calculation process.

$$\beta_0 = \left(\frac{1.40 * (K_0 - 1/8)^{.84}}{1 + 1.40 * (K_0 - 1/8)^{.84}} \right)$$ **Equation 25**

where, K_0 is the Langmuir and Blodgett's expression for modified inertia parameter.

$$K_0 = \frac{1}{8} + \frac{\lambda}{\lambda_{Stokes}} \left(K - \frac{1}{8} \right), \text{ for } K > \frac{1}{8}$$ **Equation 26**

The inertia parameter, K, can be expressed as:

$$K = \frac{\rho_w \delta^2 V}{18 d \mu_a}$$ **Equation 27**

And $\lambda / \lambda_{Stokes}$ is defined as the dimensionless droplet range parameter,

$$\frac{\lambda}{\lambda_{Stokes}} = \frac{1}{0.8388 + 0.001483 \mathrm{Re}_\delta + 0.1847 \sqrt{\mathrm{Re}_\delta}}$$ **Equation 28**

where

$$\mathrm{Re}_\delta = \frac{V \delta \rho_a}{\mu_a}$$ **Equation 29**

3.3.3.2 *Calibration Process*

It is important to note that in the Messinger Model, freezing fraction is a function of LWC. This requires a convergence loop, where an initial estimate for the chamber LWC is input into the code. Based on test experience, initial estimates for the LWC were generated based upon the particular nozzle configuration in the test chamber.

Initially, the calculation of the experimental LWC was inaccurate, producing results greater than 100 gr/m^3 for certain test conditions. A graduate research assistant in the AERTS lab, Yiqiang Han,

further developed and validated the concept. The updated code, described in paper at the upcoming 66th Annual Forum of the American Helicopter Society, has been demonstrated to accurately calculate the chamber LWC. The following figure describes the calculation process.

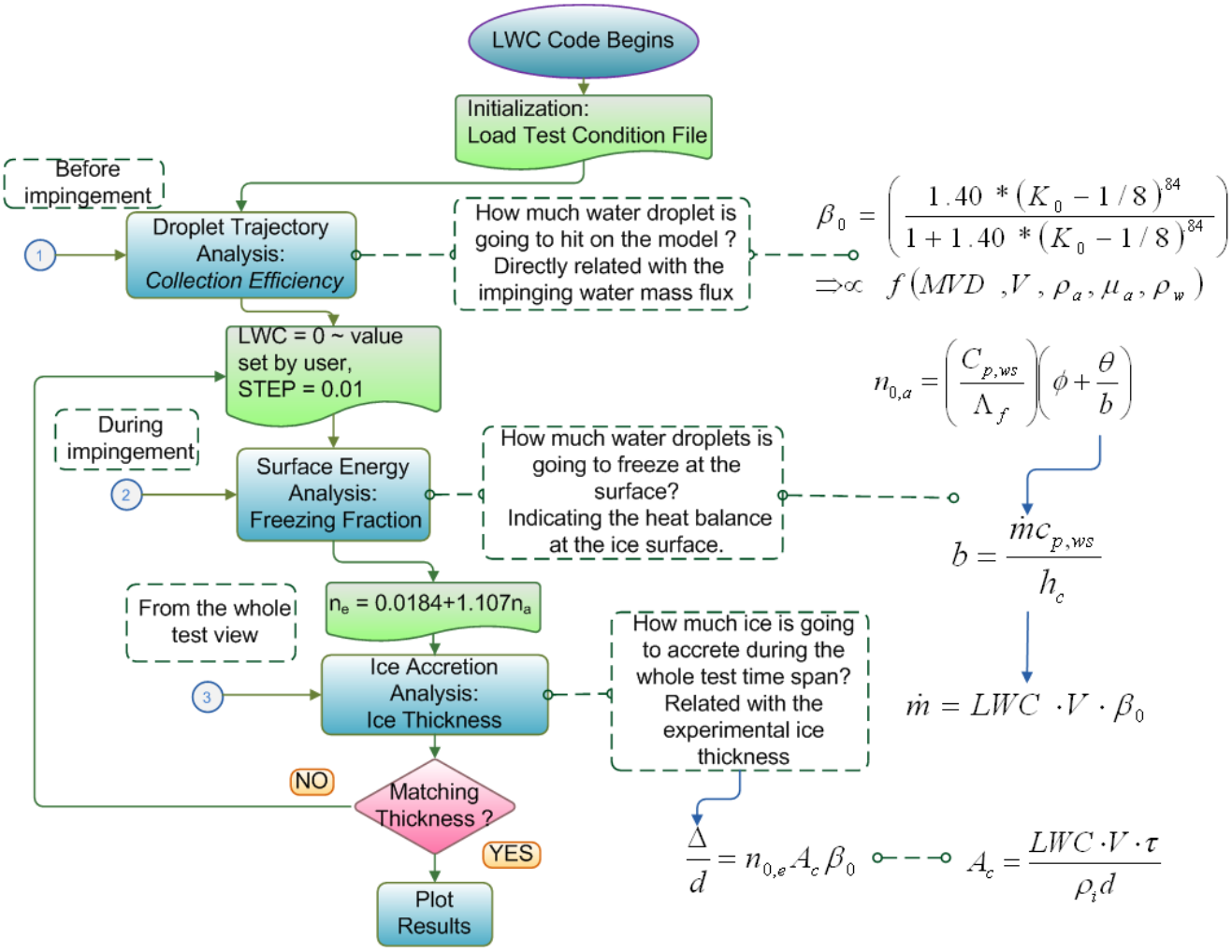


Figure 50: Experimental LWC Calculation Overview
(process and figure developed by Yiqiang Han)

The process was validated against two sets of experimental tests in the IRT. The first, shown in Figure 51, featured cylinders, while the second set of validation tests investigated NACA0012 airfoils. Error bars indicate the accepted ±15% calculation error of the LWC in the IRT.

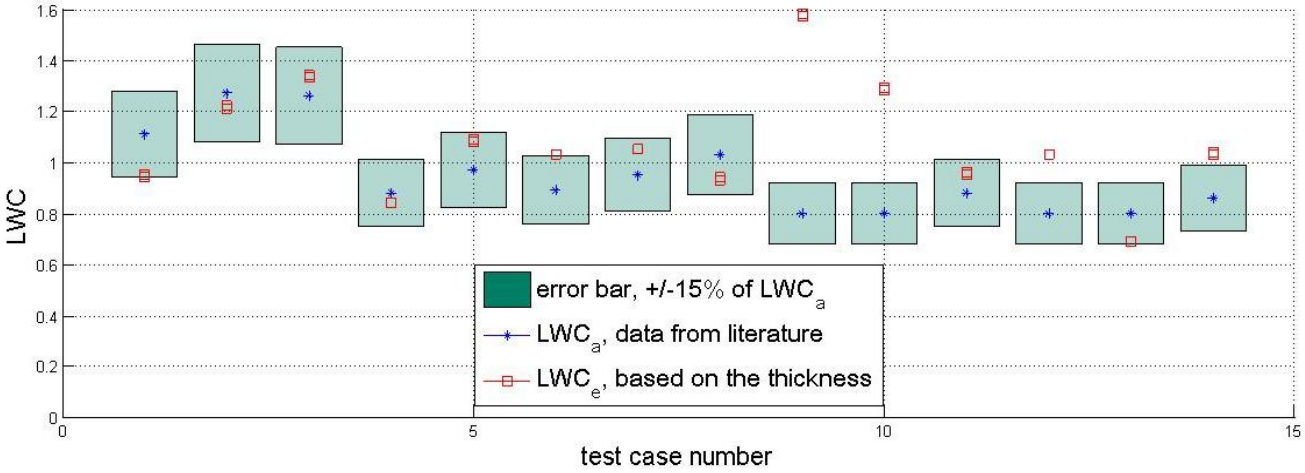


Figure 51: Cylinder LWC Calculations
Compared to Experimental Results Presented in Reference (78)
(figure developed by Yiqiang Han)

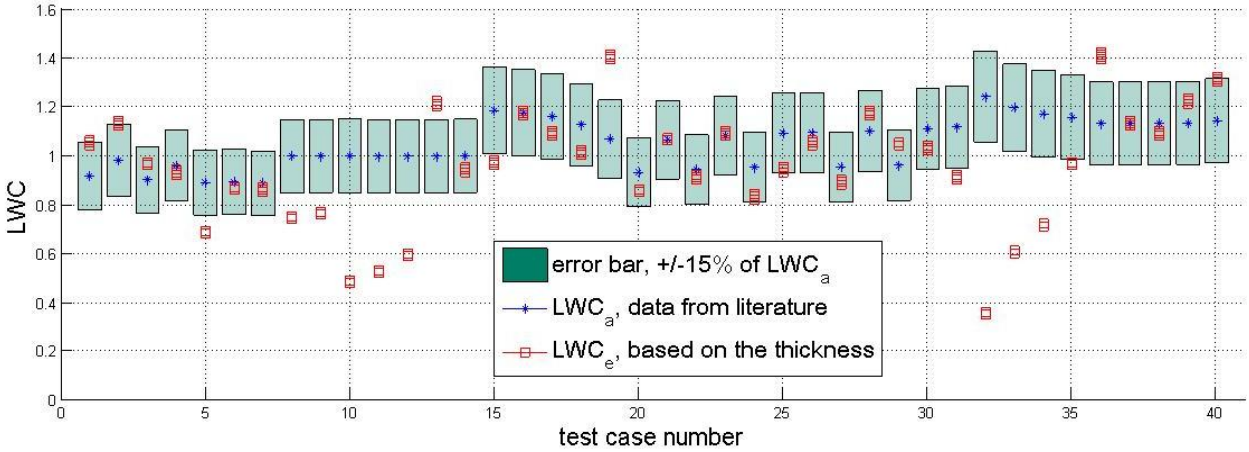


Figure 52: NACA 0012 LWC Calculations
Compared to Experimental Results Presented in Reference (77)
(figure developed by Yiqiang Han)

The code has demonstrated the ability to accurately predict most of the LWC points from literature. There are some cases that are not predicted well, but the underlying causes are understood. Minor discrepancies in the thickness measurement and the calculation of the freezing fraction may generate significant errors in the LWC calculation. For example, a 0.08 (45%) error in predicting the freezing fraction can generate a 200% error in the LWC. Also, a 0.005" (8%) error in stagnation ice thickness calculation may yield a 0.3 gr/m^3 (20%) error in LWC calculation.

3.3.4 Laser Cloud Density Instrument

3.3.4.1 *Concept*

One primary concern in the test chamber is whether the particle density is representative of natural icing conditions. Icing tunnels inject water droplets ahead of the model, and they are removed as air passes over the cooling system. In the AERTS chamber, droplets are continually injected during a test and the facility relies on accretion to the rotor, walls, floor, and ceiling to remove them from the air stream. The efficiency of this process was unknown. A secondary concern is that the particles freeze out and become solid ice crystals rather than the desired supercooled liquid droplets.

To start understanding these processes, a simple instrument was designed. The system is loosely based upon the Forward Scattering Spectormeter Probe (FSSP), commonly used to determine icing cloud parameters in icing flight trials and described in Reference (21). The primary components are a low powered laser and solar cell as shown in Figure 53. The laser shines directly on the solar cell, which registers a voltage. The solar cell is shielded from all ambient light to improve test accuracy. As particles block the laser, the voltage is generated by the laser is reduced. This output voltage gives an uncalibrated measure of cloud density.

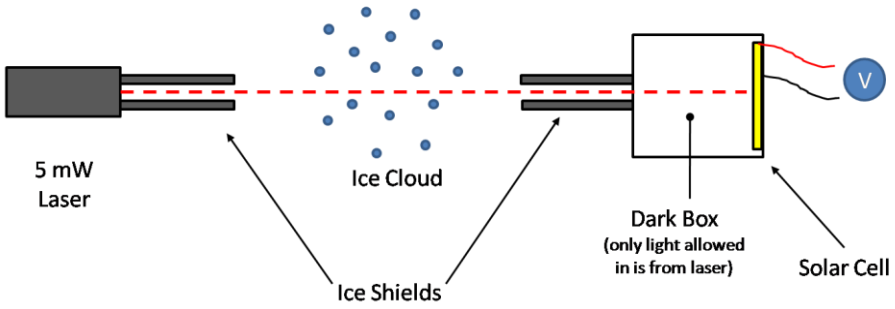


Figure 53: Laser Cloud Density Measurement Concept

A proof of concept design based around a 5 mW laser pointer was created to verify the concept. The laser was mounted in the chamber underneath the rotor as shown below. The system was separated from the walls and floor by a minimum distance of 30 inches.

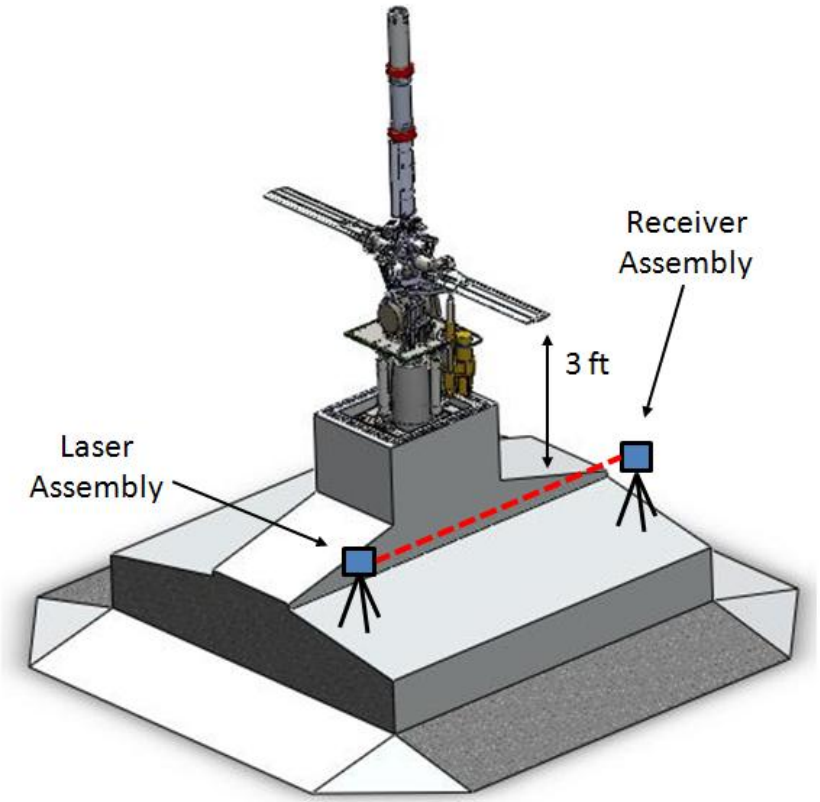


Figure 54: POC Laser Location in AERTS

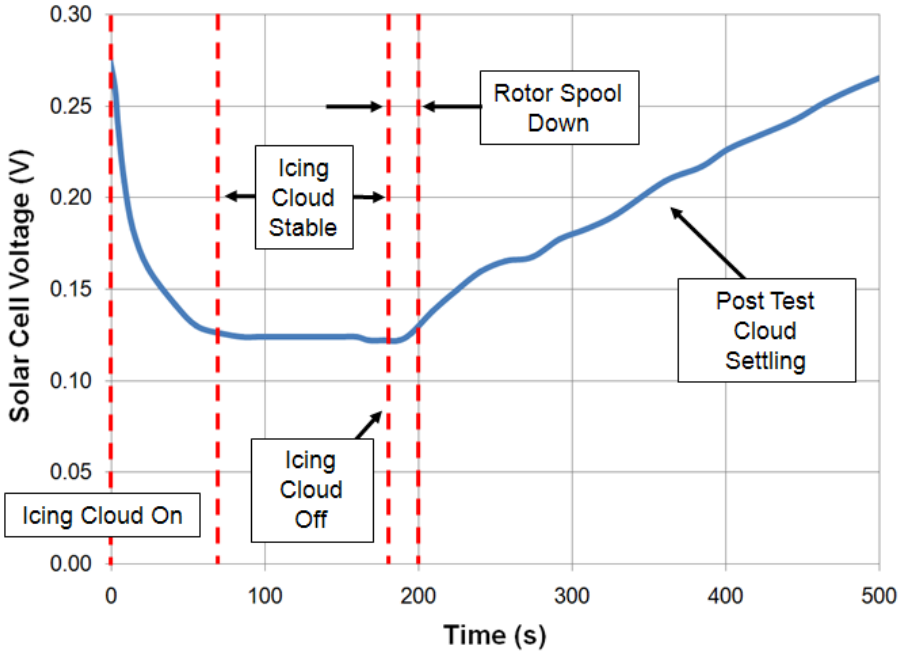


Figure 55: Proof of Concept Laser Sample Test Results

Testing in the AERTS chamber with the POC instrument showed that the cloud density increased rapidly for the first part of the testing but leveled off by 70 seconds of elapsed time as shown in Figure 55. At this time, the cloud became stable, with its density not increasing or decreasing. This occurs because the mass flow rate of water leaving the flow field balanced the mass flow rate of water added to the chamber flow field. Water leaves the flow field be accreting to the rotor, walls and floor. Lessons were also learned about the required shielding to keep ice from accreting on the laser or receiver and generate erroneous results by completely blocking the laser beam.

These tests showed that a particle removal system was not absolutely required for testing in the AERTS chamber. Eventually, one should be added to remove ice crystals, but the

current configuration produces an icing cloud with a constant particle density. If the cloud density continually increased during a test, a system of relays would be required to turn on and off various nozzles to achieve a stable cloud.

3.3.4.2 *Improved Design/Calibration*

Since the POC system worked, an improved design was implemented. A higher powered 120 mW green laser was procured to provide increased system accuracy. The laser was also qualified to operate at low temperature in high fog environments without damage or lens distortion. The laser and receiver were mounted on tripods, so they could be positioned around the chamber.

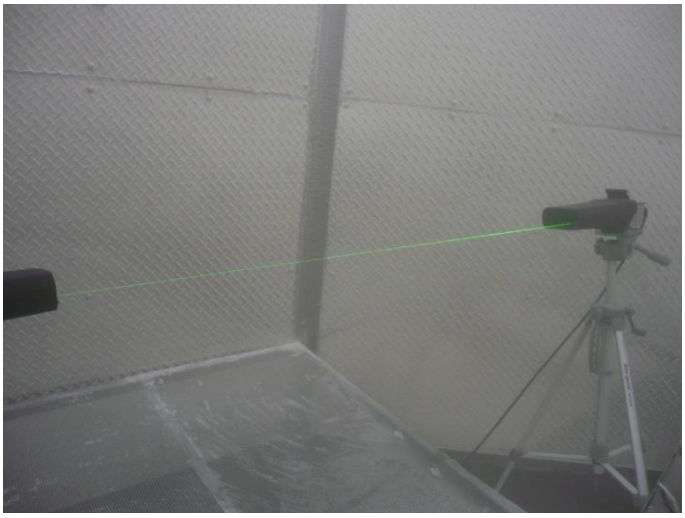


Figure 56: Improved Laser Undergoing Shakedown Tests

3.4 Icing System Shakedown Tests

3.4.1 Test Rationale and Matrix

Prior to the any ice shape validation testing, shakedown tests were conducted in the AERTS chamber. The objective of the runs was to determine the limitations of the icing system and

establish facility best practices. Running these tests was important, as the AERTS is a new facility with many unknowns. Since these were the first test cases run in the chamber, there was some difficulty in controlling precise conditions, especially LWC.

Most shakedown tests were conducted with the ARISP validation test rotor. This 7.75' diameter, 6.8" chord rotor was designed for testing up to 1200 rpm (486 ft/sec) and features a NACA0015 airfoil. APPENDIX D: Test Rotor Design contains more details of the rotor implementation. The first eight cases were accomplished with a 9' diameter, 8" chord rotor due the unavailibilty of the NACA0015 blades. The airfoil was similar to a NACA0012, but had a smaller leading edge radius. Since the rotor was originally designed for autogyros, it was limited to 300 rpm.

The maximum tip speed used during the shakedown tests was 243 ft/sec. This, combined with low collective pitch settings, kept the entire rotor well within the LEWICE validation envelope, allowing for the assumption that the icing model was relatively accurate and the unknown entity was the AERTS Facility. Test parameters are detailed in Table 12. The first few cases focused primarily on developing facility test procedures. The remainder of the cases were more structured in nature, and attempted to systematically explore the many variables in the AERTS chamber. Case 9 was a dry run to verify the new test rotor interface to the hub.

Table 12: Icing Shakedown Tests

Case	Temp	Average LWC*	MVD	Nozzle Config	Tip Speed	θ	Icing Time	Shed?	Nozzle Config Sweep	RPM Sweep	MVD Sweep	θ Sweep	Low Temperature Comparison	High Temperature Comparison
	°C	gr/m³	μm		ft/sec	°	sec							
1	-11.2	~6.0	20	All 15	141	0	180	N						
2	-13.3	~6.0	40	All 15	141	0	180	Y						
3	-14.9	~6.0	15	All 15	141	0	180	N						
4†	-14.4	~6.0	25	All 15	141	0	180	N						
5	-6.1	~6.0	20	All 15	141	0	180	Y						
6	-6.3	~6.0	20	All 15	141	0	180	Y						
7	-19.2	~6.0	10	All 15	141	0	180	N						
8	-16.4	~6.0	20	All 15	141	0	180	N						
10	-10	~6.0	20	All 15	162	2.5	150	Y						
11	-10	~4.0	20	10 Outer	162	2.5	180	N		X	X	X	X	
12	-10	~6.0	20	All 15	162	5	180	N	X					
13	-10	~4.0	20	10 Outer	162	5	180	N	X			X		
14	-10	~3.0	20	5 Outer	162	5	180	N	X					
15	-18	~4.0	20	10 Outer	162	2.5	180	N					X	
16	-10	~4.0	20	10 Outer	162	2.5	180	N		X	X			
17	-10	~4.0	20	10 Outer	243	2.5	180	N		X				
18	-10	~4.0	40	10 Outer	162	2.5	180	N			X			
19†	-10	~4.0	40	10 Outer	202	2.5	180	N						
20	-8	~4.0	20	10 Outer	202	2.5	180	N						
21	-10	~4.0	40	10 Outer	202	2.5	180	N			X			X
22	-5	~4.0	40	10 Outer	202	2.5	180	Y						X

Cases 1-8 used NACA0012 Test Rotor, all other cases used NACA0015 rotor.

*Due to ice crystals, an accurate LWC could not be determined in the chamber. Values are provided as estimates only.

†Case rejected due to droplet size control failure or nozzle water connection failure.

3.4.2 Data Recording

After each test, the stagnation ice thickness was measured with calipers and the ice shapes were recorded via tracings at specified rotor stations. For the thickness measurements, the calipers were cooled to the test temperature for a minimum of 10 minutes to ensure that they did not melt the ice during the measurement process. For the ice tracings, a slot was created in the ice at the desired tracing location with a heated steel plate. An airfoil template with a piece of graph paper taped to it was inserted into the slot and the shape transferred with a pencil as shown in Figure 57. The rime feathers generated by the nozzles during shutdown were ignored when taking ice tracings. These tracings were later digitized and compared to the experimental predictions.

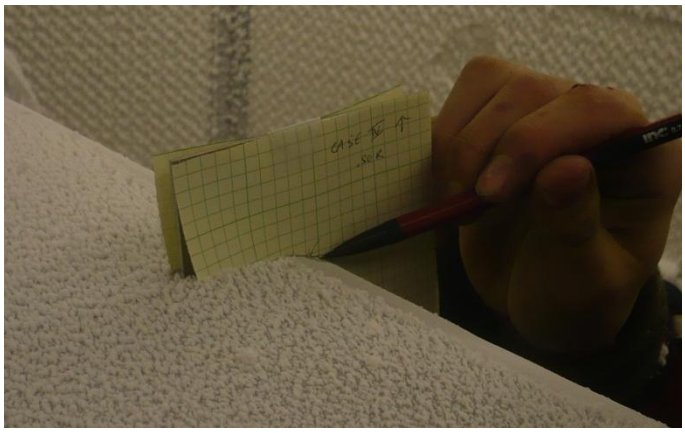


Figure 57: Ice Tracing Process

3.4.3 Ice Crystals

In general, the experimental ice accretions did not match preliminary ARISP predictions. As shown in Figure 60, the shapes were typically pointy, regardless of the icing conditions.

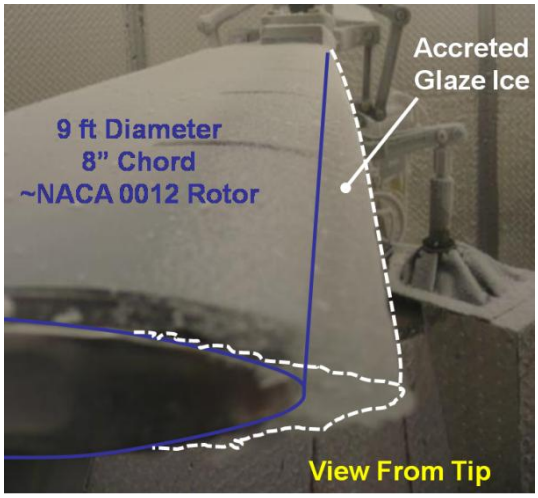


Figure 58: Case 1 Accreted Ice, NACA0012 Rotor Tip View

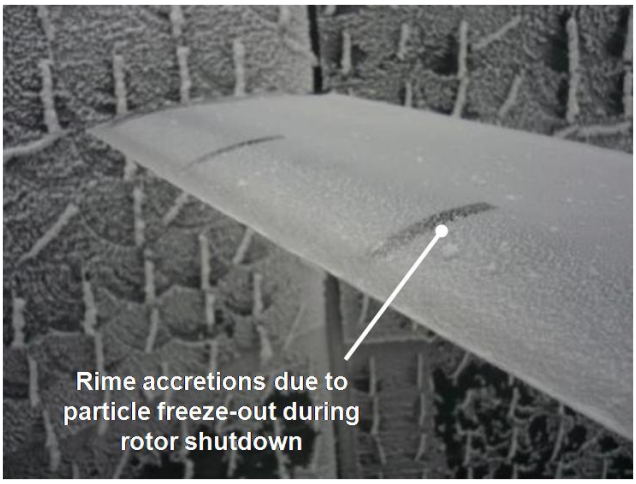


Figure 59: Case 1 Accreted Ice, NACA0012 Rotor Isometric View

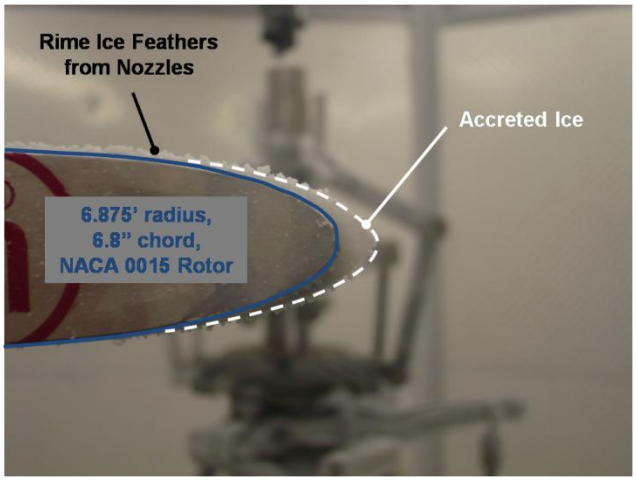


Figure 60: Case 11 Accreted Ice, NACA0015 Rotor

The ice spears created errors in the LWC calculations, because the process is dependent on an accurate measurement of the stagnation ice thickness. These errors are evident in the thickness and shape correlations as shown in Figure 61 and Figure 62.

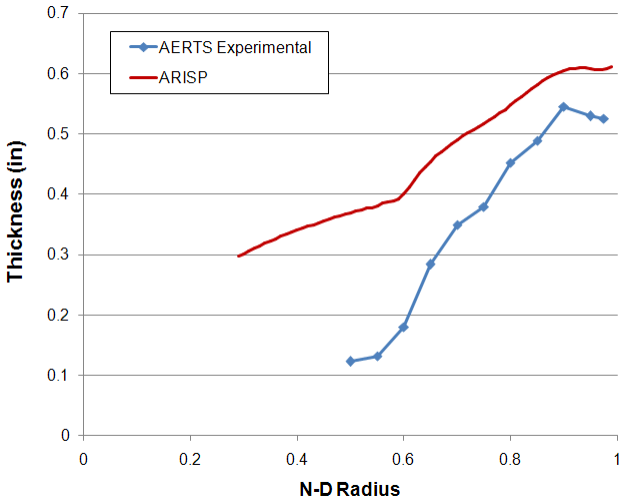


Figure 61: Case 11 Thickness Comparison

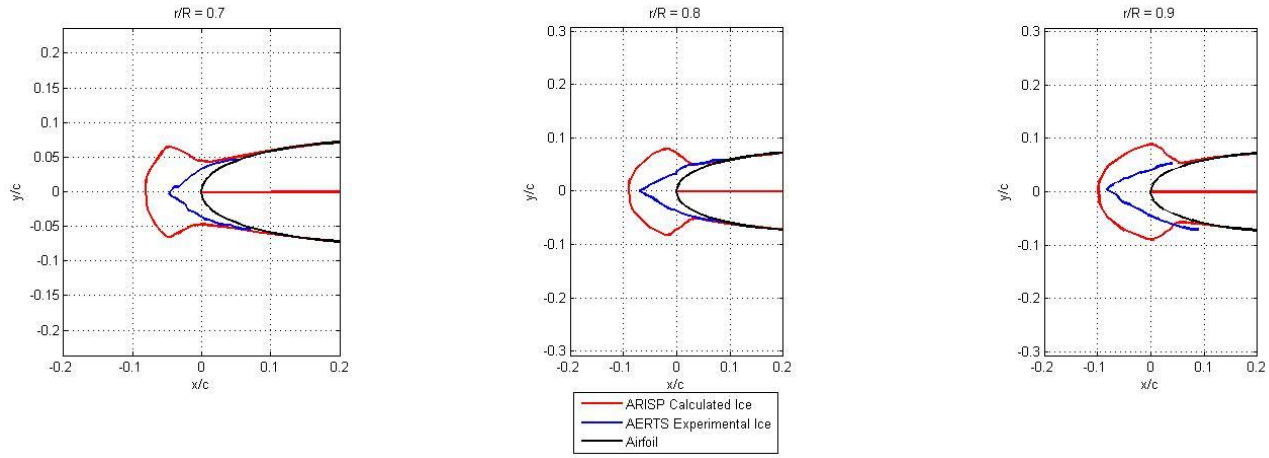


Figure 62: Case 11 Shape Comparison

The reason for the discrepancies in the thickness and shapes was ice crystals. During testing, supercooled liquid droplets are continuously added to the chamber. This may saturate it, depending on the nozzle settings and the number of active nozzles. Due to the small facility size, droplets recirculate around the ballistic wall or rotor tips after they pass through the rotor plane. Since no particle removal process is used in the facility, the droplets can freeze into solid crystals. When liquid droplets impact a crystal, the droplet is immediately crystallized, which creates a chain reaction as explained by Reference (75). Larger numbers

of particles in the chamber increase this effect and are generated when using higher air pressure inputs to the nozzle. To maintain a desired MVD at higher air pressures, water pressures need to be increased to maintain the proper pressure differential, as detailed in Figure 3. Since the water flow rate is dependent on this pressure differential, the mass of water added to the chamber increases, creating a large number of droplets. Evidence of crystals in the test chamber included significant amounts of snow on the chamber walls, rotor hub and blade upper surface as shown in Figure 63.



Figure 63: Ice Crystals on Blade Upper Surface

As demonstrated during tests of the Sikorsky S-92 in the McKinley Climatic Chamber, the crystals can then erode ice shapes, negatively affecting attempts at accretion model validation (14). This result was confirmed with IRT engineers at NASA Glenn, who had also encountered ice shape erosion as shown in Figure 64, taken from Reference (79).

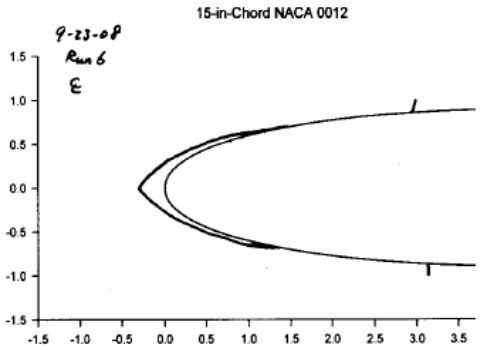


Figure 64: IRT Ice Shape Erosion

Ideally, the crystals would be removed from the chamber, but technical and financial issues prevent this. As a temporary solution, screens were added 34 inches beneath the rotor plane to capture the ice particles before they recirculated around the test chamber. The screens were built from 0.25" x 0.25" mesh, which has a high collection efficiency. Heaters were added to melt the ice accretions and prevent the screens from clogging. These were not effective due to poor contact with the screen surface, so brooms were used to clear snow from the screens after each test. A portion of the screen system is shown in Figure 65.

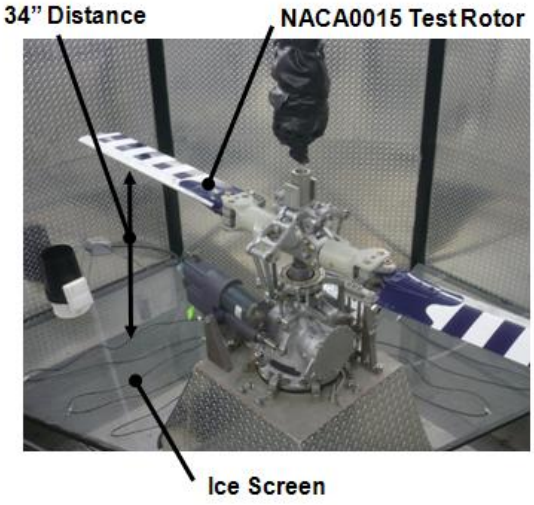


Figure 65: Anti-Crystal Screen Location

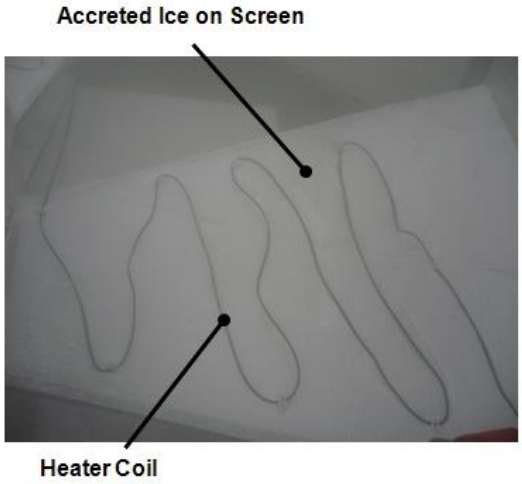


Figure 66: Ice Accretions on Anti-Crystal Screen

A better solution, found after the completion of the shakedown testing, was adjusting nozzle settings. Reducing the number of active nozzles and reducing the input air pressure significantly decreased the amount of water added to the chamber. Five nozzles were found to create a stable cloud with a low number of particles, significantly reducing the amount of crystals. With the improved nozzle settings, the ice crystal screens accrete ice at an order of magnitude lower rate as shown in Figure 67.

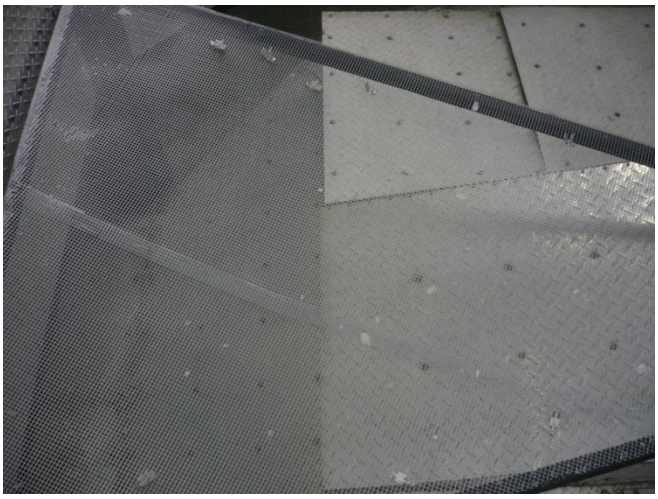


Figure 67: Reduced Ice Accretion on Screens

The maximum pressure differential to avoid crystallization problems was experimentally determined to be 23 psi. Based upon this, an operational envelope for the chamber was then established, as shown in Figure 68. If fewer nozzles are used, higher air pressure may be employed. This is preferred, because the nozzles are more stable at the higher input pressures.

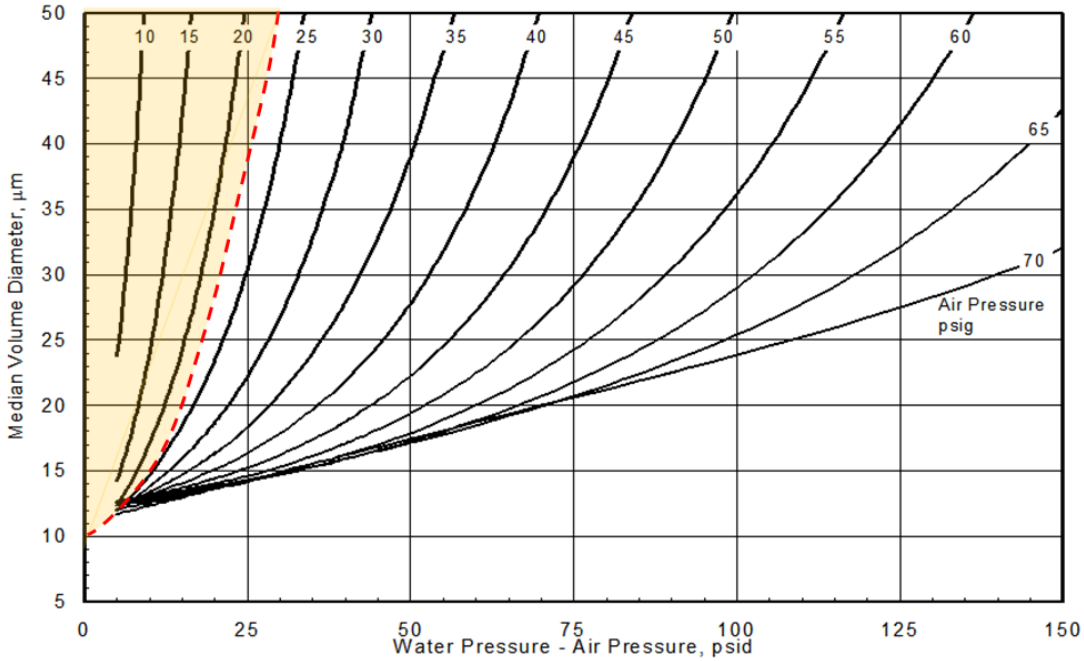


Figure 68: Refined AERTS Nozzle Operational Envelope

3.4.4 Test Chamber Sensitivity Studies

3.4.4.1 *Nozzle Setting / RPM Sensitivity*

The low RPM of the shakedown testing identified a potential nozzle placement issue. When testing with larger particle sizes at low RPMs, the high inertia of the droplets caused them to accrete to the rotor directly below the nozzles and not mix in the test chamber. Little accretion was seen elsewhere on the blade as shown in Figure 69. Using only the outer nozzles eliminated this issue.

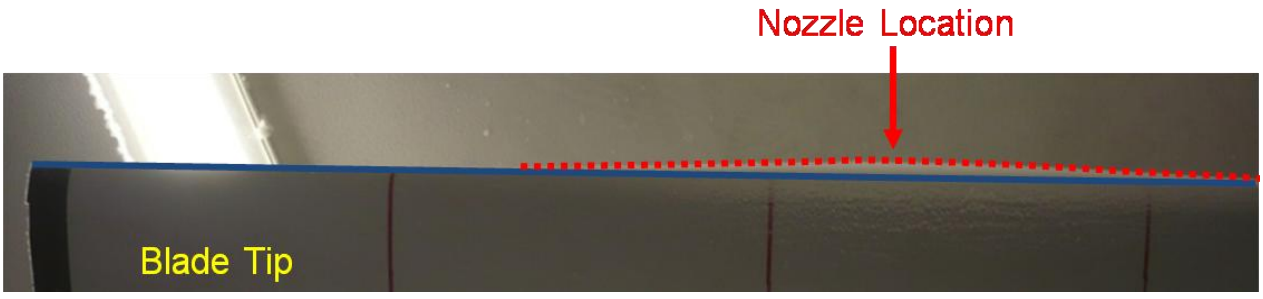


Figure 69: Case 8 Ice Accretion, NACA0012 Rotor

Various nozzle configurations and rotor velocities were tested as part of the shakedown tests. Issues with crystallization convoluted test results, so the nozzle setting sensitivity studies are not presented. It was expected that the higher rotor velocities would accrete more ice, since tip speeds are relatively low. Higher tip speeds would decrease ice accretion, as kinetic heating would lower the tip freezing fraction. The shakedown test results, provided in Figure 70, did not follow this trend as ice erosion reduced accreted ice thickness.

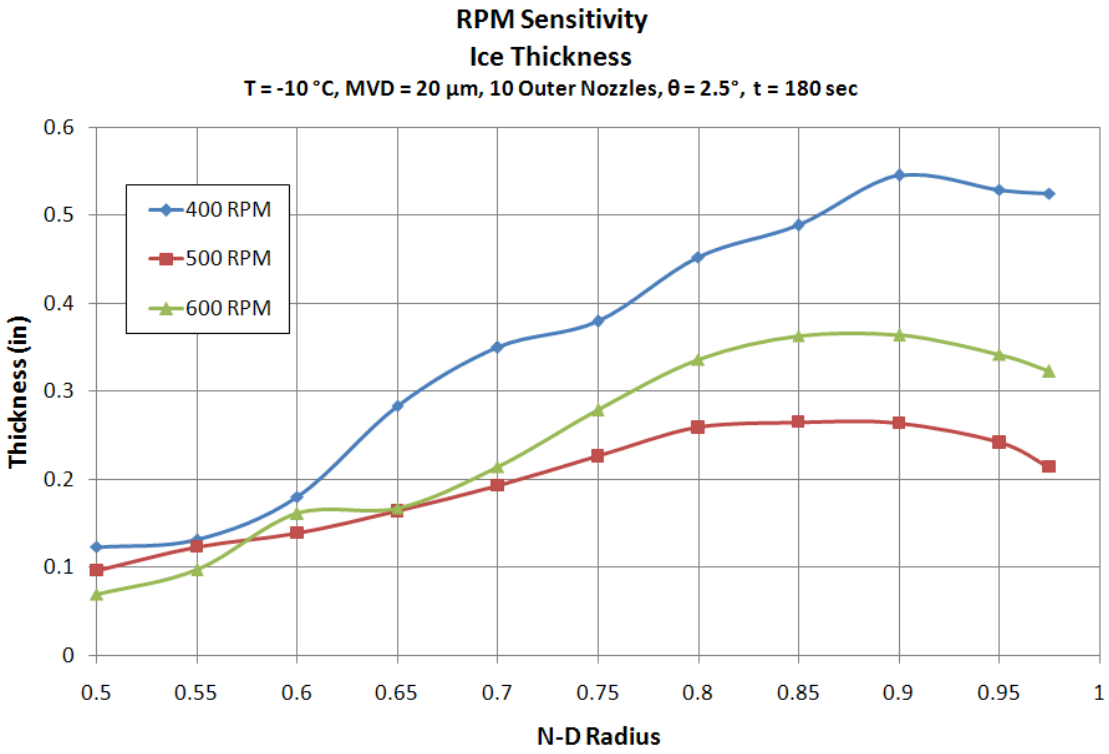


Figure 70: RPM Sensitivity

The anomaly of the low RPM testing was not significant, as nozzle settings were improved as described in Section 3.4.3. All final correlation testing will take place at higher velocities, which are more representative of rotorcraft velocities. Based upon the shakedown testing, a recommended minimum RPM of 500 was established for lifting rotors.

3.4.4.2 *Collective Pitch*

Collective pitch was another variable investigated during the testing. Higher collective pitches increased circulation in the chamber, which in theory would improve cloud mixing. As expected, the tip vortex strength increased and affected the ice accretion at the blade tips. At higher collective pitch settings, the accreted ice thickness decreased significantly as shown in Figure 71. This is due to the generation of a large tip vortex structure, which prevents the droplets from mixing in the flow. This structure can be seen Figure 40 and the effect on droplet distribution is shown analytically in Figure 46. The tip vortex also increases the chance of supercooled droplets of impacting each other and crystals. If a supercooled droplet impacts a crystal, it immediately crystallizes (12). Therefore, the tip vortex structure may have a higher density of crystals which can erode ice shapes, producing the results shown in Figure 71.

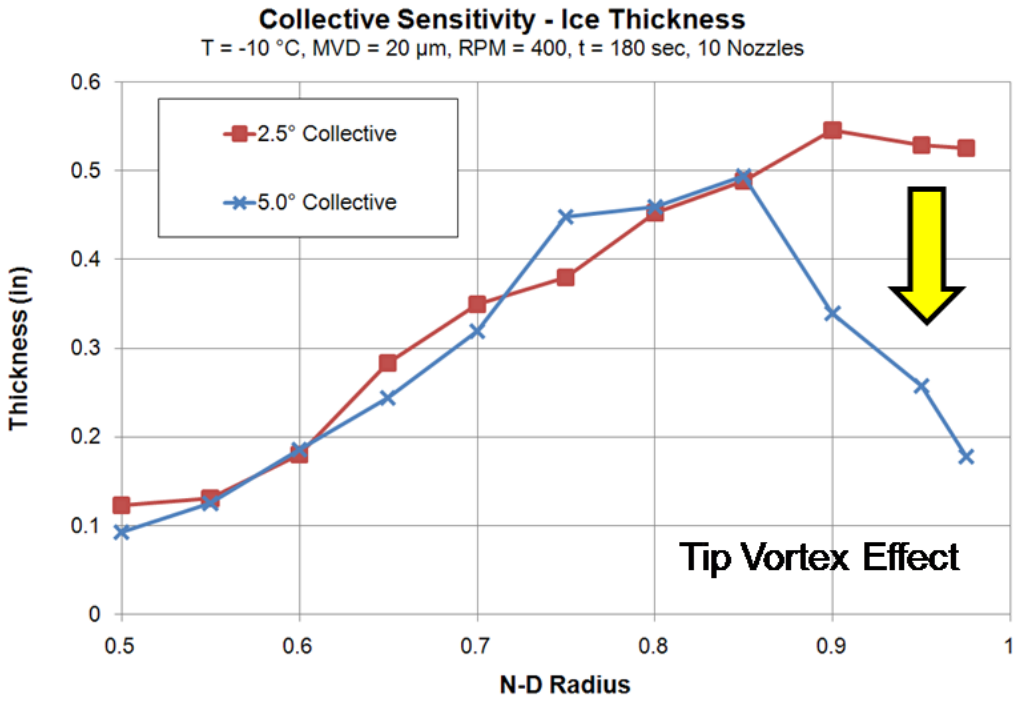


Figure 71: Collective Pitch Sensitivity

3.4.4.3 *Temperature Effects*

Temperature effects on ice accretion were explored as part of the sensitivity studies. The expected behavior of smaller accretions at colder temperatures was experimentally observed, as shown in Figure 72. The freezing fraction increases at colder temperatures, so droplets freeze instantly on contact with the rotor and forms rime ice. These accretions are typically smaller than the glaze ice accretions formed at warmer temperatures due to the lower LWC. Testing at warmer temperatures was undertaken, but shedding events complicated direct run comparisons because the stand was shut down immediately after a shedding event due to imbalance issues.

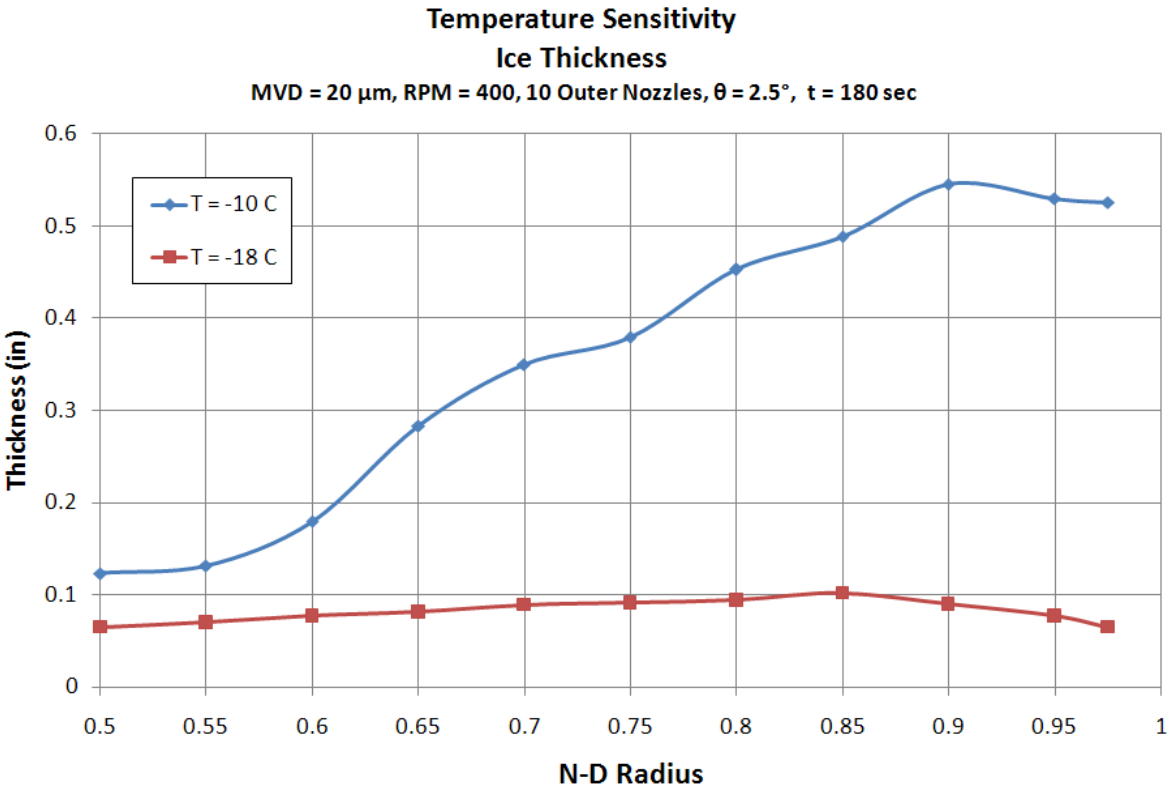


Figure 72: Temperature Sensitivity

3.4.4.4 *MVD Effects*

Droplet size effects were investigated as part of the shakedown tests. As expected, the larger droplet sizes accreted larger ice thicknesses on the rotor due to the higher LWC. Comparisons were made at two rotor velocities, as shown in Figure 73. The results indicate that the droplet size control system was functioning properly.

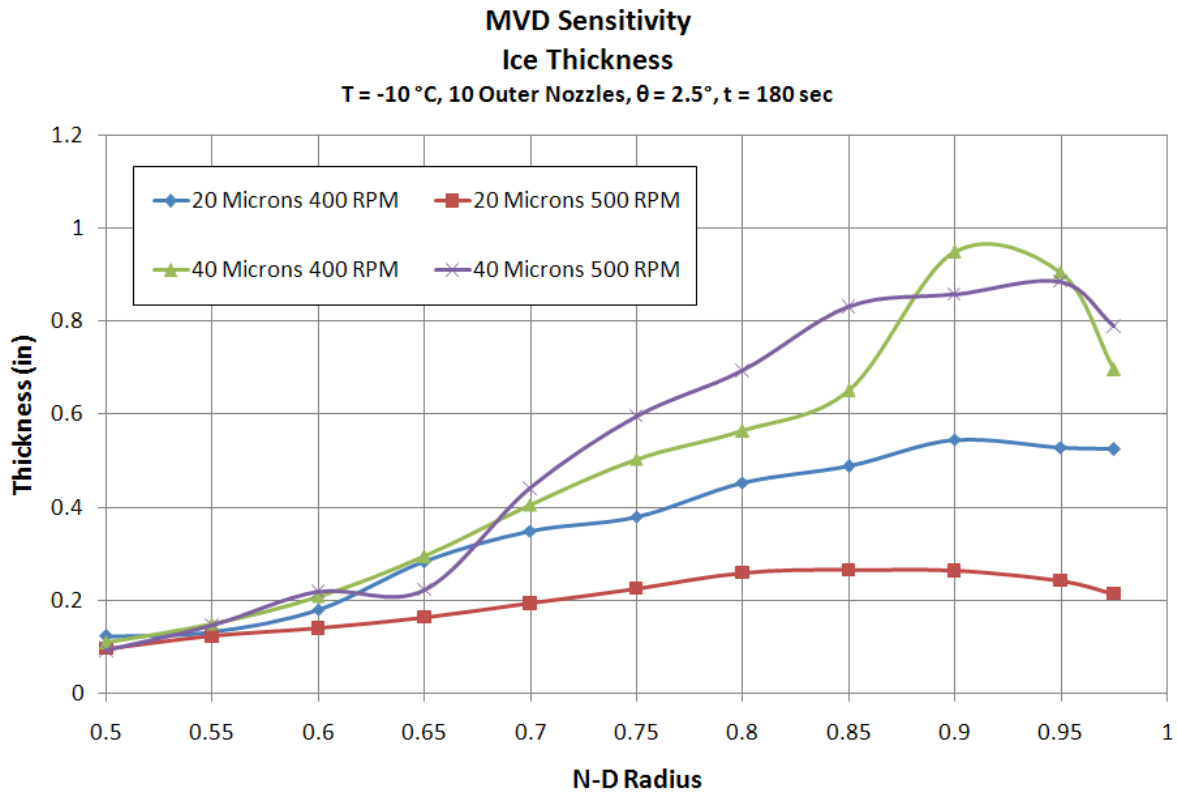


Figure 73: Droplet Size Sensitivity

3.4.4.5 *Test Repeatability*

Test repeatability was explored with Cases 5 and 6. Matching a particular test case in the AERTS is possible, but is difficult due to lack of precise temperature control. The results

from both cases were similar in terms of thickness, ice shape, and shedding when care was taken to match exact test cases. Procedures for accomplishing this matching for future tests were developed.

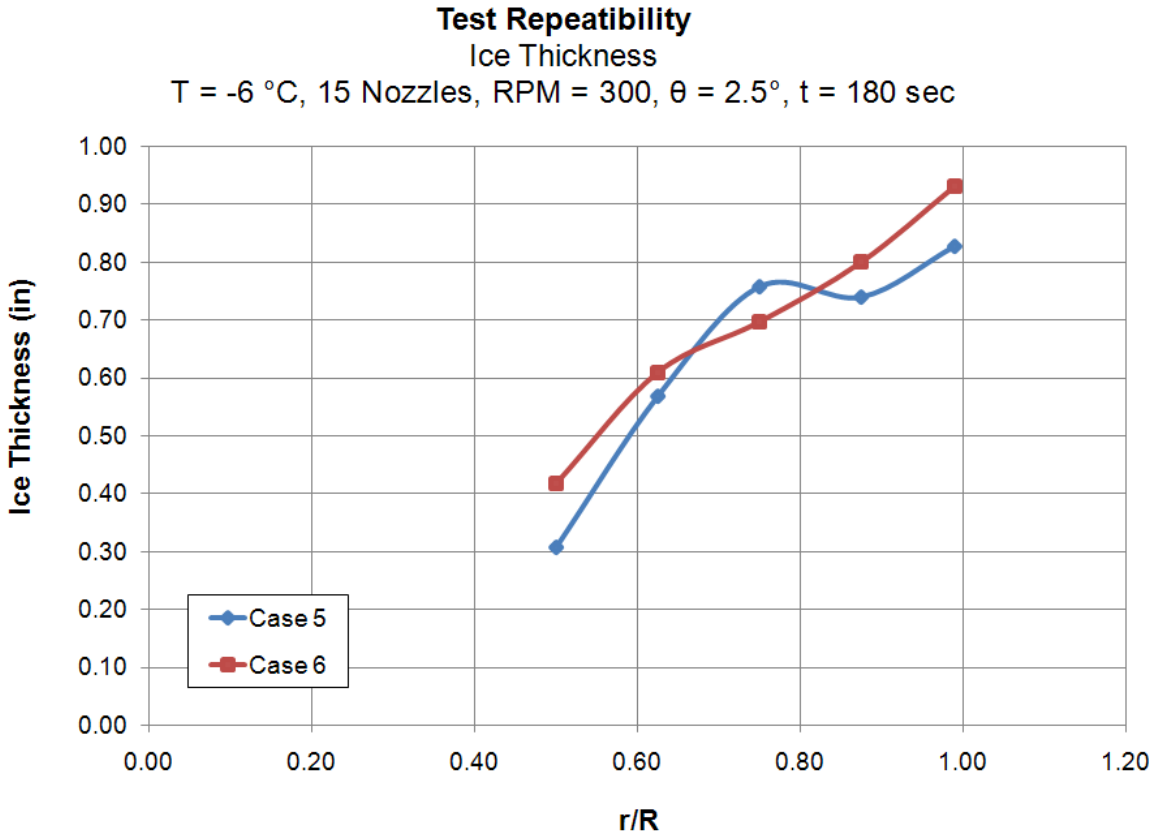


Figure 74: Case 5 and Case 6 Thickness Comparison

3.4.4.6 *Shedding*

Ice shedding was encountered on a number of the test cases. When a shedding event was detected by the load cell, the rotor was immediately shut down due to excessive rotor loads from mass imbalance. Shedding only occurred from one blade in each test, and one event is shown in Figure 75.

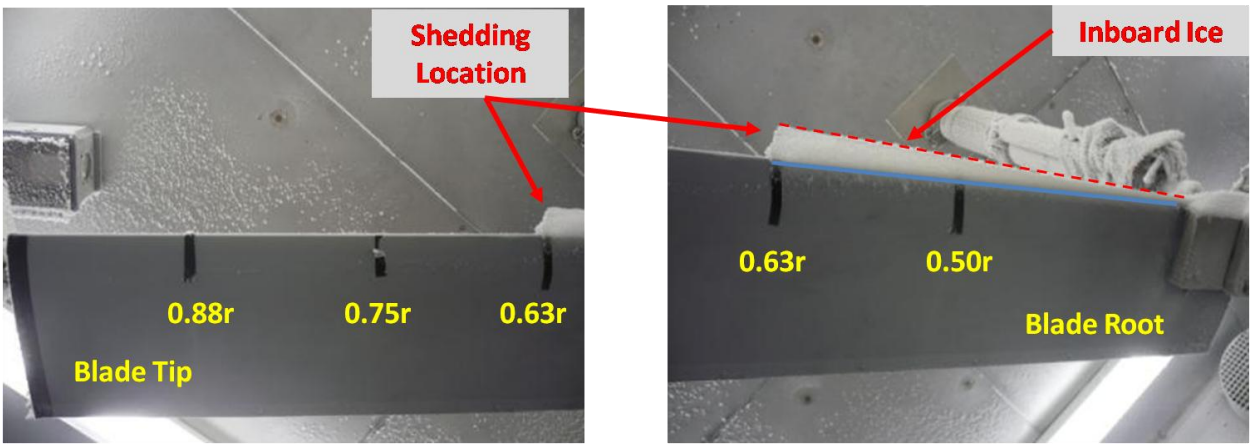


Figure 75: Case 2 Shedding Event

The nature of shedding proved that the major assumptions of the ARISP shedding module were correct. All shedding occurred during tests at high temperature and/or tests with large particles, as expected since ice cohesive and shear adhesion strengths decrease with increased temperature. Also, the shedding events started at the tip and propagated inboard as expected.

Unfortunately, none of the events were predicted by the ARISP model. The reason for this discrepancy was that significantly more ice was accreted in the AERTS chamber than predicted as shown in Figure 76. This generated up to a 4x larger ice mass, which generates higher centrifugal forces that exceed the shear adhesion and cohesive strength of the ice.

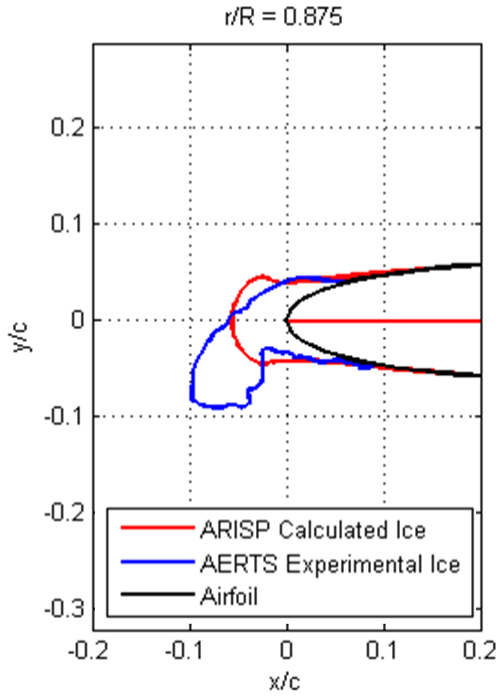


Figure 76: Case 2 Ice Tracing

3.4.5 Shakedown Test Conclusions

Overall, the shakedown tests were successful, as much was learned about the chamber. The primary result was the development of testing procedures to achieve proper ice shapes.

Crystallization issues prevented proper ice shape correlation to the ARISP model. The cause was identified as an over saturation of super cooled droplets in the test chamber. Based upon the shakedown tests, as well as additional testing on 1" diameter cylinders, it was discovered that reducing the number of active nozzles and lowering the input air pressure significantly reduced the crystals in the chamber. This improved ice shapes and also had the effect of

reducing the LWC to within the intermittent maximum conditions as specified by FAR Part 25/29 Appendix C. Table 13 summarizes the key results of the shakedown testing.

Table 13: Shakedown Test Results Overview

Cases	Primary Objective	Key Discoveries
1 – 8	Determine Basic AERTS Test Procedures	Calibration rotor utility was limited RPM must exceed 500 for proper cloud mixing
9- 22	Icing Parameter Sensitivity Studies	Reduction of operational nozzles (< 5) Reduction of nozzle input air pressure (< 23 psi) Addition of mesh screens captured crystals Collective pitch must be limited to 5°

4. Ice Shear Adhesion Strength Calculation

4.1 Background

An important issue for rotorcraft icing modeling and performance prediction is the modeling of ice shedding. The adhesion strength of ice to the rotor is a primary parameter in these calculations. Much effort has been expended to develop materials with low adhesion strength to reduce ice accretion and promote the shedding of ice that has been accreted. These materials have had various problems in application to rotorcraft, most notable is an inability to withstand erosion.

In order to quantify the performance of basic aircraft materials as well as new "icephobic" coatings, many researchers have attempted to determine the shear adhesion strength of ice. The problem, however, is that the published data varies significantly for a particular material as shown in the Table 14.

Table 14: Shear Adhesion Strength for Aluminum, T = -11˚C, **Freezer Ice, †Impact Ice*

Author/Date/Reference	Method	Aluminum Shear Adhesion Strength	
		psi	kPa
Loughborough* 1946 (80)	Pull	81	558
Stallabrass and Price† 1962 (81)	Rotating Instrumented Beam	14	97
Itagaki† 1983 (82)	Rotating Rotor	4 - 23	27 - 157
Scavuzzo and Chu† 1987(68)	Shear Window	13 - 42	90 - 290
Reich* 1994 (83)	Pull	130	896
PSU AERTS*	Pull	76	526

There are few reasons for the variation. Differences are generated in test results based upon the manner in which ice was created on the test material. Some authors have used "freezer" ice, in which liquid water is allowed to slowly freeze to the test material. This does not properly represent encounters with natural icing clouds which involves supercooled water droplets impacting an exposed surface. Other authors have studied accreted or impact ice to include the differences in ice structure relative to the freezer ice. The parameters of the test cloud are not always detailed, however.

A related problem is that not all parameters of a particular material are reported. For example, aluminum has many grades and finishes, each with a slight difference in surface structure. This has an effect on surface roughness, which has been shown by Chu and Scavuzzo to have an effect on shear adhesion strength (68). Surface roughness issues are particularly important for rotorcraft, which may operate in both erosive sand/rain environments and icing conditions. An eroded blade leading edge will have an effect on the shedding performance of the rotor.

Another cause of discrepancies is differences in test method and facilities. Various mechanical methods have been used and each test mechanism has inherent differences. In addition, icing facilities have different limitations, and some require that data ice is accreted in one location and then the ice and fixture are moved to another location for testing.

Moving the ice introduces mechanical and thermal shocks that influence data accuracy. This effect has been seen in previous testing in the AERTS Facility. The velocity limitations of most icing wind tunnels is the most important issue when applying the data to rotorcraft icing analyses.

4.2 Bending Beam Rotor Design

4.2.1 Overview

The current research has the goal of reducing data scatter contained in previous test procedures. The first way errors can be introduced into test data arises from the test method. In some test processes, the adhesion strength of ice is typically calculated in a two step process. First, ice is accreted to a test coupon in an icing facility, such as the IRT. After the ice accretion process, the samples may be transported to another rig to apply mechanical loads onto the ice to determine the shear adhesion strength. The transportation of ice from one test rig to another may introduce mechanical and/or thermal shocks that significantly influence the shear adhesion strength of ice.

The second way in which errors may be introduced is in use of appropriate test conditions. Most test data is collected in an icing wind tunnel, which generally have a maximum velocity of 475 ft/sec. This does not approach typical rotorcraft tip speeds around 700 ft/sec. Application of data taken at low velocities to the high speed analyses may introduce significant errors due to issues with heat transfer.

4.2.2 Design Requirements

In order to improve upon the previous ice adhesion strength calculations, a new test fixture was desired. The primary requirements of the new system included:

1. Determine the shear adhesion strength of ice to various materials in the AERTS Facility
2. Calculate the adhesion strength at realistic icing conditions, including icing parameters, (Temperature, LWC, MVD) and rotor parameters (Tip Speed, AoA)
3. No user intervention is permitted to avoid introducing unwanted mechanical and thermal shocks to the ice

4.2.3 Design Concept

4.2.3.1 *History*

After a reviewing the test methods described in the previous section, a test method was chosen. The design of the test mechanism is based on the work of Stallabrass and Price at the NRC Low Temperature Laboratory in Canada in 1962 (81). In this research, a specialized fixture was attached to the tip of a test rotor. The tip, shown in Figure 77, was designed to calculate the shear adhesion strength of ice to various materials.

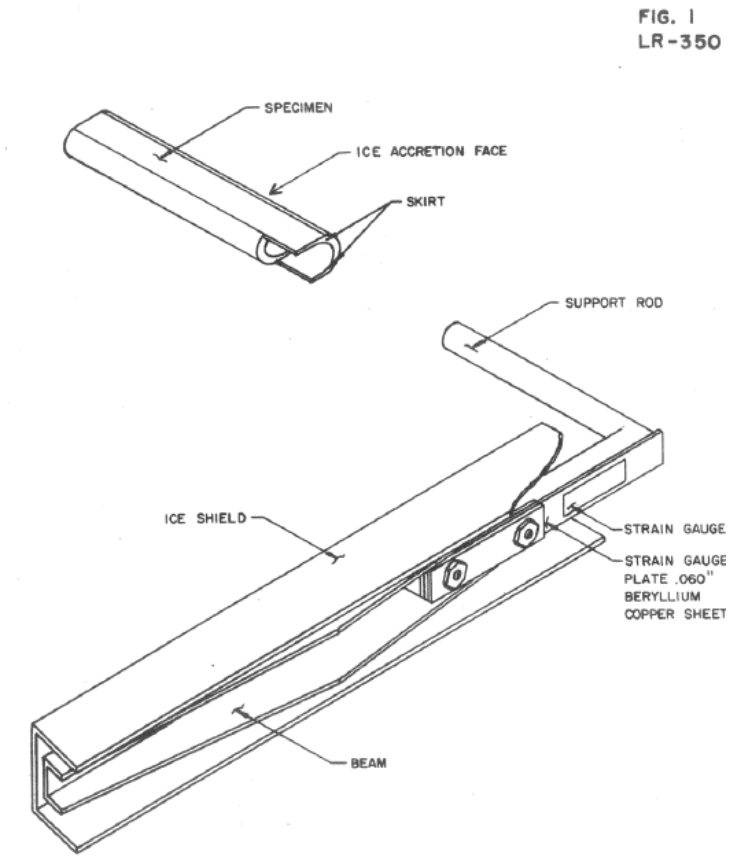


Figure 77: LR-350 Test Fixture
(adapted from Reference (81))

4.2.3.2 *Concept*

The test fixture is based around an instrumented beam that is designed to bend only in the spanwise axis. A test specimen is mounted to the tip of the test beam such that it does not interfere with the motion of the beam. When subjected to an icing environment, ice accretes on the test specimen. This additional mass increases the centrifugal force of the coupon assembly on the beam, causing it to deflect and increasing the strain read by the strain gauge mounted at the root. When ice reaches a critical thickness, it shed from the fixture, instantly reducing the strain in the beam. A schematic of this process is shown in Figure 78.

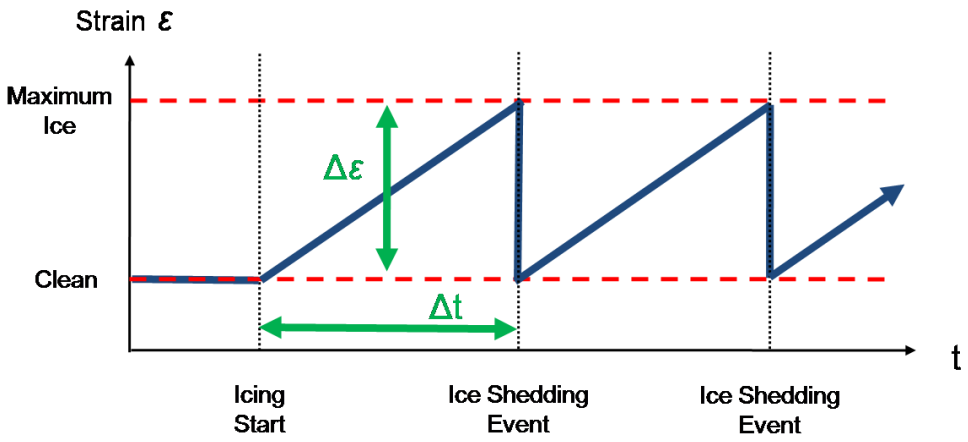


Figure 78: Shear Adhesion Strength Fixture Theory

The fixture generates two important parameters. The change in strain can be related to the ice mass on the beam. This relationship is dependent on the beam design and is explained in the following section. A second parameter is the time between ice shedding events. This parameter is a function of the specimen geometry and is important to determine how clean the specimen is after many icing cycles. If the minimum point in the sawtooth shape starts increasing, it is indicative that the ice is not fully shedding off of the test coupon.

The primary benefit of the system is the fact that ice adhesion strength of ice to various materials without user intervention. The system can be run through many icing cycles to determine an average shear adhesion value. For extended runs, special consideration must be taken in the design to ensure that ice accretes only on the test specimen and not on other portions of the test fixture. If ice bridges from the instrumented beam to the rotor, it mechanically couples the two components, influencing test results.

4.2.4 Design

4.2.4.1 *Icing*

For maximum flexibility, the test fixture should be able to generate large shear stresses at the ice/rotor interface. This gives the test fixture the ability to test materials with high shear adhesion strengths or materials degraded by sand or rain erosion. Ice accretion is highly dependent on geometry, so it must be studied to make sure that an adequate amount of ice is accreted to the coupon.

The primary parameter in these studies is RPM. At higher RPMs, the rotor encounters more icing particles in a given amount of time, but kinetic heating of rotor surfaces precludes the accretion of ice. Lower RPMs may generate more ice, but have lower centrifugal forces. Using a modified version of the ARISP code, the relative importance of these two effects were investigated to determine the optimal conditions generate the maximum shear stresses. Some results from these studies are shown below. Increasing LWC or MVD will accrete more ice and therefore generate potentially higher adhesion stresses.

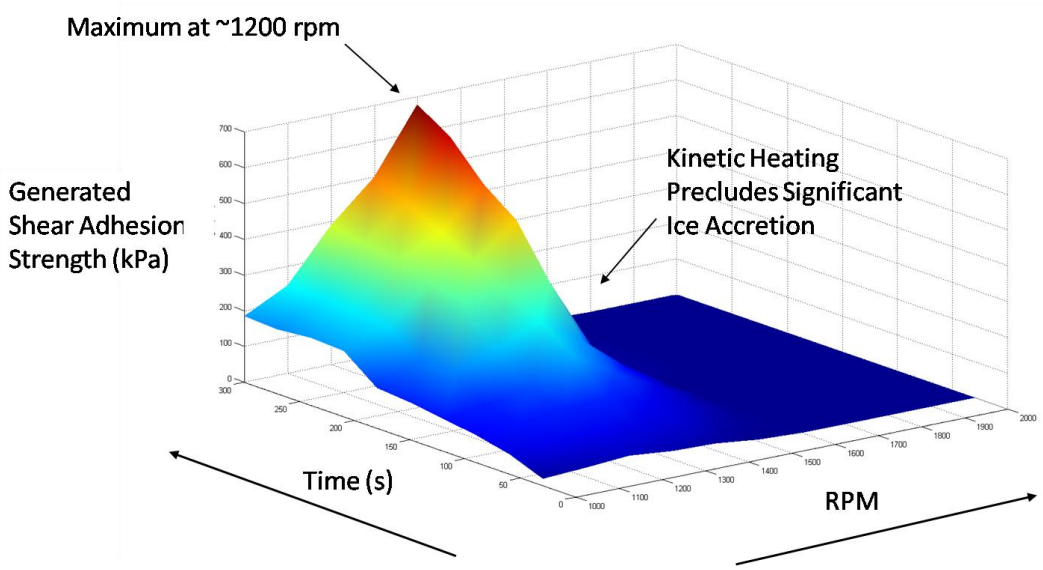


Figure 79: Shear Adhesion Optimization
T = -20°C, LWC = 1.5 gr/m³, MVD = 20 µm

Temperature has the largest effect on the ability of the test fixture to generate high shear adhesion stresses. Due to kinetic heating effects, little ice accretes at higher temperature as shown in Figure 80.

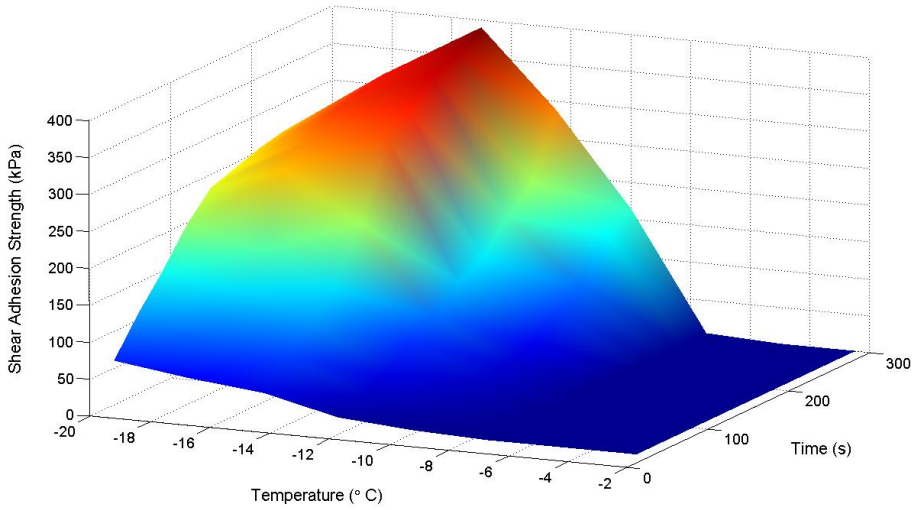


Figure 80: Temperature Effects
RPM = 1200, LWC = 1.5 gr/m³, MVD = 20 µm

Testing is therefore able to be accomplished on a variety of materials within the AERTS icing envelope. Exact testing conditions are selected for each test program, but will generally be around -10 °C.

4.2.4.2 *Mechanical*

The primary mechanical requirements for the bending beam are twofold. First, the beam must safely withstand the centrifugal forces associated with high RPMs. Secondly, the beam must generate a large enough strain field for the strain gauges to sense the accretion of ice and the important shedding event. These requirements are somewhat mutually exclusive, and required careful design to achieve.

4.2.4.2.1 *Stress Analysis*

The preliminary design for the bending beam was driven by elementary beam equations to determine suitable materials and basic geometry. From these studies, the general beam geometry was established. From a material standpoint, 6Al-4V titanium was selected because it was the only readily available material with a high enough strength to weight ratio. FEM analysis was used to model the system in detail. This analysis guided the final dimensions of the beam. A schematic of the final design beam is shown in Figure 81.

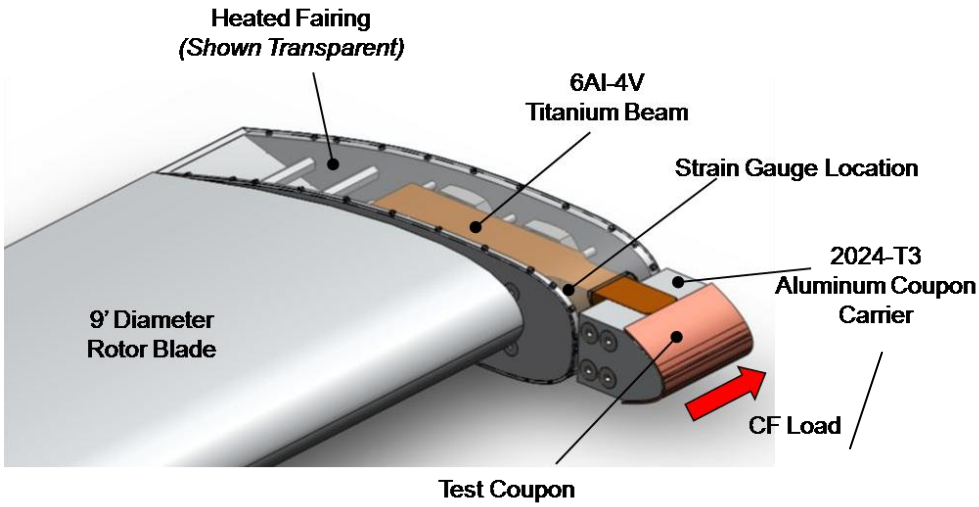


Figure 81: Schematic of Ice Shedding Rotor Beam with Fairings

The test fixture is mounted at the tip of a specialized blade. This blade is compatible with both available rotor hubs, so testing can continue if a problem arises with either system.

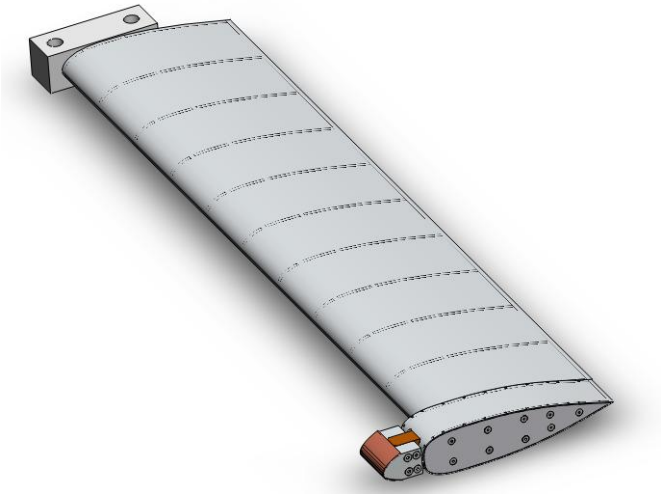


Figure 82: Single Blade Assembly with Test Fixture Mounted at Tip

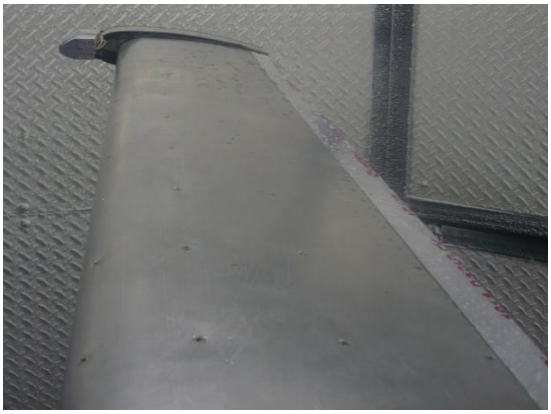


Figure 83: Test Rotor Implementation, view from hub

The beam was designed such that it provides a minimum of 10 μstrains prior ice shedding. This was determined based upon the requirement of generating a large enough signal to noise ratio for the strain gauges. This consideration was vital given the electronic

environment in which the strain gauges would be implemented. Figure 84 shows a sample output from the finite element analysis (FEA).

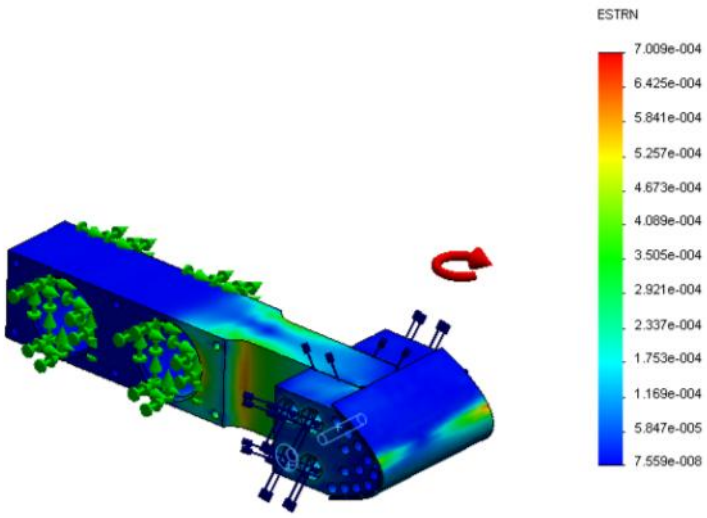


Figure 84: FEM Strain Field on the Ti Beam (1200 RPM)

While a large strain field was the primary design target, the stresses on the beam were important. Aside from being directly related to the strain field, the stresses had to be low enough for the beam to survive the high forces associated with large rpms. In the AERTS facility, every part needs to be designed to have a Factor of Safety (FOS) > 2 for the design condition. This generates a large margin for the parts to handle unanticipated loads. The following pictures show the stress and FOS distribution in the beam assembly.

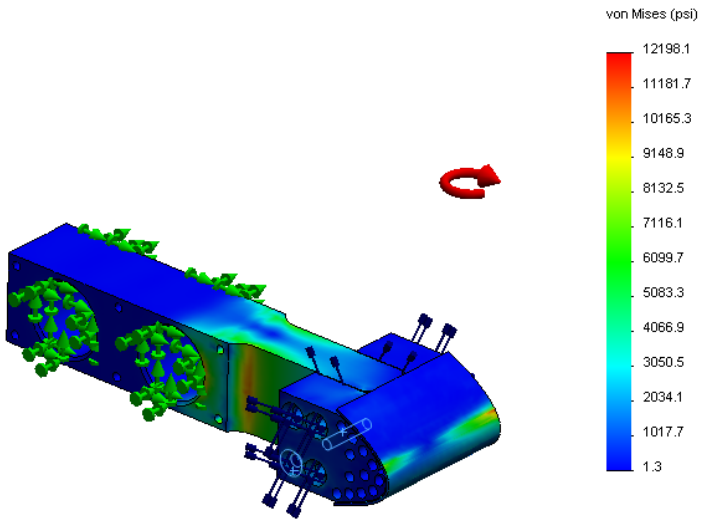


Figure 85: Beam Assembly Internal Stresses (1200 RPM)

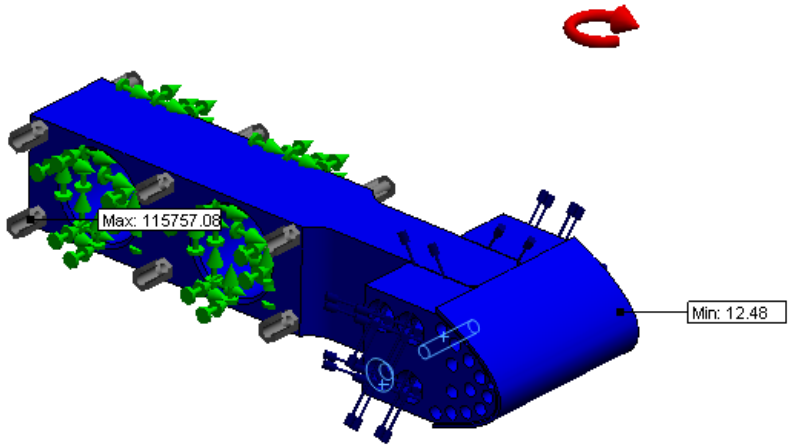


Figure 86: Beam Assembly FOS (1200 RPM)
Blue Areas: FOS > 2
Red Areas: FOS < 2

Table 15: Bending Beam Expected Strains

RPM	Expected Strains		Delta Strain
	Maximum Ice	No Ice	
800	4.67E-04	4.24E-04	4.32E-05
1000	7.30E-04	6.28E-04	1.02E-04
1200	1.05E-03	9.04E-04	1.48E-04
1500	1.63E-03	1.55E-03	7.40E-05

4.2.4.2.2 *Coupon Mounting*

For the current research, coupons are mounted to the carrier with four 4-40 flat head screws. This allows coupons to be multiple coupons to be tested with one setup. No simulated ice protection heat was required for the current research.

4.2.4.2.3 *Mounting/Fairing Design*

The beam is mounted to the rotor spars via two 1-8 316 stainless steel bolts and thread engagement is designed to withstand centrifugal loading of the system up to 2000 RPM. This secure mounting is essential because it focuses the stresses (and strains) in the area of the strain gauges.

To ensure a proper collection efficiency on the test coupon, a fairing is mounted aft of the coupon assembly. This fairing helps reduce drag on the rotor tip, important to any rotor design. The most important aspect of the fairing is that it prevents ice accretion on the beam, which would influence test results. LEWICE was used to verify the ice accretion patterns on the fairing. Additional ice protection comes from heaters mounted inside of the fairing.

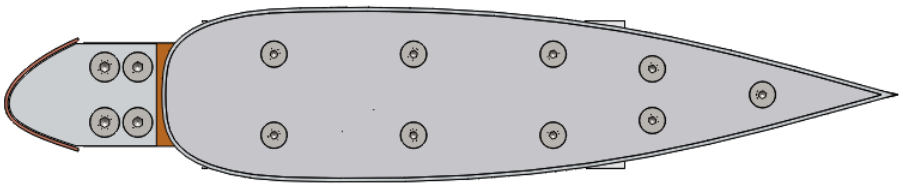


Figure 87: Fairing Design

4.2.4.3 *Heaters*

4.2.4.3.1 *Anti Ice System*

Heating mats were included on the test fixture to preclude ice bridging between the test coupon and the fixed fairing. Bridging was an issue with some early test fixtures in the AERTS facility and is a noted problem in previous ice testing. The heating mats were designed to prevent unwanted ice from interfering with readings of the strain gauges by mechanically coupling the sensitive bending beam to the fixed frame. Four mats are installed in the fairing as shown in Figure 88. All mats have a power density of 10 W/in^2, and are wired in series. Both blades receive the same heating.

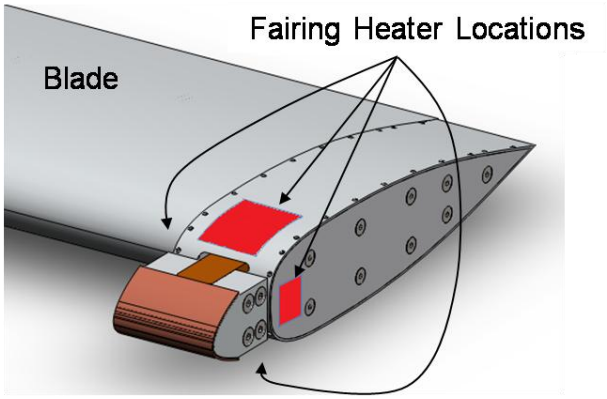


Figure 88: Fairing Ice Protection Heater Mat Locations (shown in red)
Two additional heater mats are located on the inboard side of the fairing and the bottom of the fairing

Initial testing in the AERTS chamber showed the need to anti-ice the rotor. Ice shed from the inboard blade areas, then impacted the fairing and triggered a false coupon shedding event. Twenty heaters were installed in each blade to provide an anti-icing capability, as shown in Figure 89. These heaters are 3" x 3" and have a power density of 5 W/in^2.

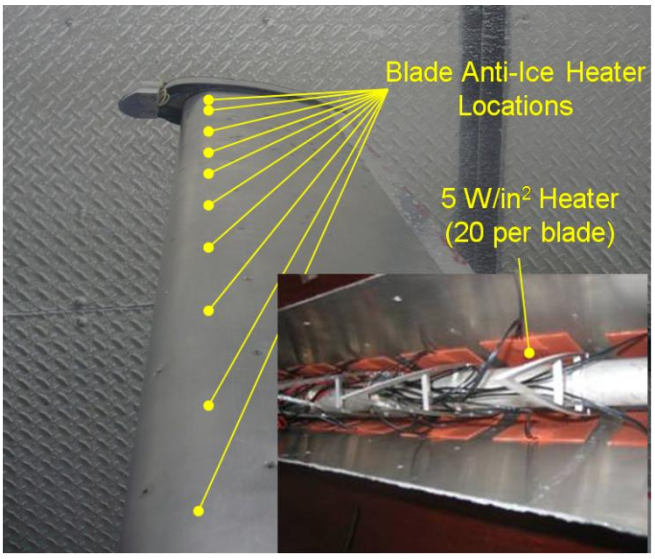


Figure 89: Shedding Rotor Anti-Ice Heaters

The anti-ice heaters provide an adequate ice protection and eliminated most of the problems with the inboard ice. Some ice, however, still accreted in the corner of the tip fairing and blade as shown in Figure 90. This problem was rectified with the addition of a 3" x 7" heater with a 10 W/in^2 power density to the external surface of each blade. The entire heater package permits safe operation of the rotor system. Since ice can only accrete to the coupon, mass imbalances are reduced.

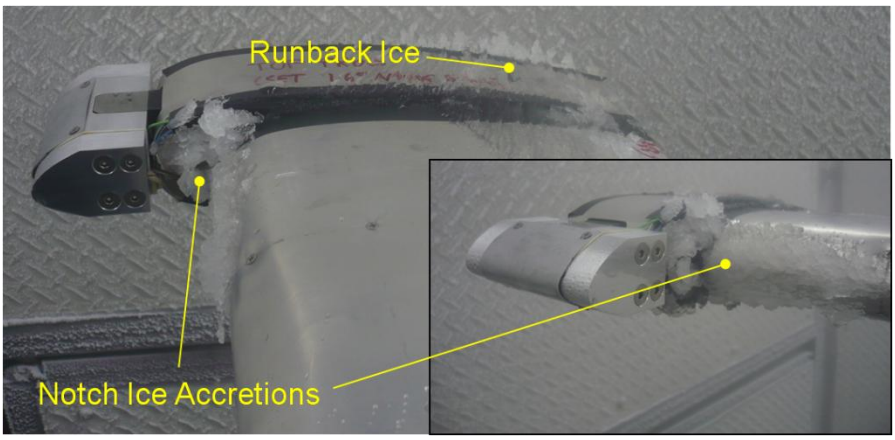


Figure 90: Notch Ice Accretions

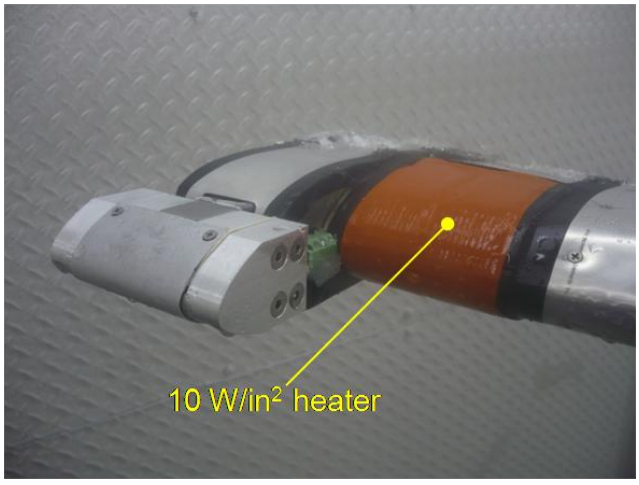


Figure 91: External Notch Heater

4.2.4.3.2 *Electrothermal Ice Protection Simulation System (EIPSS)*

In addition to heating mats on the fairings, heating coils are introduced on the aluminum coupon carrier. These heaters, which form the Electrothermal Ice Protection Simulation System (EIPSS), will be used to apply heat in a similar manner to a rotor ice protection system. This allows for studies relating heating power inputs and improvements on ice shedding capabilities of the different coatings to be conducted.

The EIPSS is formed by passing a 50 W/in heat coil through the coupon carrier. The coil provides controllable effective power density of up to 12 W/in^2, depending on the material undergoing testing. Finite element transient thermal models were used to calculate this figure. A thermocouple is mounted at the interface of the carrier and the test coupon to measure the temperature on the carrier/test coupon system during operation and provide an analytical correlation reference point. This thermocouple will assist with quantification of differences in thermal conductivity of the various coating materials while constant power input is applied during ice shedding testing.

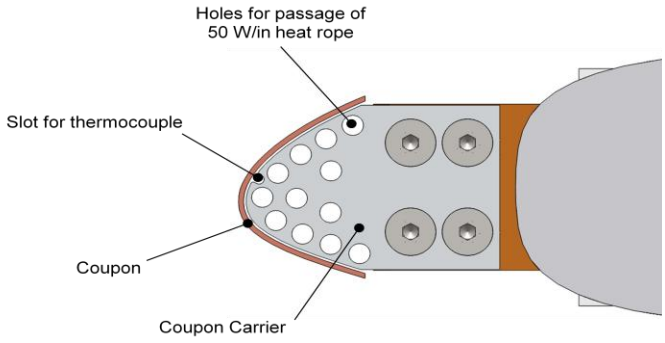


Figure 92: EIPSS Design

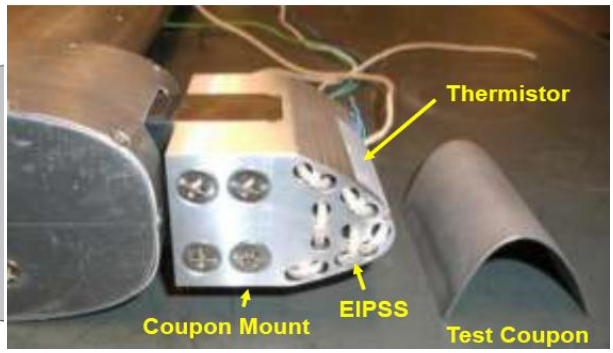


Figure 93: EIPSS Implementation

During initial icing testing in the AERTS chamber, ice accreted on the sides of the coupon, bridging to the coupon carrier and the EIPSS heat rope. An example of this is shown in Figure 94. This artificially raised the shear adhesion strength of the material. To counter this, additional fairings were added on both sides of the coupon carrier to cover the EIPSS wiring. Each fairing is anti-iced with a 25 W cartridge heater. A 0.03" layer of glass fiber insulation is mounted between the fairing and coupon carrier to reduce heat transfer from

the cartridge heaters to the coupon carrier. This prevents the heat from affecting shear accretion measurements.

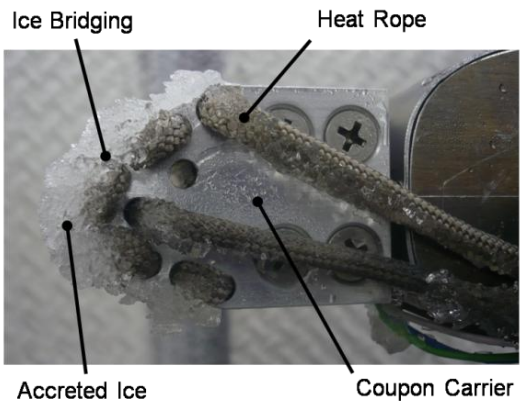


Figure 94: Ice Bridging

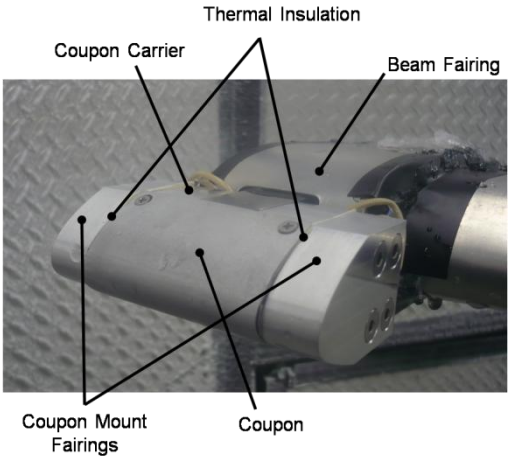


Figure 95: Side Fairings

4.3 Instrumentation

4.3.1 Configuration

A full bridge strain gauge configuration is used to measure the bending of the beam in the spanwise direction. This configuration was selected to allow for temperature compensation, increase linearity of the strain measurement, and decrease noise as compared to a half bridge configuration. The full bridge configuration also rejects all other loads, including axial, improving data accuracy.

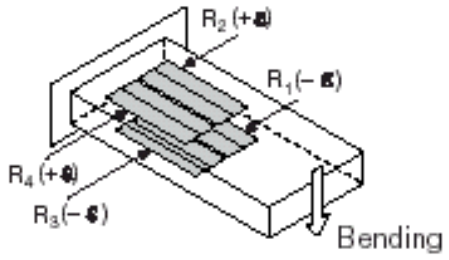


Figure 96: Full Bridge Strain Gauge Configuration Schematic

The gauges are mounted on the beam as shown in Figure 97. This locates them in the area of high strain designed into the beam. Solder pads are used to reduce stress on the gauges from attached wires and the entire assembly is coated in a layer of polyurethane to protect it from ice.

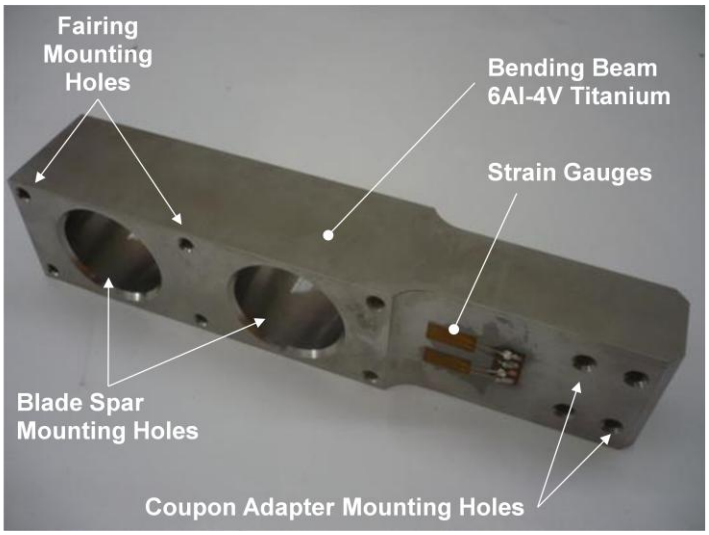


Figure 97: Strain Gauge Locations

4.3.2 Amplifier

To reduce the risk of noise on the data as transferred through the slip ring, a signal conditioners and amplifier is located at the root of each blade. These miniature amplifiers are rated for the harsh environments found in the AERTS Facility, and are calibrated for use down to -40 °C.

The primary benefit of the amplifier is that it can be programmed via a USB connection to a PC. In the programming software, the user can modify the offset and adjust the gain of the system. This allows the AERTS lab to tailor the amplification to the optimum point for the

beam assembly. The amplifier also features a small thermistor that serves as an additional temperature measurement point. This sensor, as well as the small size of the amplifier, can be seen on the amplifier data sheets shown below.

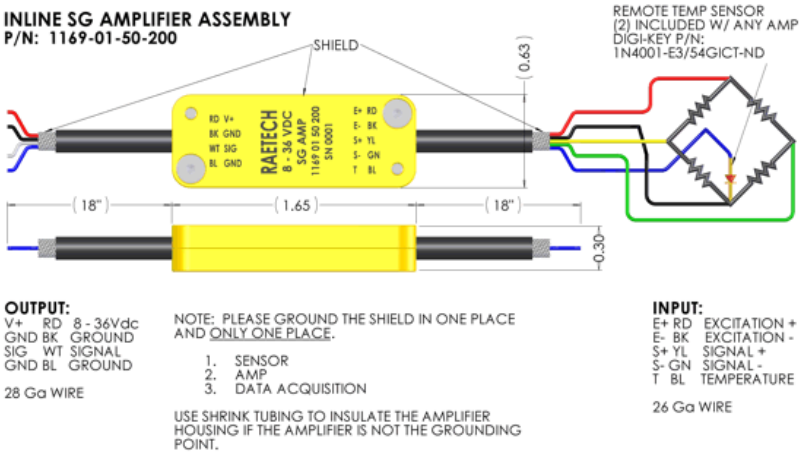


Figure 98: Strain Gauge Amplifier Schematic (Courtesy Raetech)

Table 16: Ratech 1169-01-50-100 Amplifier Specifications

Electrical Output	
Output Voltage	0.1 - 4.9 V_{DC}
Gain Range	±2 to 1150
Low Pass Filter Cutoff Frequency	80 Hz
Offset Range	0.1 - 4.9 V_{DC}
Gain Absolute Accuracy	±0.5%
Gain Temperature Stability	6ppm/°C - 15ppm/°C
Offset Temperature Stability	4ppm/°C - 10ppm/°C
Electrical Inputs	
Supply Voltage	8 to 36 V_{DC} (unregulated)
Bridge Resistance	min 175Ω
Zero Drift Compensation Range	±60 mV
Temperature	
Calibrated Range	-40 to 125°C
Functionally Tested	-80 to 200°C
Internal Measurement Accuracy	±2°C
Remote Measurement Accuracy	User Calibrated
Protection	
E+ short to ground	Continuous ≤12 V_{DC} supply
E+ short to ground	5 sec ≤40 V_{DC} supply
SIG+ Output short to ground	Continuous
Supply Voltage	Reverse Polarity Protected

4.4 Beam Calibration

4.4.1 Rotating Sensitivity Study

Prior to any testing, a non-icing sensitivity study was performed with the beam. The study was critical to validate that all components were functioning properly and that the required signal to noise ratio was achieved. Due to manufacturing issues, the rotor cannot be rotated above 380 RPM due to large 1/rev loading issues. This is far from the design RPM of 1200 RPM, so beam sensitivity needed to be investigated at the lower RPMS.

For the studies, the test rotor was rotated at various RPMS without a coupon and the strain gauge voltage response was recorded. A coupon was then adhered to the carrier with double sided tape and the tests repeated. The additional mass of the coupon and tape was 19.6 grams. Two gains were tested, as shown in Figure 99. The response of the beam is shown in Figure 100.

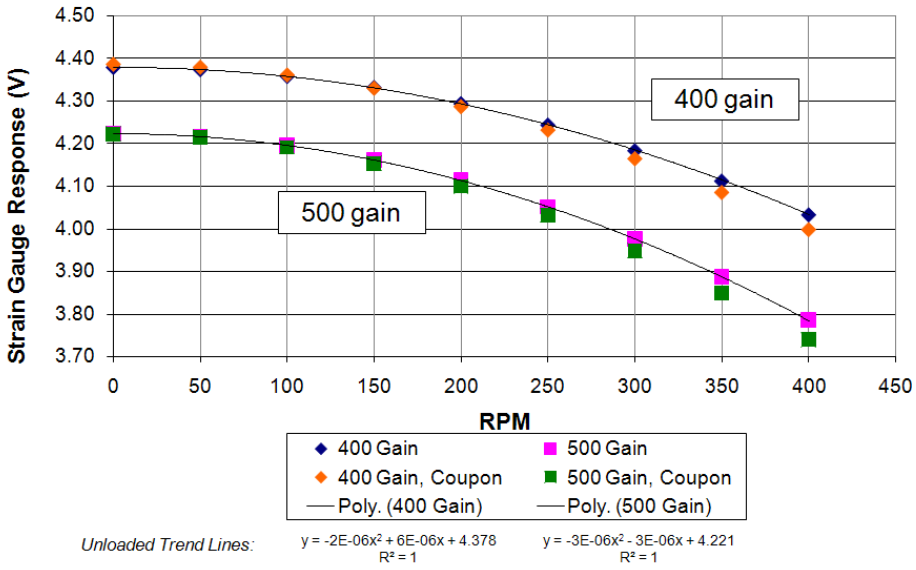


Figure 99: Bending Beam Response

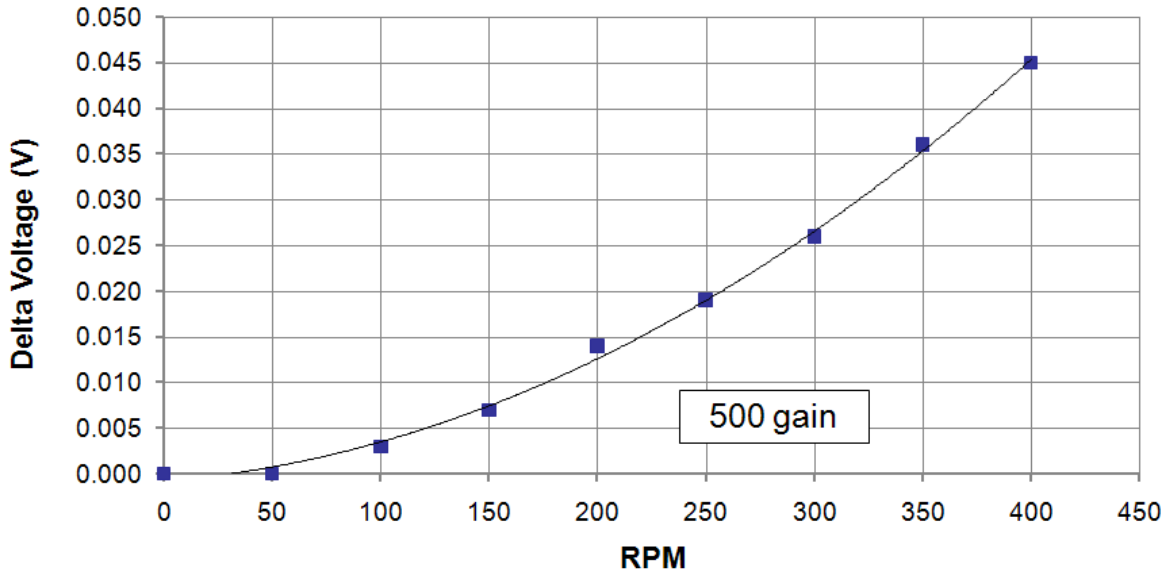


Figure 100: Coupon Sensitivity Study Loads

The results of the testing showed that the system was capable of resolving the strain of small ice accretions, even at low RPM. The gain and offset for the gauges was identified at 500 and 4.5V, respectively.

4.4.2 Beam Logic

Once the beam response was confirmed, the voltages were converted to beam load and later to shear adhesion strength. The concept of this conversion is shown in Figure 101. The process is coupon specific, as it depends on the coupon mass. Consequently, prior to coupon installation on the rotor, it must be weighed.

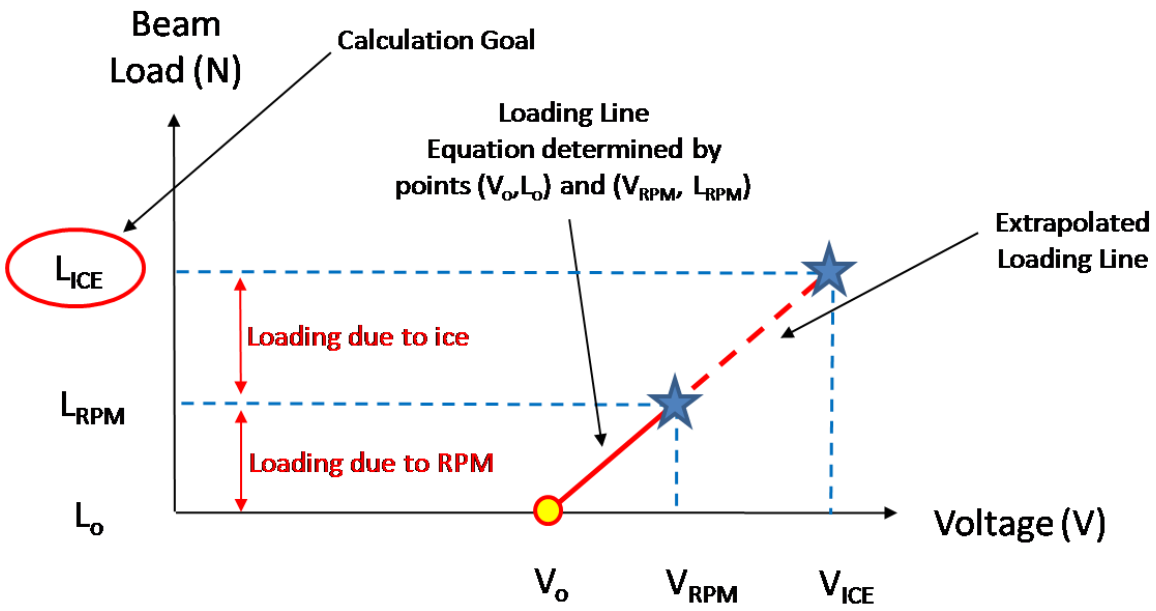


Figure 101: Beam Loading Concept

Before rotation, the 0 RPM voltage and load, (V_o and L_o respectively), are recorded. This accounts for any minimal drift in strain gauge zero load condition due to temperature. The rotor is then spooled up and the voltage, V_{RPM}, is recorded. This value is converted to load with Equation 30, where m_{assy} is the total mass of the carrier assembly, including coupon, carrier, fairing heaters and mounting bolts. The radius is determined at the center of the coupon assembly and is fixed at 1.41 m (4.625 ft).

$$L_{RPM} = m_{assy} r\Omega^2 \qquad \text{Equation 30}$$

With the points (V_o, L_o) and (V_{RPM}, L_{RPM}) identified, the equation of the line can be calculated with the common point-slope formula. Ice is then accreted to the coupon, increasing the strain gauge response. The response due additional mass is linear as indicated by Equation 30. When ice is shed, the final voltage is recorded. The beam load due to ice can be

calculated by extrapolating the previously determined loading line. This load due to ice is then converted to shear adhesion strength with Equation 31. Due to the coupon, the geometry and heater configuration, the adhesion area, A_s, is constant, especially for long icing times.

$$\tau = \frac{L_{ICE} - L_{RPM}}{A_s} \qquad \text{Equation 31}$$

A Labview computer code was developed to simplify the data recording process and provide the user with reminders to turn on anti-ice heaters. Shedding events are identified based upon differential voltages between subsequent time points. All data is output to a .txt file for post processing.

4.5 Adhesion Strength Calculation Test Results

4.5.1 Test Coupon

The test coupon for the current research was cut from the aft portion of a scrap Schweizer 269C rotor blade and was constructed from 2024 Aluminum. The paint and anti-corrosion anodize layers were removed with a commercial bead-blasting machine, leaving a bare aluminum surface that matched the leading edge of the icing trials test rotor. The matching of the materials and surface finish was critical in the shedding events. The coupon was attached to the carrier with four countersunk 4-40 screws as shown in Figure 95. These screws may influence the test results by providing additional area for ice to accrete, but are necessary for the system to be able to test multiple coupons.

4.5.2 Test Matrix

Due to AERTS facility scheduling issues, only a limited test matrix was attempted. These tests were part of the system shakedown for an AATD/Boeing sponsored test program. Temperature, the most important icing parameter, was investigated as shown by the test cases described in Table 17. The LWC was not measured, but was approximately 6 gr/m^3. All testing occurred at 350 RPM.

Table 17: Ice Adhesion Test Matrix

Case	Start Temp	End Temp	Avg Temp	MVD	Time
	°C			µm	min
1	-9.8	-5.9	-7.8	40	7.8
2	-13.7	-8.5	-11.1	40	7.7
3	-8.9	-5.6	-7.2	40	7.8
4	-5.1	-4.9	-5.0	40	4.7

4.5.3 Test Results

Figure 102 shows a typical strain gauge response during an icing test. The rotor spool up and stabilization at the test RPM are not shown for clarity. As can be seen, the shedding event is clearly evident.

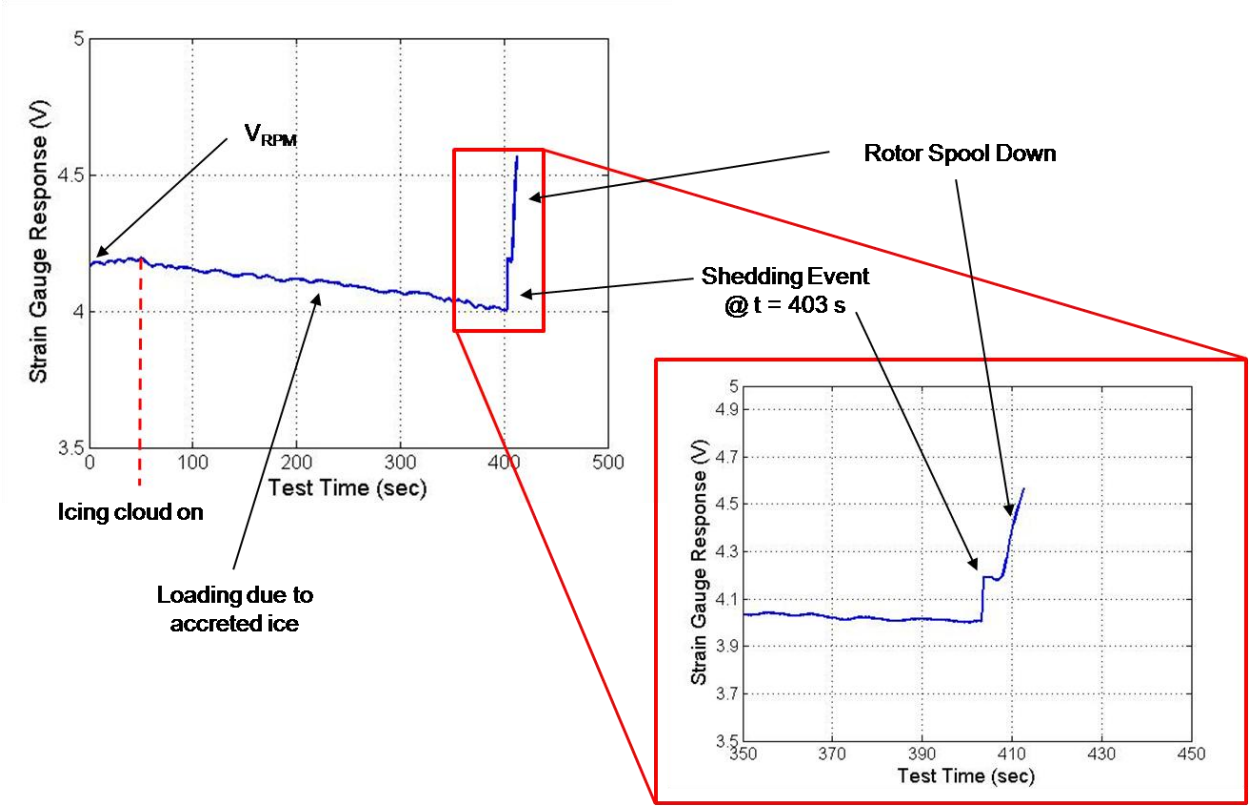


Figure 102: Typical Test Strain Gauge Trace

The results from the shear adhesion testing are shown in Figure 103. The long duration of the test significantly increased the temperature in the test chamber, so the average temperature was reported. Only a single point was generated in Case 1 due to issues with the strain gauge wiring. The expected trend of increasing adhesion strength with decreasing temperature was not observed due to issues with Case 2. The validity of the test point was therefore called into question, and a failure of the ice control system was found upon reviewing the data.

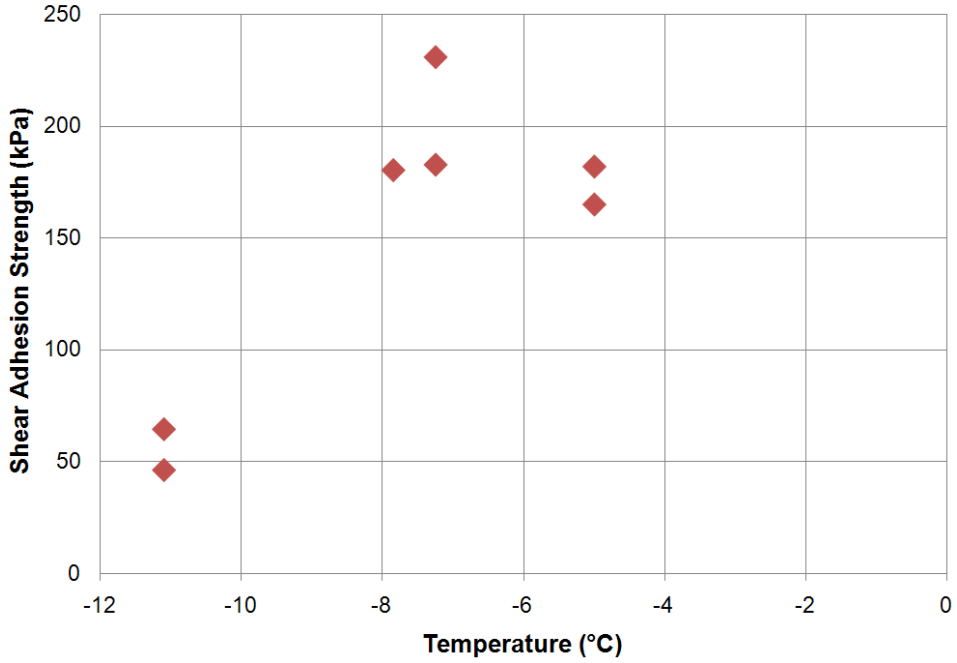


Figure 103: AERTS Shear Adhesion Strength Results

If Case 2 is removed, the data can be represented with the following equation, where T_c is the static temperature in Celsius.

$$\tau = -28.33 * T_C \text{ (in kPa)} \qquad \text{Equation 32}$$

The test results were compared against values in literature. Data from Loughborough (80), Stallabrass and Price (81), Itagaki (82), Scavuzzo and Chu (68) and Reich (83) are plotted in the following figure. As shown in Figure 104, the AERTS data points lie directly in the range of the published data. The test data shows a higher adhesion strength than the work of Stallabrass and Price, which the rotor test concept is based upon. A primary cause for this

discrepancy is the screws used to mount the coupon to the rotor system. Also, material surface roughnesses may be different. This parameter was not measured for the current test coupons and was not reported in most previous literature.

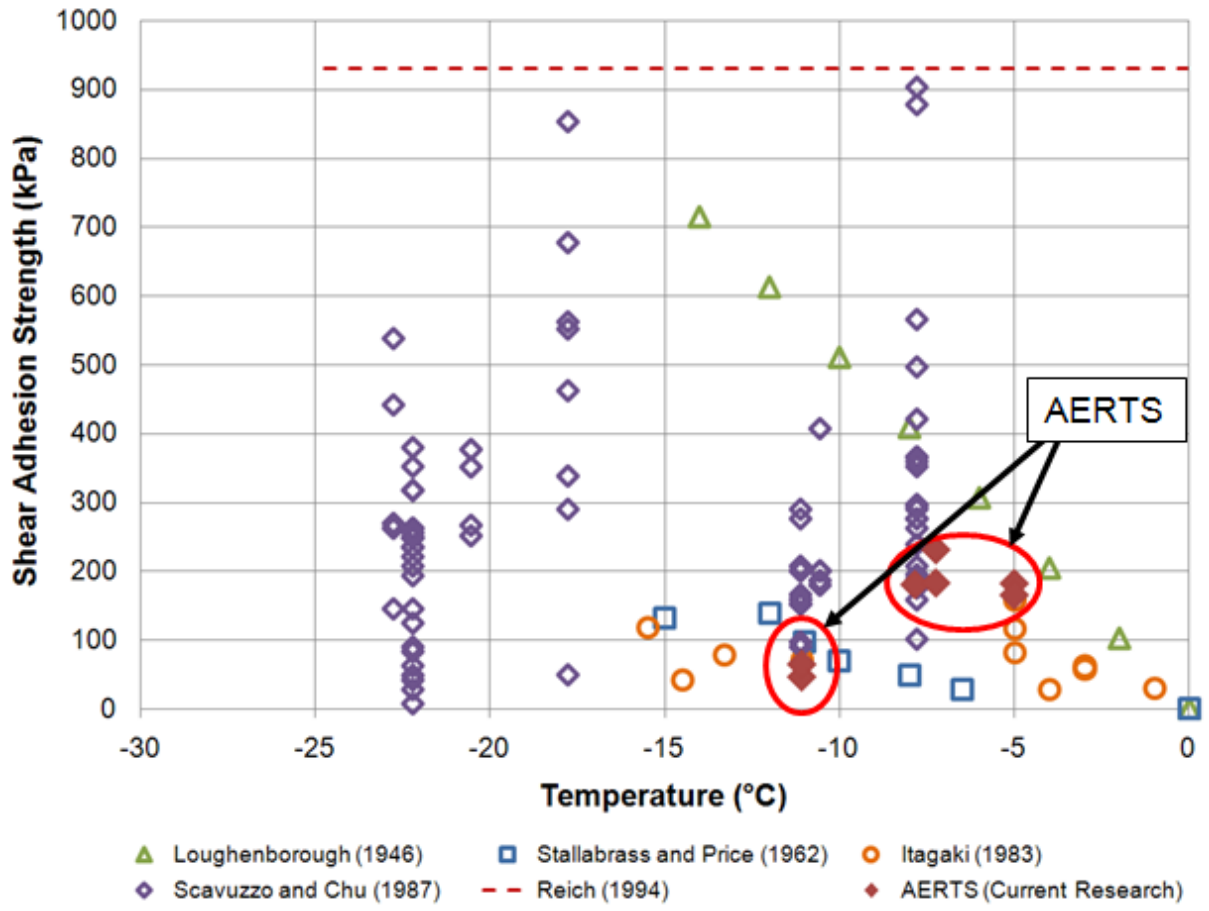


Figure 104: AERTS Shear Adhesion Strength Results Comparison

5. Experimental Investigation of ARISP Code

5.1 Testing Overview

5.1.1 Objective

A series of icing experiments were undertaken in the AERTS chamber. The tests explored a wide range of icing parameters, including temperature, droplet size and icing severity. The primary objective was to validate the ARISP model under realistic icing conditions. Correlations between the accreted ice thickness, impingement limits, performance degradation and torque were made to understand limitations of the icing model.

The secondary objective of the tests was the continued development of the AERTS facility operational envelope. Since the AERTS facility has only recently achieved an initial operational capability, the icing system limitations are not fully understood. Many tests are required to calibrate the laboratory.

5.1.2 Test Rotor

For the ARISP model validation, shortened Schweizer 269C blades were used. This 7.75' diameter, 6.8" chord rotor was designed for testing up to 1200 rpm (486 ft/sec) and features a NACA0015 airfoil. Appendix D of this thesis contains more details of the rotor design and integration to the AERTS rotor hub. Figure 105 shows the facility test configuration. For all tests, the slip ring assembly was disconnected to reduce wear on the slip ring brushes.

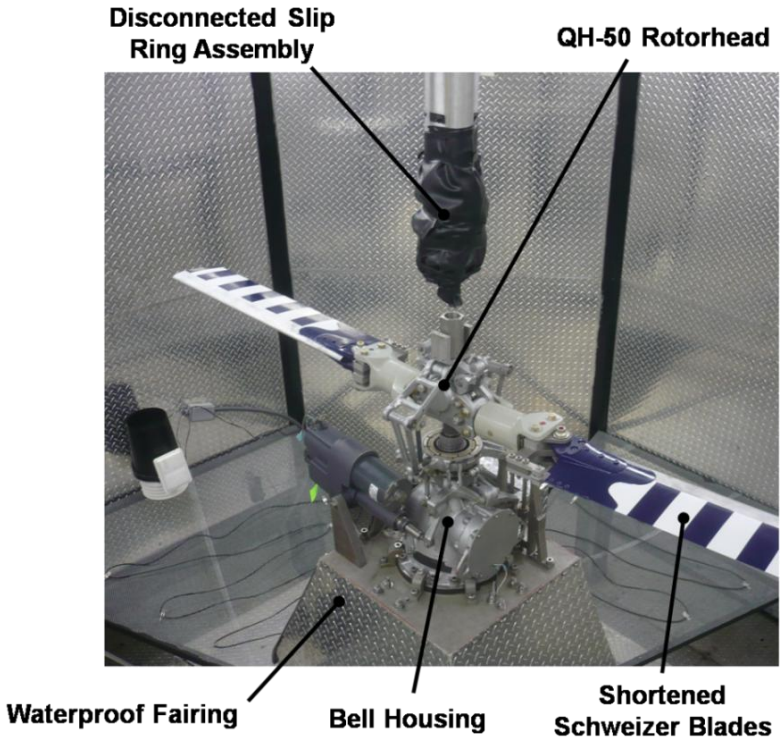


Figure 105: AERTS Configuration for ARISP Validation

For the current test rotor, 2.5° of collective pitch was considered to be the minimum due to the -2.13° twist of the blades. A slightly positive induced velocity across the entire rotor plane was desired to help draw the supercooled droplets through the rotor plane. The low pitch maintained sectional blade angles of attack low, allowing for accurate flowfield calculations by the flowfield solver.

5.1.3 Test Matrix

Results from the icing shakedown tests, described in Section 3.4, were incorporated into the final test matrix. The test conditions, shown in Table 18, were selected to explore variations in temperature and droplet size and icing severity. A summary of the conditions is presented

in Table 19. Graphical descriptions of the test points are shown in Figure 106 and Figure 107.

Table 18: ARISP Validation Test Matrix

Case	Average Temperature	MVD	Outboard LWC	Nozzle Configuration	RPM	Collective	Time	Shedding
	°C	µm	gr/m³			°	sec	
24	-10.8	25	2.1	5 Outer	500	2.5	180	N
25	-10.5	25	1.6	5 Outer	600	2.5	180	N
26	-10.9	35	1.2	5 Outer	600	2.5	180	N
27	-10.5	35	2.3	5 Outer	600	2.5	180	N
28	-15.6	15	2.2	5 Outer	600	2.5	180	N
29	-5.1	35	6.5	5 Outer	600	2.5	180	N
31	-5.8	35	3.0	5 Outer	600	2.5	215	Y
32	-9.9	35	2.8	5 Outer	600	2.5	300	N
33	-4.7	20	6.9	5 Outer	600	2.5	352	Y
34	-6.5	35	3.7	5 Outer	600	2.5	272	Y
35	-8.0	25	1.2	4 Outer	600	2.5	320	Y
36	-5.2	25	1.3	4 Outer	600	2.5	427	Y
37	-5.1	15	2.2	4 Outer	600	2.5	313	Y
38	-3.2	25	1.3	4 Outer	600	2.5	304	Y
39	-3.1	15	2.2	4 Outer	600	2.5	391	Y
40	-14.2	15	1.2	3 Outer	600	2.5	180	N
41	-12	15	0.9	3 Outer	600	2.5	180	N
43	-8.6	15	1.2	3 Outer	600	2.5	180	Y
44	-10.1	15	1.3	3 Outer	600	2.5	180	N
*Cases 23, 30 and 42 were rejected due to icing system control problems								

Table 19: ARISP Validation Test Matrix Summary

Parameter	Minimum	Maximum
Temperature (°C)	-15.6	-3.1
MVD (µm)	15	35
LWC (gr/m³)	0.9	6.5
Icing Time (s)	144	427

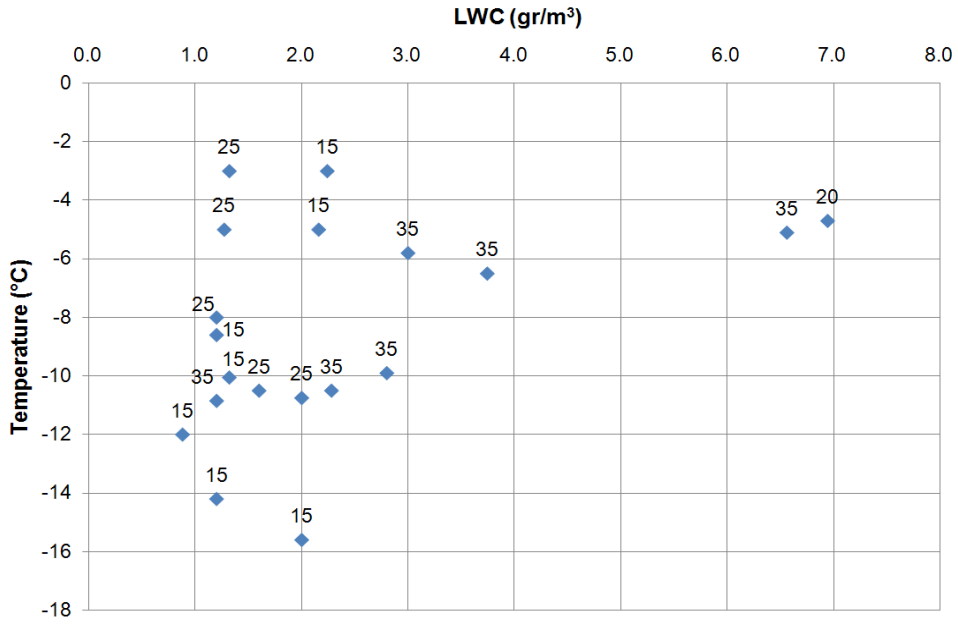


Figure 106: ARISP Validation Test Points, Temperature vs LWC
(Particle size in μm labeled next to each point)

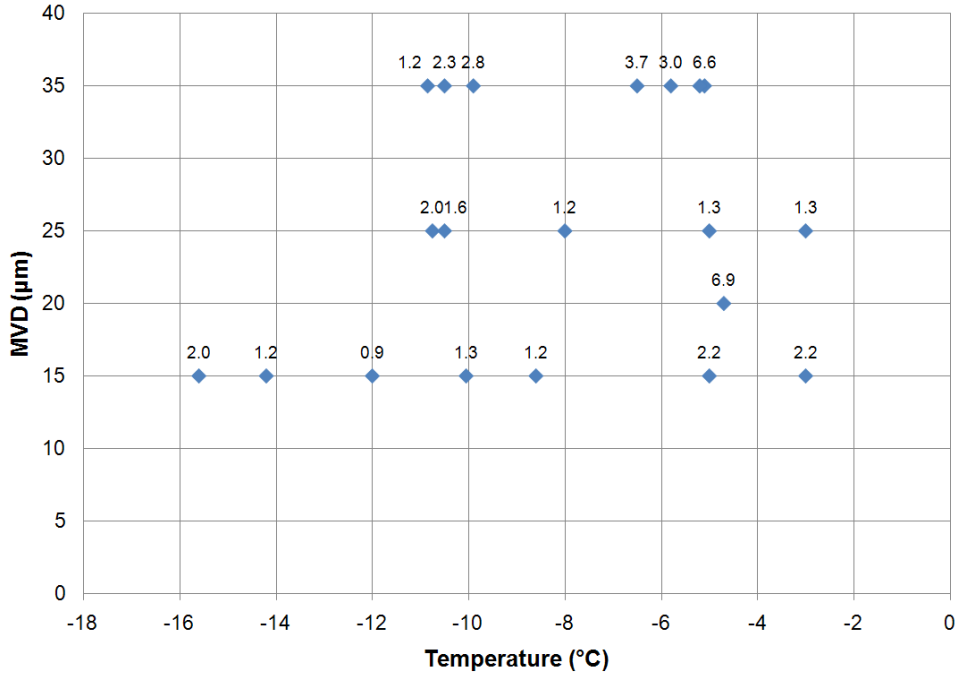


Figure 107: ARISP Validation Test Points, MVD vs Droplet Size
(LWC in gr/m³ labeled next to each point)

Each test point was intended to lie within the intermittent maximum icing conditions as described in the FAR Part 25/29 Appendix C and shown in Figure 7. This would ensure that the combination of icing parameters generated a realistic icing cloud. The icing system can produce clouds with significantly higher icing severities than can be encountered in the atmosphere, so the FAR definitions of icing environments were used to provide limitations for the AERTS system. The intermittent maximum conditions were selected over the continuous maximum conditions because they have higher icing severities and shorter exposure times, ideal for the AERTS facility.

The most difficult icing parameter to control was the chamber temperature. As detailed in Section 3.3.1.1, the cooling fans were required to be shut down which causes the chamber temperature to increase during the icing tests. Due to this, the starting temperature must be below the desired test temperature. All results are presented for the average test temperature.

As described in Section 3.3.3, the chamber LWC was not monitored during each test, but rather calculated based upon the post test stagnation ice accretion measurements. The input pressures and number of active nozzles were altered for each test to produce the desired LWC. Due to an incomplete understanding of the facility, the test LWC was not exactly as intended, and sometimes was outside of the Appendix C guidelines as shown in the following figures. Lower particle sizes were easier to produce and control at low LWC.

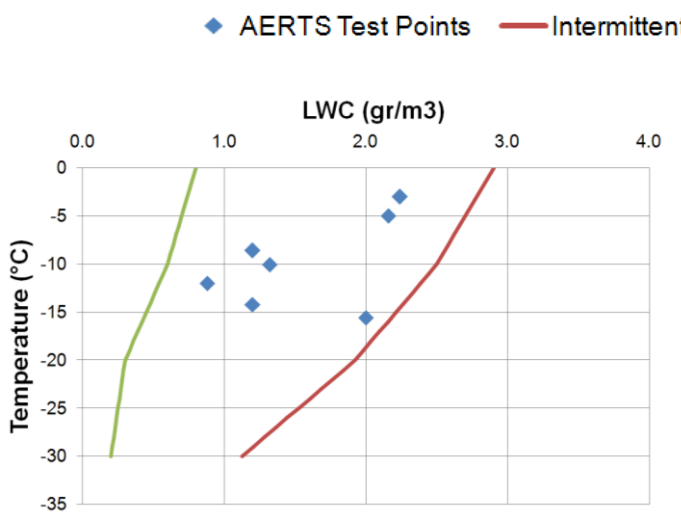


Figure 108: 15 µm Droplet Size Cases

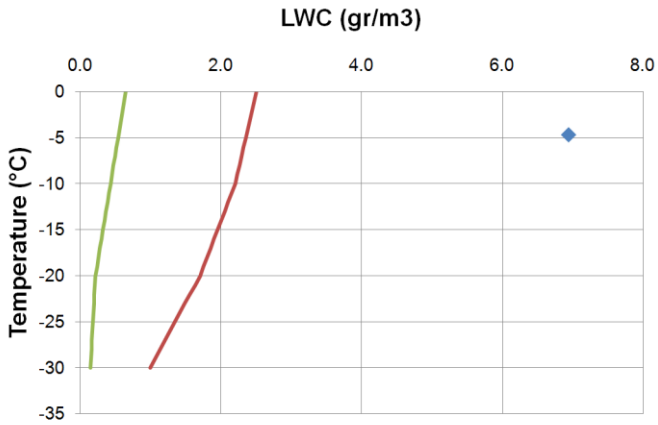


Figure 109: 20 µm Droplet Size Cases
(note x-axis scale change)

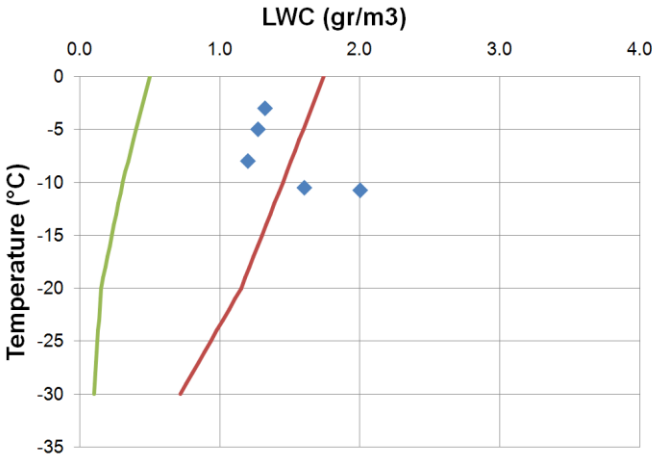


Figure 110: 25 µm Droplet Size Cases

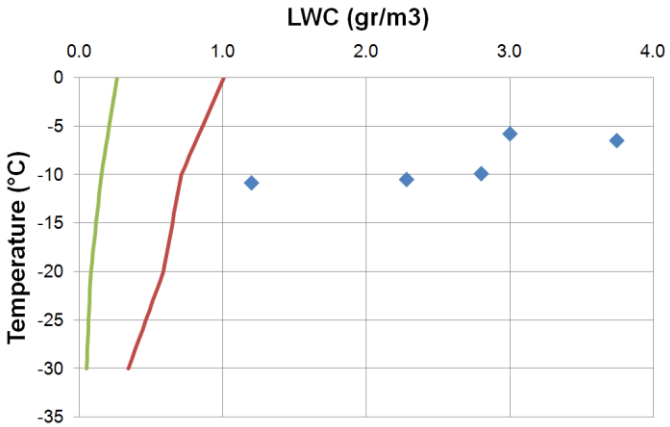


Figure 111: 35 µm Droplet Size Cases

Most tests were run for a standard time of 180 seconds to provide a consistent icing time for shape and thickness comparisons. Other cases were run until a shedding event occurred, at which time the test stand was immediately shut down. A quick reduction in rotor RPM was important because of high 1/rev loads from the mass imbalance. Also, the shutdown prevented the opposite blade from shedding, preserving the accreted ice for post test measurements.

5.2 Recorded Data

During each test, data was recorded regarding the icing conditions and other pertinent test parameters. Variables recorded included test case number, test date, nozzle configuration, collective pitch, rotor RPM, icing time, test start temperature, test end temperature, and droplet size dispersion. Rotor shaft torques were recorded every 15 seconds. Forces and moments measured by the load cell were monitored, but not recorded unless a significant deviation from nominal test parameters occurred. Due to the high work load of monitoring the icing cloud, checking the rotor loads and recording test information, at least two operators were required.

Post test documentation included detailed photographs of the accreted ice. Pictures were taken from standard angles and vantage points to allow for direct comparisons of the accreted ice. Samples of photographs are provided in the following figures.

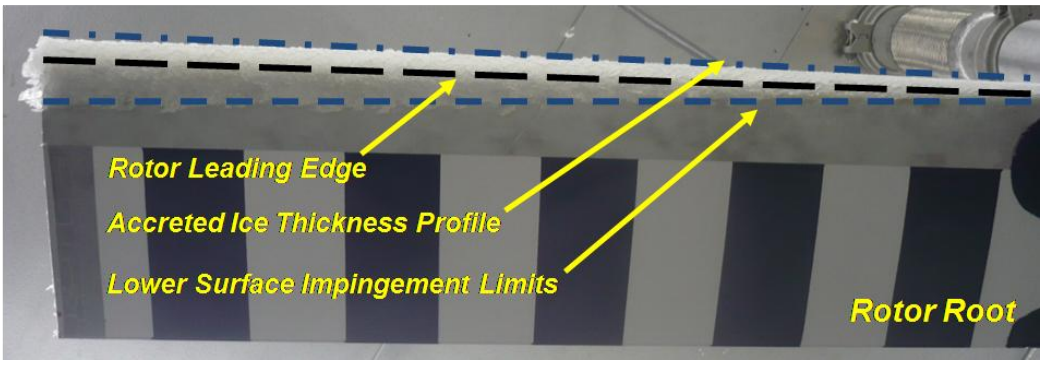


Figure 112: Case 23 Thickness Profile

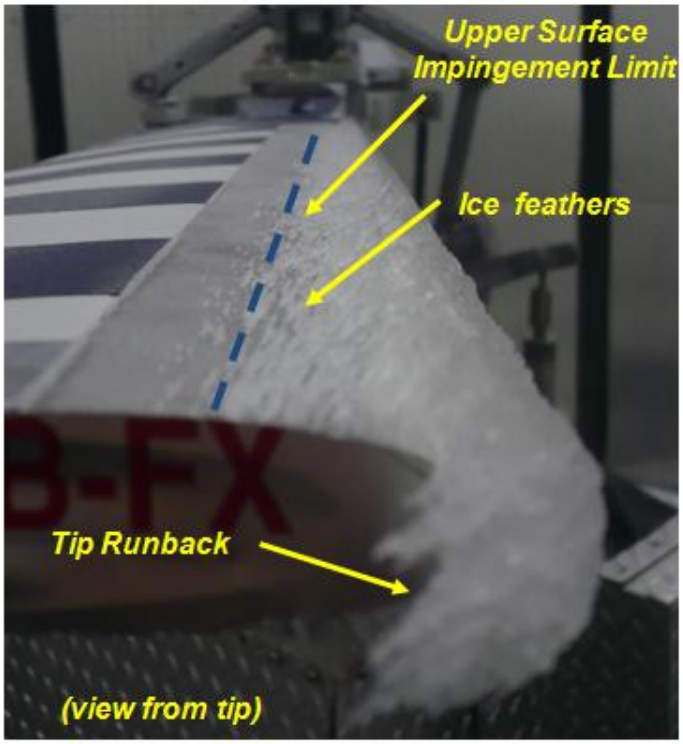


Figure 113: Case 23 Ice Accretion

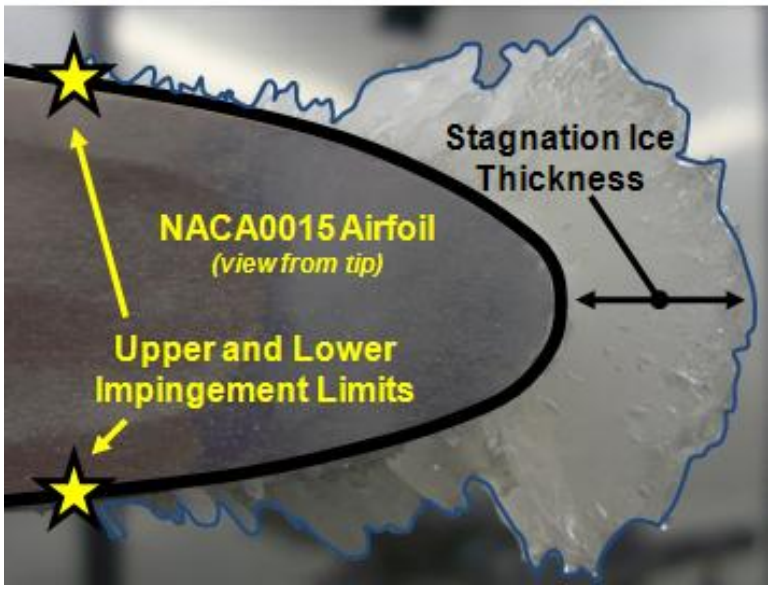


Figure 114: Case 23 Tip Ice Shape
(tip runback removed with heated plate)

Measurements of the accreted ice stagnation thickness were made at 11 blade stations with a pair of calipers. The calipers were left in the test chamber for the duration of the tests so

they would be cooled to the chamber temperature and not melt the ice during the measurement process. Painted stripes on the test blades provided standard measurement locations, starting at 0.50r. These thicknesses were used to calculate the LWC for the test case as described in Section 3.3.3.

After the thickness measurements were completed, ice tracings were taken at standard locations. For cases 24 through 34, tracings were taken at 0.7r, 0.8r and 0.9r. For cases 35 through 44, an additional tracing was made at 0.6r. To make the tracings, a heated aluminum plate was used to melt a slot in the accreted ice at the desired station. A template of the airfoil leading edge shape with a piece of graph paper attached to it was inserted into the slot and the ice profile transferred to it.

At the conclusion of data recording, the ice was melted off the blades with an electric heat gun. Paper towels were used to remove liquid water from the blade surface before it refroze. A thorough cleaning of the blades was important to ensure that each test started at the same blade surface condition.

5.3 Experimental Thickness Trends

Prior to correlations with the ARISP model, trends in the AERTS systems were examined. The trends supersede those explored in the icing shakedown tests because the crystallization issues were significantly reduced . In the cases selected for the thickness trend comparison,

no shedding occurred. Efforts were made to maintain constant test parameters, with the exception of the one variable under study. This was not always possible, as identified in the captions of the following figures. Also, not all cases could be directly compared to each other because of the icing exposure time. Since the shedding events occurred at various times and the stand was shut down immediately after the shedding event, each case had a different exposure time.

5.3.1 RPM Sensitivity

Figure 115 shows that comparison between two cases with similar icing conditions and the test rotor operating at different RPMs. The tip speeds for Cases 24 and 25 were 203 and 243 ft/sec respectively. As indicated by the figure, there was not a significant difference in the stagnation ice accretion.

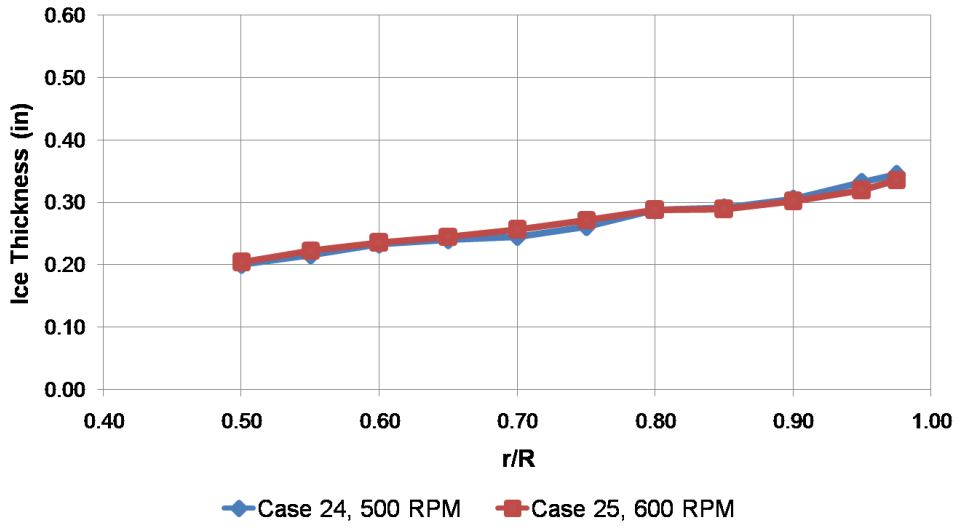


Figure 115: RPM Sensitivity
T ≈ -10°C, LWC ≈ 1.6 – 2.1 gr/m³

It was expected that Case 25 would have a higher ice accretion due to the increased number of impacts with supercooled droplets. At these low tip speeds, kinetic heating is not a

significant effect. The differences in cloud LWC prevented a clear comparison, as the there was a 0.5 gr/m^3 difference in icing intensity between the two cases.

5.3.2 Temperature Sensitivity

Effects of temperature were also investigated as part of the sensitivity studies. In Figure 116, the differences in stagnation thickness are shown for two cases in which the LWC was held relatively constant.

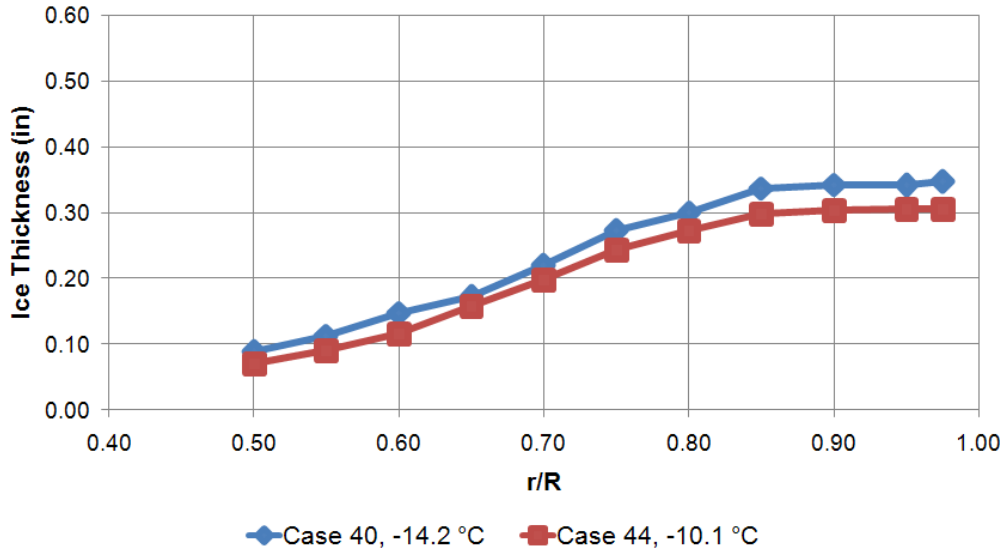


Figure 116: Temperature Sensitivity
LWC ≈ 1.2 – 1.3 gr/m^3, MVD = 15 μm

In natural ice encounters, the ice accretions are smaller at lower temperatures due to the lower LWC and MVD. In the AERTS chamber, however, icing conditions can be generated that prevent these trends from being observed. Comparisons are difficult because the ice type is different between the two cases. At the colder temperatures, rime ice extends further outboard than at warmer temperatures. This ice, with a prominent spear, increases the ice stagnation thickness, making comparisons to glaze ice difficult.

5.3.3 LWC Sensitivity

As shown in Figure 117, increased LWC yielded an increased stagnation thickness. This was not unexpected, since the LWC is calculated based upon the thickness. As experience grows with the laboratory, nozzle settings required to achieve particular cloud densities will be identified.

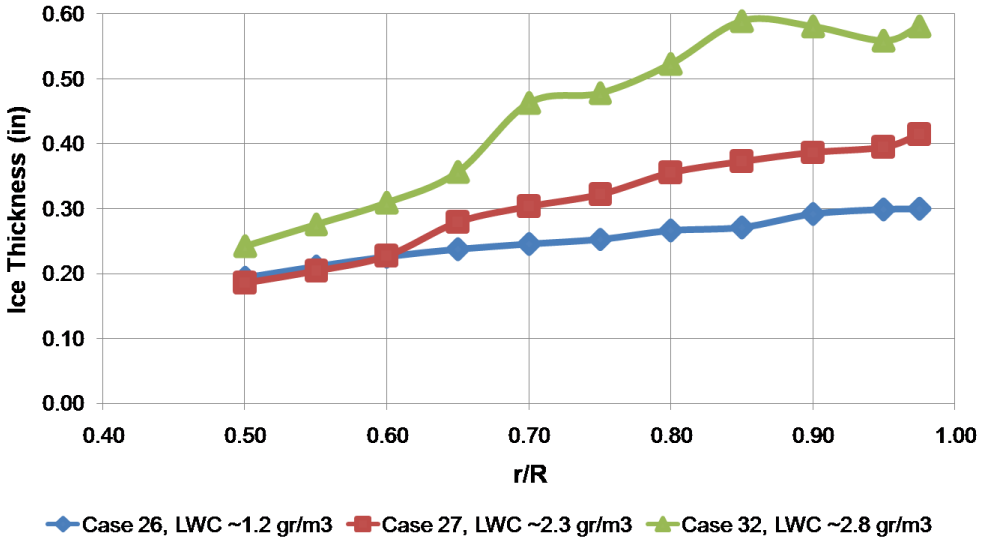


Figure 117: LWC Sensitivity
T ≈ -10°C, MVD = 35 μm

5.3.4 MVD Sweep

Particle size has an important effect on the accretion thickness. The larger particles, with higher inertia, are less sensitive to the flowfield inside of the chamber. As shown in Figure 118, larger particles produce a more even ice accretion across the rotor span.

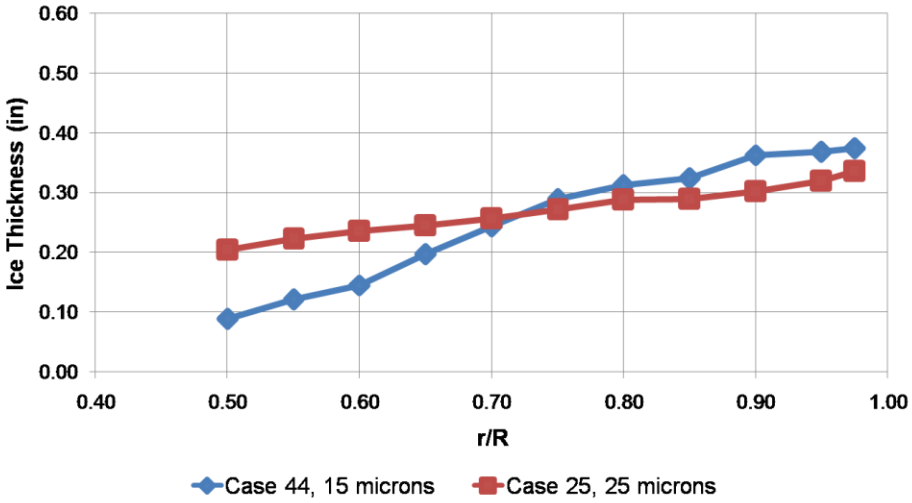


Figure 118: MVD Sensitivity
T ≈ -10°C, LWC ≈ 1.3 – 1.6 gr/m³

5.4 Liquid Water Content Calculations

Based upon the accreted stagnation ice thickness measurements, the LWC distribution across the rotor span was calculated. The results of the calculation are shown in Figure 119. The LWC calculation code did not always converge due to low accreted ice thicknesses, especially at inboard blade stations. If this occurred, the LWC was interpolated between two converged points.

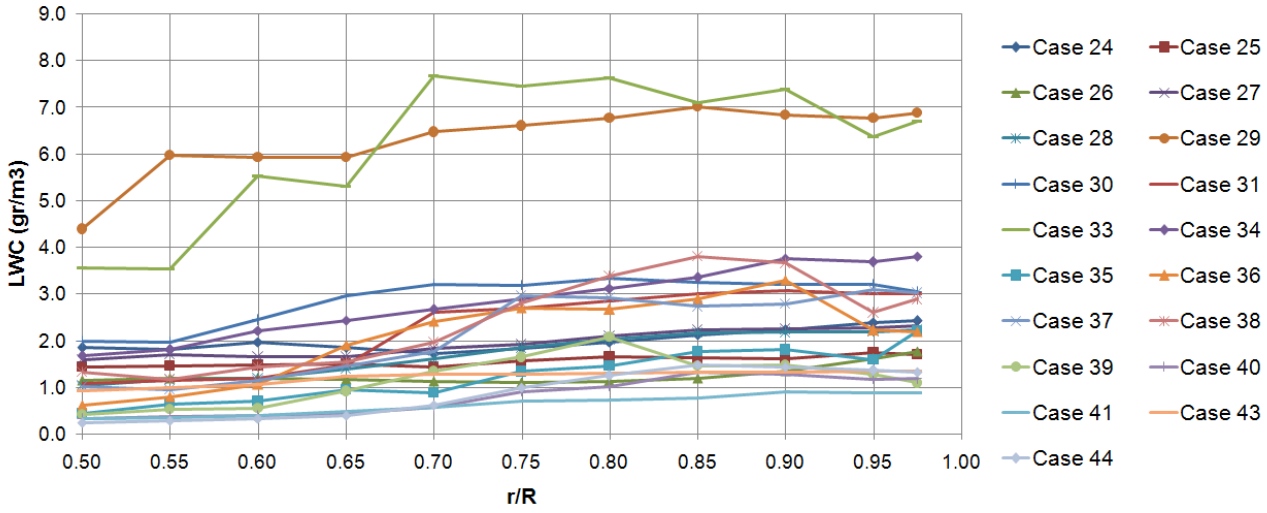


Figure 119: Spanwise LWC Distribution

It is not important to distinguish between the different cases in the chart. The takeaway message from the figure is that gradients from the blade root to the tip existed in the LWC distributions. The rotor tip LWC was an average of 147% higher than the midspan value. It is thought that this trend due to the small size of the facility and not necessarily representative of icing clouds on free rotors. The gradients do not pose problems with the correlations to the ARISP model because it can handle the distributions, but they must be taken into account when interpreting the test results.

5.5 Ice Shape Correlations

Icing and rotor system parameters, including the LWC gradient, were input into the ARISP model and predicted ice shapes were calculated for each case. Experimental shapes were generated from hand tracings. These tracings were taken at 0.7r, 0.8r and 0.9r for all test cases. For cases 35+, an additional tracing was made at 0.6r. Many cases showed favorable comparisons, as shown in Figure 120.

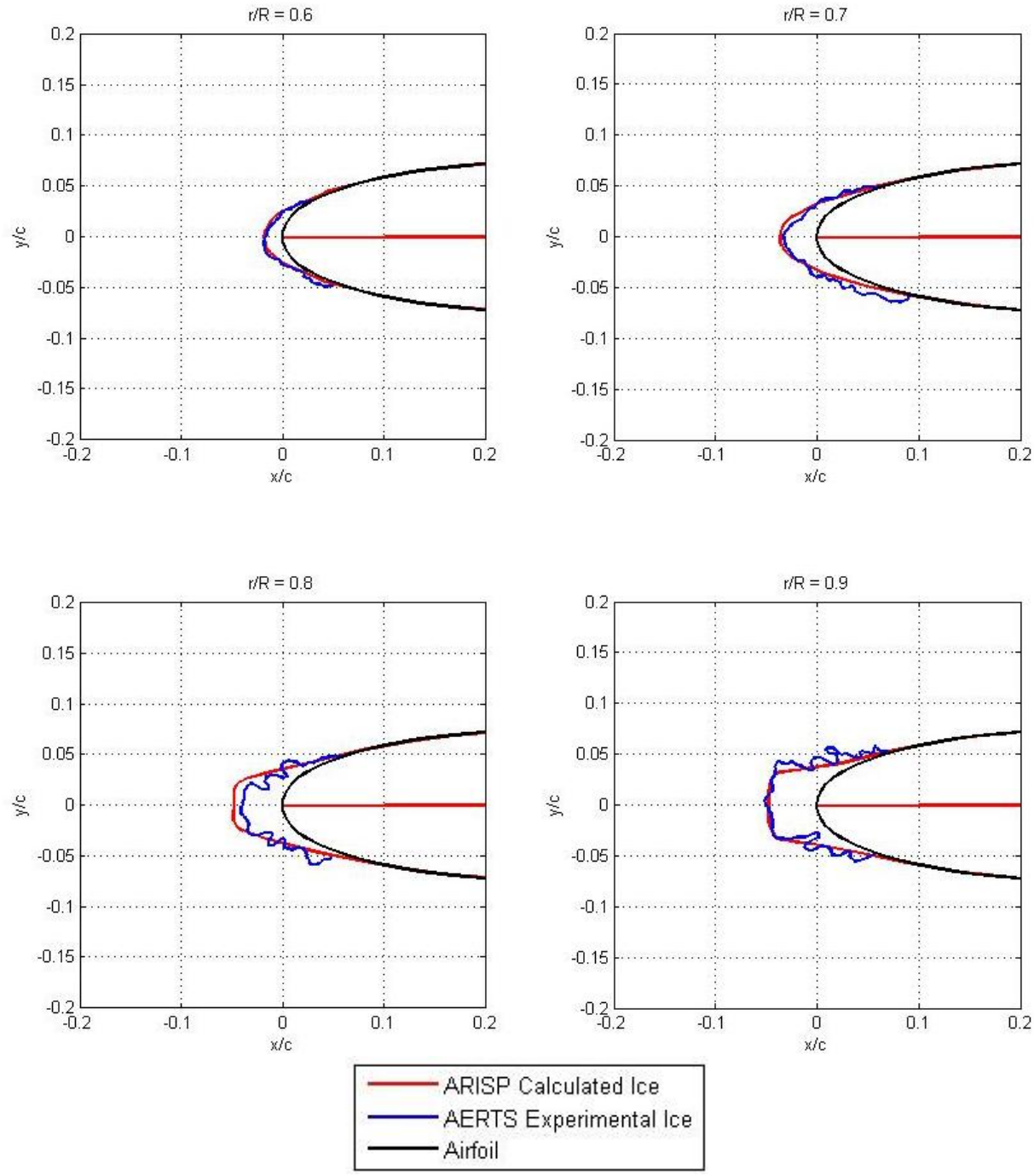


Figure 120: Case 44 Ice Tracings

Qualitative comparisons were made between the experimental and calculated shapes due to the difficulties in making quantitative evaluations. A three-tiered system was used, with the following figures providing examples from each class. The primary criteria for the evaluation included impingement limits, thickness and ice cross sectional area. Very good correlations were given a green rating. Fair correlations were assigned a yellow rating, while poor correlations were labeled with a red rating. While green ratings were desired for all cases, the yellow correlation indicates that the model predicted shape trend accurately but did not capture all of the ice details. Examples of each tier are given in Figure 121-123.

The results from this comparison are given in Table 20. Most of the ice shapes were generally close to the predications, with the best correlations occurring at low LWC and small MVD. Improvements in test procedures increased test accuracy in the later cases. Errors arise from the fact that the temperature, MVD and LWC parameters of the icing cloud are either not directly measured or controlled. Also, difficulties in extracting detailed ice tracings and a tedious digitizing process increase correlation errors. The accretions with a yellow or green rating provide accurate enough ice shapes for the evaluation of ice protection systems and evaluation of ice accretion protective materials. Red correlations need to be further investigated.

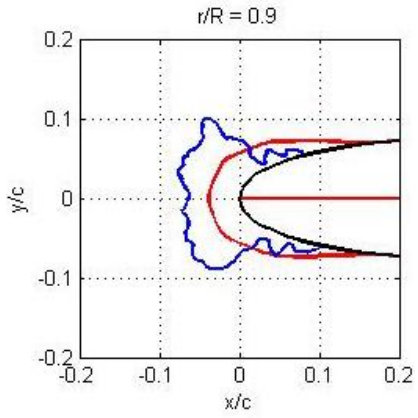


Figure 121: Red - Case 29, 0.9r

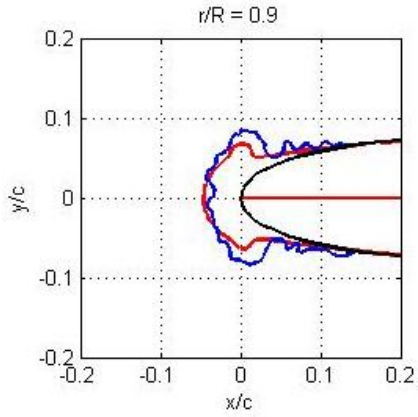


Figure 122: Yellow - Case 24 0.9r

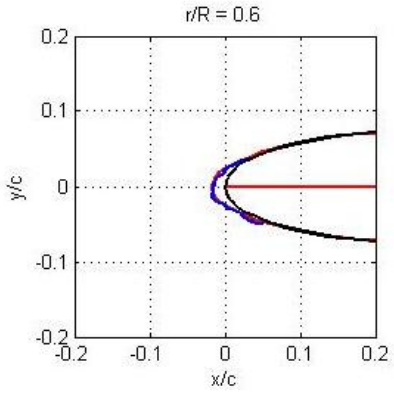


Figure 123: Green - Case 25, 0.8r

Table 20: Shape Correlation Results

Case	Station			
	0.6r	0.7r	0.8r	0.9r
24	-	Y	Y	G
25	-	Y	G	Y
26	-	Y	Y	R
27	-	Y	Y	Y
28	-	R	R	R
29	-	Y	R	R
31	-	G	G	G
32	-	G	G	G
33	-	R	R	Y
34	-	Y	G	G
35	G	G	Y	Y
36	R	R	R	R
37	G	G	Y	Y
38	G	G	Y	Y
39	G	G	G	G
40	G	G	G	G
41	G	G	Y	Y
43*	G	G	-	-
44	G	G	G	G

**Dual blade shedding prevented tracings from being taken*

The results from Table 20 are summarized in Figure 124. It is clear that overall correlation was better at inboard stations, with 89% of the cases matching the ARISP model at 0.6r. Correlations neat the rotor tip were not as favorable, with 22% of experimental ice accretions significantly deviated from the ARISP predictions. Overall, the ARISP model is accurate, with 50% of the test cases receiving a green rating. If the yellow ratings are included, the model accuracy increases to over 81%.

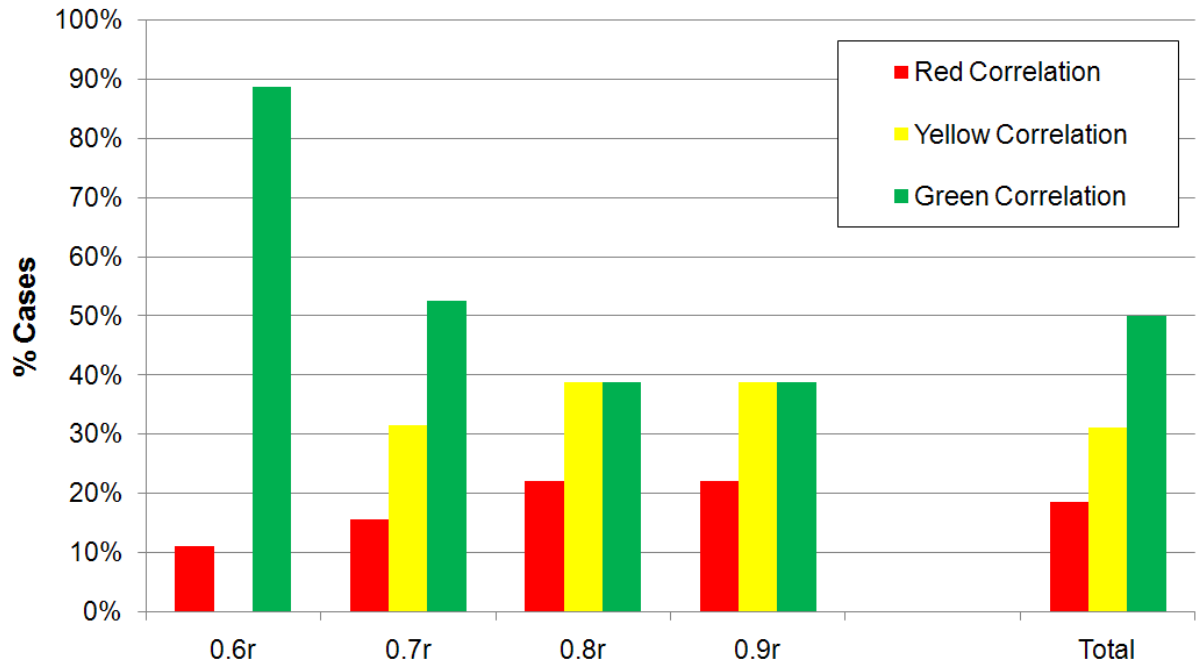


Figure 124: Shape Correlation Summary

5.6 Impingement Limit Correlations

The impingement limits were calculated for each condition based upon the ice tracings and numerous post test pictures. Due to the nozzle location above the rotor plane, the impingement limits were difficult to measure after a test. Since the icing cloud is not instantly removed from the test chamber, additional accretions are made during rotor spool

down. As the cloud settles in the room, particles may settle to the upper surface of the blade making the upper impingement limits difficult to determine. These accretions were not included in the impingement limit calculations. Glaze ice feathers, as shown in Figure 125, were included in the impingement limit calculations.

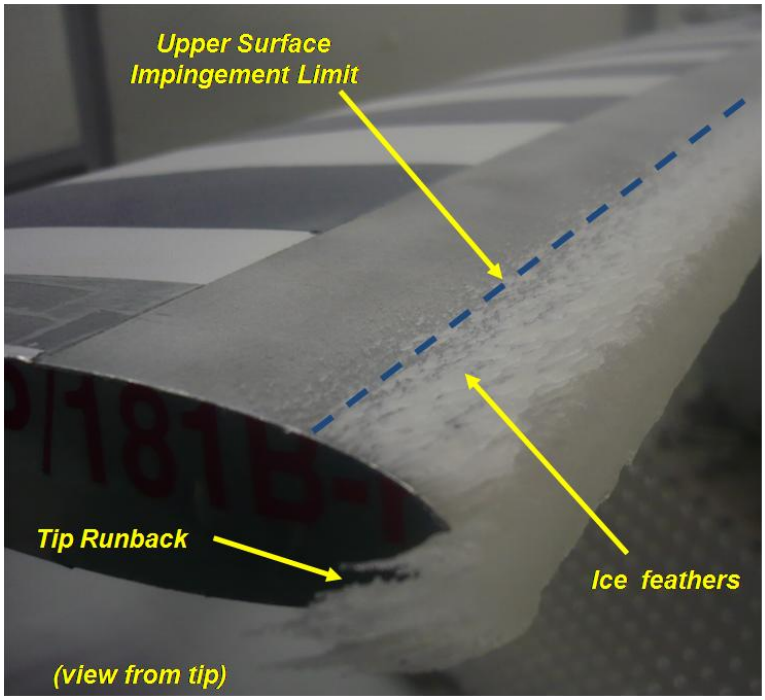


Figure 125: Case 43 Glaze Feathers

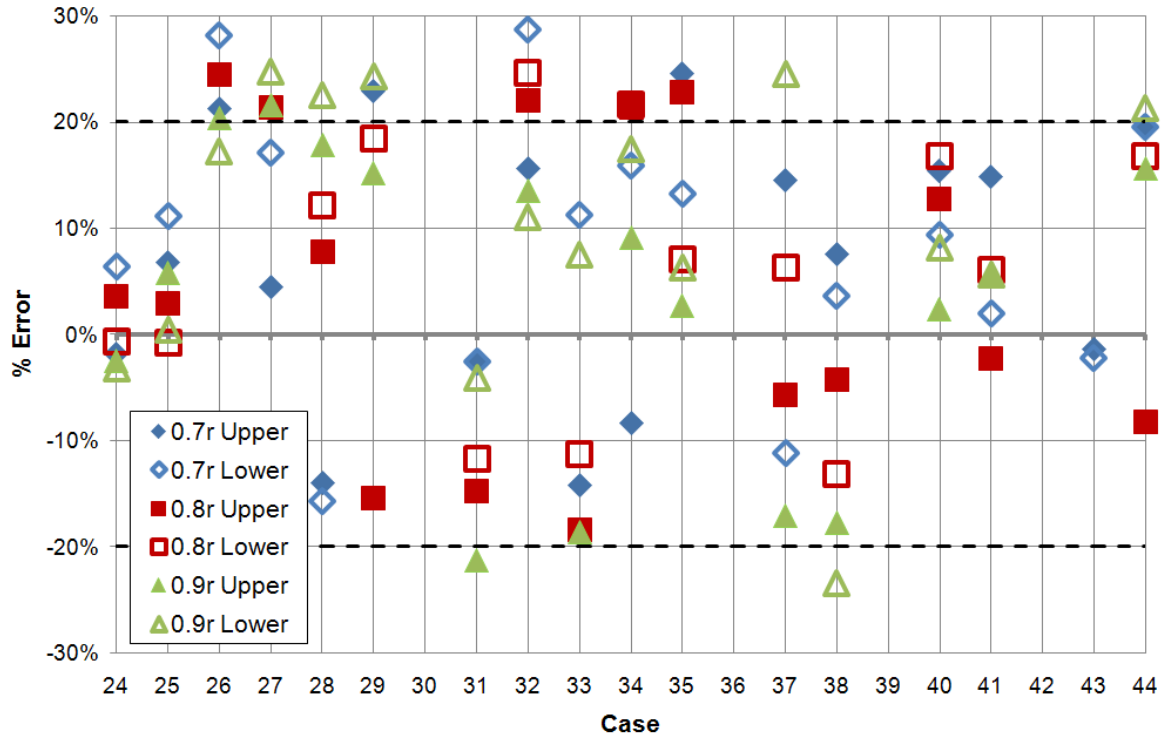


Figure 126: Impingement Limit Correlation Results

As shown in Figure 126, the impingement limits were generally within 20% of the predicted results. Since impingement limits are primarily controlled by airfoil geometry and droplet size, this indicates that the AERTS droplet size control system is relatively accurate. Larger droplets impact further aft on the airfoil surfaces due to their higher inertia. Errors in the correlation may arise from the ice tracing process. Due to the ice feathers, the exact location of the upper and lower impingement limits was sometimes difficult to determine.

5.7 Performance Degradation Correlations

During the tests, rotor performance was monitored. Due to the low collective pitch, rotor thrust was not significant (< 30 lbf), so shaft torque was the primary metric for determining

the effect of ice on rotor performance. The torque sensor was calibrated by the manufacturer prior to use.

Prior to any icing tests, the torque requirements for the hub alone and clean blades were determined as a function of shaft RPM. For each test point, the motor was rotated for at least 30 seconds to allow bearings and seals to warm up to their operating temperatures. These results are shown in Figure 127.

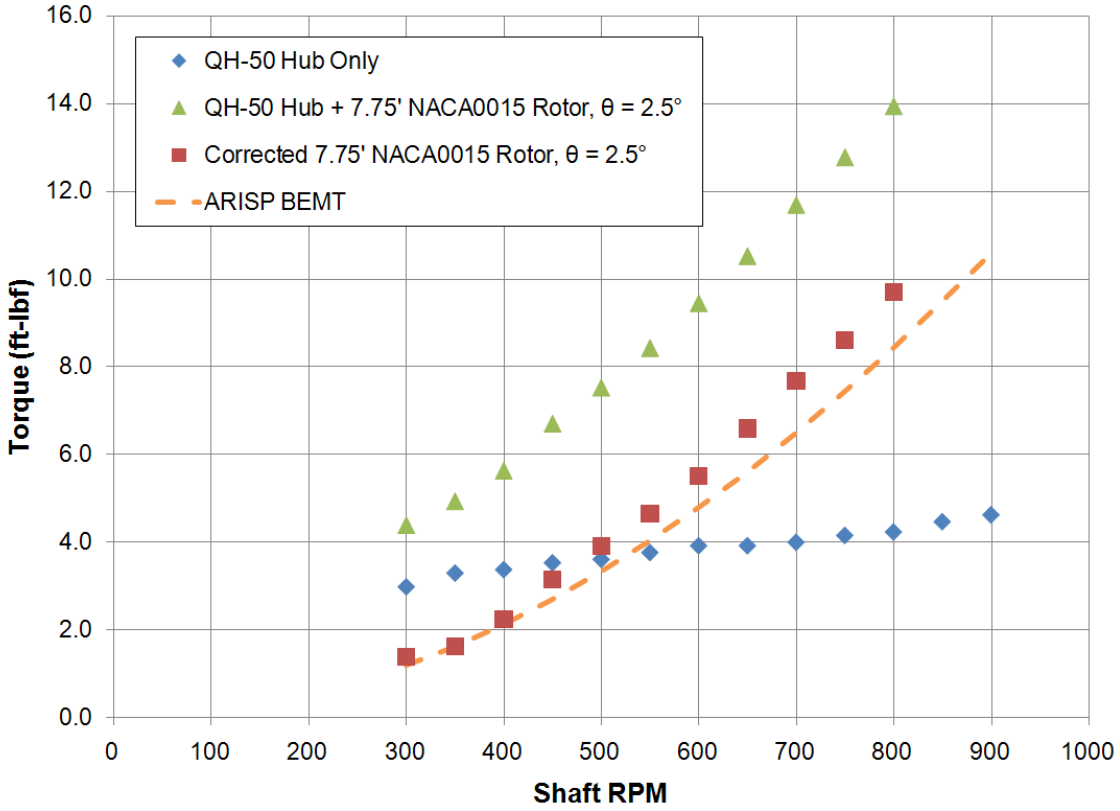


Figure 127: Torque Requirements

The hub alone calculations were needed to separate the effects of the various seals and bearings in the new QH-50 hub installation from the blades. The experimentally determined rotor-alone torque compared favorably with the BEMT predictions, with a 13% error at 600

RPM. The error includes a 10% increase in drag coefficients due to the rough blade surface and is generally good considering that the BEMT model does not capture 3D flow effects. The corrected clean blade calculations served as a reference point at the start of each icing test. If the torque required at the test RPM exceeded the clean blade torque, the blades still had residual ice from the previous icing test and required additional cleaning.

During the icing trials, the torque was recorded at 15 second intervals. Due to the high LWC and small rotor chord, the torque increased significantly during the tests. The small rotor had a higher collection efficiency than a full scale rotor and therefore accreted ice quickly. The following figure shows a typical torque increase. For most testing in the AERTS, a rough approximation of sectional torque increase is ΔC_d /s=~0.00011.

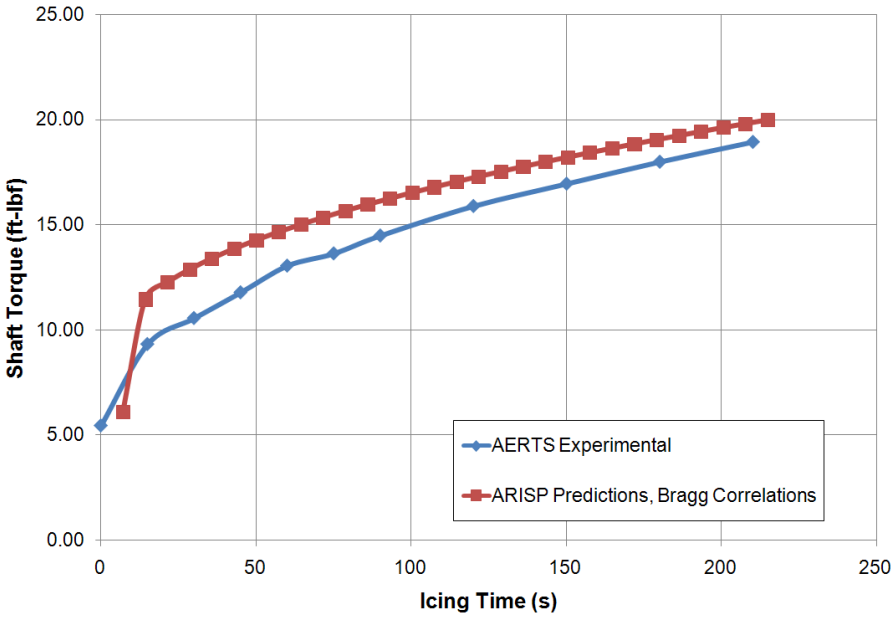


Figure 128: Case 31 Torque Increase

During an icing encounter, first the blade accretes a layer of frost. This generates the large initial increase in sectional airfoil drag from increased blade surface roughness. As ice exposure time increases, the drag increases at a lower rate. This is due to the ice shape maturation process, in which the stagnation thickness tends to increase. This has a smaller effect on the flowfield around the airfoil than increased ice horn height. Figure 129 shows the increase in sectional drag coefficient for Case 31. Due to the low collective pitch, there is not a strong spanwise variation in angle of attack. This generates a small spanwise gradient in C_d.

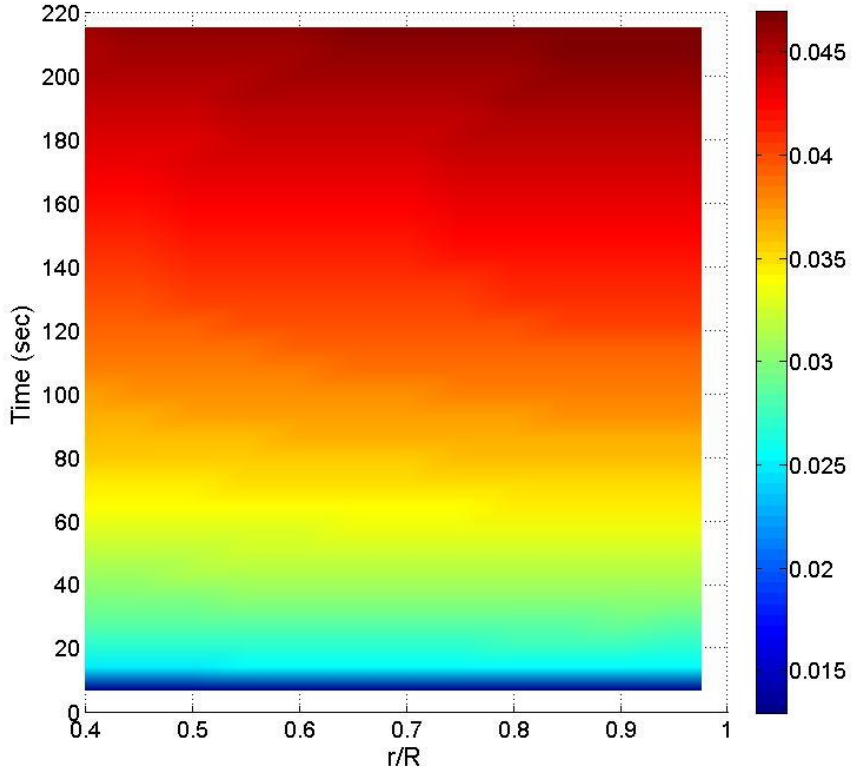


Figure 129: Case 31 C_d Distribution

For each case, the experimental torque profile was compared against the ARISP predictions. All ARISP torque calculations utilized the empirical correlations developed by Bragg due to the low rotor tip speeds. Testing at higher tips speeds will employ Flemming's coefficients. As shown in Figure 130, the coefficients generally underpredicted the torque increase for each test. This underpredicion of the torque increase was in contrast to previous testing, where Bragg's equations have shown a tendency to overpredict the drag increment (51).

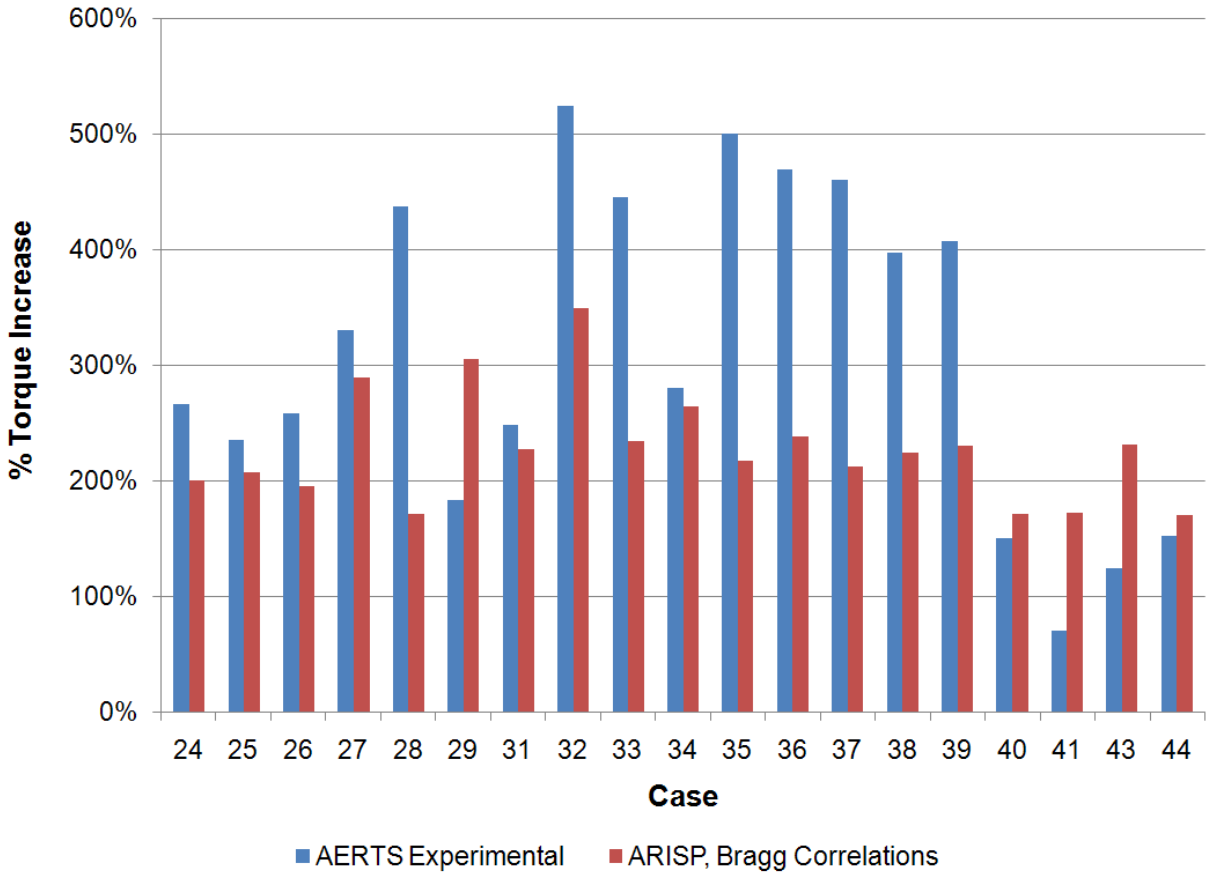


Figure 130: Torque Correlations

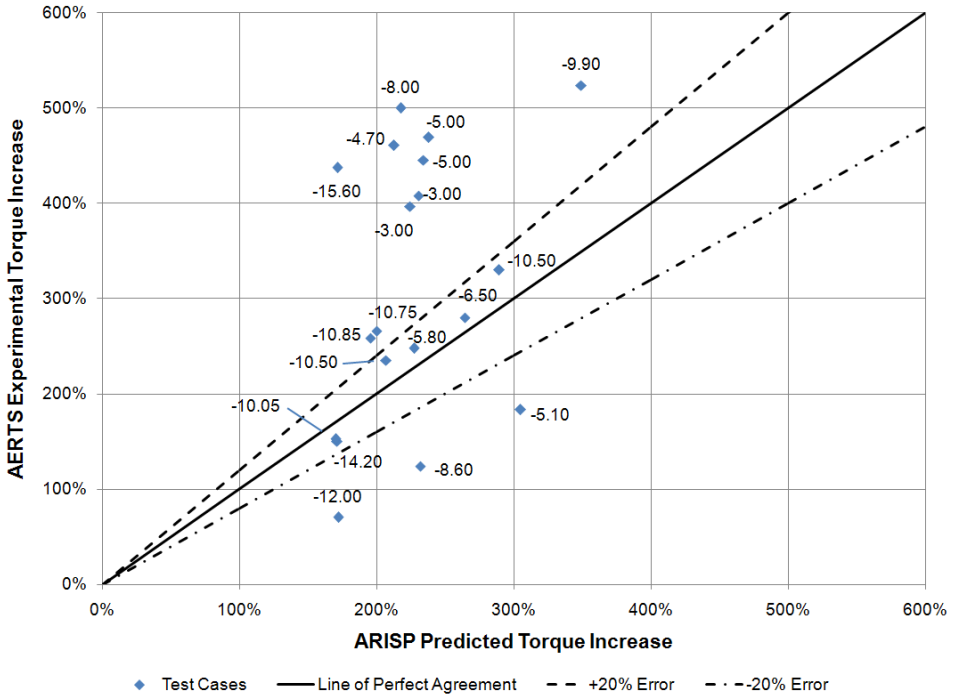


Figure 131: Torque Correlations
(Average test temperature in °C indicated next to each point)

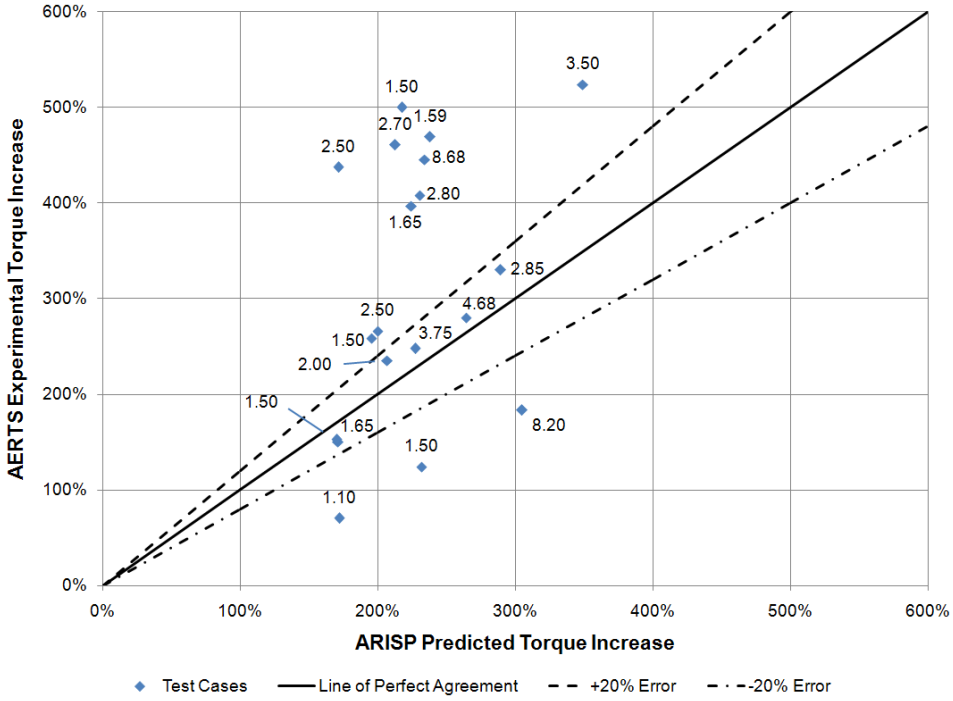


Figure 132: Torque Correlations
(LWC in gr/m³ indicated next to each point)

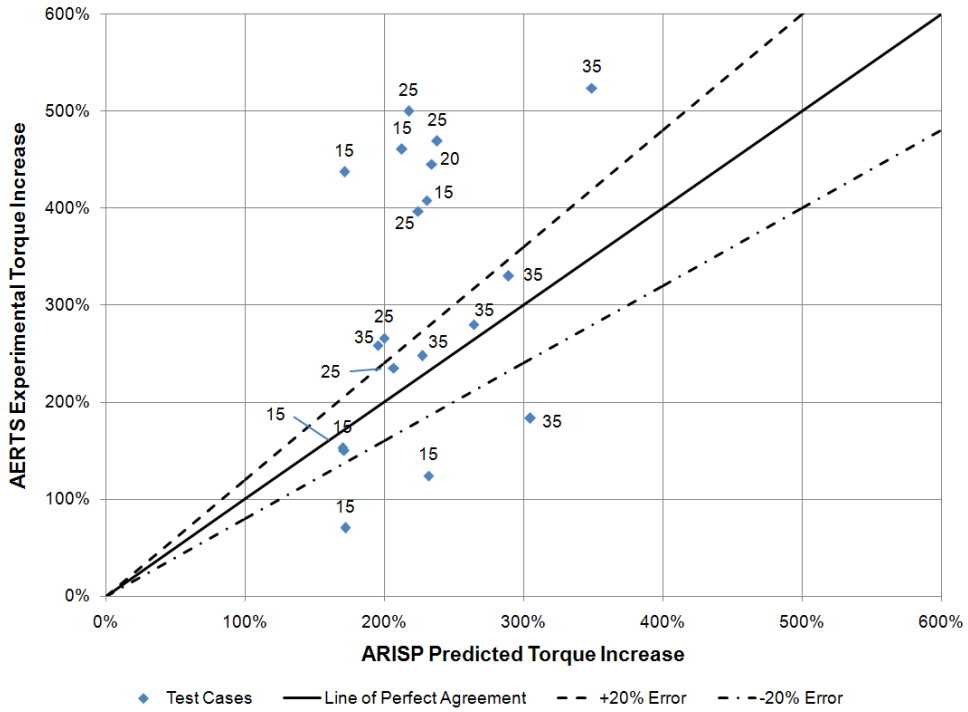


Figure 133: Torque Correlations
(MVD in μm indicated next to each point)

The discrepancy is most likely related to the inaccuracies in icing condition measurement. Most of the discrepant points were tested at higher temperatures. At warmer temperatures, small deviations in static temperature may generate significant differences in accreted ice shape. In addition, a 15% error in LWC measurement can yield a 3.4% error in the calculation of the iced C_d. At the rotor tips, which generate the majority of the torque requirements, the LWC is less certain than inboard regions, which may yield a large error in the rotor torque measurements.

5.8 Shedding Correlations

5.8.1 General Test Results

Shedding characteristics were investigated to determine suitability of the model in predicting such events. Most icing experiments were run for three minutes for ice shape and thickness comparison purposes, but to increase the number of shedding events, some cases were run until a shedding event occurred. Due to temperature control limitations in the test chamber, all cases were undertaken at warmer temperatures to improve test validity. When shedding was detected with the load cell, the rotor was immediately shut down due to large 1/rev imbalances. A clearly audible bang from the ice impacting the ballistic wall was also an indicator of shedding. In almost all cases (except Case 43), ice shed only from one blade, which was ideal for validation purposes because the second blade retained a record of the accreted ice. The following figures show some results of the shedding testing.

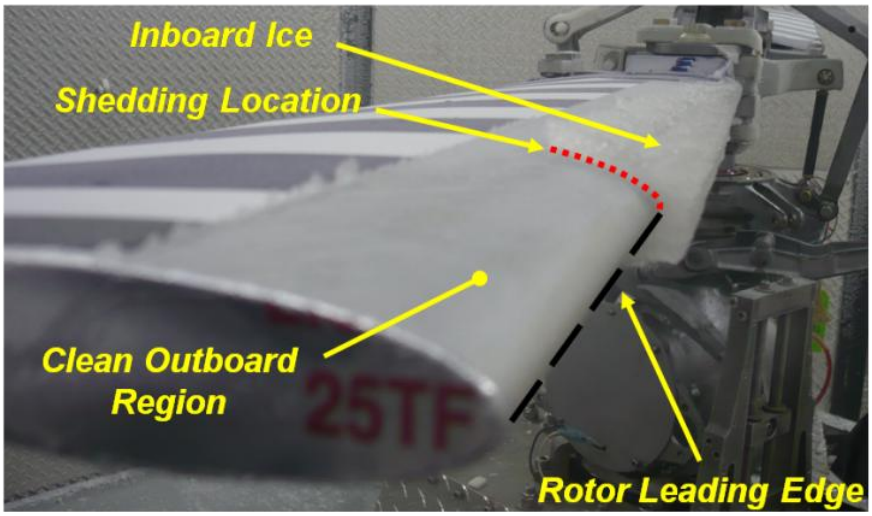


Figure 134: Case 34 Ice Accretion, Showing Shedding

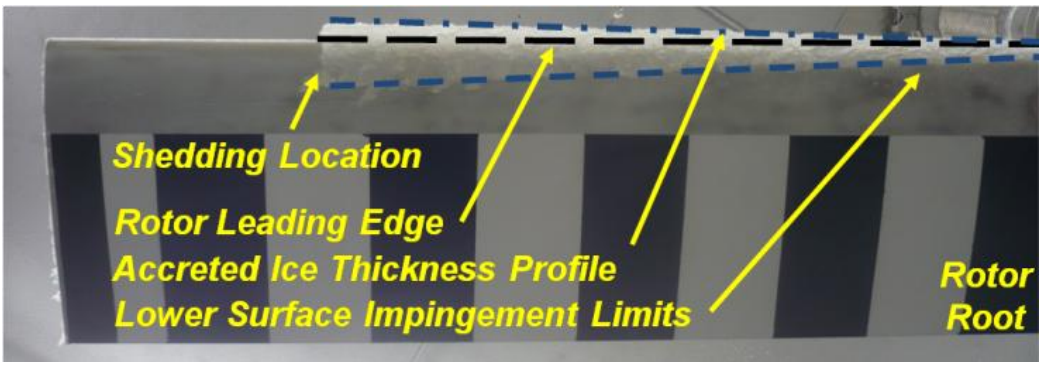


Figure 135: Case 34 Thickness Profile, Showing Shedding Location

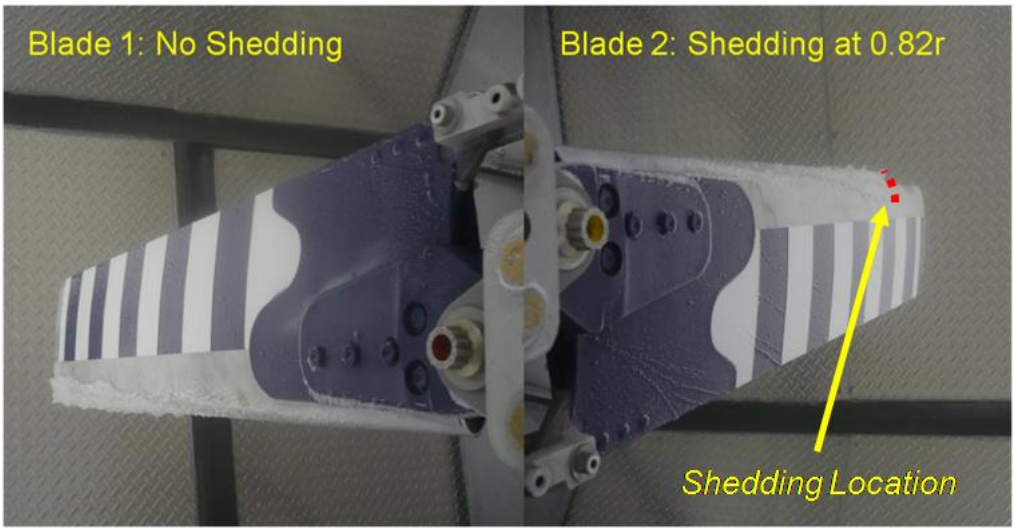


Figure 136: Case 39 Shedding, view from hub

At no point during the testing was there a ballistic danger from shed ice pieces. The ice sometimes had kinetic energy exceeding 1000 Joules, which created impact marks on the wall after each test. A sample of these marks is shown in Figure 137, the usual consequence was shattering of ice accreted on the wall. If the ice impacted near the edge of the aluminum panel, some local buckling resulted. At no time did the ice breach the wall.

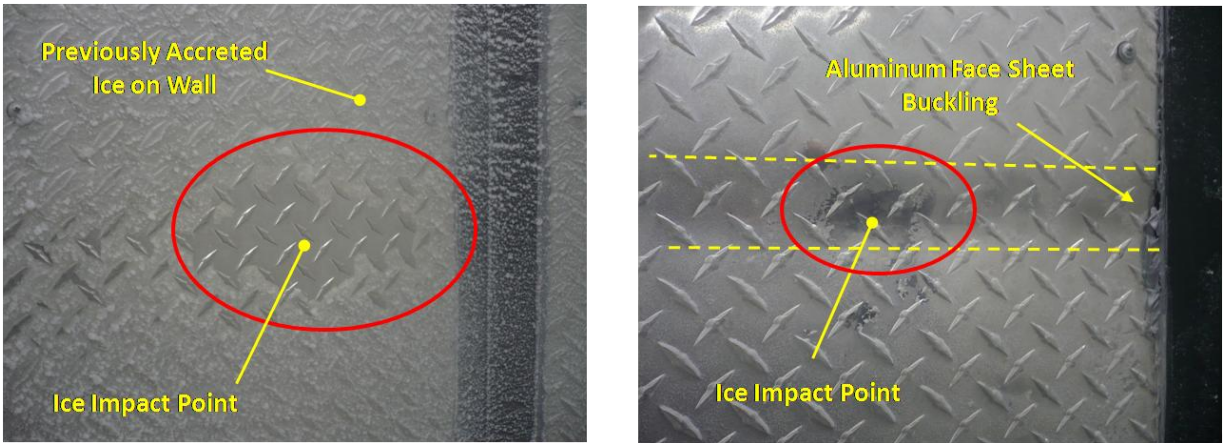


Figure 137: Shedding Damage

It was expected that ice accreted with larger particle sizes and warmer temperatures would shed before cases with smaller particles and colder temperatures, due to the lower ice shear adhesion and cohesive strengths. This was not observed experimentally, as shown in Figure 138. Due to the range of icing severities seen in the shedding events, colored circles identify test points with similar LWCs for comparison purposes. Instead, colder temperatures generally shed earlier due to the larger ice accretions, which created higher centrifugal debonding forces.

Little variation of shedding station was observed for the test cases. All shedding occurred at approximately 0.8r as indicated by Figure 139. Blade 1, identified with a black tip stripe, shed ice 88% of the test cases. No significant differences between the two blades could be identified and nothing particular was noted about the 0.8r station that would influence shedding behavior. No effort was made to characterize the blade surface. An optical profilometer is recommended to determine blade surface roughness in future work.

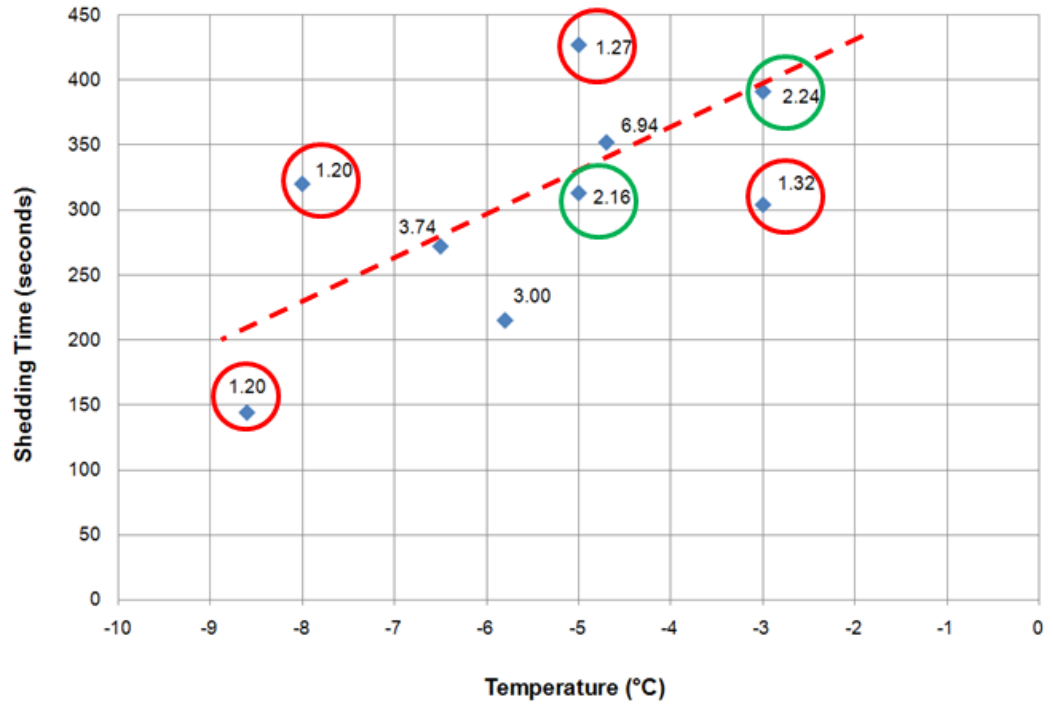


Figure 138: Shedding Time
(LWC in gr/m³ indicated next to each point)

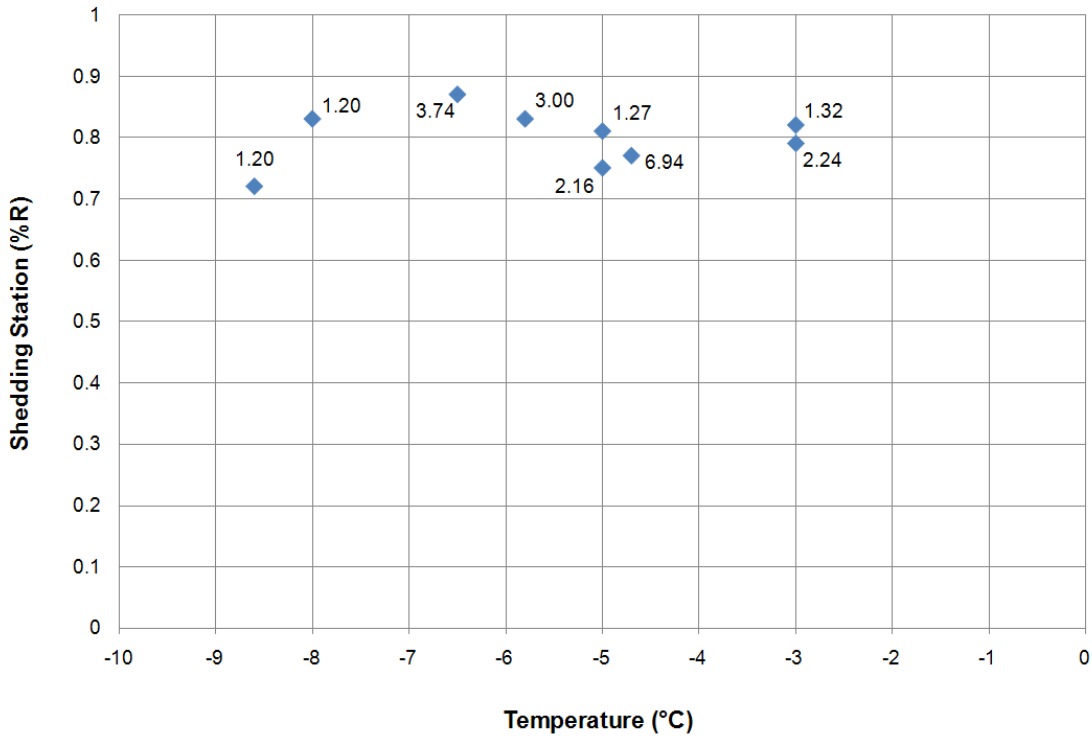


Figure 139: Shedding Station Correlations
(LWC in gr/m³ indicated next to each point)

5.8.2 Experimental Correlations

The shedding module of the ARISP model was run against the cases to verify the ice shedding locations and times. A total of 40 blade stations were used to provide sufficient resolution for the shedding calculations. Due to scatter in the test data from the shear adhesion rig, the shear adhesion strength results from Reich (83) were used in the model.

Initial correlations were not favorable, as the ice mass predictions near the rotor tip are not good as shown in Table 20. The ice shear adhesive and cohesive areas are critical to the proper prediction of shedding behavior. If the experimental and predicted shapes do not match, the experimental and predicted shedding events will also not match. When compared to experimental results in the IRT and full scale ice shapes on the UH-1H, the model generally over predicts the ice accretions. Correlations to experiments in the AERTS chamber are better, due to the lower tip speed of the test rotor, which is within the LEWICE validation envelope. This is important as shedding calculations rely on the ice geometry. Since the ARISP model over predicts ice accretions near the blade tip, the shedding frequency is also over predicted. Additional problems arise from the tip runback, shown in Figure 113, which further increases the mass of accreted ice at the tip and promotes early shedding.

A correction factor was therefore necessary to adjust the cohesive area at the tip sections to account for the over prediction of ice near the blade tips. This also affected the ice mass, as it is calculated based upon the cohesive area reported by LEWICE. The shear adhesion area was not modified, as the impingement limits were generally predicted accurately. Based upon the experiments, the correction was computed by comparing the cross sectional areas of the predictions and experiments. For the cases that shed, the correction factors are shown in Figure 140. Since ice tracings were only taken at 0.7r, 0.8r and 0.9r for every case, the corrections were based upon photographic evidence of ice shapes.

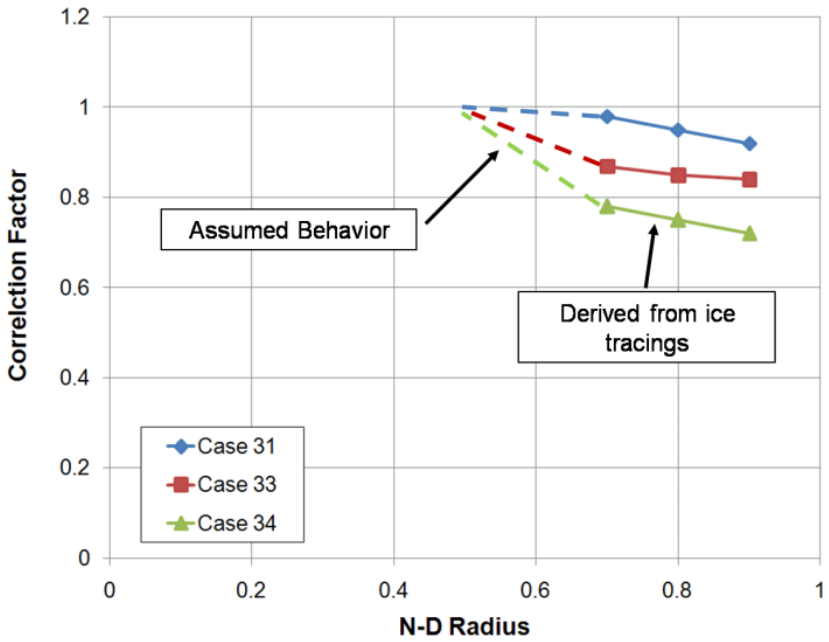


Figure 140: Shedding Correction Factor

This correction factor affects both the time and location of shedding events as shown in Figure 141 and Figure 142. The total number of shedding events is reduced as well as the total mass of ice released from the rotor as detailed in Table 21.

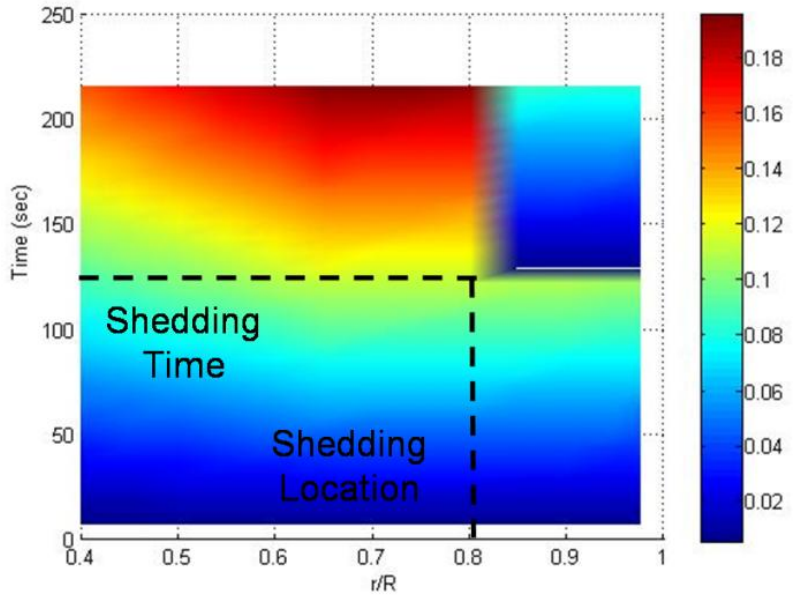


Figure 141: Case 31 Ice Thickness, No Corrections

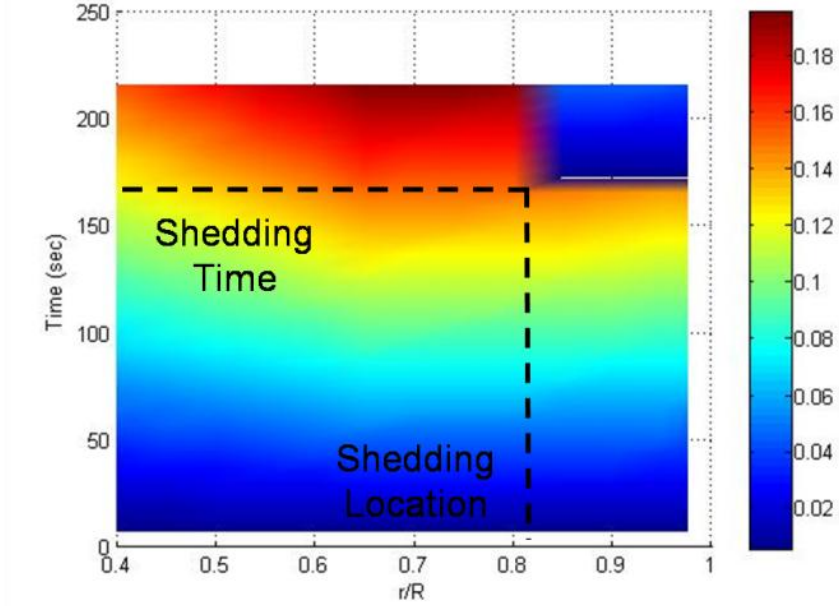


Figure 142: Case 31 Ice Thickness, Corrections

Table 21: Shedding Results

Case	Shedding Event	AERTS		ARISP No Corrections		ARISP Corrections	
		Time (s)	%R	Time (s)	%R	Time (s)	%R
31	1	215	0.82	130	0.82	160	0.81
33	1	352	0.77	100	0.93	150	0.92
	2			210	0.64	300	0.72
	3			320	0.93		
34	1	272	0.87	70	0.97	120	0.90
	2			150	0.73	220	0.71
	3			210	0.97		
35	1	320	0.83	85	0.99	123	0.99
	2			107	0.92	151	0.84
	3			160	0.76	268	0.69
	4			245	0.99		
	5			266	0.92		
	6			309	0.61		
36*	1	427	0.81	170	0.91		
	2			270	0.68		
37	1	313	0.75	94	0.88	135	0.94
	2			167	0.69	188	0.72
	3			261	0.88		
38	1	304	0.79	91	0.95	101	0.97
	2			132	0.76	132	0.88
	3			203	0.95	203	0.73
	4			243	0.66		
39	1	391	0.82	183	0.95	248	0.97
	2			261	0.80	300	0.84
43†	1	144	0.72	125	0.96		
*Significant errors were encountered in ice shape predictions, preventing meaningful correction factor calculation							
†Both blades shed, preventing the calculation of the correction factor							

The correction factors improved the shedding time and location predictions. Some error remains as shown in Table 22. In all cases, the experimental shedding event occurred after the prediction. In addition, multiple events are still expected based upon the model. Many more test points are required before the appropriate correction factors are created, but the

basis for the shedding analysis has been demonstrated. Further work is also required to accurately determine ice properties.

Table 22: Shedding Prediction Errors

Error in Shedding Predictions		
Case	Time	Station
31	25.58%	1.22%
33*	14.77%	6.49%
34*	19.12%	18.39%
35†	16.25%	16.87%
37*	40.00%	4.00%
38†	33.32%	7.59%
39*	23.32%	2.44%

*Second shedding event
†Third shedding event

6. Summary, Conclusions and Recommendations for Future Work

6.1 Summary

6.1.1 Background and Motivation

A critical operational problem for rotorcraft is flight in icing conditions. Ice accretes on critical components of rotorcraft when combinations of temperatures close to freezing and high supercooled water concentrations are encountered (12). Accreted ice can severely reduce aircraft performance, rendering safe flight impossible. The effects of a highly modified flowfield resulting from accreted ice include slightly reduced sectional lift coefficient and modified pitching moment characteristics, as described in References (9) - (11). The most significant effect of icing, however, is the increase in blade profile drag (84). Ice does not typically accrete in a uniform coating on the rotor; rather a rough and sometimes jagged structure is formed at the leading edge, causing premature flow separation and a considerable reduction in rotor lift/drag ratio. The torque can increase rapidly as described in Reference (55), potentially reaching the transmission or engine limits (12). This trend has been observed both in flight and in experiments in icing wind tunnels (85), (13).

Another problem of rotorcraft icing is due to the high rotational speed of the rotor. The system generates high centrifugal forces on accreted ice, especially near the tips. This provides a natural de-icing capability, which is theoretically beneficial for helicopter. The problem arises when the ice does not shed in a symmetric fashion from all blades. Uneven

ice accretion causes rotor imbalance and subsequent severe vibrations. The transmission or engine torque limits can be exceeded due to the fluctuating drag, preventing the pilot from maintaining certain flight conditions (12). Furthermore, researchers have identified problems with damage to the fuselage, engines, empennage and tail rotors from high velocity shed particles. Larger, multiengine vehicles, especially those of tandem rotor configuration, are more susceptible to this hazard (6).

The icing problem is compounded by the fact that helicopters are more susceptible to icing than a fixed wing aircraft of similar gross weight. Due to performance and mission limitations, operations occur almost exclusively at low altitudes, where the atmosphere may contain supercooled water droplets (8). Furthermore, rotorcraft operations are typically time sensitive. The hovering capability of the vehicle allows it to participate in search and rescue, medical evacuation and critical supply missions, in which the urgency requires rotorcraft to fly in Instrument Meteorological Conditions (IMC). This proximity of rotorcraft and high concentration of supercooled water generates a high probability of icing, as described in Reference (7).

Significant effort has been invested in understanding ice accretion phenomena to rotor systems to improve rotorcraft flight safety. In recent years, effort has been focused on simulations, so that the expensive icing flight trial duration can be reduced (75). Ice accretion on rotors and its subsequent effect on helicopter performance cannot be analyzed

in the same manner used to explain ice accretion on the leading edges of a fixed wing aircraft (6). Rotor icing predictions are made more complex by periodic changing of angle of attack and velocity. In addition, spanwise surface temperature differences and thinner airfoils make all rotor blade icing analysis challenging. If these items are properly accounted for, the resulting model can provide insight into the effects of various conditions and designs on the icing performance on rotor configurations. Correct shape prediction allows for the development of icing flight envelopes and for the proper sizing and implementation of ice protection systems. Finally, the model can be directly applied to the certification process, for interpolation or extrapolation of icing flight trial results (75).

Current efforts to model rotor ice accretion are limited either by significant code complexity or more importantly by a lack of available validation data. The lack of data may be a result of the existence of few icing facilities focusing on rotorcraft research. Ice protection system and other component level testing can be accomplished at a number of icing wind tunnels, but these tests are often limited by tunnel velocity and the fact that the centrifugal forces inherent to rotor rotation are not represented. With the closing of the Canadian National Research Council (NRC) Helicopter Spray Rig, only two primary facilities remain. Complete rotor icing studies can be undertaken in NASA Glenn's Icing Research Tunnel (IRT), but model rotor diameters are limited to 6 ft by the test section (73). Full scale studies may be accomplished with the Helicopter In-flight Spray System (HISS), but detailed icing shapes are difficult to acquire. Test aircraft need to descend through layers of warm air and

may shed ice prior to landing. This shedding, which aside from being inherently dangerous, eliminates the possibility of ice shape data collection.

6.1.2 ARISP Model

In this thesis, a rotor icing model was developed to analyze the ice accretion on a rotor in hovering flight. The model is based around a coupling of NASA's LEWICE ice accretion code with pre and post processors. The program takes a quasi-2D approach and is broken into four major modules. The first module calculates rotor performance via Blade Element Momentum Theory (BEMT). Based upon the particular icing conditions and rotor geometry, required LEWICE input files are generated and LEWICE is run in the second module. The third module calculates the ice properties and evaluates ice shedding. Finally, the fourth module calculates performance degradation due to ice with a series of empirical equations derived from wind tunnel experiments. Individual modules were validated against published data.

The entire model was validated to both small scale rotor tests in an icing wind tunnel and full scale tests in an outdoor icing stand. Shape correlations were generally good, especially at the inboard rotor stations. Accreted ice mass at the outboard stations tended to be over predicted. This is due to the 2D approach of the model, which does not account for the three dimensional flowfield and centrifugal effects on rotor ice accretions.

6.1.3 AERTS Facility

A new facility was developed to test ice accretion behavior of various rotor systems. The facility is designed to generate an accurate icing cloud around test rotor. The primary mission of the facility is to evaluate new ice protection system concepts and test new "ice phobic" materials. Details and limitations of the facility are described in Appendices A – C.

Computational Fluid Dynamics (CFD) was used to understand the complex flowfield inside of the small test chamber. The objective of the modeling was not to exactly calculate the flowfield, but rather to explore the effects of various rotor and nozzle configurations. Particle trajectory analysis was used to understand the paths of the supercooled droplets.

A new calculation method was developed to experimentally determine the chamber Liquid Water Content (LWC), which is a critical icing parameter. No sensor is available for the AERTS laboratory due to cost and flowfield issues. Therefore icing cloud severity is calculated experimentally based upon the stagnation ice accretion after each test, which requires an accurate calculation of the freezing fraction.

Sensitivity studies were undertaken to test the newly constructed icing facility. A series of test cases were run to explore the operational limitations of the icing system. Problems with particle crystallization and subsequent erosion of accreted ice shapes were identified and

corrected with ice screens and improved nozzle settings. The limits of the facility icing envelope were established for all subsequent testing.

6.1.4 Ice Shear Adhesion Test Fixture

An important parameter in the prediction of shedding events is the shear adhesion strength of ice to various materials. In order to measure this value, a new ice adhesion fixture was designed, manufactured and calibrated. The system, based upon the work of Stallabrass and Price, is designed to measure the shear adhesion strength of ice under realistic conditions, including impact ice and high velocity rotation. Thermal and mechanical shocks, inherent to other testing mechanisms, are eliminated by mounting the fixture on the tip of a specialized rotor.

Extensive shakedown tests of the specialized rotor were conducted, which identified the need of additional heaters to prevent undesired ice from influencing test data. A limited test matrix was accomplished with the fixture. The shear adhesion test results were found to be within the range of published literature.

6.1.5 Experimental of the ARISP Model in the AERTS Facility

A large matrix of test points were run to validate the ARISP icing model inside of the AERTS test chamber. Test points were selected to explore the effects of various parameters on accretion, performance degradation and shedding. Most points were inside of the

intermittent maximum icing envelope specified by the FAA in FAR Part 29 Appendix C. A modified Schweizer 269C test rotor was employed in all testing.

Overall shape correlations were good, with the ARISP model overpredicting the accretions near the rotor tips. Impingement limits were found to be within 20% of predicted values. Problems with determining the exact icing conditions in the test chamber, along with the temperature rise during each test complicated ice shape correlations. For each test case, the ARISP model was run with the best estimate of icing conditions in the test chamber.

Rotor performance was monitored during the icing experiments and compared to analytical predictions. The general torque profile for each test was matched well, as the initial torque rise due to an increase in surface roughness was shown in the predictions. Performance correlations were generally good, with experimental torque rises within 20% of predicted values when using Bragg's coefficients.

Multiple shedding events were encountered during testing in the AERTS Facility. Initial shedding correlations were presented, which did not match ARISP predictions due to the over prediction of ice accretions in the tip sections. The shedding predictions were improved with the application of correction factors derived from the difference in ice cross sectional areas at the tip.

6.2 Conclusions

6.2.1 ARISP Model

A new icing model, based upon the NASA LEWis ICE accretion code (LEWICE), was developed to explore rotor icing phenomena. The model has the ability to predict ice accretions on rotor systems, calculate performance degradation and investigate shedding. Based upon some correlations to published data, the following conclusions can me made:

- When compared to small scale and full scale rotor ice accretions, a rotor icing analysis model based upon LEWICE generates accurate ice shapes at inboard blade regions. The model tended to over predict ice accretions at the rotor tips, by approximately 20% of ice mass. The reasons for the difference may be due to the centrifugal effects of the rotor on the icing cloud LWC distribution. Also, the quasi two dimensional analysis, combined with the panel nature of LEWICE, does not accurately model the complex flowfield at the rotor tips.
- The ARISP model over predicts ice extent at the rotor tips when compared to a simplified model of blade tip kinetic heating. ARISP predicted ice accretions to the rotor an average of 16% further outboard for the AERTS test rotor from the simplified model.

6.2.2 AERTS Facility

A new icing facility, the Adverse Environment Rotor Test Stand (AERTS), was developed at the Pennsylvania State University. The facility is focused on rotorcraft and is intended to study new ice protection systems. The following paragraphs indicate conclusions from the development from the AERTS icing chamber:

- Numerical studies of the AERTS facility were required due the small overall dimensions of the test chamber. Velocity profiles and stream tubes indicated a strong tip vortex structure, which influenced the droplet trajectories in the chamber. The vortex structure is responsible for the generation of gradients in icing cloud density.
- A new method was developed to experimentally determine the Liquid Water Content (LWC) in the test chamber since no sensors are applicable to the AERTS configuration. The calculation process, based upon the post test stagnation ice accretion thickness measurements, was able to determine the analytical LWC within ±15% of the experimental LWC for the 68% of the 54 validation cases. The calculation method is requires an accurate the stagnation ice thickness measurement and freezing fraction determination because small errors in either can create large errors (>150%) in the LWC calculation.
- Sensitivity studies were required to evaluate the AERTS icing system. The effects of temperature, nozzle settings, droplet size, collective pitch and rotor RPM on stagnation ice accretion were investigated and facility limitations were established for the test rotor. These limitations include a minimum of 500 RPM for adequate cloud mixing and a maximum collective pitch of 5°. Temperature increases during testing were sometimes significant, exceeding 2°C.
- A cloud density measuring instrument was designed and implemented. Testing results show that the icing cloud stabilizes after approximately 70 seconds, indicating that rate

of water injection to the test chamber is balanced by ice accretion to the rotor and other surfaces in the test chamber.

- Ice crystallization was a significant problem in the small test chamber due to ice shape erosion. The addition of ice crystal collection screens and reduced water pressure input to the nozzles significantly reduced the amount of ice crystals in the facility. The maximum number of 5 active nozzles and the maximum nozzle input air pressure limit was determined to be 23 psi to reduce crystal formation.

6.2.3 Ice Shear Adhesion Test Rotor

A test fixture was developed to accurately determine the shear adhesion strength of impact ice to various materials, an important parameter for shedding analyses. The test fixture is mounted at the tip of a specialized rotor, so ice accretion physics at realistic flow velocities can are represented. No user intervention is required during a test so mechanical and thermal shocks to the ice are precluded. Conclusions that can be drawn from this work include:

- Initial testing with the specialized rotor indicated that careful application of anti-icing heaters were required to prevent distortion of test data from ice bridging.
- The shear adhesion strength of a 2024 aluminum coupon was calculated for a limited number of test points. Comparisons of the data to literature from a variety test methods indicate that the test fixture is accurate. More data points are required to fully characterize the material, but all data points were within the range of published data.

6.2.4 Experimental of the ARISP Model in the AERTS Facility

A series of icing experiments were conducted in the AERTS facility and ice shapes, impingement limits, performance degradation and shedding behavior was compared to ARISP model predictions . These tests represented the first attempt at ice shape correlation in the new test chamber. Conclusions from a series of 21 icing test experiments include the following:

- When operating the facility within the limitations established by the sensitivity studies, test points can be generated inside of the intermittent maximum icing envelope as established by FAR Part 25/29 Appendix C. Ice crystallization problems were significantly reduced with proper nozzle settings and ice collection screens.
- Liquid water content gradients were observed across the span of the rotor. The blade tip generally saw a 147% increase in LWC over the midspan value. It is thought that this is a facility effect, not a general rotorcraft icing trend, but further investigation is required.
- A qualitative grading scale was used to evaluate the accuracy of the ice shape predictions due to the complexities of the ice accretions. 50% of the ice tracings were given a green rating, indicating that the predicted and experimental shapes matched very well. Over 81% of the ice tracings were given a yellow or green rating, signifying that the model can be used for general icing trend analysis. Uncertainties in determination of icing parameters significantly impacted model shape correlations.

- Accurate prediction of impingement limits is critical to the sizing of ice protection systems. The ARISP model predicted most upper and lower impingement limits within 20% of experimental values. Correlations were hampered by the tedious and potentially inaccurate ice tracing process, which does not provide an accurate record of the impingement limits.
- Experimental torque increases due to ice accretion were generally within 20% of predicted values when using Bragg's correlations. The initial rapid torque rise due to increased blade surface roughness was captured. It was critical to calculate increases in blade sectional C_d relative to the clean airfoil performance due to the short intervals of icing time in the ARISP model. Most of the outlying test cases were outside of the icing envelope used to develop Bragg's correlations.
- A number of test cases involved shedding events. The ARISP model shedding predictions did not match the experimental shedding time and location due to an over prediction of rotor tip ice. This created an artificially high ice mass, which with increase centrifugal forces shed earlier than observed experimentally. All experimental shedding events occurred as a single large event, unlike the small multiple events predicted by the model. Correction factors were derived based on ice shape discrepancies to improve shedding correlations. When second and third shedding events were considered, the average error in shedding time was reduced to 25%, while the shedding station prediction was improved to 8%.

- Based upon the current testing, a rotor icing model based upon LEWICE has shown that it can be accurate enough for engineering level analyses. Further work is required to improve shedding correlations, but the foundation of a useful icing model has been established.

6.3 Future Work

6.3.1 Icing Model

6.3.1.1 Model Design

The ARISP model can be improved with a detailed examination of the ice accretion at the rotor tips. If the shape correlations can be improved, the accuracy of the rotor performance and shedding calculations would increase. Effort should be focused on centrifugal effects on the ice accretion, especially for glaze and beak ice, and quantifying the tip flowfield.

6.3.1.2 Shedding Implementation

Further exploration of the shedding characteristics is required. More development of ice properties needs to be undertaken. With the specialized shear adhesion test rotor, many more data points can be collected regarding the adhesive properties of ice to various materials. Variations in material surface roughness, inherent to sand and rain erosion as well as icing cloud parameters should be considered. Effort should also be invested into accurately establishing the cohesive strength of ice.

Once the ice parameters are better established, the current shedding analysis can be extended by considering non-square tip shapes. Most new rotor tip designs feature a combination of sweep, taper, dihedral and anhedral for performance and noise improvements. The current shedding assumptions cannot accommodate this geometry.

A second extension of the current analysis is the addition of a tracking module to model the trajectories of shed ice pieces, which pose a ballistic danger to rotorcraft. Simple projectile motion equations would be sufficient to model the problem. The difficulty lies in an accurate determination of the drag coefficient for the irregular shed pieces of ice.

6.3.1.3 Model Validation

A larger test matrix is required to fully investigate the model. Future work should emphasize the completion of a test matrix with icing clouds within the Part 25/29 Appendix C continuous maximum icing envelope. This requires an evenly distributed icing cloud with a low LWC. The tests will validate modifications to the ice accretion model at the rotor tips.

6.3.2 AERTS Laboratory

6.3.2.1 Temperature Control

An improved temperature control system needs to be implemented in the AERTS facility. Currently, the chamber temperature can increase significantly (> 5 °C in some cases) over the duration of a test run. This problem complicates all aspect of icing testing, especially ice shape correlations and the determination of ice properties, as small deviations in temperature

have a large influence on ice accretion physics. Data points in the colder areas of the icing envelope are especially hard to collect due to temperature deviations. Typical icing facilities maintain the static temperature within 1 °C during testing to minimize errors in ice accretion, so this capability should be matched in the AERTS facility.

Auxiliary cooling systems may be required to increase the system temperature stability during tests. The systems must not disturb the icing cloud and remain efficient even with ice accreting on critical components.

6.3.2.2 Icing Cloud Sensors

Sensors need to be implements in the AERTS facility to directly measure droplet size (MVD) and liquid water content (LWC). Currently these icing cloud properties are not measured or experimentally determined, respectively. The LWC calculation process is only applicable to rotor geometry with known stagnation collection efficiencies and is cannot be applied when ice is shed from the rotor.

A LWC sensor based upon a laser system has been demonstrated in the AERTS facility, as described in Section 3.3.4. The concept has been proven, but significant effort is required to calibrate the system for accurate calculations of the LWC in the test chamber. Other sensor types are possible, but the laser system is preferred because it can be positioned around the chamber to measure LWC gradients.

The development of soot slide detectors has started. This simple droplet size sensor is used as a backup to more expensive systems in flight trials. Work is required to calibrate the system and implement it in the facility. The system needs to be rotated to provide adequate velocity for particle impacts. Care should be taken in the design of the system so that a representative sample of the icing cloud is collected.

6.3.2.3 Nozzle Configuration

The current CFD studies should be expanded to evaluate improvements in nozzle positioning. Since most tests will only require less than 5 nozzles, the injection points are critical to the establishment of a uniform icing cloud in the facility. The primary concern is the reduction of the spanwise LWC gradient in the facility due to the tip vortices developed by the rotor. The nozzle exit velocity may be able to counter the formation of this vortex and promote recirculation around the ballistic wall. The studies should be verified with experimental verification.

6.3.2.4 Ice Shape Measurement

The current hand tracing of ice shapes produces a record of the accreted ice for comparisons to models, but is tedious and does not record 3D characteristics of the ice. Furthermore, the digitizing process is time consuming. Work on a new laser scanning system has been started, but the processes for recording shapes needs to be developed. The optical properties of ice, especially glaze ice, do not readily lend themselves to accurate scanning. The lasers either pass directly through the accretions or otherwise diffract enough to not accurately record the

shape. Therefore, a surface treatment, such as baby power, must be used to create a surface that does not reflect the lasers. This treatment application process must be consistent and repeatable to produce accurate scans.

6.3.2.5 Rotor Balancing

Improved test safety is required, especially for experiments that require operation at high rotor RPM. Rotor design and construction should emphasize rotor tracking and balance requirements. Stability analyses should be undertaken for all new designs and accommodations for balance weights should be made in the blade tips. New 1/rev sensors should be installed on the test stand to allow for rapid identification of discrepant blades.

7. Works Cited

1. **Lederer, Jerome "Jerry".** Flight Saftey Foundation (FSF) Lecture. 1942.

2. **Leary, William M.** *We Freeze to Please: A History of NASA's Icing Research Tunnel and Quest for Flight Saftey .* s.l. : The NASA History Series, 2002. NASA SP-2002-4226.

3. **Smith, Wesley L.** Weather Problems Peculiar to the New York-Chicago Airway. *Monthly WeatherReview.* 1929, 57, pp 503-506.

4. NTSB – Aviation Accident Database. [Online] National Transporation Saftey Board. http://www.ntsb.gov/ntsb/query.asp.

5. **Peck, Lindamae, Ryerson, Charles and Martel, C. James.** *Army Aircraft Icing.* s.l. : US Army Corps of Engineers Engineering Research and Development Center Cold Regions Research and Engineering Laboratory, 2002. ERDC/CRREL TR-02-13.

6. Inflight Icing and the Helicopter. *Flight Saftey Foundation Helicopter Saftey.* November/December, 1990, Vol. 16, 6.

7. *The Influence of Ice Accretion Physics on the Forecasting of Aircraft Icing Conditions.* **Hansman Jr., R. John.** s.l. : Thrid International Conference onthe Aviation Weather System, 1989. N90-20928.

8. *Investigation into the Effectiveness of Certification and Operational Icing Procedures for Helicopters .* **Simpson, M.P. and Render, P.M.** Reno, NV : 36th Aerospace Sciences Meeting and Exhibit, 1998. AIAA-1998-750.

9. *A Wind Tunnel Study of Icing Effects on a Business Jet Airfoil.* **Addy, Harold E., et al.** Reno, NV : 41st Aerospace Sciences Meeting and Exhibit, 2003. AIAA 2003-727.

10. *Unsteady Aerodynamic Measurments on an Iced Airfoil.* **Gurbacki, H. M. and Bragg, M. B.** Reno, NV : 40th AIAA Aerospace Sciences Meeting and Exhibit , 2002. AIAA 2002-9241.

11. *Aerodynamic Simulations of Airfoils with Upper-Surface Ice-Shapes.* **Kumar, Satish and Loth, Eric.** 2, s.l. : Journal of Aircraft, 2001, Vol. 38.

12. **Gent, R.W., Dart, N.P and Candsdale, J.T.** Aircraft Icing. *Philosophical Transactions of the Royal Society of London Series.* 2000, Vol. 358.

13. *Icing Research Tunnel Test of a Model Helicopter Rotor.* **Miller, Thomas L. and Bond, Thomas H.** s.l. : Prepared for the 45th Annual Forum and Technology Display sponsored by the American Helicopter Society, 1989. NASA TM-101978.

14. *Artificial Icing Tests of the S-92A Helicopter in the McKinley Climatic Laboratory.* **Flemming, Robert J, Alldridge, Phillip and Doeppner, Ronald.** s.l. : American Institute of Aeronautics and Astronautics, Inc, 2004. AIAA-2004-0737.

15. *The UH-1H Helicopter Icing Flight Test Program: An Overview.* **Shaw, Robert J. and Richter, G. Paul.** Reno, NV : 23rd Aerospace Sciences Meeting, 1985. AIAA 1985-338.

16. **Scavuzzo, R.J., Chu, M.L and Kellackey, C.J.** Impact Ice Stress in Rotating Airfoils. *Journal of Aircraft.* 7, 1991, Vol. 28.

17. *The Effect of Atmospheric Droplet Size Distribution on Aircraft Ice Accretion.* **Hansman Jr., R. John.** s.l. : AIAA 22nd Aerospace Sciences Meeting, 1984. AIAA 84-0108.

18. **Lankford, Terry T.** *Aviation Weather Handbook.* New York : McGraw-Hill, 2000. ISBN 0-07-136103-0.
19. **Cao, Yihua, Zhang, Qiang and Sheridan, John.** *Numerical Simulation of Ice Accretion on Airfoils.* Adelaide, Austrailia : XXII ICTAM, August 2008.
20. **Tinetti, A. and Korkan, K.** Generic Icing Effects on Forward Flight Performance of a Model Helicopter Rotor. *Vertica.* 1989, Vol. 13, 1.
21. *Aircraft Icing Handbook, Volumes I-III.* Atlantic City International Airport, NJ : FAA Techical Center, 1991. DOT/FAA/CT-88/8-1, AD-A238 039.
22. **Gray, V.H. and Von Glahn, U.H.** *Aerodynamic Effects Caused by Icing of an Unswept NACAA004.* s.l. : NACA Technical Note, 1958.
23. **Ramage, Andrew.** *Apache Icing Clearances.* Rotorcraft Operations in Icing Conditions Conference, London : Westland Helicopters Limited, 2006.
24. **Hansman Jr., R. John and Turnock, Stephen R.** *Invesitgation of Surface Water Behavior During Glaze Ice Accretion.* Reno, NV : 26th Aerospace Sciences Meeting, 1988. 88-0015.
25. *SmaggIce User Manual.* Cleveland, OH : NASA Glenn Research Center, 2006.
26. **Flemming, Robert J. and Lednicer, David A.** *High Speed Ice Accretion on Rotorcraft Airfoils.* s.l. : NASA STI Branch, 1985. NASA CR 3910.
27. *Communications Structures.* **Smith, Brian W.** London : Thomas Telford Publishing, 2007. 978-0-7277-3400-6.
28. JAR/FAR Part 29 Appendix C. *Icing Certification.* s.l. : Federal Aviation Administration.
29. *Aircraft Aerodynamic Effects Due to Large-Droplet Ice Accretions.* **Bragg, M.B.** 1996. AIAA 96-0932.
30. *AIRA ASME 2007 Engine Harmonization Working Group Activities Report.* 2007.
31. *Supercooled Large Droplets in Icing Conditions.* **Jentink, H.W.** Nice, France : European Geophysical Society, 1998. National Aerospace Laboratory (NLR) NLR-TP-98216.
32. *Icing Caused by Large Supercooled Drops.* **Politovich, M.K.** 9, pp 856-868, s.l. : Journal of Applied Meteorology, 1989, Vol. 28.
33. **Oleskiw, M.** A Review of 65 Years of Aircraft In-Flight Icing Research at NRC. *Canadian Aeronautics and Space Journal.* 2001, Vol. 47, 3.
34. **Perkins, Porter J.** *Icing Frequencies Experienced During Climb and Descent by Fighter-Interceptor Aircraft.* July 1958. NACA TN 4314.
35. **Flemming, Robert J., Britton, Randall K. and Bond, Thomas H.** *Role of Wind Tunnels and Computer Codes in the Certification and Qualification of Rotorcraft for Flight in Forecast Icing.* Amsterdam, The Netherlands : Twentieth European Rotorcraft Forum , 1994. NASA TM 106747.
36. **Angers, Sandy.** Nice Ice, Baby. *Boeing Frontiers.* March 2008.
37. *Goodrich Icing Wind Tunnel Overview, Improvements and Capabilities.* **Herman, Edward.** Reno, NV : 44th American Institute of Aeronautics and Astronautics Aerospace Sciences Meeting and Exhibit, 2006. AIAA 2006-862.
38. *Development of the Cox Icing Research Facility.* **Al-Khalil, K., Salamon, L. and Tenison, G.** Reno,NV : 36th AIAA Aerospace Sciences Meeting and Exhibit, 1998. AIAA 98-0097.

39. *Calibration of the CIRA IWT in the High Speed Configuration.* **Ragni, A., et al.** Reno, NV : 43rd AIAA Aerospace Sciences Meeting and Exhibit, 2005. AIAA 2005-471.
40. *Icing and Deicing Test on a 1/4-scale Rotor in the ONERA S1MA Wind Tunnel.* **Gufford, D.P.** s.l. : AIAA 24th AerAIAAospace Sciences Meeting, 1986. AIAA 86-480.
41. **Bailey, D.L., Samolewicz, J.J. and MacPhail, D.C.** *Description of the Spray Rig Used to Study Icing on Helicopters in Flight.* National Research Laboratories, Ottawa, Canada : Division of Mechanical Engineering, Low Temperature Laboratory, 1960. LR-186A, File M4-A3-R26.
42. **Peterson, Andrew A. and Oldenburg, John R.** *Spray Nozzle Investigation for the Improved Helicopter Icing Spray System (IHISS).* Reno, NV : AIAA 28th Aerospace Sciences Meeting, 1990. AIAA-90-0666.
43. Objective Helicopter Icing Spray System (OHISS). [Online] US Army Program Execution Office for Simulation, Training and Instrumentation, December 1, 2008. [Cited: March 22, 2009.] http://www.peostri.army.mil/PRODUCTS/OHISS/.
44. *A Three-Dimensional Water Droplet Trajectory and Impingement Analysis Program.* **Saeed, Farooq, et al.** Toronto, Ontario, Canada : 23rd AIAA Applied Aerodynamics Conference, 2005. AIAA 2005-4838.
45. *Use of Ice Accretion Computer Codes - Certification Authorities.* **Hill, Eugene G.** s.l. : NATO/RTO Ice Accretion Simulation Evaluation W/S, 2000.
46. **Wright, William B.** *User Manual for the NASA Glenn Ice Accretion Code LEWICE (Version 2.2.2).* Cleveland, OH : s.n., 2002. NASA/CR-2002-211793.
47. *State-of-the-Art Aircraft Icing and Anti-Icing Simulation.* **Saeed, Farooq.** 25-27, Cleveland, OH : 25th Annual ARA Congress, 2000, Vols. 2000-2002.
48. **Langmuir, I. and Blodgett, K.B.** *A Mathematical Investigation of Water Droplet Trajectories.* s.l. : US Army Air Forces (Contract Number W-33-038-ac-9151 with General Electric Co.), 1946. TR 5418.
49. **Glauert, M.** *A Method of Constructing the Paths of Raindrops of Different Diameters Moving in the Neighbourhood of (1) a Circular Cylinder, (2) an Aerofoil Placed in a Uniform Stream of Air; and Determination of the Rate of Deposit of the Drops on the Surface and the .* s.l. : Aeronautical Research Council, UK, Percentage of Drops Caught,1940. R&M 2025.
50. *Validation Results of LEWICE 2.0.* **Wright, William B. and Rutkowski, Adam.** s.l. : NASA STI Program Office, 1999. CR 19990298690.
51. *Analytical Determination of Propeller Performance Degradation Due to Ice Accretion.* **Miller, T. L., Korkan, K. D. and Shaw, R. J.** 11, s.l. : Journal of Aircraft, 1987, Vol. 24.
52. *Development of an Analytical Method to Predict Helicopter Main Rotor Performance in Icing Condition.* **Britton, Randall K.** 1992. NASA CR 189110.
53. **Hartman, P, et al.** *Prediction of Ice Accumulation and Airfoil Performance Degradation: A Boeing - CIRA Research Collaboration.* Phoenix, AZ : American Helicopter Society 62nd Annual Forum Proceedings , May 2006.

54. *Ice Accretion Prediction on Helicopter Rotor Blade in Hover Flight.* **Zanazzi, G., et al.** Seville, Spain : SAE Aricraft and Engine Icing International Conference, 2007. SAE Paper 2007-01-3309.
55. *Spinning Rotor Blade Tests in Icing Wind Tunnel.* **Fortin, G. and Perron, J.** San Antonio, TX : 1st AIAA Atmospheric and Space Environments Conference, 2009. AIAA 2009-4260.
56. **Leishman, J. Gordon.** *Principles of Helicopter Aerodynamics, Second Edition.* New York : Cambridge University Press, 2000.
57. **Hassan, Ahmed.** *Predicted Characteristics of a NACA0015 Airfoil Having a 25% Integral-Type Trailing Edge Flap.* s.l. : NASA Langley Research Center, 1999. CR-1999-209328.
58. **Cheeseman, I.C. and Bennet, W.E.** *The Effect of the Ground on a Helicopter Rotor in Forward Flight.* s.l. : ARC R&M 3021, 1955.
59. **Rossow, Veron J.** *Effect of Ground and/or Ceiling Planes on Thrust of Rotors in Hover.* 1985. NASA TM 86754.
60. **Prouty, Raymond W.** *Helicopter Performance, Stability and Control.* Malabar, FL : Krieger Publishing Company, 1995.
61. **Landgrebe, Anton and Bellinger, E. Dean.** *Experimental Investigation of Model Variable Geometry and Ogee Tip Rotors.* 1974. NASA CR 2275.
62. *Influence of Aerodynamic Forces in Ice Shedding.* **Scavuzzo, R. J., Chu, M. L. and Ananthaswamy, V.** 3, s.l. : Journal of Aircraft, 1994, Vol. 31.
63. **Ashton, George D. (editor).** *River and Lake Ice Engineering.* Highlands Ranch, CO : Water Resources Publishers, 1986 (Second Printing 2004). ISBN 0-918834-59-4.
64. **Macklin, W.C.** The Density and Structure of Ice Formed by Accretion. *Quarterly Journal of the Royal Meterological Society.* 88, 1962, Vols. pgs 30-50.
65. *Comparison of Droplet Size Measurments by Three Methods.* **Stallabrass, J.R.** s.l. : Proceedings of the Third International Workshop on Atmospheric Icing to Structures, 1986. Paper 3.3.
66. **Jones, Kathleen F.** The Density of Natural Ice Accretions Related to Non-Dimensional Icing Parameters. *Quarterly Journal of the Royal Meteorlogical Society.* 116, 1990, pages 477-496.
67. *Tests of the Performance of Coatings for Low Ice Adhesion.* **Anderson, David N. and Reich, Allen D.** Reno, NV : 35th AIAA Aerospace Sciences Meeting and Exhibit , 1997. AIAA 97-0303.
68. *Structural Properties of Impact Ices Accreted on Aircraft Structures.* **Scavuzzo, R. J. and Chu, M. L.** Akron, Ohio : The University of Akron , 1987. NASA CR 179580.
69. *Adhesive Shear Strength of Ice.* **Chu, M. C. and Scavuzzo, R. J.** 11, s.l. : AIAA Journal, 1991, Vol. 29.
70. *Tensile Properties of Impact Ices.* **Chu, M.L. and Scavuzzo, R.J.** Reno, NV : AIAA 30th Aerospace Sciences Meeting, 1992. AIAA 1992-0883.
71. **Bragg, M. B.** *Rime Ice Accretion and its Effects on Airfoil Performance.* s.l. : NASA, 1982. NASA CR 165599.

72. **Flemming, Robert and Lednicer, David.** *High Speed Ice Accretion on Rotorcraft Airfoils.* 1985. NASA CR 3910.
73. *A Review of Ice Accretion Data from a Model Rotor Icing Test and Comparison with Theory.* **Britton, Randall K. and Bond, Thomas H.** Reno, NV : 29th Aerospace Sciences Meeting Sponsored by the American Institute of Aeronautics and Astronautics, 1991. AIAA 91-0661.
74. **Lee, John D., Harding, Rorry and Palko, Richard L.** *Documentation of Ice Shapes on the Main Rotor of a UH-1H Helicopter in Hover.* 1984. NASA CR 168332.
75. *Further Studies of Helicopter Rotor Ice Protection Accretion and Protection.* **Gent, R.W., Markiewicz, R.H. and Cansdale, J.T.** 3, s.l. : Vertica, 1987, Vol. 11. pp 473-492.
76. **Dassault Systèmes Solidworks Corporation.** *SolidWorks Flow Simulation 2009 Techincal Reference.* 2009.
77. *Evaluation of and Validation of the Messinger Freezing Fraction.* **Anderson, David N. and Tsao, Jen-Ching.** Reno, NV : AIAA 41st Aerospace Sciences Meeting and Exhibit, 2003. AIAA 2003-1218 (also NASA CR 2005-213852).
78. **Anderson, David N.** *Evaluation of Constant-Weber-Number Scaling for Icing Tests.* 1996. NASA TM 107141.
79. *Experimental Evaluation of Stagnation Point Collection Efficiency of the NACA 0012 Swept Wing Tip.* **Tsao, J. and Kreeger, R.** San Antonio, TX : 1st AIAA Atmospheric and Space Environments Conference, 2009. AIAA 2009-4125.
80. *Reduction of Adhesion of Ice to De-Icer Surfaces.* **Loughborough, D. and Hass, E.** 126, s.l. : Journal of Aeronautical Sciences, 1946, Vol. 13.
81. **Stallabrass, J.R. and Price, R. D.** *On the Adhesion of Ice to Various Materials.* Ottawa, CA : National Research Laboratory, 1962. LR-350, M4-A3-R26.
82. **Itagaki, K.** *Self-Shedding of Accreted Ice from High Speed Rotors.* s.l. : US Army Cold Regions Research and Engineering Laboratory, 1983. ASME 83-WA/HT-68.
83. *Interface Influences Upon Ice Adhesion to Airfoil Materials.* **Reich, A.** Reno, NV : 32nd American Institue of Aeronautics and Astronautics Aerospace Sciences Meeting and Exhibit, 1994. AIAA 94-0714.
84. **Bragg, M. B.** *Rime Ice Accretion and Its Effect on Airfoil Performance.* 1982. NASA CR 165599.
85. *The Design, Fabrication and Testing of Simulated Ice Shapes for the S-92A Helicopter.* **Flemming, Robert J., et al.** s.l. : American Institute of Aeronautics and Astronautics, 2004. AIAA 2004-736.
86. **Papadakos, Peter.** Personal Correspondance . 2009.
87. **Johnson, Wayne.** *Helicopter Theory.* Princeton, New Jersey : Princeton University Press, 1980.
88. **Leishman, J. Gordon.** *The Helicopter Thinking Forward, Looking Back.* College Park, Maryland : College Park Press, 2007.

89. *Invesitgation of Ultrasonic Ice Protection System for Helicopter Rotor Blades.* **Palacios, Jose L.** American Helicopter Society 64th Annual Forum Montreal, Canada : American Helicopter Society International, 2007.
90. **Strallabrass, J.R.** Some Aspects of Helicopter Icing. *Canadian Aeronautical Journal.* October 1957, Vol. 3, 8.
91. **Anderson, David N.** *Acceptable Tolerances for Matching Icing Similarity Parameters in Scaling Applications.* 2003. NASA CR 2003-211822.
92. *Aircraft Performance Sensitivity to Icing Cloud Conditions.* **Campbell, Scot E., et al.** Reno, NV : s.n., 2007. 45th AIAA Aerospace Sciences Meeting and Exhibit AIAA 2007-86.
93. **Bragg, Michael B.** *A Study of Ice Accretion Physics to Improve the Ice Accretion on Airfoils.* 2001. Final Report NAG3-1998.
94. *Helicopter Rotor Icing Protection Systems.* **Coffman, H.J.** 2, s.l. : Journal of the American Helicopter Society, 1987, Vol. 32.
95. *The Past Twenty Years of Icing Research and Development at Sikorsky Aircraft.* **Flemming, Robert.** Reno, NV : 40th Aerospace Sciences Meeting and Exhibition. AIAA 2002-0238.
96. *De-Icing Equipment for Helicopters.* **Hermes, H.** s.l. : AEG-Telefunken Program, No. 2, 1970.
97. *A History of Ice Protection System Development at Sikorsky Aircraft.* **Flemming, Robert J.** Chicago : FAA In-Flight Icing/Ground De-Icing International Conference and Exhibition, 2003. 2003-01-2092.
98. *Effect of Curvature in the Numerical Simulation of an Electrothermal De-Icer Pad.* **Huang, J.R., Keith, Theo G. and De Witt, Kenneth J.** 1, s.l. : Journal of Aircraft, January - February 1995, Vol. 32.
99. *Development and Demonstration of Low Power Electrothermal De-icing System.* **Botura, Galdemir C., Sweet, Dave and Flosdorf, David.** Reno, NV : 43rd AIAA Aerospace Sciences Meeting and Exhibit, 2005. AIAA 2005-1460.
100. *Design of an Advanced Pneumatic Deicer for the Composite Rotor Blade.* **Weisend, Jr., Norbert A.** 10, s.l. : Journal of Aircraft, October 1989, Vol. 26.
101. *Flight Tests of the Helicopter Pneumatic Deicing System.* **Haworth, L.A. and Graham, M.S.** Williamsburg, VA : AHS National Specialists' Meeting on Helicopter Test Technology, October 1984.
102. *Icing Tunnel Tests of an Engine Inlet and High Speed Wings.* **Zumwalt, G.W.** Reno, NV : AIAA 23rd Aerospace Sciences Meeting, 1985. AIAA-85-0466.
103. *The Structural Dynamics of Electro-Impulsive De-Icing on the Lear Fan Kevlar Composite Leading Edge.* **Paramasivam, T. and Zumwalt, Glen W.** s.l. : SAE International, April 1985. 850917.
104. *Investigation of the Certification and Operational Procedures for Turboprop Aircraft in Icing.* **Simpson, M.P. and Render, P.M.** s.l. : The Aeronautical Journal, October 1999. pp 449 - 454.

105. **Magenheim, Bertram.** *Demonstration of the Microwave Ice Protection Concept.* s.l. : Mechanics Research, Inc, May 1978. USAAMRDL-TR-77-34, Eustis Directorate, US Army Air Mobility Research and Development Laboratory.
106. *High Frequency Microwave Anti-De-Icing System for Carbon Reinforced Airfoil Structures.* **Feher, Lambert E. and Thumm, M.** s.l. : Proceedings of SPIE, 2001, Vol. 4371. 0277-786X.
107.
108. *Vibratory Ice Protection for Helicopter Rotor Blades.* **Lemont, L.E. and Upton, H.** June 1978. USAAMRDL-TR-77-29.
109. *Tensile Properties of Impact Ices.* **Chu, M. L.** Reno, NV : 30th AIAA Aerospace Sciences Meeting and Exhibit, 1992. AIAA 92-0883.
110. **Gray, Vernon H. and von Glahn, Uwe H.** *Aerodynamic Effects Caused by Icing on an Unswept NACA 65A004 Airfoil.* s.l. : NASA STI. NASA TN 4155.
111. **Scavuzzo, J. R. and Price, R. D.** *On the Adhesion of Ice to Various Materials.* Ottawa, CA : National Research Laboratory, 1962. LR-350, M4-A3-R26.
112. *Generic Icing Effects on Forward Flight Performance of a Model Helicopter Rotor.* **Tinetti, Ana F. and Korkan, Kenneth D.** 1, s.l. : Vertica, 1989, Vol. 13. pp 63-85.
113. *Icephobic Material Centrifuge Adhesion Test.* **LaForte, C. and Beisswenger, A.** Montreal : International Workshop on Atmospheric Icing on Structures (IWAIS XI), 2005.
114. **Landgrebe, Anton J. and Bellinger, E. Dean.** *Experimental Investigation of Model Variable Geometry and Ogee Tip Rotors.* 1974. NASA CR-2275.
115. **Ide, Robert F. and Oldenburg, John R.** *Icing Cloud Calibration of the NASA Glenn Icing Research Tunnel.* 2001. NASA TM 2001-210689.
116. **Dassault Systèmes SolidWorks Corp.** *Solidworks Flow Simulation 2009 Technical Reference.* s.l. : Dassault Design Systems, 2009.

A. APPENDIX A: AERTS Facility Limitations

The following tables describe the limitations of the AERTS Facility as of April 2010.

Table 23: AERTS Rotor System Limitations

Parameter	Value	Notes
Motor Power (HP)	84 max	Unlimited run time
	120 max	3 minute run limit
Motor Torque (in-lbf)	696 max	Unlimited run time
	995 max	3 minute run limit
RPM	400 to 1000	400 RPM required for adequate cloud mixing; 1000 RPM test stand safety limit
Rotor Tip Radius (ft)	2 to 4.5	Minimum radius required to reduce effects of hub and support structure on icing cloud; maximum radius dictated by ballistic wall geometry
Blade Grip Radius (ft)	0.94	
Blade Grip CF Load (lbf)	14,000	Based upon QH-50 hub design loads. Includes required facility FOS.
Hub Precone (°)	3	
Hub Flap (Teeter) Range (°)	-12 to +12	Limited by teeter bumpers
Collective Pitch (°)	-2 to +12	Controlled with linear actuator.
Lateral Cyclic Pitch (°)	-5 to +5	Controlled with linear actuator. Longitudinal cyclic pitch is locked out.

Table 24: AERTS Icing System Limitations

Parameter	Value	Notes
Active Nozzles	1 to 15	Nozzles are arranged in 2 concentric rings in chamber ceiling, with 5 nozzles in inner ring and 10 nozzles in outer ring. Any combination of nozzles can be used for each test.
Temperature (°C)	Ambient to -25	Chamber cooling system is shut down during each test to avoid disrupting icing cloud. Temperature increases during each test due to kinetic friction of the rotor and warm water inputs to the chamber.
MVD (μm)	10 to 50	Not directly measured. Droplet size based upon NASA Standard nozzle calibration tables. Water and air pressure input control system maintains droplet size input to the test chamber ± 2 μm. Larger particle sizes are possible, but are outside the calibrated range of the nozzles.
LWC (gr/ m^3)	1 to 5	Not directly measured. Controlled by number of active nozzles and input pressures and calculated after each test based upon accreted ice thickness. Not all MVD/LWC points are possible based upon chamber limitations.
Icing Time (seconds)	30 to 240	Approximately 30 seconds is required for the nozzles to stabilize and for the cloud properly to mix in the chamber. Icing duration limit is based on the requirement to maintain static temperature in chamber ± 1 °C from desired point.
Water Input Temperature (°C)	20 to 71	Water input temperature can be varied to properly supercool droplets at all temperatures.
Water Purity (ppm impurities)	1 to 3	Reverse Osmosis Purification System generates water with 2 MΩ resistance (6" probe separation)

Table 25: AERTS Instrumentation Limitations

Parameter	Value	Notes
CCD Cameras	3	Two cameras monitor entire test chamber; third camera is focused on rotor tip.
Static Temperature Sensors	5	Thermistors are positioned around test chamber to monitor internal temperature. A commercial weather station measures temperature as well as pressure, humidity and rainfall rates.
Slip Ring Power Channels	24 max	Each channel is rated up to 15A, 800V
Slip Ring Signal Channels	24 max	Each channel is rated up to 2A, 100V
6 Axis Load Cell Forces (lbf)	Thrust: 875	Measurement range. Sensor can safely handle 8x reported loads.
	Lat/Long: 300	
6 Axis Load Cell Moments (in-lbf)	Pitch/Roll/Yaw: 1800	
Shaft Torque Sensor (in-lbf)	1500	Measurement range.
Available Output Channels	8	Each channel is rated at 0 - 10V, 10 ksps. Current is based upon DAQ module usages, but typically ~ 10 mA. Used for triggering relays etc.
Available Generic Input Channels	38 (+4)	Each channel is rated at 250 ksps. Four channels are reserved for temperature (thermocouple) measurements.
Available Strain Gauge Modules	3	Used only for fixed frame measurements
Strain Gauge Amplifiers	2	Strain gauge amplifiers are used for on blade measurements. The full bridge completion units are mounted at the blade roots and amplify signals prior to them entering the rotor hub to improve overall S/N ratio.
Accelerometer Conditioning Circuits	2	Accelerometer signals are conditioned and amplified prior to being read by Labview

B. APPENDIX B: AERTS Facility Icing System

B.1. Overview

The Vertical Lift Research Center of Excellence at the Pennsylvania State University has developed a new facility for rotorcraft icing research. Achieving an initial operational capability in November 2009, the Adverse Environment Rotor Test Stand (AERTS) is designed to generate an accurate icing cloud around test rotor. The primary mission of the facility is to evaluate new ice protection system concepts and test new "ice phobic" materials. An overview of the facility is given in Figure 143.

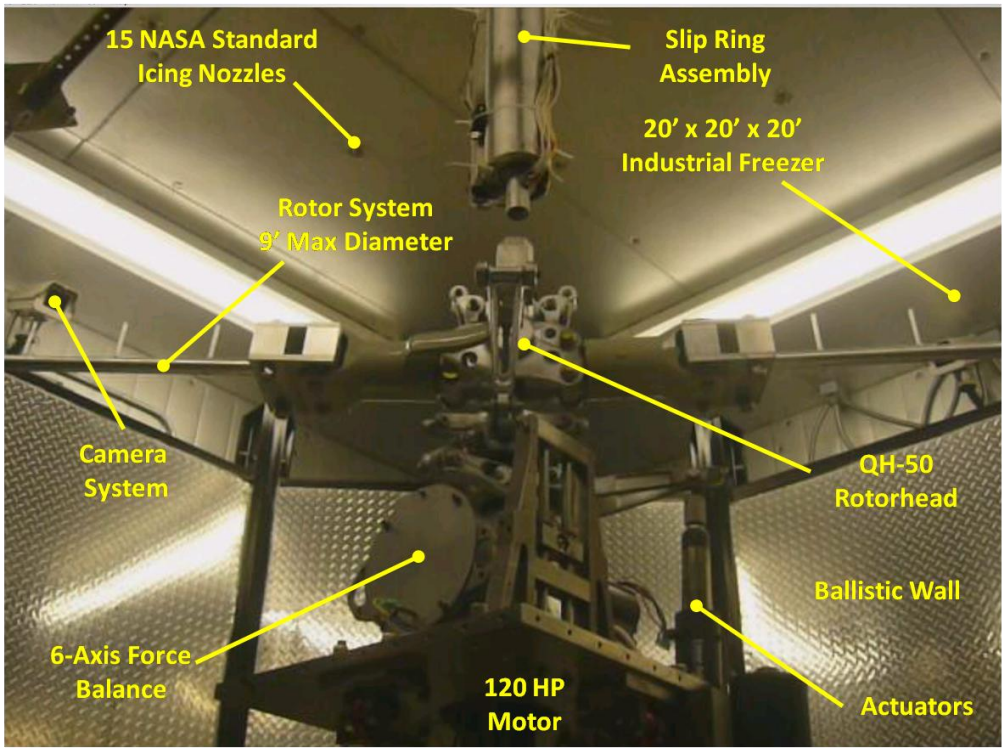


Figure 143: AERTS Facility Overview

B.2. Test Chamber

The AERTS test chamber consists of an industrial freezer, measuring approximately 20 ft x 20 ft x 20 ft internally. The freezer features 4 inch thick insulated walls and a robust cooling system that can maintain a constant internal temperature as low as -13 °F (-25 °C). Approximately 5 hours are required to cool the chamber from room temperature down to -4 °F (-20 °C).

The freezer is cooled by an evaporator assembly and two large fans circulate cooled air throughout the chamber. Whenever the icing cloud is active, the cooling fans are required to be shut down. The high CFM fans distort the icing cloud by blowing it to the rear of the chamber. More importantly, the fans draw supercooled droplets through the evaporator cooling assembly, clogging the inlet with ice and reducing the system efficiency. Ice also accretes to the cooling fans blades, which eventually shed the ice, potentially damaging the unit. To avoid these problems, the fans are shut down whenever the icing system was active via a switch in the control panel.

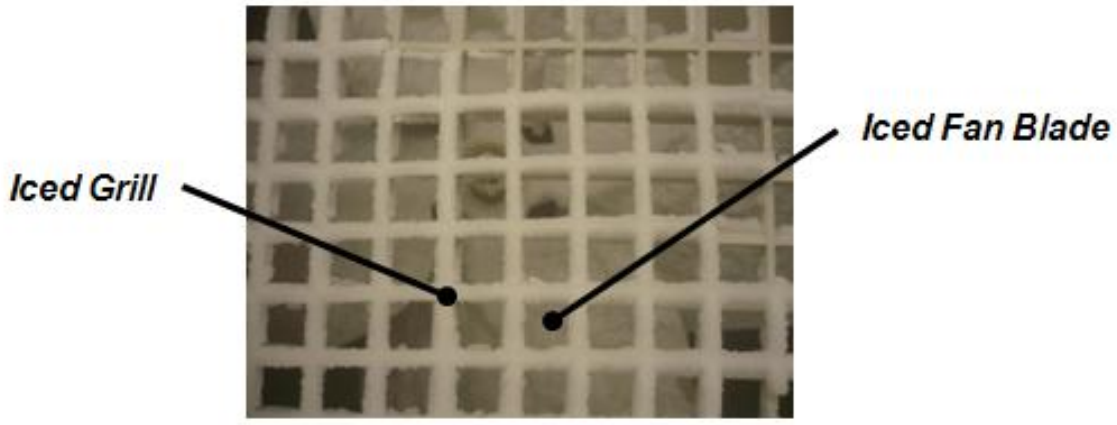


Figure 144: Close-up of Evaporator Fans, Showing Large Ice Accretions

Modifications to the freezer and cooling system for icing tests were minimal and consisted mainly of holes in the walls for wires. Additional features were added to the freezer, such as a watertight floor, drainage system and ballistic wall. These features were installed by two undergraduate research assistants, Michael Jostes and Zachary Weyant.

B.3. Icing System

B.3.1. Icing Nozzles

The icing system is based around 15 NASA Standard icing nozzles, graciously donated to the facility by NASA Glenn. Similar nozzles are used in the Icing Research Tunnel (IRT), the world's largest icing research facility at NASA Glenn. They operate by aerosolizing water with a precise combination of input water and air pressures. Figure 145 shows the approximately 3 inch long nozzle outside of its housing. The nozzles are capable of accurately producing droplets with mean volumetric diameters (MVD) between 10 and 50 microns by precisely controlling the input pressure of both the water and air. Larger particle sizes are possible, but are not required for most icing testing.

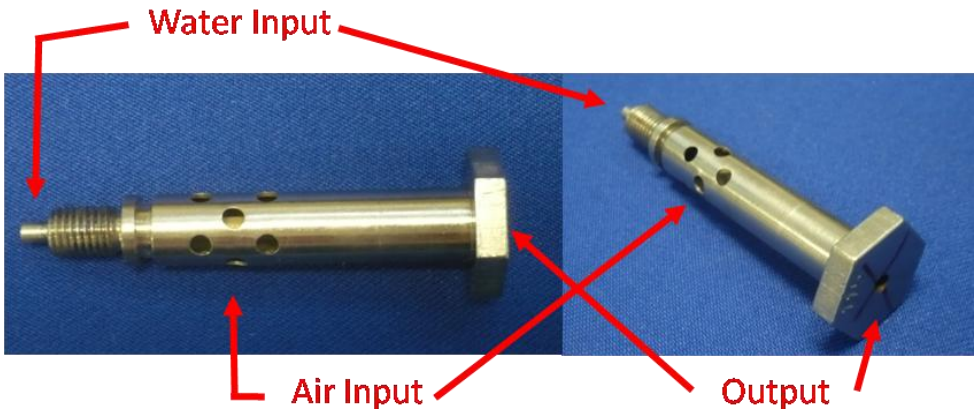


Figure 145: NASA Standard Ice Nozzle

The nozzles are installed in the ceiling of the icing chamber in two concentric circles. The inner ring consists of five nozzles, while the outer ring is made up of two arms of five nozzles each. The last nozzle on each arm is instrumented with 0-5V pressure gauges on both the water and air input lines. This pressure measurement is used to determine the droplet size on the calibration charts.

Dividing the nozzles into three separate arms allows for more precise control of the icing cloud. The nozzles are more stable when operating at higher water and air input pressures, so it is desirable to utilize them in this regime whenever possible. Higher pressures increases the water input into the AERTS chamber, generating higher LWCs. When attempting to generate a cloud with lower LWC, it is therefore advantageous to shut down one or more of the operating arms to reduce overall water input, while retaining nozzle operational stability.

B.3.2. Air System

B.3.2.1. *Compressor*

The air system was designed to provide accurate and consistent air pressure to the icing nozzles. The primary driver for the system was the large volume of air required by the nozzles to aerosolize water droplets. Each nozzle requires approximately 15 CFM to provide a stable icing cloud.

This large mass flow rate required a large air compressor. Various options were explored, including purchasing a new compressor and plumbing an auxiliary output line from the Aerospace Engineering Department's compressor and air storage system. Both options would have been prohibitively expensive, while the second option would be operationally difficult because compressor time would have to be shared with the jet noise laboratory.

Therefore, a third option was pursued: rehabilitating an old air compressor. The compressor is a 25 HP Ingersol Rand Model 25 formerly used by the Philadelphia Naval Shipyard. The unit was donated to Penn State and eventually moved to University Park, where it was positioned as the backup compressor for the jet noise laboratory. Due to its relatively small capacity, it was never plumbed into the system, rendering it available for the AERTS laboratory, albeit with some maintenance.

The compressor is connected to the shop air system, which provides a relatively constant 70 psi supply, depending on load, to all of the labs in Hammond Building. Adding the two air supplies together allows for the nozzle CFM requirements to be met. During the calibration of the AERTS facility, it was found that a constant nozzle input pressure of 35 psi could be maintained with all 15 nozzles in operation and 45 psi with 5 nozzles running. This does not cover the entire operational range of the nozzles, but provides enough capability to generate most droplet sizes.

A 200 psi tank attached to the compressor can be pressurized to provide short bursts of higher pressure, but this will most likely not be used. The tank however, has a pressure switch connected to the compressor controller, which automatically shuts the compressor off when the internal pressure reaches a preset level. If the tank pressure drops below the threshold, the compressor starts up again. This system has the benefit of allowing the compressor to be left on during between test runs without risk of damage to the nozzles. Finally, filters and dryers are mounted on the lines before and after the storage tank to remove any contaminants from the air system before it reaches the nozzles.

B.3.2.2. *Control System*

A master electric ball valve, activated from the control room, turns on and off the air system. The valve takes approximately 3 seconds to fully open or close, so no large transient air pressure spikes reach the nozzles, potentially damaging them. LEDs indicate the position of the valve to the operator. During operation, the air pressure is controlled by a set of pneumatic relays, which have valves to release excess air pressure. The relays are installed just before the manifolds and are then in turn activated by a 0-10V output from the Labview system. This allows the air pressure in the nozzles to be adjusted in 5 psi increments. Tighter modulation of the system is possible, but use of the facility has shown that this is not required.

B.3.3. Water System

B.3.3.1. *Equipment*

The water system is generally similar to the air system, but is a little more complicated due to the water purification system. Tap water contains various impurities and chemical additives that affect ice properties, such as density and shear and cohesion strengths. Aerosolized water particles injected into the chamber by the nozzles can to stick to small impurities, forming larger droplets. Finally, the purification system removes contaminants that can clog the icing nozzles.

The water purification system has two stages. Tap water enters the equipment area of the lab and is initially treated by a commercial grade water softener. This conditions the water prior to it entering the Reverse Osmosis (RO) water filtration system. The RO system can clean water to 1 ppm contaminants at a rate of 1 gal/min. Due the flow rates anticipated with the rain experiments, a large storage system was implemented. A total of 920 gallons of purified water can be stored in three tanks, which provides about 20 minutes of run time while using the rain nozzle at design rain rates. The primary issue with the system is that the purified water is somewhat corrosive and requires either specialized plastic or stainless steel plumbing.

A small water draw pump is placed at the output of the tanks to move the water up to the hot water heater, located on top of the equipment loft. A control system helps maintain a

constant pressure input to the rest of the water system regardless of the tank level. The heater is essential to increase the water temperature so that it does not freeze in the nozzles. A computer controlled mixing valve allows the temperature to be modified from the control room while maintaining required flow rates. The heater would otherwise be a system choke point, reducing system flow rate.

The final step in the process is the variable rpm boost pump. This final stage in the water system before the manifold assembly increases the system pressure to required levels. A Labview controlled vector drive modulates the pump RPM as required to maintain nozzle pressures.

B.3.4. Manifold System

The air and water systems have separate manifolds to distribute air and water pressure to the nozzles. A series of ball valves control which combination of nozzles are active, including the primary NASA nozzle arms, impinging ice jet (IIJ) and rain nozzle. For most icing tests, only five nozzles are required to generate an even cloud of sufficient density. Extra air and water manifold positions allow for future system growth.

Table 26: Manifold Positions

Position	Water Manifold	Air Manifold
1	Ice Nozzle Inner Arm	Ice Nozzle Inner Arm
2	Ice Nozzle Outer Arm 1	Ice Nozzle Outer Arm 1
3	Ice Nozzle Outer Arm 2	Ice Nozzle Outer Arm 2
4	Impinging Ice Jet	Impinging Ice Jet
5	Vacuum System	Auxiliary 1
6	Auxiliary 2	Auxiliary 2
7	Air Input from Water Manifold	Air Output to Water Manifold
8	Rain Nozzle	Auxiliary 3

B.3.5. Nozzle Freezing Problem

An important part of the system is the connection between the air and water manifolds. The nozzles, mounted in the ceiling of the chamber, can easily freeze due to moisture in the water line. During normal system operation, this is not an issue because the warm water does not freeze in the nozzles. During breaks between tests, however, any bit of moisture left in the nozzles will freeze, rendering the them useless until they are taken out of their mounts and warmed up to melt the ice.

This problem is solved with a twofold process. At the conclusion of an icing run, the water input to the nozzles is turned off. When this occurs, a small vacuum system is turned on, sucking any additional water out of the nozzles. This process helps prevent excess water from dripping out of the nozzles and potentially damaging the ice shapes on the rotor.

The second part of the solution is a connection between the air and water manifolds. Even though the vacuum system removes most of the water from the nozzle, the few drops that remain can freeze. When the vacuum process is complete, a switch in the control panel operates an electric valve, which connects the air and water manifolds. Since the air system is still active, air is forced through the entire nozzle and blows all water out, preventing the nozzles from freezing. Prior to the subsequent testing, the air and water manifolds are disconnected, allowing them to operate independently.

B.3.6. Icing System Schematic

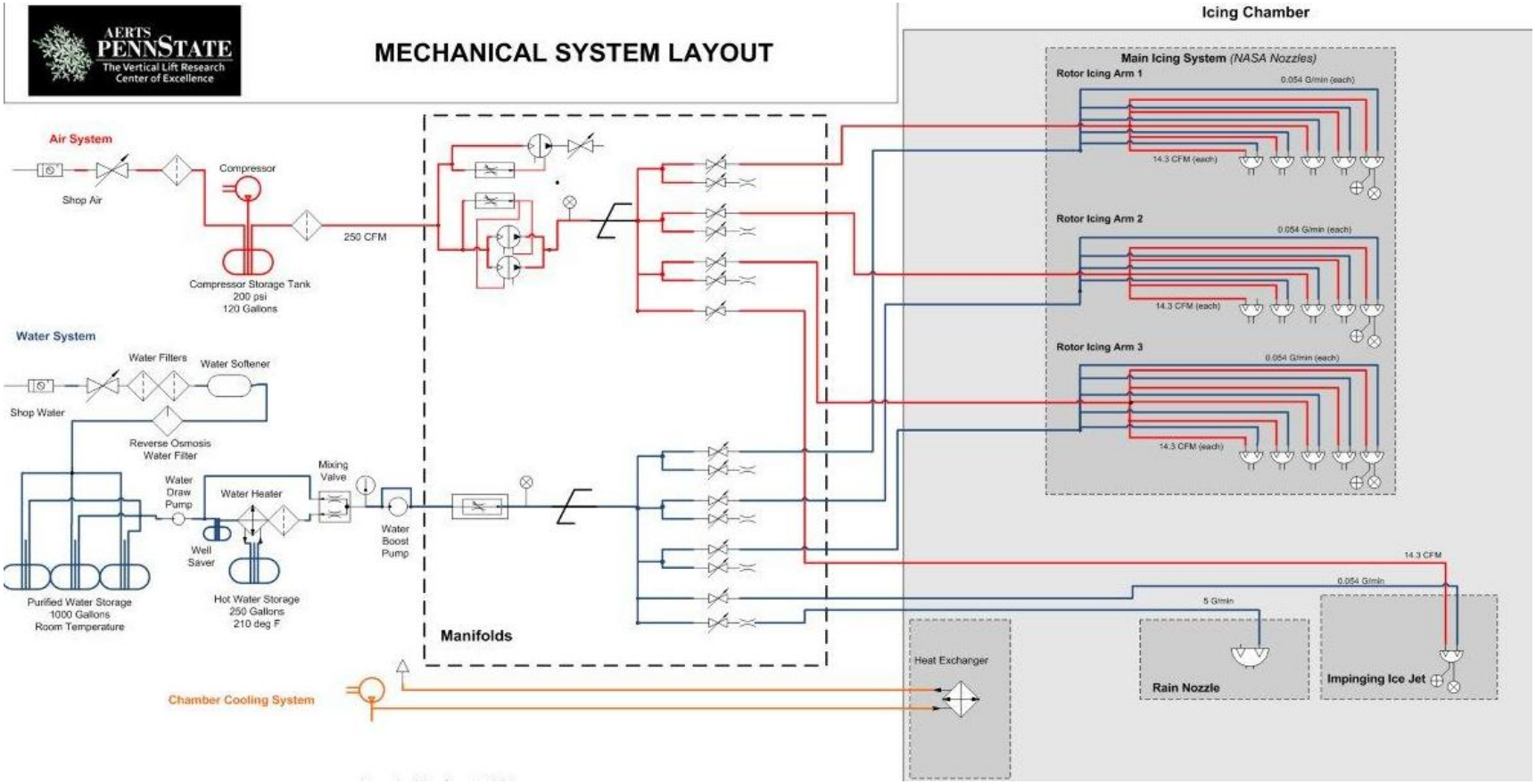


Figure 146: AERTS Water and Air System Layout

C. APPENDIX C: AERTS Drive System and Rotor Hub

C.1. Rotor Drive Components

C.1.1. Motor/Gearbox

The AERTS rotor is powered by a 120 HP water-cooled electric motor and gearbox mounted in a steel casting. The Boeing Company donated the system to Penn State in 1999. It was originally built in 1962 for use in the Boeing Philadelphia V/STOL wind tunnel for powered model tests of tilt rotors. As such, the motor and gearbox assembly can rotate 40 degrees within the confines of the stand support structure to test tilt rotor transition performance. For the AERTS application, the test stand is locked in a vertical position with steel brackets. Figure 147 shows the primary components of the stand.

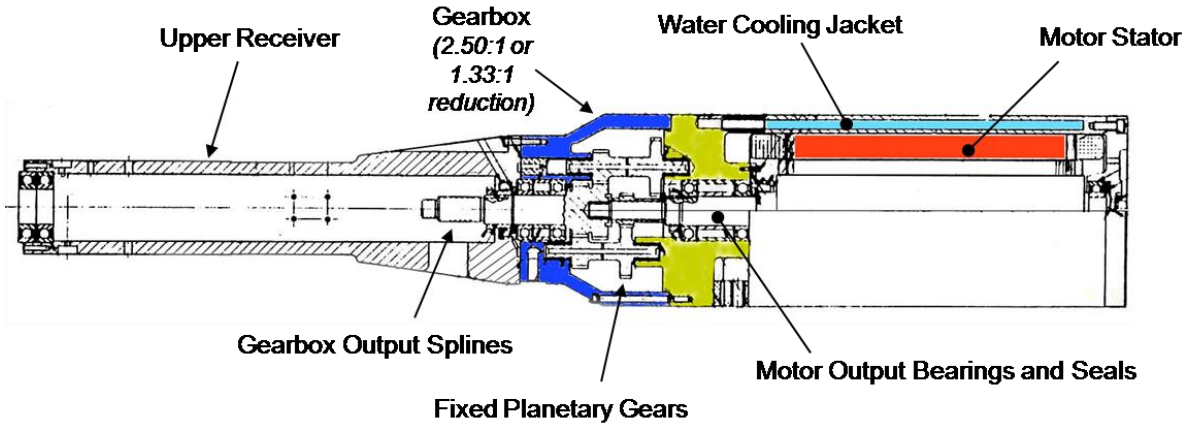


Figure 147: AERTS Drive System Components

Figure 148 details the test stand motor capabilities. With the current controller, the motor is capable of rotating a rotor at a maximum of 2000 rpm. Torque limits are often reached before this as indicated on the chart. High drag blades, such as bat rotors, are limited to 600 rpm.

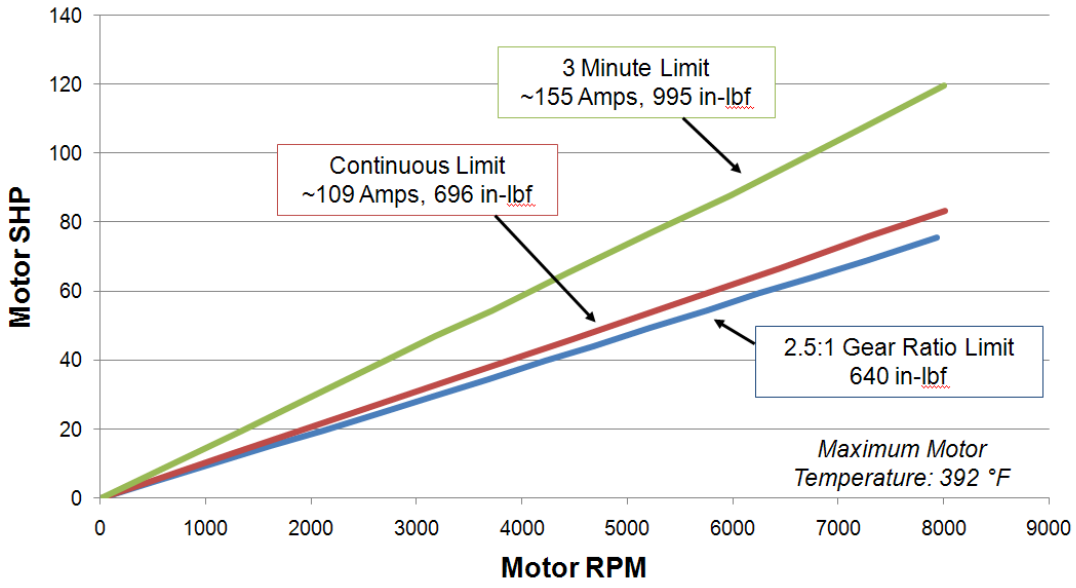


Figure 148: AERTS Motor Capabilities

C.1.2. Auxiliaries

Auxiliaries are installed in the control room to provide adequate cooling for the motor and gearbox. The gearbox is cooled by a constant 2 gal/min flow of Paradene 55 hydraulic fluid. A water cooled heat exchanger maintains the hydraulic fluid temperature below 100°F. The flow rate of the closed loop system needs to be carefully monitored to maintain proper fluid levels in the gearbox. Motor temperature is regulated with a 2-ton water chiller. The process fluid consists of 30% (by volume) of inhibited glycol to prevent it from freezing in motor cooling jacket.

C.2. Initial Testing Configuration

C.2.1. Hub

Initial calibration and testing of the AERTS chamber did not require collective and cyclic control for the rotor, so a relatively simple hub design could be used. An existing block hub, left over from previous testing was modified to increase centrifugal load limits and overall test safety.

C.2.2. Stand

As with the hub, existing components of the test stand were modified for AERTS initial testing. The primary concern was bending moments applied the rotorshaft, since the original Boeing hub components could not be used to properly support the shaft. A large stiff outer shaft was originally installed for previous testing, but improper mounting caused problems with the rotorshaft disengaging from the transmission. The outer shaft support, which mounted two taper roller bearings to support shaft bending loads, was modified to be rigidly attached to the rotorstand with new adapter parts. In addition bearing retainers were added as well as parts were designed to properly mount the slip ring and waterproof the entire assembly. These parts can be seen in the figure below.

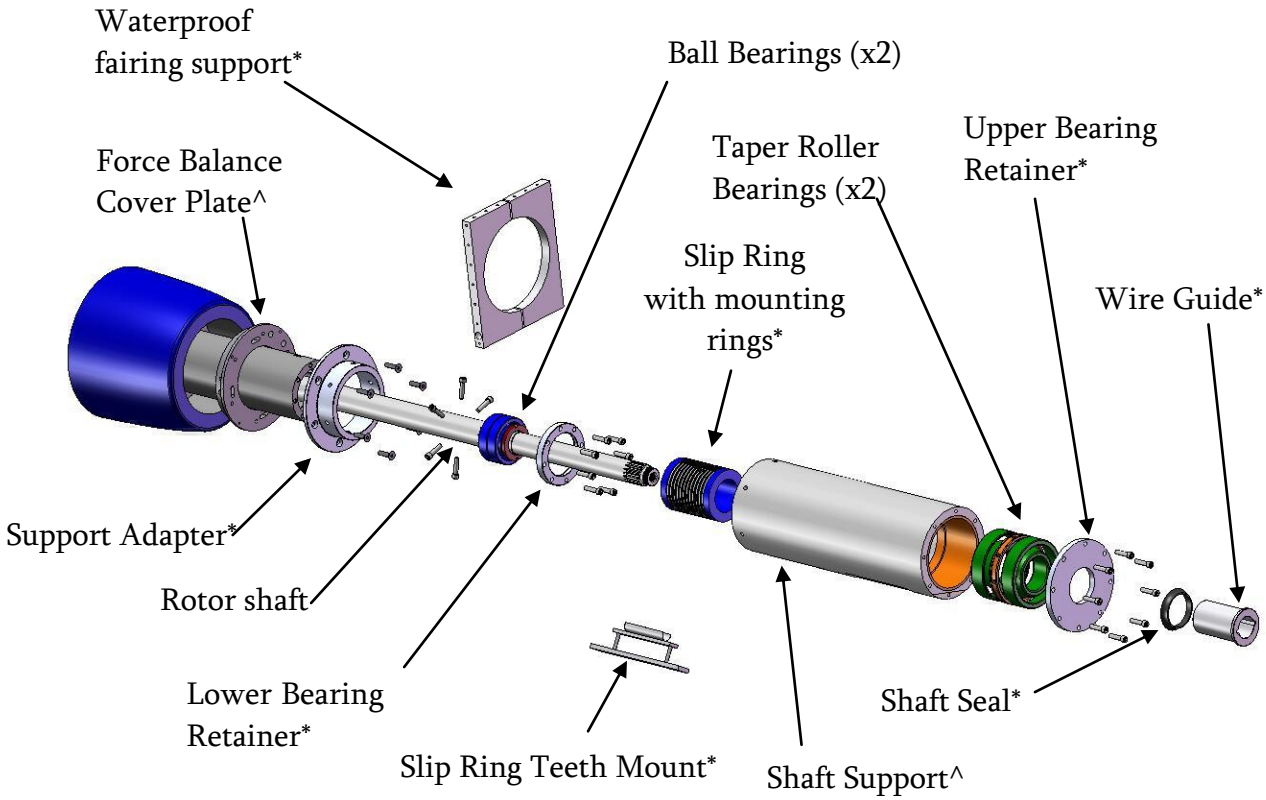


Figure 149: Rotor Stand Components (New Slip Ring Mounts not Shown)
* = New Parts, ^ Modified Part

C.3. New Rotorhead Requirements

The stand modifications worked well for initial AERTS testing requirements, but limited the application of the AERTS stand to more advanced projects. Test plans for various programs required a rotor hub capable of mounting various test blades, while retaining the ability to control the blades. An articulated, three bladed rotorhead was provided by Boeing with the test stand. The rotorhead was missing some components and did not have the capability of rotating large blades, so a larger hub was sought. Furthermore, the hydraulic actuators were not reliable or repeatable, so a new control system was required as well. The primary requirement for the new rotorhead was remote collective pitch capability. Due to the small

room size, cyclic pitch was considered as optional. Other parameters, shown below, were considered more guidelines than rote requirements, since the final test stand capabilities were going to be based around the rotor hub.

Table 27: Rotorhead Requirements

Parameter	Type	Range	Units
Number of Blades	Required	2	
Collective Pitch Capability	Required	-2 to 15	Degrees
Cyclic Pitch	Optional	Lateral: -5 to 5 Longitudinal: -5 to 5	Degrees
Type	Required	Teetering	
Gust Locks	Optional		
Blade Grip Diameter	Required	Max 2.5	ft
Rotor Weight	Required	5 – 25	lbs
Rotor RPM	Required	0 - 2000	

The final requirement for the new system was that its installation was to be completely reversible. As part of the risk reduction effort, no modifications were made to the existing structure. This allows for the reversion to the previous configuration at any time.

C.4. QH-50 Rotorhead Selection

Various avenues were consulted to find an appropriate rotorhead for the AERTS stand. Initially, a student designed assembly was to be used, but time, cost, and most importantly, safety concerns precluded this. Only basic design studies were completed before the focus was shifted to other options.

The second avenue pursued was using a production helicopter rotor procured from a salvage yard. Helicopters with teetering main rotors and large teetering tail rotors were identified. Aircraft junkyards were canvassed to locate a serviceable Bell 206 rotorhead. This rotorhead was selected because it was met most of the AERTS requirements as well as having large parts availability. Other Bell aircraft such as the Model 222 and 230 corporate helicopters and A/UH-1 military aircraft were considered as well, but availability was lower and the hubs were somewhat oversized for the AERTS application. Pricing and concern over parts safety were the primary hurdles with this approach. An example price was $2250 for a Bell 47 tail rotor trunion and grips (p/n 047-641-104-003) overhauled in 1974. This cost did not include other components such as a rotor shaft, swashplate, pitch links and torque scissors.

During conversations regarding locating an OH-58A/B/C rotorhead from the Corpus Christi Army Depot, the option of using a Gyrodyne QH-50D rotorhead surfaced. The QH-50D, shown in Figure 150, was designed as the first armed unmanned helicopter to operate from the decks of US Navy Destroyers (86). Production of the aircraft ceased in 1969 and the aircraft were used in active Navy service until 1970. The Japanese Maritime Self-Defense Force (JMSDF) used a number of aircraft before their retirement in 1976. Some units were used as target tugs and eventually targets the White Sands Missile Range. The aircraft were finally retired in 2001, a testament to their robust construction. The aircraft featured a coaxial rotor design, so the only the lower rotor system was used for the AERTS stand.

After discussions with Peter Papadakos, President of the Gyrodyne Helicopter Historical Foundation, it was decided to use the QH-50 rotor components. The foundation graciously donated low time rotorhead components, including an upper transmission "bell" housing, swashplate system, lower controls, upper rotorhub, and rotorshaft. For the AERTS chamber, an upper rotor trunion assembly is used with all of the rotor lower rotor controls to provide the correct rotation direction. Later, a second set of components and a set of blades were donated to the laboratory for use for spare parts.

Table 28: QH-50 Rotorhead Parameters

Parameter	Value	Units
Type	Teetering	-
N_b	2	-
Rotor Diameter	20	ft
Nominal Rotor RPM	610	-
Wooden Blade Weight (each)	52	lbs
Fiberglass Blade Weight (each)	28	lbs
Pitch Range	-2 to +18	deg



Figure 150: QH-50D DASH

The rotorhub has two additional features that are advantageous for use in the icing chamber. The first is the fact that the aircraft was designed for shipboard service. This environment required that the aircraft have extensive corrosion prevention applied to prolong the life of the vehicle. This corrosion prevention will be used to great advantage in the AERTS chamber, where tests will continuously expose the rotor components to moisture.

The second advantage of the QH-50 hub for application to the AERTS lab also comes from its original shipboard use. Gust locks were installed on the rotorhead to control teetering at low rpms and when parked on the deck. Gusts of wind from the sea and unsteady wake off the irregular ship superstructure across the destroyer flight deck could be potentially dangerous to the aircraft during engine startup, when centrifugal forces are too small to stabilize rotor motion. This blade sail was especially important for coaxial rotor systems, where rotors from the upper and lower rotors could impact, destroying the blades and some components of the vehicle. The QH-50 gust locks were designed to disengage at 400 rpm. For the AERTS applications, the gust locks help stabilize the blades at low RPM. The locks also prevent blade motion while taking ice measurements, which is especially important for the 3D laser scanning system.

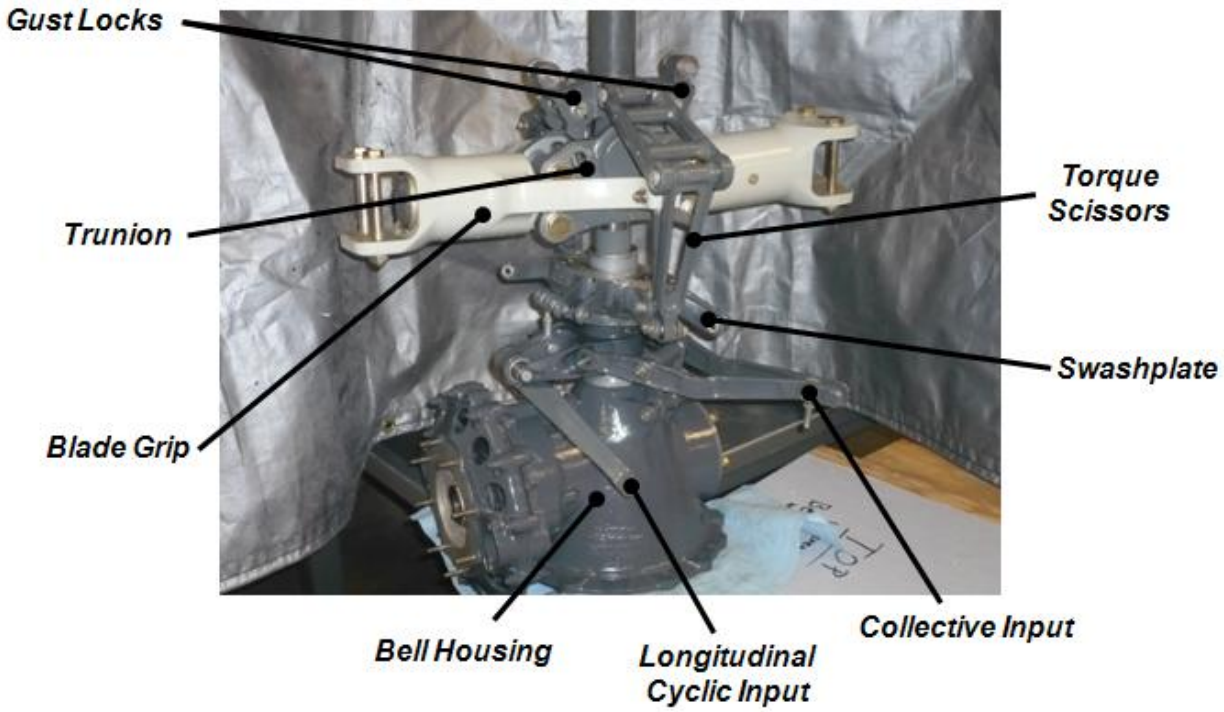


Figure 151: QH-50 Rotorhead Assembly as Received from Gyrodyne

C.5. Drivetrain Integration

Installing the new rotorhead required some significant changes to the test stand drive train. The QH-50 rotorshaft was both shorter in length and had a larger diameter than the original Boeing supplied shaft. A shaft assembly approximately 9" long was needed to connect the new rotor shaft to the existing gearbox output splines because no modifications were to be made to the test stand structure. In addition to extending the driveline, the new assembly was required to mount a rotating torque sensor and requisite bearings to locate all components. The original design, shown in the figures below, utilized 4 needle bearings in addition to the two torque sensor ball bearings. A taper roller bearing mounted in the bell housing located the upper portion of the rotor shaft. While the concept was sound, manufacturing tolerance stackup resulted in excessive runout and subsequent 1/rev shaft vibrations. Runout measurements exceeding 0.04" were recorded on the upper spline adapter.

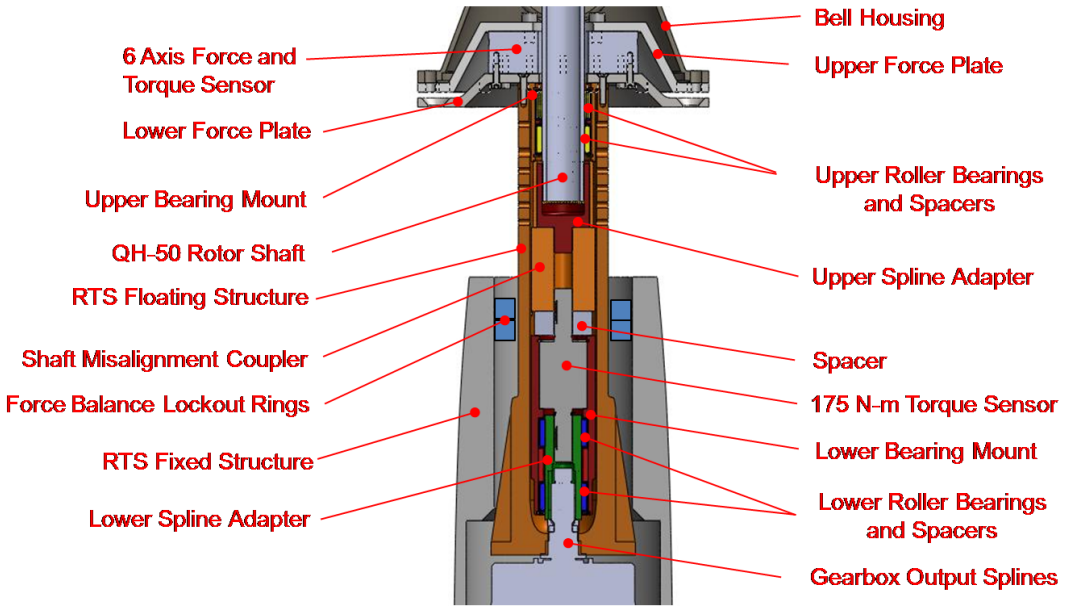


Figure 152: Lower Driveline Initial Component Arrangement

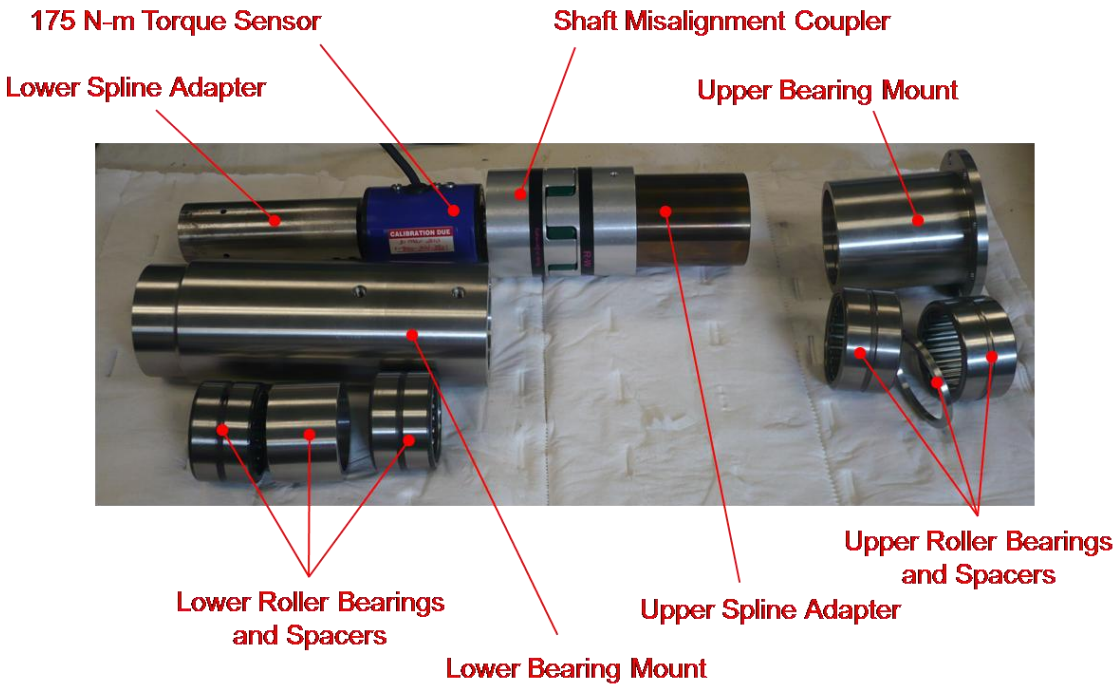


Figure 153: Lower Driveline Components

The solution to the runout problem was adding additional roller and thrust bearings to the assembly. Some parts were remanufactured to reduce assembly tolerance buildup. The entire assembly was shimmed as required and built up in stages to reduce runout. The shaft misalignment coupler was removed due to excessive motion. The final runout measurement was less than 0.005" at the upper spline adapter, resulting in the reduction of an order of the 1/rev vibration magnitude.

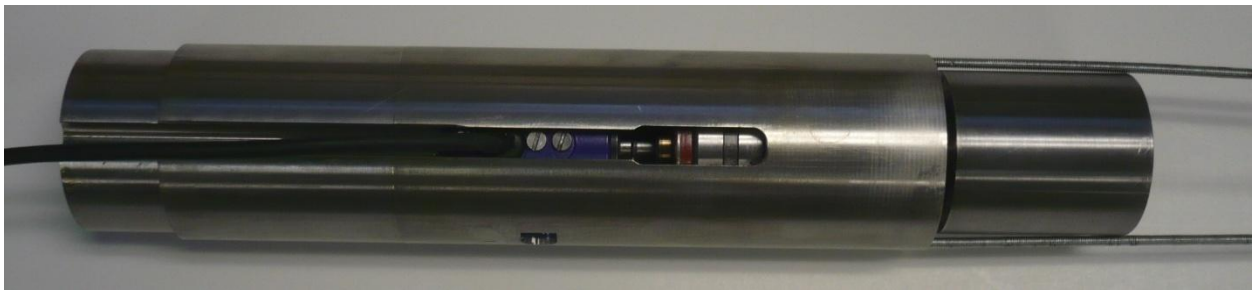


Figure 154: Revised Lower Driveline Assembly

A second issue encountered with the lower drive assembly. Originally, the parts were designed to stack up and locate the upper roller bearing mounted on the rotor shaft in its outer race located in the bell housing. The bearing is separated for ease of assembly and the two parts needed to be exactly located to properly center the shaft in the housing. Initial run in tests indicated that the two parts were not located correctly. A new bearing assembly consisting of a thrust bearing and adapter corrected this issue. In addition, the hollow rotor shaft was sealed at both ends and filled with transmission fluid. This fluid lubricates the two bearings to prolong their life. The fluid is kept warm by a heater mounted in the bell housing and an overflow tube is provided.

C.6. Controls Integration

A central requirement to the new rotorhead installation was the addition of collective pitch control. Using the original QH-50 servo system was judged to be too complex, as the vehicle used a single four axis system controlled by a system of vacuum tubes. Most of the aircraft electrical system would have to replicated, especially the 400 Hz AC power supply, and a new control system would be required to use the servos.

A requirements review was then completed for the linear actuators. Only the collective and lateral cyclic controls were needed because adding longitudinal cyclic was redundant for a hover stand. It was decided to significantly oversize the actuators to handle unanticipated

loads from the current rotor system and safely handle future projects. Safety was the primary concern, and this is reflected in a number of the requirements in the matrix below.

Table 29: Actuator Requirements Matrix

Requirement	Collective Actuator	Lateral Cyclic Actuator	Notes
Blocked Force	200 lbf	200 lbf	Base analysis included calculating the pitching moment of largest blade AERTS could safely rotate
Peak Force	1200 lbf	1200 lbf	6x blocked force
Extension Rate	0.25 in/sec	4 in/sec*	No particular requirement for collective, but oscillating icing experiments might be run in the lab at a later date so a fast actuator was desired
Stroke	9.2 inches	7.5 inches	Sized via control linkage and actuator placement analysis
Type	Acme Screw		Acme screw can support a load with the actuator unpowered. This is a safety feature that prevents unwanted motion if power is lost to the actuators.
Control System	Independent Controls		Safety feature to prevent unwanted actuator motion if Labview computer fails.
Potentiometers	Yes		Required so that Labview can read actuator position
Limit Switches	Yes		Safety feature to prevent over extension
Operating Environment	-30 °C to ambient temperature Actuator needs to be sealed		Actuators will be mounted inside of the test chamber

Based upon these requirements, actuators from Joyce Dayton were selected. For installation and maintenance simplicity, two versions of the same actuator type were procured. The collective actuator had a stroke of 12" and the collective actuator had a stroke of 8".

The longitudinal cyclic was locked out at the zero input position. A structure permits the vertical sliding of the swashplate assembly in response to collective input without any movement of the longitudinal cyclic. This structure can be seen in Figure 155.

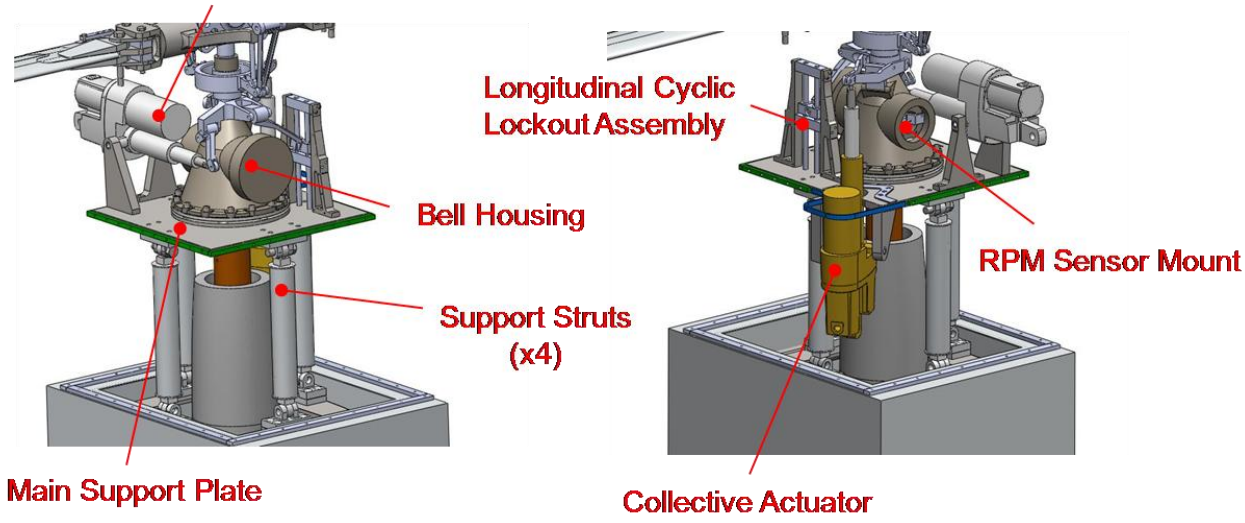


Figure 155: Actuator Implementation Models

C.7. Structural Integration

To support all of the new rotor components, a new mounting plate and struts was designed. This 0.75" thick 304 stainless steel plate is mounted to the formerly floating RTS inner structure. Four adjustable legs connect the plate to the large fixed steel casting to stiffen the structure. The legs are constructed from 3" diameter 6061 aluminum tubes with 304 stainless steel plug inserts at each end with adjustable ball joint rod ends. The struts are connected to the main plate and fixed structure via 304 stainless steel clevises. The entire structure is overbuilt for safety. Each leg assembly was analyzed with Solidworks Simulation finite element analysis (FEA) software and can support 2000 lbf with a significant Factor of Safety.

The actuators are mounted to the with 1" thick 304 stainless steel brackets. Bushings allow them to pivot as required due to the geometry of the QH-50 control system. The struts and the area between the support plate and rotor stand is covered with a waterproof fairing as shown in Figure 156.

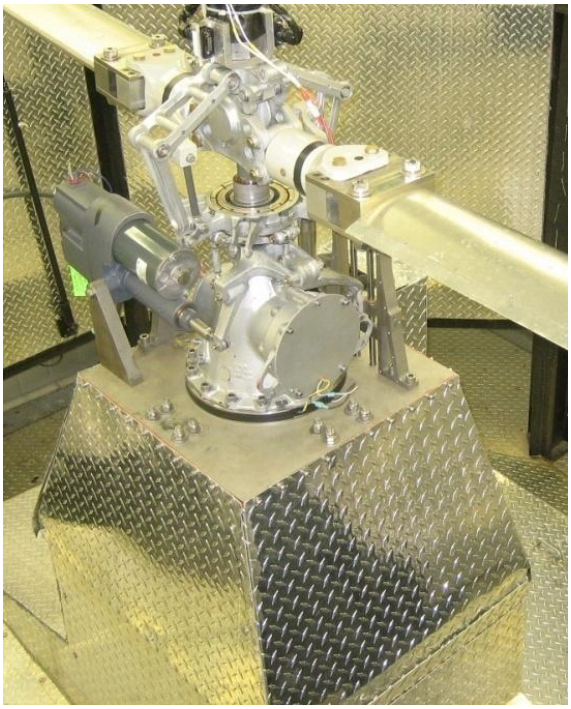


Figure 156: Waterproof Fairing

C.8. Slip Ring Assembly

The AERTS requires a number of slip ring channels to fulfill its mission as a research test stand testing various ice protection systems and monitor on blade sensors. The original Boeing Rotor Test Stand featured a 50-channel signal slip ring built into the rotorshaft. The system was unusable because the leads were significantly damaged and the rings could not carry the high voltages associated with ultrasonic ice protection systems and high currents

associated with electrothermal systems. Therefore, a new ring was sought with the following specifications.

Table 30: Primary Slip Ring Specifications

Parameter	Ideal	Acceptable	Rationale
Signal Channel Count	24	12	Need to monitor 2 full bridge strain gauges, 2 thermocouples, plus appropriate shielding
Signal Channel Voltage Rating	250 V	100 V	
Signal Channel Current Rating	5 A	2 A	
Power Channel Count	24	12	Need to run a 6 zone electrothermal ice protection system
Power Channel Voltage Rating	800 V	500 V	Need to run d_{15} based ultrasonic ice protection system
Power Channel Current Rating	20 A	10 A	Need to run a 6 inch span 27 W/in^2 electrothermal system
RPM	2000	1500	Do not want slip rings to limit testing RPM

Early in the procurement process, it was obvious that no new slip ring would fit in the lower shaft area. Also, the channel requirements could not be met with a single ring without excessive cost. After some studies, the only possible location for the rings was suspending them from the ceiling, above the rotor. An interlaced assembly of multiple slip rings was also required.

A search of available slip rings led to the selection of Fabricast Type 29824 rings. The company was able to modify a standard model to allow it to operate at 2000 rpm. Three rings were procured: two 12 channel high powered rings and a single 24 channel signal slip

ring. The signal ring has four silver graphite brushes and solid coin silver rings to reduce electronic noise. The power slip rings have every other channel replaced by delrin spacers to increase voltage limits. The differences between the two rings can be seen in Figure 157.

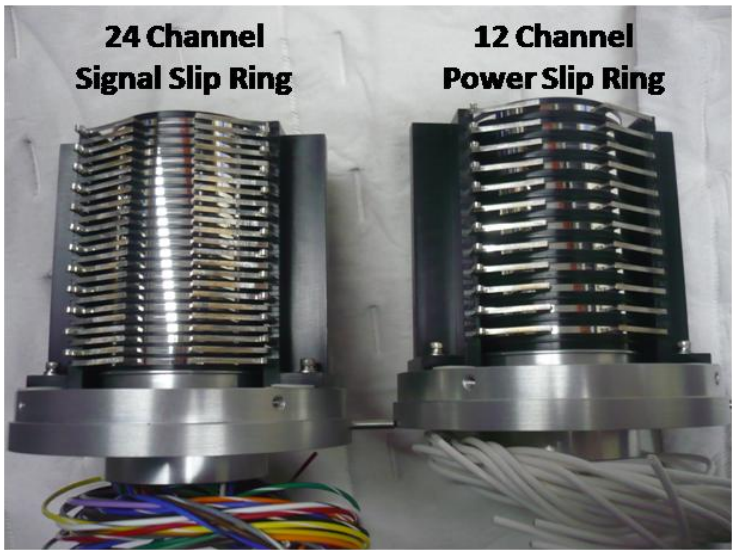


Figure 157: Slip Rings

The slip rings are daisy chained together on a common shaft. This requires a specific wiring system to keep the rotating and fixed frames separated. As shown in Figure 158, fixed frame wiring enters the top of the assembly and is wired into each slip ring. Signals are passed to the rotating frame, which is in the central rotating pipe. A bracket at the bottom of the assembly mounts the waterproof connectors.

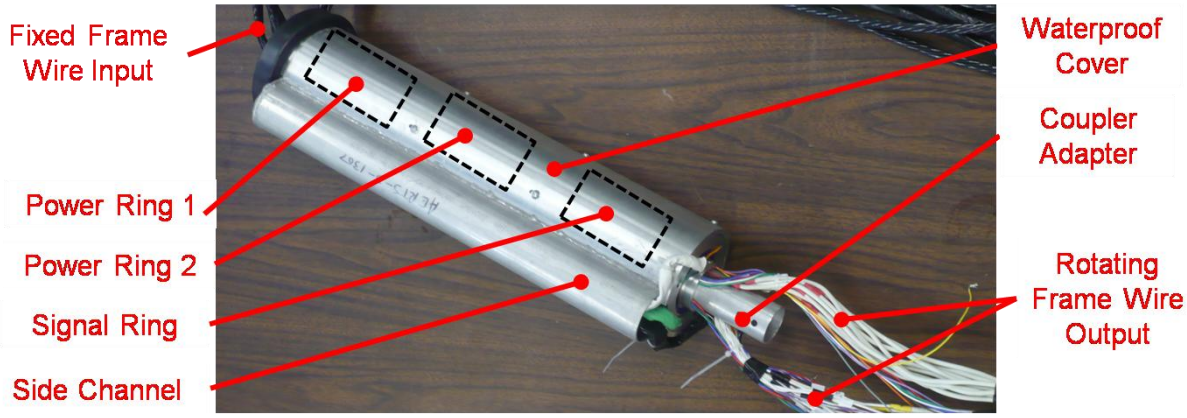


Figure 158: Slip Ring Configuration

These rings are mounted in a waterproof assembly suspended from the ceiling. The original QH-50 rotorshaft was shortened and the "Jesus" nut was relocated. In addition to locking the rotorhead in place, the nut is the adapter between the rotorshaft and slip ring assembly, keeping them rotating together. The assembly is dynamically isolated from the rotor system with the use of a flexible pipe fitting, normally used in ceiling expansion joints. Two Victaulic couplers provide easy installation and removal of the assembly, while preventing water from entering the system. Testing showed that a second isolation system was required, so a bellows shaft coupling was added just above the rotorhead. The combination of the two couplings relieves the slip ring bearings from excessive load and allows for both parallel and angular misalignments between the slip ring assembly and rotorshaft. Mounts for 4 waterproof DB-15 connectors are provided on the rotorhead.

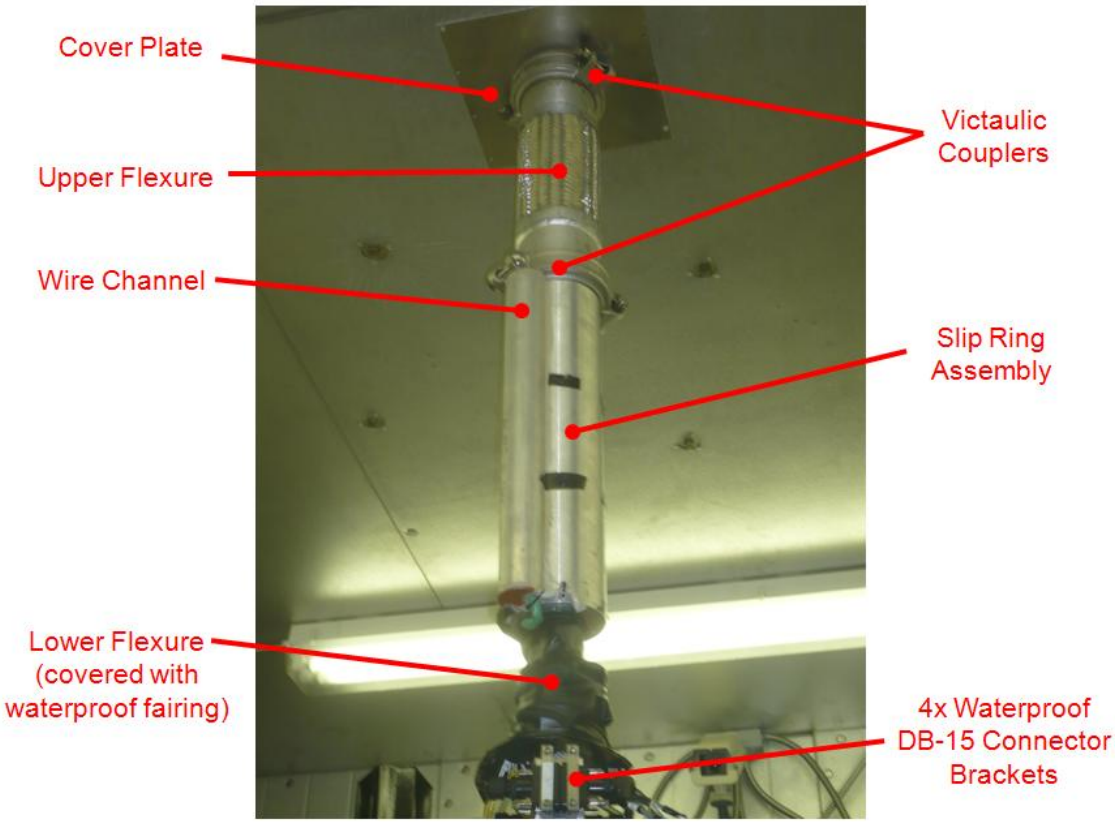


Figure 159: Slip Ring Installation in AERTS Chamber

C.9. Rotor Sensors

As part of the new rotor head installation, explained in Section C, additional sensors were installed in the facility to monitor rotor operation. These sensors are needed to ensure safe operation of the facility as well as to provide measurements of rotor performance degradation due to ice accretion.

C.9.1. Six Axis Force and Torque Sensor

An important part of the new rotorhead installation was the addition of a new force balance. For the AERTS application, the balance was primarily intended to monitor the rotor for

safety purposes and not for explicit rotor performance testing due to the small size of the test chamber. The primary performance parameter monitored by the facility is shaft torque.

Two different options were explored when designing the force balance. The first was a reconditioning of the old Boeing balance. The balance was formed by 4 sensing rigs and an integrated thrust/torque sensor. The entire motor, gearbox and rotor support system were part of the active side of the force balance, as shown in the Figure 160.

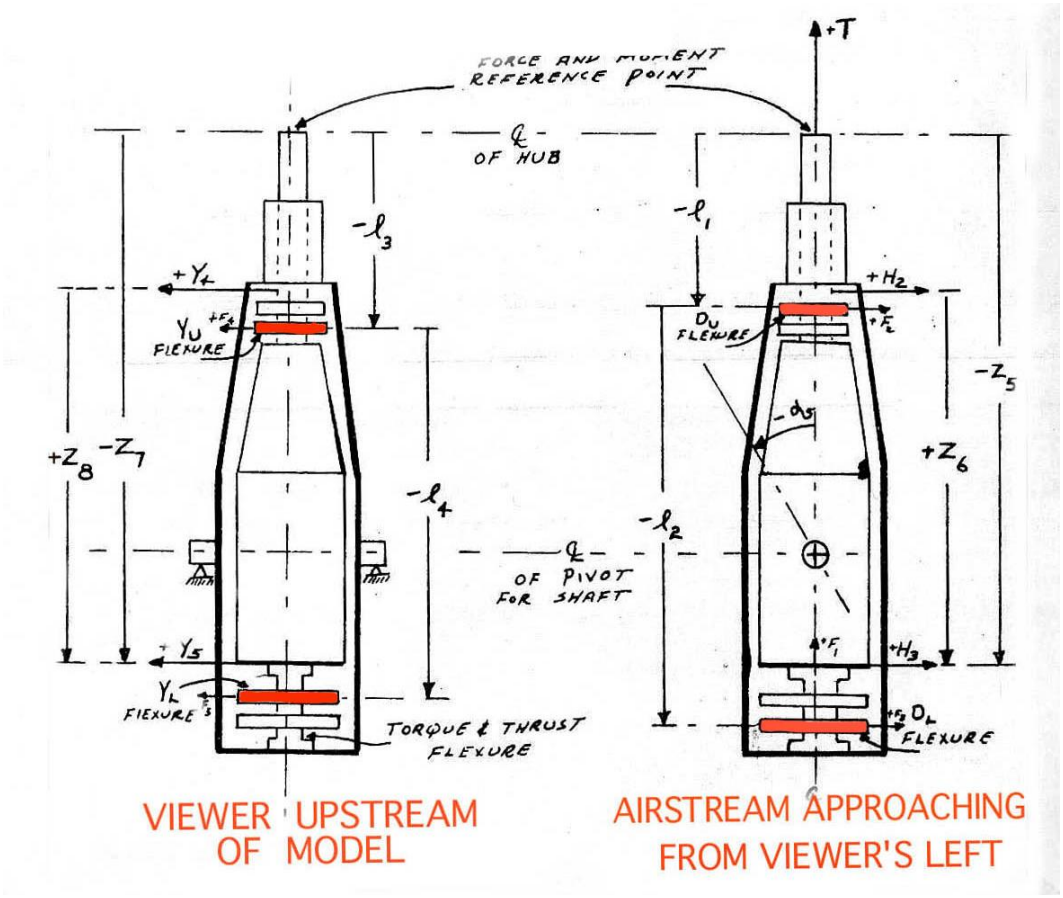


Figure 160: Previous Boeing RTS Force Balance

After some exploration, this option was rejected due of the lack of documentation on the design and wiring. The strain gauges were also not thermally compensated, which is a vital requirement for the icing facility. Finally, the integrated thrust/torque sensor was unserviceable and a replacement could not be procured. To improve stand rigidity, the old balance was locked out by removing some of the sensing rings and replacing them with solid stainless steel plates.

The second option was designing a new system. Ideally, an intershaft system would be used, but this system was cost prohibitive. Implementing a system of load cells and calibrating the system was judged to be too complex. Therefore, the new force balance was designed around a 6-axis force and torque sensor. A custom Omega 160 sensor from ATI Industrial Automation was mounted in the base of the bell housing. The sensor, originally intended for industrial robots, is capable of resolving three forces and three moments with high precision.

Table 31: Force Balance Sensing Range and Resolution

Force/Torque	Description	Sensing Range	Resolution
Fx	Lateral Force	300 (lbf)	±5/68 (lbf)
Fy	Longitudinal Force	300 (lbf)	±5/68 (lbf)
Fz	Thrust	875 (lbf)	±5/34 (lbf)
Tx	Pitching Moment	1800 (lbf-in)	±5/16 (lbf-in)
Ty	Rolling Moment	1800 (lbf-in)	±5/16 (lbf-in)
Tz	Yaw Moment	1800 (lbf-in)	±5/16 (lbf-in)

Due to the size of the Omega 160, it could only be mounted in the bottom of the bell housing without requiring significant modifications to the test stand. The rotor shaft runs through a central cavity in the sensor. Special mounting plates transmit forces to the sensor from the

rotorhead via the taper rolling bearing and bell housing. A small Faraday cage screen was installed to reduce electromagnetic interference from the motor while not mechanically interfering with the sensor. Heaters are mounted in the bell housing to keep it within a calibrated temperature range. The central location of the sensor allows it to read all forces from the rotor. However, there is some mechanical coupling as both the rotor shaft and actuators connect the "active" components to the fixed components.

C.9.2. Shaft Torque Sensor

As part of the new rotorhead installation, a shaft torque sensor was added inside the lower stand in the drive line. The sensor is a key part in the adapter assembly between the reduction gearbox and the QH-50 rotor hub. The Cooper LXT 971 was selected primarily due to its small form factor, so it could be integrated in the lower portion of the test stand. Also, the sensor was capable of operating in the harsh environments typical of the AERTS test stand. The model procured is capable of reading a maximum of 175 N-m (129 ft-lbf) of torque with a resolution of ±1%, which spans the operational capability of the AERTS motor.

C.9.3. Stand Accelerometers

Two accelerometers were installed onto the stand to monitor oscillatory motions of the test stand. The sensors were required to increase test stand safety to ensure modes were not encountered. One accelerometer was placed on the lower rotor stand to monitor oscillations

of the support structure. This was important to monitor because the test stand was originally developed to test tilt rotor vehicles in the Boeing V/STOL wind tunnel. As such, the stand can tilt around an axis up to 40° to allow transitional flight performance to be monitored. For the AERTS application, the stand was locked in a vertical position and additional support legs were added. This additional structure modifies the stand properties and must be monitored.

The second accelerometer was installed on the bell housing, above the force and torque sensor, which has a lower stiffness than the rest of the test stand. This placement allows it to monitor the dynamics of the upper component. Both accelerometers are installed in the same plane, so comparisons can be made between them to determine the source of vibration issues. This property of the accelerometers was used to determine the location of a 1/rev force in the lower rotor stand.

As with other sensors, electromagnetic noise was an issue. To reduce noise to a minimum, proper coaxial cables connected the accelerometers directly to a MEPTS-9000 signal conditioner and amplifier.

C.9.4. Rotor RPM Sensor

In order to monitor the rotor shaft rpm, a Honeywell 3050A13 VRS passive rpm sensor was installed in the upper housing. The sensor is similar to a standard Hall Effect rpm sensor in

the fact that it uses the magnetic field of a ferrous target, such as a gear tooth, passing in front of its sensing face to generate a pulsed signal. The Honeywell sensor, however, is designed for operation in environments with high electronic noise. It achieves this by operating in a similar fashion to an electric generator and creates an electric current which is less susceptible to electronic noise than the voltage output of a Hall Effect sensor. As the rpm varies, the output varies in amplitude and shape. Labview programming reduces this to a reliable measurement of rpm. The sensor did not require an extensive calibration. Only a functional check with the flash gun was required to ensure that the sensor was operating properly.

D. APPENDIX D: Test Rotor Design

D.1. ARISP Test Rotor

D.1.1. Requirements

In order to properly validate the ARISP model, a test rotor was required. The rotor has some unique requirements, detailed in Table 32. Ideally, the blades would be commercial off the shelf (COTS), and would not require a new design.

Table 32: Validation Rotor Requirements

Parameter	Range	Justification
Radius	3 – 4 ft	Need to fit rotor in the AERTS Facility with suitable wall clearance, while providing enough blade length to perform ice measurements
Chord	< 8”	Need small chord for its increased ice catch efficiency to reduce test times
Tip Speed	0 – 750 ft/sec	Need to cover operational tip speed envelope of modern rotorcraft.
Rotor Erosion Cap Material	Aluminum, Stainless Steel, Titanium	Need isotropic material to compare shear adhesion forces calculated in AERTS facility to published data
Airfoil	NACA0012, NACA0015	Need generic airfoil to compare ice shapes and performance to published data
Root End Fitting	Vertical Pins	Need to be easily adaptable to the QH-50 rotor hub

D.1.2. Rotor Selection

Based upon these requirements, a survey of production rotorcraft rotors was undertaken. Tail rotors were of particular interest because they typically operate at the high RPMs inherent to the AERTS chamber and have a small size. The final selection, however, was primarily driven by rotor availability.

Schweizer Aircraft corporation graciously donated a set of cut Schweizer 269C main rotor blades. The blades were retired from operational use because they were close to exceeding their operational limit of 5500 flight hours, but were in good condition. The specifications of the blades are given in Table 33. Schweizer cut the blades to form a 7.75' diameter main rotor for testing in the AERTS chamber. This diameter was chosen to reduce chamber aerodynamic effects while providing enough area to measure ice accretions.

Table 33: Schweizer Main Rotor Specifications

Parameter	Schweizer 269C Application	PSU AERTS Application	Units
Radius	13.42	3.87	ft
Chord	6.8		in
Solidity	0.040	0.092	-
Tip Speed	662	200 – 608	ft/sec
RPM	471	500 - 1500	-
Twist	-8.7	-2.1	degrees
Airfoil	NACA 0015		-
LE Material	2024-T3 Aluminum		-
Weight	~1.33		lb/ft

The primary benefit of the Schweizer blades is the aluminum construction and NACA 0015 airfoils. These allow results in the AERTS chamber to be published because of the non-proprietary nature of the two entities.

D.1.3. Blade Integration

D.1.3.1. *Blade Root Stress*

The most important requirement for using the Schweizer blades was safety. Much effort was made to ensure that the blades would be properly mounted to reduce the risk of any safety related accidents. Initial stress analyses focused on the blades themselves. Based upon

centrifugal force calculations and conversations with the blade manufacturer, the blades could be safely rotated up to 1500 RPM (608 ft/sec) with significant margin. This was considered to be the upper limit of the operational envelope, but most testing would not exceed 1000 RPM. Due to the nature of the AERTS chamber, a limit of 5° of collective pitch was applied, so thrust forces and resultant moments could be neglected in first order analyses. The root stress of approximately 9600 lbf was well within the QH-50 hub design loads, which were determined to be 17,000 lbf based upon a review of the original QH-50 wooden blade design and analysis report and internal calculations. This report was provided by Peter Papadokos, president of the Gyrodyne Helicopter Historical Foundation.

D.1.3.2. *Dynamic Analysis*

The Schweizer 269 aircraft was used as a primary trainer. In their original application, the blades were designed to provide pilots with simple handling characteristics. Due to the significant shortening of the blades for the AERTS application, a dynamics analysis was conducted to locate flapping, torsional and lag natural frequencies of the blades. A simple flapping finite element analysis was undertaken, with the flapping frequencies plotted in Figure 161. As expected, shortening of the blades increased flapping and stiffness. All required blade properties were calculated in Solidworks.

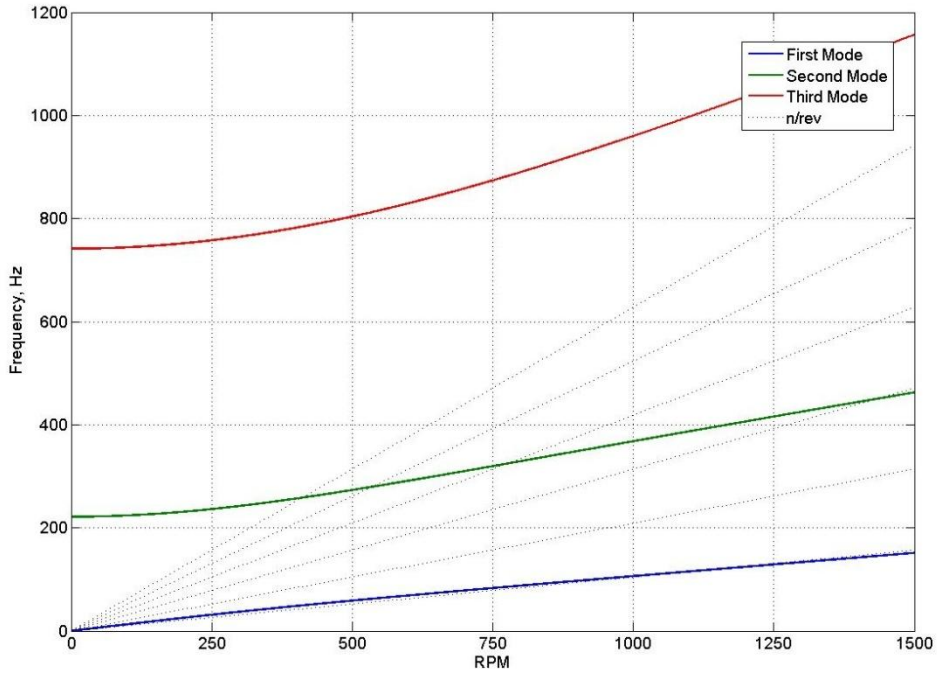


Figure 161: Flapping Response of Schweizer Blades

Finite element studies were performed in Solidworks Simulation package, focusing on the lead-lag and torsional motions of the blade. For all calculations, the blade was fixed at the root end. This approximated the final mounting configuration and assumed that the hub trunion and blade grip assembly was rigid. The calculation results, provided in Figure 162 and Figure 163, show that the resonance frequencies of the shortened blades were far from the n/rev operational frequencies. Also, little changes in the resonance frequencies did not vary at significantly with RPM.

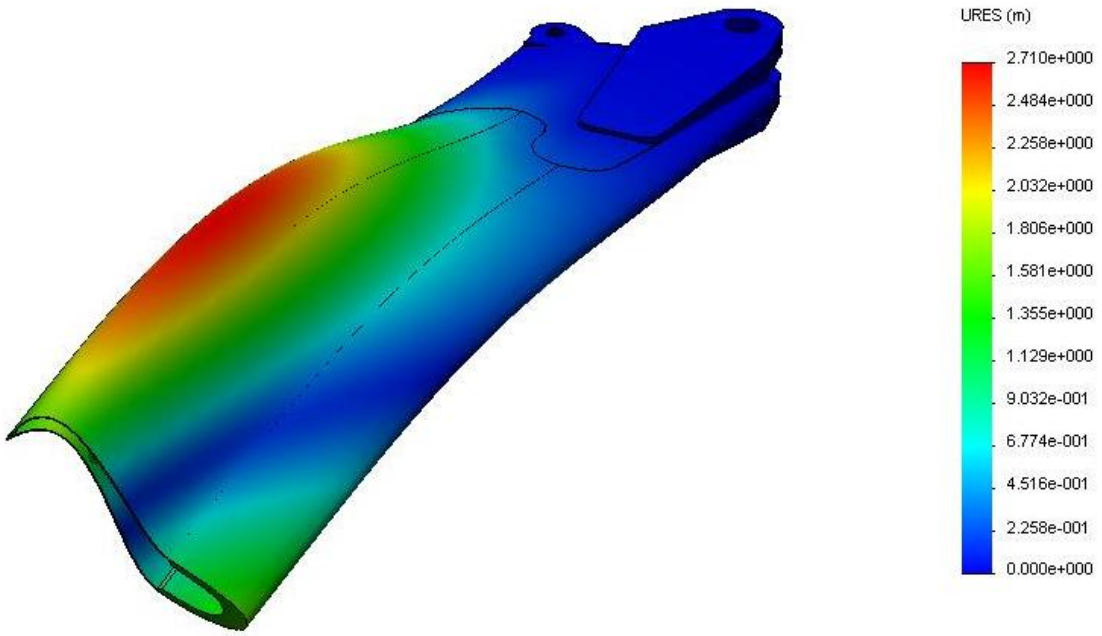


Figure 162: Sample FEA Analysis of Modified Schweizer Rotor
600 RPM, Torsional Mode (255.85 Hz), Cantilevered Rotor Blade

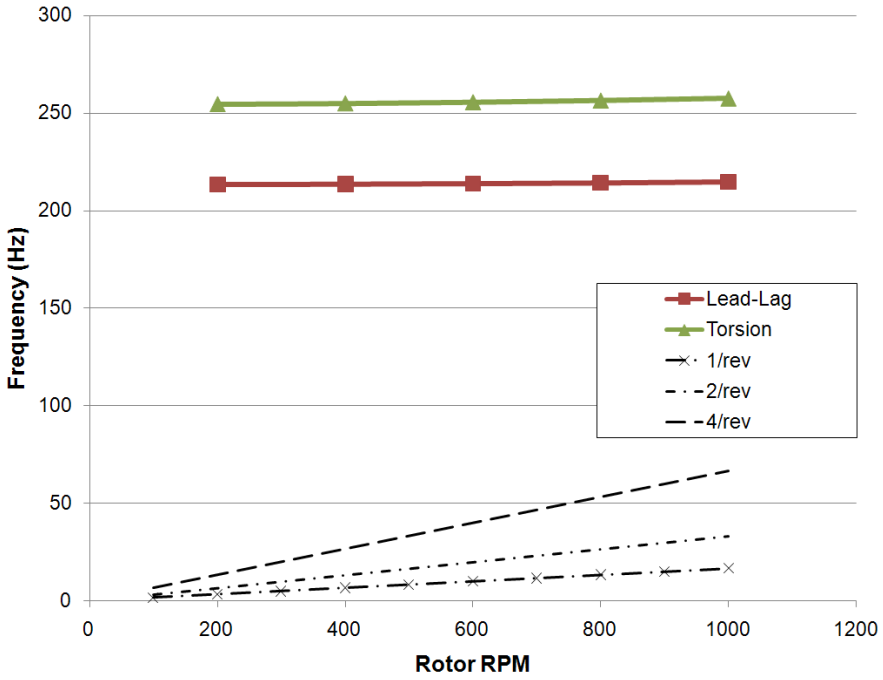


Figure 163: FEA Analysis of Modified Schweizer Rotor Results

A detailed analysis of pitch divergence and flutter stability was not required because the blade cross sectional center of gravity is positioned forward of the aerodynamic center as shown in Figure 164. Based upon the work of Johnson, this precludes potential instabilities (87).

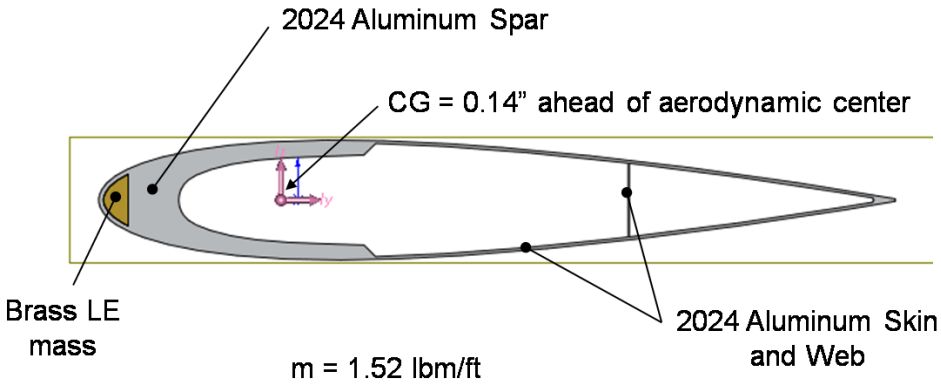


Figure 164: Blade Cross Sectional Properties

D.1.3.3. *Mounting System*

Once the blades were judged to be safe to operate within a limited testing envelope in the AERTS facility, a mounting system was designed. The Schweizer blades featured a single vertical pin mounting scheme, and therefore needed adapter plates to be mount on dual vertical pin QH-50 hub.

The plate assembly, shown in Figure 165, was designed to mount the blades with the ¼ chord on the pitch axis of the rotor hub. Three plates are used to transfer loads from the blade retention bolt to the QH-50 hub. The Schweizer supplied retention bolts were retained for the AERTS application due to their strength. The material for the adapter plates

was donated by Olympic Tool and Machine Corporation and machining time and heat treating was donated by Folsom Tool and Mold Corporation. The plates were machined from 15-5 PH Stainless Steel and then heat treated to 34-38 Rockwell C scale. This increased the plate tensile strength to 250 ksi, improving test safety. This provides a tensile strength of approximately 220 ksi, with a measure of corrosion resistance. For the final installation, 0.005" thick shims waterjetted from 316 stainless steel plates to ensure a proper fit in the blade grip.

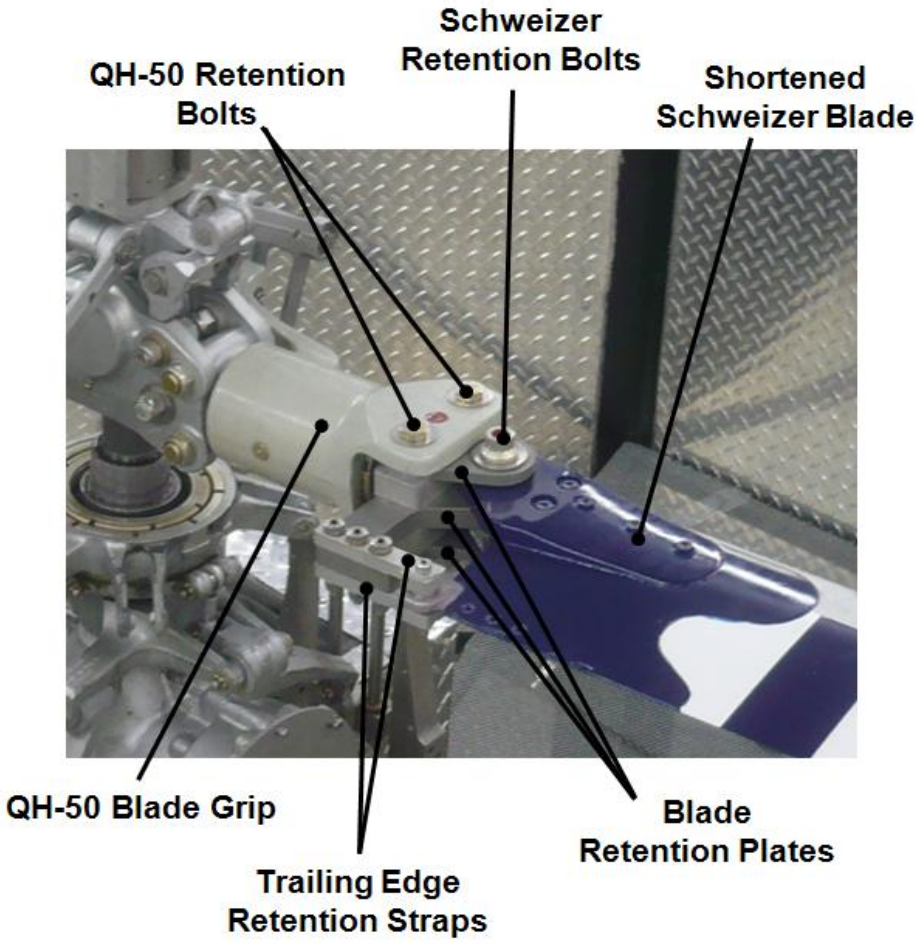


Figure 165: Schweizer Blade Mounting Plates

Since the 269 blades were designed for a three bladed articulated rotorhead, a large lag damper was provided at the trailing edge. A fixed trailing edge grip was designed to mount the blades on the teetering QH-50 hub, which has no provisions for a lag damper. Due to the blade construction, centrifugal loads at the lag damper are less than 10% of those at the primary root fitting. The exact load carrying capability of the internal structure is not known, but detailed conversations with the blade manufacturer indicated that the arrangement was acceptable, especially for a limited number of research runs.

Stress analyses were accomplished with Solidworks Simulation software. A mesh with 462,680 nodes and 301,874 elements was used. The blade grip adapter, shown in gold in the following figures, was assumed to be rigid.

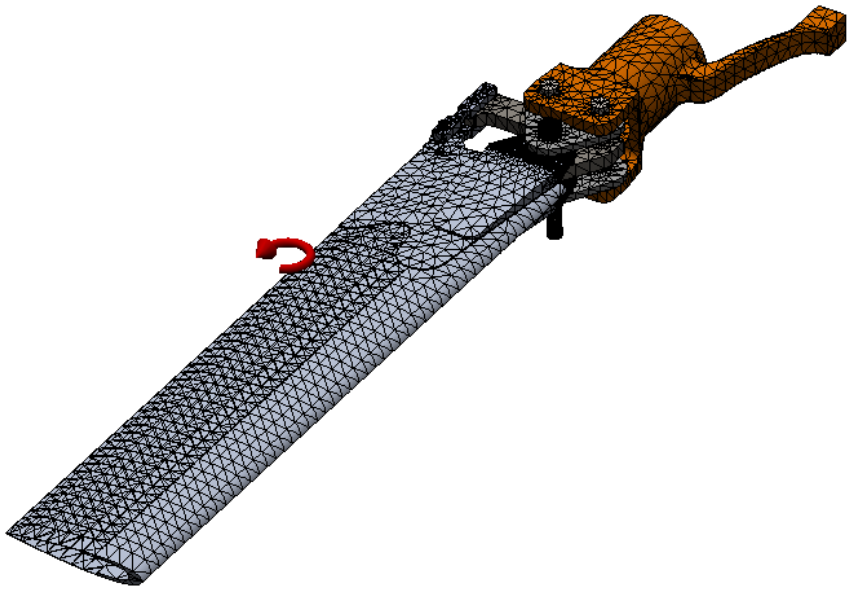


Figure 166: FEM Mesh

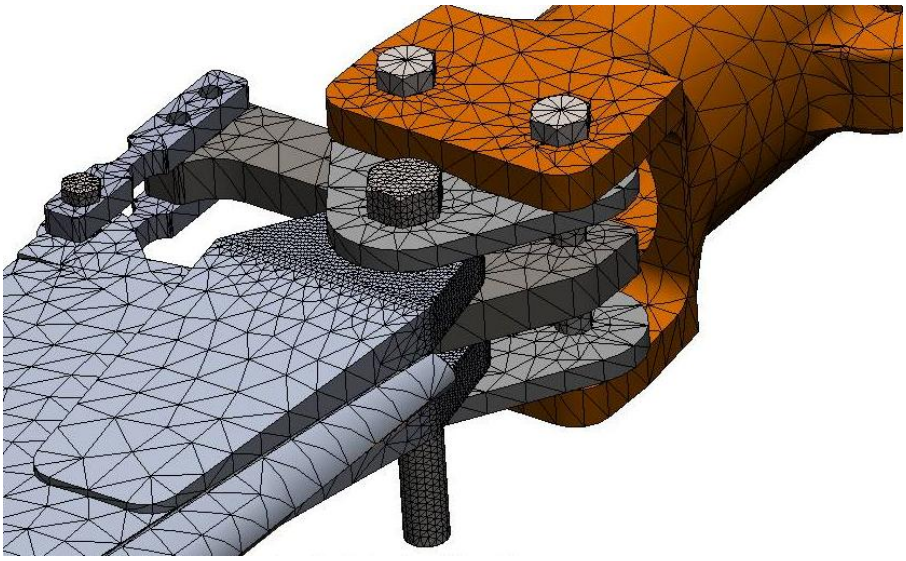


Figure 167: FEM Mesh, Root Closeup

Some results are below in Figure 168 for a test case of 1000 RPM, the maximum RPM for the present research.

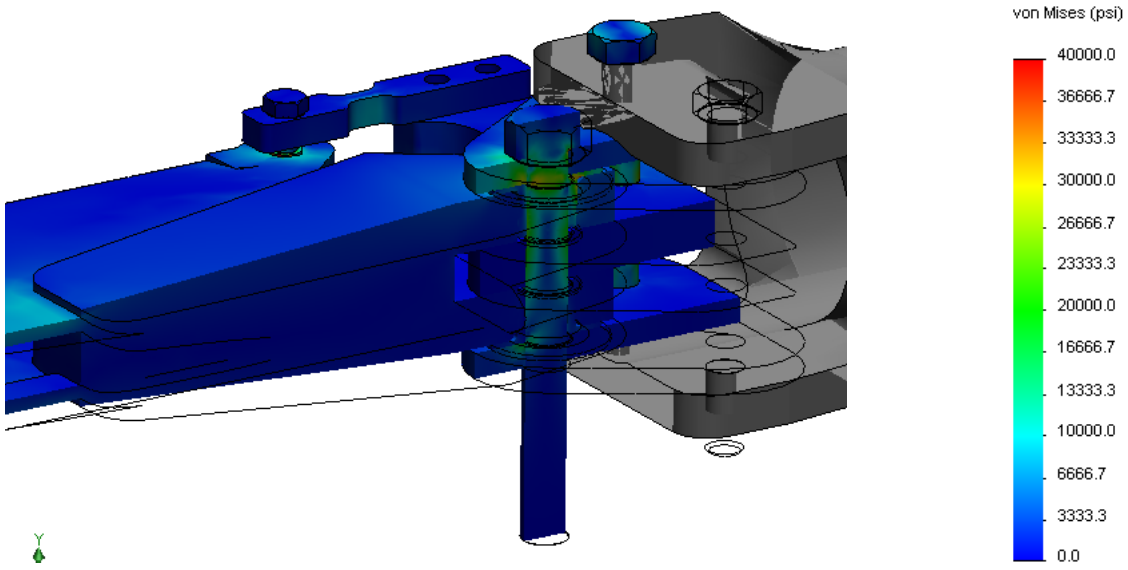


Figure 168: Stress Distribution in Schweizer 269 Blade Assembly

As expected, the highest concentration of loads was contained in the primary retention bolt. The loads, however, were within the safety limits of the system, as shown by Figure 169.

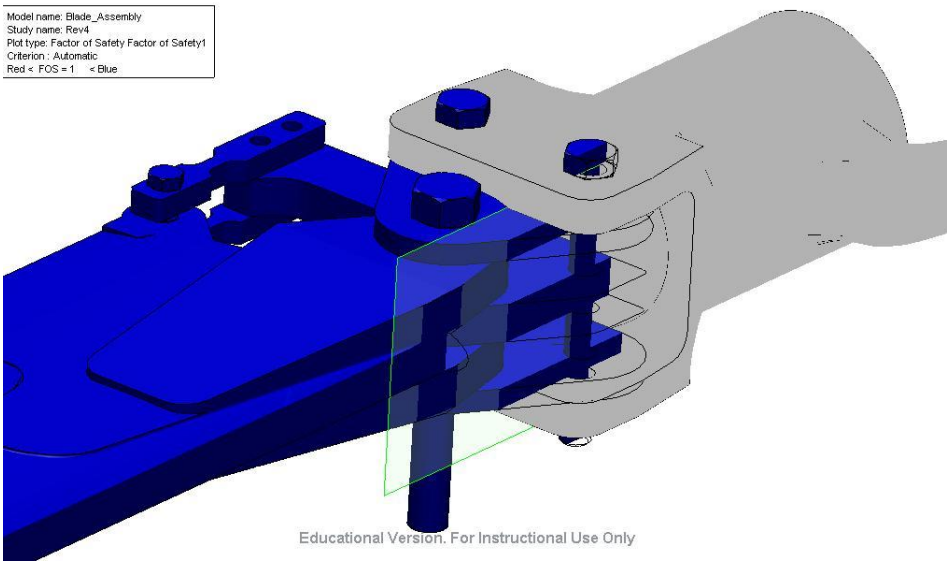


Figure 169: FOS Distribution in Schweizer 269 Blade Assembly
Red = FOS < 1, Blue = FOS > 1

D.1.4. Blade Paint

As shown in Figure 170, the blades were painted in blue and white stripes. Each stripe is 2.33" wide, which corresponds to 5% of the blade radius. The stripes are laid out such that they are a convenient reference points at which to measure ice thicknesses. At the tip, an additional 0.025r wide stripe is added, to identify each blade. Blade 1 has a black stripe, while Blade 2 features a silver stripe.

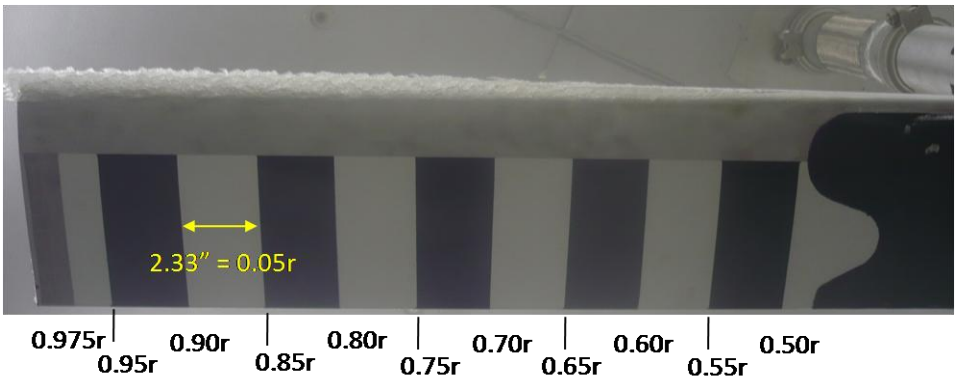


Figure 170: Blade Stripes

D.1.5. Tip Caps

When the test blades were cut from a full length Schweizer 269C rotor, an exposed open tips was created. The standard blade tip cap could not be used because it was riveted in place and featured integral tip masses. Therefore, a simple solution was implemented to prevent water from entering the blade and creating mass imbalance issues. Standard aluminum tape was installed over the open ends of the blades. The tape was lightweight, so hub loading was not a concern. Issues with the tape debonding during shedding events were resolved with improved installation procedures. The tape can be seen in Figure 171.

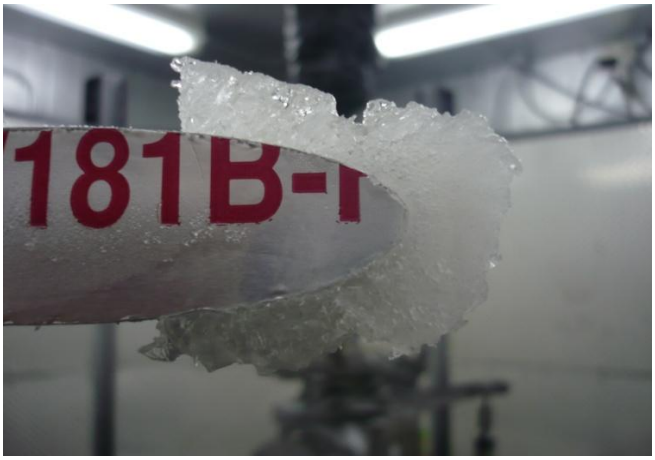


Figure 171: Rotor Tip Tape Cover